

Model for “URu2Si2”

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General Condition

- Basis type: jml
- SAMB selection:
 - Type: [Q, G]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_g , E_u]
 - Spin (s): [0, 1]
- Atomic selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_g , E_u]
 - Spin (s): [0, 1]
- Site-cluster selection:
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_g , E_u]
- Bond-cluster selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , B_{1g} , B_{1u} , B_{2g} , B_{2u} , E_g , E_u]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: false

Group and Unit Cell

- Group: SG No. 139 D_{4h}^{17} $I4/mmm$ [tetragonal]
- Associated point group: PG No. 139 D_{4h} $4/mmm$ [tetragonal]
- Unit cell:
 - $a = 4.12600$, $b = 4.12600$, $c = 9.58000$, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 90.0$
- Lattice vectors (conventional cell):
 - $\mathbf{a}_1 = [4.12600, 0.00000, 0.00000]$
 - $\mathbf{a}_2 = [0.00000, 4.12600, 0.00000]$
 - $\mathbf{a}_3 = [0.00000, 0.00000, 9.58000]$

- Plus sets:
+ $[0, 0, 0]$, + $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1 0\}$	2	$\{2_{001} 0\}$	3	$\{4^+_{001} 0\}$	4	$\{4^-_{001} 0\}$	5	$\{2_{010} 0\}$
6	$\{2_{100} 0\}$	7	$\{2_{110} 0\}$	8	$\{2_{1-10} 0\}$	9	$\{-1 0\}$	10	$\{m_{001} 0\}$
11	$\{-4^+_{001} 0\}$	12	$\{-4^-_{001} 0\}$	13	$\{m_{010} 0\}$	14	$\{m_{100} 0\}$	15	$\{m_{110} 0\}$
16	$\{m_{1-10} 0\}$								

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$Q_0(A_{1g})$	A_{1g}	0	Q, T	-	-	1
2	$Q_2(A_{1g})$	A_{1g}	2	Q, T	-	-	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
3	$Q_4(A_{1g}, 1)$	A_{1g}	4	Q, T	1	-	$\frac{\sqrt{21}(x^4 - 3x^2y^2 - 3x^2z^2 + y^4 - 3y^2z^2 + z^4)}{6}$
4	$Q_4(A_{1g}, 2)$	A_{1g}	4	Q, T	2	-	$-\frac{\sqrt{15}(x^4 - 12x^2y^2 + 6x^2z^2 + y^4 + 6y^2z^2 - 2z^4)}{12}$
5	$G_5(A_{1g})$	A_{1g}	5	G, M	-	-	$\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
6	$\mathbb{Q}_6(A_{1g}, 1)$	A_{1g}	6	Q, T	1	-	$\frac{\sqrt{2}(2x^6 - 15x^4y^2 - 15x^4z^2 - 15x^2y^4 + 180x^2y^2z^2 - 15x^2z^4 + 2y^6 - 15y^4z^2 - 15y^2z^4 + 2z^6)}{8}$
7	$\mathbb{Q}_6(A_{1g}, 2)$	A_{1g}	6	Q, T	2	-	$-\frac{\sqrt{14}(x^6 - 15x^4z^2 + 15x^2z^4 + y^6 - 15y^4z^2 + 15y^2z^4 - 2z^6)}{8}$
8	$\mathbb{G}_7(A_{1g})$	A_{1g}	7	G, M	-	-	$-\frac{\sqrt{231}xyz(x-y)(x+y)(3x^2 + 3y^2 - 10z^2)}{4}$
9	$\mathbb{G}_0(A_{1u})$	A_{1u}	0	G, M	-	-	1
10	$\mathbb{G}_2(A_{1u})$	A_{1u}	2	G, M	-	-	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
11	$\mathbb{G}_4(A_{1u}, 1)$	A_{1u}	4	G, M	1	-	$\frac{\sqrt{21}(x^4 - 3x^2y^2 - 3x^2z^2 + y^4 - 3y^2z^2 + z^4)}{6}$
12	$\mathbb{G}_4(A_{1u}, 2)$	A_{1u}	4	G, M	2	-	$-\frac{\sqrt{15}(x^4 - 12x^2y^2 + 6x^2z^2 + y^4 + 6y^2z^2 - 2z^4)}{12}$
13	$\mathbb{Q}_5(A_{1u})$	A_{1u}	5	Q, T	-	-	$\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$
14	$\mathbb{G}_6(A_{1u}, 1)$	A_{1u}	6	G, M	1	-	$\frac{\sqrt{2}(2x^6 - 15x^4y^2 - 15x^4z^2 - 15x^2y^4 + 180x^2y^2z^2 - 15x^2z^4 + 2y^6 - 15y^4z^2 - 15y^2z^4 + 2z^6)}{8}$
15	$\mathbb{G}_6(A_{1u}, 2)$	A_{1u}	6	G, M	2	-	$-\frac{\sqrt{14}(x^6 - 15x^4z^2 + 15x^2z^4 + y^6 - 15y^4z^2 + 15y^2z^4 - 2z^6)}{8}$
16	$\mathbb{Q}_7(A_{1u})$	A_{1u}	7	Q, T	-	-	$-\frac{\sqrt{231}xyz(x-y)(x+y)(3x^2 + 3y^2 - 10z^2)}{4}$
17	$\mathbb{G}_8(A_{1u}, 1)$	A_{1u}	8	G, M	1	-	$\frac{\sqrt{33}(x^8 - 14x^6y^2 - 14x^6z^2 + 35x^4y^4 + 35x^4z^4 - 14x^2y^6 - 14x^2z^6 + y^8 - 14y^6z^2 + 35y^4z^4 - 14y^2z^6 + z^8)}{8}$
18	$\mathbb{G}_1(A_{2g})$	A_{2g}	1	G, M	-	-	z
19	$\mathbb{G}_3(A_{2g})$	A_{2g}	3	G, M	-	-	$-\frac{z(3x^2 + 3y^2 - 2z^2)}{2}$
20	$\mathbb{Q}_4(A_{2g})$	A_{2g}	4	Q, T	-	-	$\frac{\sqrt{35}xyz(x-y)(x+y)}{2}$
21	$\mathbb{G}_5(A_{2g}, 1)$	A_{2g}	5	G, M	1	-	$\frac{z(15x^4 + 30x^2y^2 - 40x^2z^2 + 15y^4 - 40y^2z^2 + 8z^4)}{8}$
22	$\mathbb{G}_5(A_{2g}, 2)$	A_{2g}	5	G, M	2	-	$\frac{3\sqrt{35}z(x^2 - 2xy - y^2)(x^2 + 2xy - y^2)}{8}$
23	$\mathbb{Q}_6(A_{2g})$	A_{2g}	6	Q, T	-	-	$-\frac{3\sqrt{7}xyz(x-y)(x+y)(x^2 + y^2 - 10z^2)}{4}$
24	$\mathbb{G}_7(A_{2g}, 1)$	A_{2g}	7	G, M	1	-	$-\frac{z(35x^6 + 105x^4y^2 - 210x^4z^2 + 105x^2y^4 - 420x^2y^2z^2 + 168x^2z^4 + 35y^6 - 210y^4z^2 + 168y^2z^4 - 16z^6)}{16}$
25	$\mathbb{G}_7(A_{2g}, 2)$	A_{2g}	7	G, M	2	-	$-\frac{\sqrt{231}z(x^2 - 2xy - y^2)(x^2 + 2xy - y^2)(3x^2 + 3y^2 - 10z^2)}{16}$
26	$\mathbb{Q}_1(A_{2u})$	A_{2u}	1	Q, T	-	-	z

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
27	$\mathbb{Q}_3(A_{2u})$	A_{2u}	3	Q, T	-	-	$-\frac{z(3x^2+3y^2-2z^2)}{2}$
28	$\mathbb{G}_4(A_{2u})$	A_{2u}	4	G, M	-	-	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
29	$\mathbb{Q}_5(A_{2u}, 1)$	A_{2u}	5	Q, T	1	-	$\frac{z(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4)}{8}$
30	$\mathbb{Q}_5(A_{2u}, 2)$	A_{2u}	5	Q, T	2	-	$\frac{3\sqrt{35}z(x^2-2xy-y^2)(x^2+2xy-y^2)}{8}$
31	$\mathbb{G}_6(A_{2u})$	A_{2u}	6	G, M	-	-	$-\frac{3\sqrt{7}xy(x-y)(x+y)(x^2+y^2-10z^2)}{4}$
32	$\mathbb{Q}_7(A_{2u}, 1)$	A_{2u}	7	Q, T	1	-	$-\frac{z(35x^6+105x^4y^2-210x^4z^2+105x^2y^4-420x^2y^2z^2+168x^2z^4+35y^6-210y^4z^2+168y^2z^4-16z^6)}{16}$
33	$\mathbb{Q}_7(A_{2u}, 2)$	A_{2u}	7	Q, T	2	-	$-\frac{\sqrt{231}z(x^2-2xy-y^2)(x^2+2xy-y^2)(3x^2+3y^2-10z^2)}{16}$
34	$\mathbb{G}_8(A_{2u}, 1)$	A_{2u}	8	G, M	1	-	$\frac{3\sqrt{715}xy(x-y)(x+y)(x^2-2xy-y^2)(x^2+2xy-y^2)}{16}$
35	$\mathbb{Q}_2(B_{1g})$	B_{1g}	2	Q, T	-	-	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
36	$\mathbb{G}_3(B_{1g})$	B_{1g}	3	G, M	-	-	$\sqrt{15}xyz$
37	$\mathbb{Q}_4(B_{1g})$	B_{1g}	4	Q, T	-	-	$\frac{\sqrt{5}(x-y)(x+y)(x^2+y^2-6z^2)}{4}$
38	$\mathbb{G}_5(B_{1g})$	B_{1g}	5	G, M	-	-	$\frac{\sqrt{105}xyz(x^2+y^2-2z^2)}{2}$
39	$\mathbb{Q}_6(B_{1g}, 1)$	B_{1g}	6	Q, T	1	-	$-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$
40	$\mathbb{Q}_6(B_{1g}, 2)$	B_{1g}	6	Q, T	2	-	$\frac{\sqrt{42}(x-y)(x+y)(x^4-9x^2y^2-5x^2z^2+y^4-5y^2z^2+5z^4)}{8}$
41	$\mathbb{G}_7(B_{1g}, 1)$	B_{1g}	7	G, M	1	-	$\frac{\sqrt{91}xyz(3x^4-5x^2y^2-5x^2z^2+3y^4-5y^2z^2+3z^4)}{2}$
42	$\mathbb{G}_7(B_{1g}, 2)$	B_{1g}	7	G, M	2	-	$-\frac{\sqrt{77}xyz(3x^4-20x^2y^2+10x^2z^2+3y^4+10y^2z^2-6z^4)}{4}$
43	$\mathbb{G}_2(B_{1u})$	B_{1u}	2	G, M	-	-	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
44	$\mathbb{Q}_3(B_{1u})$	B_{1u}	3	Q, T	-	-	$\sqrt{15}xyz$
45	$\mathbb{G}_4(B_{1u})$	B_{1u}	4	G, M	-	-	$\frac{\sqrt{5}(x-y)(x+y)(x^2+y^2-6z^2)}{4}$
46	$\mathbb{Q}_5(B_{1u})$	B_{1u}	5	Q, T	-	-	$\frac{\sqrt{105}xyz(x^2+y^2-2z^2)}{2}$
47	$\mathbb{G}_6(B_{1u}, 1)$	B_{1u}	6	G, M	1	-	$-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
48	$\mathbb{G}_6(B_{1u}, 2)$	B_{1u}	6	G, M	2	-	$\frac{\sqrt{42}(x-y)(x+y)(x^4-9x^2y^2-5x^2z^2+y^4-5y^2z^2+5z^4)}{8}$
49	$\mathbb{Q}_7(B_{1u}, 1)$	B_{1u}	7	Q, T	1	-	$\frac{\sqrt{91}xyz(3x^4-5x^2y^2-5x^2z^2+3y^4-5y^2z^2+3z^4)}{2}$
50	$\mathbb{Q}_7(B_{1u}, 2)$	B_{1u}	7	Q, T	2	-	$-\frac{\sqrt{77}xyz(3x^4-20x^2y^2+10x^2z^2+3y^4+10y^2z^2-6z^4)}{4}$
51	$\mathbb{Q}_2(B_{2g})$	B_{2g}	2	Q, T	-	-	$\sqrt{3}xy$
52	$\mathbb{G}_3(B_{2g})$	B_{2g}	3	G, M	-	-	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
53	$\mathbb{Q}_4(B_{2g})$	B_{2g}	4	Q, T	-	-	$-\frac{\sqrt{5}xy(x^2+y^2-6z^2)}{2}$
54	$\mathbb{G}_5(B_{2g})$	B_{2g}	5	G, M	-	-	$-\frac{\sqrt{105}z(x-y)(x+y)(x^2+y^2-2z^2)}{4}$
55	$\mathbb{Q}_6(B_{2g}, 1)$	B_{2g}	6	Q, T	1	-	$\frac{\sqrt{462}xy(x^2-3y^2)(3x^2-y^2)}{16}$
56	$\mathbb{Q}_6(B_{2g}, 2)$	B_{2g}	6	Q, T	2	-	$\frac{\sqrt{210}xy(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4)}{16}$
57	$\mathbb{G}_7(B_{2g}, 1)$	B_{2g}	7	G, M	1	-	$\frac{\sqrt{6006}z(x-y)(x+y)(x^2-4xy+y^2)(x^2+4xy+y^2)}{32}$
58	$\mathbb{G}_7(B_{2g}, 2)$	B_{2g}	7	G, M	2	-	$\frac{\sqrt{42}z(x-y)(x+y)(15x^4+30x^2y^2-80x^2z^2+15y^4-80y^2z^2+48z^4)}{32}$
59	$\mathbb{G}_2(B_{2u})$	B_{2u}	2	G, M	-	-	$\sqrt{3}xy$
60	$\mathbb{Q}_3(B_{2u})$	B_{2u}	3	Q, T	-	-	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
61	$\mathbb{G}_4(B_{2u})$	B_{2u}	4	G, M	-	-	$-\frac{\sqrt{5}xy(x^2+y^2-6z^2)}{2}$
62	$\mathbb{Q}_5(B_{2u})$	B_{2u}	5	Q, T	-	-	$-\frac{\sqrt{105}z(x-y)(x+y)(x^2+y^2-2z^2)}{4}$
63	$\mathbb{G}_6(B_{2u}, 1)$	B_{2u}	6	G, M	1	-	$\frac{\sqrt{462}xy(x^2-3y^2)(3x^2-y^2)}{16}$
64	$\mathbb{G}_6(B_{2u}, 2)$	B_{2u}	6	G, M	2	-	$\frac{\sqrt{210}xy(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4)}{16}$
65	$\mathbb{Q}_7(B_{2u}, 1)$	B_{2u}	7	Q, T	1	-	$\frac{\sqrt{6006}z(x-y)(x+y)(x^2-4xy+y^2)(x^2+4xy+y^2)}{32}$
66	$\mathbb{Q}_7(B_{2u}, 2)$	B_{2u}	7	Q, T	2	-	$\frac{\sqrt{42}z(x-y)(x+y)(15x^4+30x^2y^2-80x^2z^2+15y^4-80y^2z^2+48z^4)}{32}$
67	$\mathbb{G}_{1,1}(E_g)$	E_g	1	G, M	-	1	x
68	$\mathbb{G}_{1,2}(E_g)$					2	$-y$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
69	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$\sqrt{3}yz$
70	$\mathbb{Q}_{2,2}(E_g)$					2	$\sqrt{3}xz$
71	$\mathbb{G}_{3,1}(E_g, 1)$	E_g	3	G, M	1	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
72	$\mathbb{G}_{3,2}(E_g, 1)$					2	$\frac{y(3x^2-2y^2+3z^2)}{2}$
73	$\mathbb{G}_{3,1}(E_g, 2)$	E_g	3	G, M	2	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
74	$\mathbb{G}_{3,2}(E_g, 2)$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$
75	$\mathbb{Q}_{4,1}(E_g, 1)$	E_g	4	Q, T	1	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
76	$\mathbb{Q}_{4,2}(E_g, 1)$					2	$\frac{\sqrt{35}xz(x^2-z^2)}{2}$
77	$\mathbb{Q}_{4,1}(E_g, 2)$	E_g	4	Q, T	2	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
78	$\mathbb{Q}_{4,2}(E_g, 2)$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
79	$\mathbb{G}_{5,1}(E_g, 1)$	E_g	5	G, M	1	1	$\frac{x(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4)}{8}$
80	$\mathbb{G}_{5,2}(E_g, 1)$					2	$\frac{y(-15x^4+40x^2y^2-30x^2z^2-8y^4+40y^2z^2-15z^4)}{8}$
81	$\mathbb{G}_{5,1}(E_g, 2)$	E_g	5	G, M	2	1	$\frac{3\sqrt{35}x(y^4-6y^2z^2+z^4)}{8}$
82	$\mathbb{G}_{5,2}(E_g, 2)$					2	$\frac{3\sqrt{35}y(-x^4+6x^2z^2-z^4)}{8}$
83	$\mathbb{G}_{5,1}(E_g, 3)$	E_g	5	G, M	3	1	$\frac{\sqrt{105}x(2x^2y^2-2x^2z^2-y^4+z^4)}{4}$
84	$\mathbb{G}_{5,2}(E_g, 3)$					2	$\frac{\sqrt{105}y(x^4-2x^2y^2+2y^2z^2-z^4)}{4}$
85	$\mathbb{Q}_{6,1}(E_g, 1)$	E_g	6	Q, T	1	1	$\frac{3\sqrt{7}yz(10x^2y^2-10x^2z^2-y^4+z^4)}{4}$
86	$\mathbb{Q}_{6,2}(E_g, 1)$					2	$\frac{3\sqrt{7}xz(-x^4+10x^2y^2-10y^2z^2+z^4)}{4}$
87	$\mathbb{Q}_{6,1}(E_g, 2)$	E_g	6	Q, T	2	1	$\frac{\sqrt{462}yz(3y^4-10y^2z^2+3z^4)}{16}$
88	$\mathbb{Q}_{6,2}(E_g, 2)$					2	$\frac{\sqrt{462}xz(3x^4-10x^2z^2+3z^4)}{16}$
89	$\mathbb{Q}_{6,1}(E_g, 3)$	E_g	6	Q, T	3	1	$\frac{\sqrt{210}yz(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4)}{16}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
90	$\mathbb{Q}_{6,2}(E_g, 3)$					2	$\frac{\sqrt{210}xz(x^4 - 16x^2y^2 + 2x^2z^2 + 16y^4 - 16y^2z^2 + z^4)}{16}$
91	$\mathbb{G}_{7,1}(E_g, 1)$	E_g	7	G, M	1	1	$\frac{x(16x^6 - 168x^4y^2 - 168x^4z^2 + 210x^2y^4 + 420x^2y^2z^2 + 210x^2z^4 - 35y^6 - 105y^4z^2 - 105y^2z^4 - 35z^6)}{16}$
92	$\mathbb{G}_{7,2}(E_g, 1)$					2	$\frac{y(35x^6 - 210x^4y^2 + 105x^4z^2 + 168x^2y^4 - 420x^2y^2z^2 + 105x^2z^4 - 16y^6 + 168y^4z^2 - 210y^2z^4 + 35z^6)}{16}$
93	$\mathbb{G}_{7,1}(E_g, 2)$	E_g	7	G, M	2	1	$\frac{\sqrt{231}x(10x^2y^4 - 60x^2y^2z^2 + 10x^2z^4 - 3y^6 + 15y^4z^2 + 15y^2z^4 - 3z^6)}{16}$
94	$\mathbb{G}_{7,2}(E_g, 2)$					2	$\frac{\sqrt{231}y(3x^6 - 10x^4y^2 - 15x^4z^2 + 60x^2y^2z^2 - 15x^2z^4 - 10y^2z^4 + 3z^6)}{16}$
95	$\mathbb{G}_{7,1}(E_g, 3)$	E_g	7	G, M	3	1	$\frac{\sqrt{6006}x(y^6 - 15y^4z^2 + 15y^2z^4 - z^6)}{32}$
96	$\mathbb{G}_{7,2}(E_g, 3)$					2	$\frac{\sqrt{6006}y(-x^6 + 15x^4z^2 - 15x^2z^4 + z^6)}{32}$
97	$\mathbb{G}_{7,1}(E_g, 4)$	E_g	7	G, M	4	1	$\frac{\sqrt{42}x(48x^4y^2 - 48x^4z^2 - 80x^2y^4 + 80x^2z^4 + 15y^6 + 15y^4z^2 - 15y^2z^4 - 15z^6)}{32}$
98	$\mathbb{G}_{7,2}(E_g, 4)$					2	$\frac{\sqrt{42}y(-15x^6 + 80x^4y^2 - 15x^4z^2 - 48x^2y^4 + 15x^2z^4 + 48y^4z^2 - 80y^2z^4 + 15z^6)}{32}$
99	$\mathbb{Q}_{1,1}(E_u)$	E_u	1	Q, T	-	1	x
100	$\mathbb{Q}_{1,2}(E_u)$					2	y
101	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$\sqrt{3}yz$
102	$\mathbb{G}_{2,2}(E_u)$					2	$-\sqrt{3}xz$
103	$\mathbb{Q}_{3,1}(E_u, 1)$	E_u	3	Q, T	1	1	$\frac{x(2x^2 - 3y^2 - 3z^2)}{2}$
104	$\mathbb{Q}_{3,2}(E_u, 1)$					2	$\frac{y(-3x^2 + 2y^2 - 3z^2)}{2}$
105	$\mathbb{Q}_{3,1}(E_u, 2)$	E_u	3	Q, T	2	1	$\frac{\sqrt{15}x(y^2 - z^2)}{2}$
106	$\mathbb{Q}_{3,2}(E_u, 2)$					2	$\frac{\sqrt{15}y(x^2 - z^2)}{2}$
107	$\mathbb{G}_{4,1}(E_u, 1)$	E_u	4	G, M	1	1	$\frac{\sqrt{35}yz(y^2 - z^2)}{2}$
108	$\mathbb{G}_{4,2}(E_u, 1)$					2	$\frac{\sqrt{35}xz(-x^2 + z^2)}{2}$
109	$\mathbb{G}_{4,1}(E_u, 2)$	E_u	4	G, M	2	1	$\frac{\sqrt{5}yz(6x^2 - y^2 - z^2)}{2}$
110	$\mathbb{G}_{4,2}(E_u, 2)$					2	$\frac{\sqrt{5}xz(x^2 - 6y^2 + z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
111	$\mathbb{Q}_{5,1}(E_u, 1)$	E_u	5	Q, T	1	1	$\frac{x(8x^4 - 40x^2y^2 - 40x^2z^2 + 15y^4 + 30y^2z^2 + 15z^4)}{8}$
112	$\mathbb{Q}_{5,2}(E_u, 1)$					2	$\frac{y(15x^4 - 40x^2y^2 + 30x^2z^2 + 8y^4 - 40y^2z^2 + 15z^4)}{8}$
113	$\mathbb{Q}_{5,1}(E_u, 2)$	E_u	5	Q, T	2	1	$\frac{3\sqrt{35}x(y^4 - 6y^2z^2 + z^4)}{8}$
114	$\mathbb{Q}_{5,2}(E_u, 2)$					2	$\frac{3\sqrt{35}y(x^4 - 6x^2z^2 + z^4)}{8}$
115	$\mathbb{Q}_{5,1}(E_u, 3)$	E_u	5	Q, T	3	1	$\frac{\sqrt{105}x(2x^2y^2 - 2x^2z^2 - y^4 + z^4)}{4}$
116	$\mathbb{Q}_{5,2}(E_u, 3)$					2	$\frac{\sqrt{105}y(-x^4 + 2x^2y^2 - 2y^2z^2 + z^4)}{4}$
117	$\mathbb{G}_{6,1}(E_u, 1)$	E_u	6	G, M	1	1	$\frac{3\sqrt{7}yz(10x^2y^2 - 10x^2z^2 - y^4 + z^4)}{4}$
118	$\mathbb{G}_{6,2}(E_u, 1)$					2	$\frac{3\sqrt{7}xz(x^4 - 10x^2y^2 + 10y^2z^2 - z^4)}{4}$
119	$\mathbb{G}_{6,1}(E_u, 2)$	E_u	6	G, M	2	1	$\frac{\sqrt{462}yz(3y^4 - 10y^2z^2 + 3z^4)}{16}$
120	$\mathbb{G}_{6,2}(E_u, 2)$					2	$\frac{\sqrt{462}xz(-3x^4 + 10x^2z^2 - 3z^4)}{16}$
121	$\mathbb{G}_{6,1}(E_u, 3)$	E_u	6	G, M	3	1	$\frac{\sqrt{210}yz(16x^4 - 16x^2y^2 - 16x^2z^2 + y^4 + 2y^2z^2 + z^4)}{16}$
122	$\mathbb{G}_{6,2}(E_u, 3)$					2	$\frac{\sqrt{210}xz(-x^4 + 16x^2y^2 - 2x^2z^2 - 16y^4 + 16y^2z^2 - z^4)}{16}$
123	$\mathbb{Q}_{7,1}(E_u, 1)$	E_u	7	Q, T	1	1	$\frac{x(16x^6 - 168x^4y^2 - 168x^4z^2 + 210x^2y^4 + 420x^2y^2z^2 + 210x^2z^4 - 35y^6 - 105y^4z^2 - 105y^2z^4 - 35z^6)}{16}$
124	$\mathbb{Q}_{7,2}(E_u, 1)$					2	$\frac{y(-35x^6 + 210x^4y^2 - 105x^4z^2 - 168x^2y^4 + 420x^2y^2z^2 - 105x^2z^4 + 16y^6 - 168y^4z^2 + 210y^2z^4 - 35z^6)}{16}$
125	$\mathbb{Q}_{7,1}(E_u, 2)$	E_u	7	Q, T	2	1	$\frac{\sqrt{231}x(10x^2y^4 - 60x^2y^2z^2 + 10x^2z^4 - 3y^6 + 15y^4z^2 + 15y^2z^4 - 3z^6)}{16}$
126	$\mathbb{Q}_{7,2}(E_u, 2)$					2	$\frac{\sqrt{231}y(-3x^6 + 10x^4y^2 + 15x^4z^2 - 60x^2y^2z^2 + 15x^2z^4 + 10y^2z^4 - 3z^6)}{16}$
127	$\mathbb{Q}_{7,1}(E_u, 3)$	E_u	7	Q, T	3	1	$\frac{\sqrt{6006}x(y^6 - 15y^4z^2 + 15y^2z^4 - z^6)}{32}$
128	$\mathbb{Q}_{7,2}(E_u, 3)$					2	$\frac{\sqrt{6006}y(x^6 - 15x^4z^2 + 15x^2z^4 - z^6)}{32}$
129	$\mathbb{Q}_{7,1}(E_u, 4)$	E_u	7	Q, T	4	1	$\frac{\sqrt{42}x(48x^4y^2 - 48x^4z^2 - 80x^2y^4 + 80x^2z^4 + 15y^6 + 15y^4z^2 - 15y^2z^4 - 15z^6)}{32}$
130	$\mathbb{Q}_{7,2}(E_u, 4)$					2	$\frac{\sqrt{42}y(15x^6 - 80x^4y^2 + 15x^4z^2 + 48x^2y^4 - 15x^2z^4 - 48y^4z^2 + 80y^2z^4 - 15z^6)}{32}$

Table 3: dimension = 24

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ \frac{3}{2}, \frac{3}{2}; d\rangle @U(1)$	1	$ \frac{3}{2}, \frac{1}{2}; d\rangle @U(1)$	2	$ \frac{3}{2}, -\frac{1}{2}; d\rangle @U(1)$	3	$ \frac{3}{2}, -\frac{3}{2}; d\rangle @U(1)$	4	$ \frac{5}{2}, \frac{5}{2}; d\rangle @U(1)$
5	$ \frac{5}{2}, \frac{3}{2}; d\rangle @U(1)$	6	$ \frac{5}{2}, \frac{1}{2}; d\rangle @U(1)$	7	$ \frac{5}{2}, -\frac{1}{2}; d\rangle @U(1)$	8	$ \frac{5}{2}, -\frac{3}{2}; d\rangle @U(1)$	9	$ \frac{5}{2}, -\frac{5}{2}; d\rangle @U(1)$
10	$ \frac{5}{2}, \frac{5}{2}; f\rangle @U(1)$	11	$ \frac{5}{2}, \frac{3}{2}; f\rangle @U(1)$	12	$ \frac{5}{2}, \frac{1}{2}; f\rangle @U(1)$	13	$ \frac{5}{2}, -\frac{1}{2}; f\rangle @U(1)$	14	$ \frac{5}{2}, -\frac{3}{2}; f\rangle @U(1)$
15	$ \frac{5}{2}, -\frac{5}{2}; f\rangle @U(1)$	16	$ \frac{7}{2}, \frac{7}{2}; f\rangle @U(1)$	17	$ \frac{7}{2}, \frac{5}{2}; f\rangle @U(1)$	18	$ \frac{7}{2}, \frac{3}{2}; f\rangle @U(1)$	19	$ \frac{7}{2}, \frac{1}{2}; f\rangle @U(1)$
20	$ \frac{7}{2}, -\frac{1}{2}; f\rangle @U(1)$	21	$ \frac{7}{2}, -\frac{3}{2}; f\rangle @U(1)$	22	$ \frac{7}{2}, -\frac{5}{2}; f\rangle @U(1)$	23	$ \frac{7}{2}, -\frac{7}{2}; f\rangle @U(1)$		

• U : 'U' site-cluster

* bra: $\langle \frac{3}{2}, \frac{3}{2}; d|, \langle \frac{3}{2}, \frac{1}{2}; d|, \langle \frac{3}{2}, -\frac{1}{2}; d|, \langle \frac{3}{2}, -\frac{3}{2}; d|, \langle \frac{5}{2}, \frac{5}{2}; d|, \langle \frac{5}{2}, \frac{3}{2}; d|, \langle \frac{5}{2}, \frac{1}{2}; d|, \langle \frac{5}{2}, -\frac{1}{2}; d|, \langle \frac{5}{2}, -\frac{3}{2}; d|, \langle \frac{5}{2}, -\frac{5}{2}; d|$

* ket: $|\frac{3}{2}, \frac{3}{2}; d\rangle, |\frac{3}{2}, \frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{5}{2}; d\rangle, |\frac{5}{2}, \frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, -\frac{5}{2}; d\rangle$

* wyckoff: 2a

$$\boxed{z1} \quad Q_0^{(c)}(A_{1g}) = Q_0^{(a)}(A_{1g})Q_0^{(s)}(A_{1g})$$

$$\boxed{z2} \quad Q_2^{(c)}(A_{1g}) = Q_2^{(a)}(A_{1g})Q_0^{(s)}(A_{1g})$$

$$\boxed{z3} \quad Q_4^{(c)}(A_{1g}, 1) = Q_4^{(a)}(A_{1g}, 1)Q_0^{(s)}(A_{1g})$$

$$\boxed{z4} \quad Q_4^{(c)}(A_{1g}, 2) = Q_4^{(a)}(A_{1g}, 2)Q_0^{(s)}(A_{1g})$$

$$\boxed{z5} \quad Q_2^{(1,-1;c)}(A_{1g}) = Q_2^{(1,-1;a)}(A_{1g})Q_0^{(s)}(A_{1g})$$

$$\boxed{z6} \quad Q_4^{(1,-1;c)}(A_{1g}, 1) = Q_4^{(1,-1;a)}(A_{1g}, 1)Q_0^{(s)}(A_{1g})$$

$$\boxed{\text{z7}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \mathbb{Q}_0^{(1,1;a)}(A_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_2^{(1,1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,1;a)}(A_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z200}} \quad \mathbb{Q}_4^{(c)}(A_{2g}) = \mathbb{Q}_4^{(a)}(A_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z201}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z202}} \quad \mathbb{G}_1^{(1,0;c)}(A_{2g}) = \mathbb{G}_1^{(1,0;a)}(A_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z203}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(A_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z375}} \quad \mathbb{Q}_2^{(c)}(B_{1g}) = \mathbb{Q}_2^{(a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z376}} \quad \mathbb{Q}_4^{(c)}(B_{1g}) = \mathbb{Q}_4^{(a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z377}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z378}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z379}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}) = \mathbb{Q}_2^{(1,1;a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z380}} \quad \mathbb{G}_3^{(1,0;c)}(B_{1g}) = \mathbb{G}_3^{(1,0;a)}(B_{1g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z566}} \quad \mathbb{Q}_2^{(c)}(B_{2g}) = \mathbb{Q}_2^{(a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z567}} \quad \mathbb{Q}_4^{(c)}(B_{2g}) = \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z568}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z569}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z570}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,1;a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z571}} \quad \mathbb{G}_3^{(1,0;c)}(B_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z745}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z746}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z747}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z748}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z749}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z750}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z751}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z752}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z753}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z754}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z755}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z756}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z757}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z758}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z759}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z760}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z761}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z762}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z763}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z764}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

• **U : 'U' site-cluster**

* bra: $\langle \frac{3}{2}, \frac{3}{2}; d |, \langle \frac{3}{2}, \frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, \frac{5}{2}; d |, \langle \frac{5}{2}, \frac{3}{2}; d |, \langle \frac{5}{2}, \frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, -\frac{5}{2}; d |$

* ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

* wyckoff: **2a**

$$\boxed{\text{z104}} \quad \mathbb{Q}_5^{(c)}(A_{1u}) = \mathbb{Q}_5^{(a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z105}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}) = \mathbb{Q}_5^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z106}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{1u}) = \mathbb{Q}_5^{(1,0;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z107}} \quad \mathbb{G}_2^{(c)}(A_{1u}) = \mathbb{G}_2^{(a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z108}} \quad \mathbb{G}_4^{(c)}(A_{1u}, 1) = \mathbb{G}_4^{(a)}(A_{1u}, 1)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z109}} \quad \mathbb{G}_4^{(c)}(A_{1u}, 2) = \mathbb{G}_4^{(a)}(A_{1u}, 2)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z110}} \quad \mathbb{G}_2^{(1,-1;c)}(A_{1u}) = \mathbb{G}_2^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z111}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 1) = \mathbb{G}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z112}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 2) = \mathbb{G}_4^{(1,-1;a)}(A_{1u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z113}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{1u}, 1) = \mathbb{G}_6^{(1,-1;a)}(A_{1u}, 1) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z114}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{1u}, 2) = \mathbb{G}_6^{(1,-1;a)}(A_{1u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z115}} \quad \mathbb{G}_2^{(1,0;c)}(A_{1u}) = \mathbb{G}_2^{(1,0;a)}(A_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z116}} \quad \mathbb{G}_4^{(1,0;c)}(A_{1u}, 1) = \mathbb{G}_4^{(1,0;a)}(A_{1u}, 1) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z117}} \quad \mathbb{G}_4^{(1,0;c)}(A_{1u}, 2) = \mathbb{G}_4^{(1,0;a)}(A_{1u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z118}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) = \mathbb{G}_0^{(1,1;a)}(A_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z119}} \quad \mathbb{G}_2^{(1,1;c)}(A_{1u}) = \mathbb{G}_2^{(1,1;a)}(A_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z120}} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 1) = \mathbb{G}_4^{(1,1;a)}(A_{1u}, 1) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z121}} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 2) = \mathbb{G}_4^{(1,1;a)}(A_{1u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z279}} \quad \mathbb{Q}_1^{(c)}(A_{2u}) = \mathbb{Q}_1^{(a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z280}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) = \mathbb{Q}_3^{(a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z281}} \quad \mathbb{Q}_5^{(c)}(A_{2u}, 1) = \mathbb{Q}_5^{(a)}(A_{2u}, 1) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z282}} \quad \mathbb{Q}_5^{(c)}(A_{2u}, 2) = \mathbb{Q}_5^{(a)}(A_{2u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z283}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = \mathbb{Q}_3^{(1,-1;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z284}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 1) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 1) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z285}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 2) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z286}} \quad \mathbb{Q}_1^{(1,0;c)}(A_{2u}) = \mathbb{Q}_1^{(1,0;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z287}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}) = \mathbb{Q}_3^{(1,0;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z288}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 1) = \mathbb{Q}_5^{(1,0;a)}(A_{2u}, 1) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z289}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 2) = \mathbb{Q}_5^{(1,0;a)}(A_{2u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z290}} \quad \mathbb{Q}_1^{(1,1;c)}(A_{2u}) = \mathbb{Q}_1^{(1,1;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z291}} \quad \mathbb{Q}_3^{(1,1;c)}(A_{2u}) = \mathbb{Q}_3^{(1,1;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z292}} \quad \mathbb{G}_4^{(c)}(A_{2u}) = \mathbb{G}_4^{(a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z293}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{2u}) = \mathbb{G}_4^{(1,-1;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z294}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{2u}) = \mathbb{G}_6^{(1,-1;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z295}} \quad \mathbb{G}_4^{(1,0;c)}(A_{2u}) = \mathbb{G}_4^{(1,0;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z296}} \quad \mathbb{G}_4^{(1,1;c)}(A_{2u}) = \mathbb{G}_4^{(1,1;a)}(A_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z471}} \quad \mathbb{Q}_3^{(c)}(B_{1u}) = \mathbb{Q}_3^{(a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z472}} \quad \mathbb{Q}_5^{(c)}(B_{1u}) = \mathbb{Q}_5^{(a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z473}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}) = \mathbb{Q}_3^{(1,-1;a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z474}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{1u}) = \mathbb{Q}_5^{(1,-1;a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z475}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}) = \mathbb{Q}_3^{(1,0;a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z476}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{1u}) = \mathbb{Q}_5^{(1,0;a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z477}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{1u}) = \mathbb{Q}_3^{(1,1;a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z478}} \quad \mathbb{G}_2^{(c)}(B_{1u}) = \mathbb{G}_2^{(a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z479}} \quad \mathbb{G}_4^{(c)}(B_{1u}) = \mathbb{G}_4^{(a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z480}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{1u}) = \mathbb{G}_2^{(1,-1;a)}(B_{1u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z481}} \quad & \mathbb{G}_4^{(1,-1;c)}(B_{1u}) = \mathbb{G}_4^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z482}} \quad & \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 1) = \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z483}} \quad & \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 2) = \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z484}} \quad & \mathbb{G}_2^{(1,0;c)}(B_{1u}) = \mathbb{G}_2^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z485}} \quad & \mathbb{G}_4^{(1,0;c)}(B_{1u}) = \mathbb{G}_4^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z486}} \quad & \mathbb{G}_2^{(1,1;c)}(B_{1u}) = \mathbb{G}_2^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z487}} \quad & \mathbb{G}_4^{(1,1;c)}(B_{1u}) = \mathbb{G}_4^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z650}} \quad & \mathbb{Q}_3^{(c)}(B_{2u}) = \mathbb{Q}_3^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z651}} \quad & \mathbb{Q}_5^{(c)}(B_{2u}) = \mathbb{Q}_5^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z652}} \quad & \mathbb{Q}_3^{(1,-1;c)}(B_{2u}) = \mathbb{Q}_3^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z653}} \quad & \mathbb{Q}_5^{(1,-1;c)}(B_{2u}) = \mathbb{Q}_5^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z654}} \quad & \mathbb{Q}_3^{(1,0;c)}(B_{2u}) = \mathbb{Q}_3^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z655}} \quad & \mathbb{Q}_5^{(1,0;c)}(B_{2u}) = \mathbb{Q}_5^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z656}} \quad & \mathbb{Q}_3^{(1,1;c)}(B_{2u}) = \mathbb{Q}_3^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z657}} \quad & \mathbb{G}_2^{(c)}(B_{2u}) = \mathbb{G}_2^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z658}} \quad & \mathbb{G}_4^{(c)}(B_{2u}) = \mathbb{G}_4^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z659}} \quad & \mathbb{G}_2^{(1,-1;c)}(B_{2u}) = \mathbb{G}_2^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z660}} \quad & \mathbb{G}_4^{(1,-1;c)}(B_{2u}) = \mathbb{G}_4^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z661}} \quad & \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 1) = \mathbb{G}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{Q}_0^{(s)}(A_{1g})
\end{aligned}$$

$$\boxed{\text{z662}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 2) = \mathbb{G}_6^{(1,-1;a)}(B_{2u}, 2) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z663}} \quad \mathbb{G}_2^{(1,0;c)}(B_{2u}) = \mathbb{G}_2^{(1,0;a)}(B_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z664}} \quad \mathbb{G}_4^{(1,0;c)}(B_{2u}) = \mathbb{G}_4^{(1,0;a)}(B_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z665}} \quad \mathbb{G}_2^{(1,1;c)}(B_{2u}) = \mathbb{G}_2^{(1,1;a)}(B_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z666}} \quad \mathbb{G}_4^{(1,1;c)}(B_{2u}) = \mathbb{G}_4^{(1,1;a)}(B_{2u}) \mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z1071}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u) = \frac{\sqrt{2} \mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1072}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u) = \frac{\sqrt{2} \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1073}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1) = \frac{\sqrt{2} \mathbb{Q}_{3,1}^{(a)}(E_u, 1) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1074}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1) = \frac{\sqrt{2} \mathbb{Q}_{3,2}^{(a)}(E_u, 1) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1075}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2) = \frac{\sqrt{2} \mathbb{Q}_{3,1}^{(a)}(E_u, 2) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1076}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2) = \frac{\sqrt{2} \mathbb{Q}_{3,2}^{(a)}(E_u, 2) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1077}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 1) = \frac{\sqrt{2} \mathbb{Q}_{5,1}^{(a)}(E_u, 1) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1078}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 1) = \frac{\sqrt{2} \mathbb{Q}_{5,2}^{(a)}(E_u, 1) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1079}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 2) = \frac{\sqrt{2} \mathbb{Q}_{5,1}^{(a)}(E_u, 2) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1080}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 2) = \frac{\sqrt{2} \mathbb{Q}_{5,2}^{(a)}(E_u, 2) \mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\begin{aligned}
\boxed{\text{z1081}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 3) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1082}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 3) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1083}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1084}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1085}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1086}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1087}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1088}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1089}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1090}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1091}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 3) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1092}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 3) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1093}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u) &= \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1094}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u) &= \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}
\end{aligned}$$

$$\boxed{\text{z1095}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1096}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1097}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1098}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1099}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1100}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1101}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1102}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1103}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1104}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1105}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1106}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1107}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1108}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\begin{aligned}
\boxed{\text{z1109}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1110}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1111}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1112}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) &= \frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1113}} \quad \mathbb{G}_{4,1}^{(c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1114}} \quad \mathbb{G}_{4,2}^{(c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1115}} \quad \mathbb{G}_{4,1}^{(c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1116}} \quad \mathbb{G}_{4,2}^{(c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1117}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1118}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1119}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1120}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1121}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2} \\
\boxed{\text{z1122}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}
\end{aligned}$$

$$\boxed{\text{z1123}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1124}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1125}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1126}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1127}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1128}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1129}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1130}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1131}} \quad \mathbb{G}_{4,1}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1132}} \quad \mathbb{G}_{4,2}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1133}} \quad \mathbb{G}_{4,1}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1134}} \quad \mathbb{G}_{4,2}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1135}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1136}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1137}} \quad \mathbb{G}_{4,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1138}} \quad \mathbb{G}_{4,2}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1139}} \quad \mathbb{G}_{4,1}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z1140}} \quad \mathbb{G}_{4,2}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

• $\mathbb{U} : \text{'U' site-cluster}$

- * bra: $\langle \frac{5}{2}, \frac{5}{2}; f |, \langle \frac{5}{2}, \frac{3}{2}; f |, \langle \frac{5}{2}, \frac{1}{2}; f |, \langle \frac{5}{2}, -\frac{1}{2}; f |, \langle \frac{5}{2}, -\frac{3}{2}; f |, \langle \frac{5}{2}, -\frac{5}{2}; f |, \langle \frac{7}{2}, \frac{7}{2}; f |, \langle \frac{7}{2}, \frac{5}{2}; f |, \langle \frac{7}{2}, \frac{3}{2}; f |, \langle \frac{7}{2}, \frac{1}{2}; f |, \langle \frac{7}{2}, -\frac{1}{2}; f |, \langle \frac{7}{2}, -\frac{3}{2}; f |, \langle \frac{7}{2}, -\frac{5}{2}; f |, \langle \frac{7}{2}, -\frac{7}{2}; f |$
- * ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$
- * wyckoff: **2a**

$$\boxed{\text{z10}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z11}} \quad \mathbb{Q}_2^{(c)}(A_{1g}) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z12}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1) = \mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z13}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 2) = \mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z14}} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 1) = \mathbb{Q}_6^{(a)}(A_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z15}} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 2) = \mathbb{Q}_6^{(a)}(A_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z16}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z17}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z18}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z19}} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 1) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z20}} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 2) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z21}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}) &= \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z22}} \quad \mathbb{Q}_2^{(1,1;c)}(A_{1g}) &= \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z23}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 1) &= \mathbb{Q}_4^{(1,1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z24}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 2) &= \mathbb{Q}_4^{(1,1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z25}} \quad \mathbb{G}_5^{(1,0;c)}(A_{1g}) &= \mathbb{G}_5^{(1,0;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z204}} \quad \mathbb{Q}_4^{(c)}(A_{2g}) &= \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z205}} \quad \mathbb{Q}_6^{(c)}(A_{2g}) &= \mathbb{Q}_6^{(a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z206}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}) &= \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z207}} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{2g}) &= \mathbb{Q}_6^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z208}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{2g}) &= \mathbb{Q}_4^{(1,1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z209}} \quad \mathbb{G}_1^{(1,0;c)}(A_{2g}) &= \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z210}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}) &= \mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z211}} \quad \mathbb{G}_5^{(1,0;c)}(A_{2g}, 1) &= \mathbb{G}_5^{(1,0;a)}(A_{2g}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z212}} \quad \mathbb{G}_5^{(1,0;c)}(A_{2g}, 2) &= \mathbb{G}_5^{(1,0;a)}(A_{2g}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z381}} \quad \mathbb{Q}_2^{(c)}(B_{1g}) &= \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z382}} \quad \mathbb{Q}_4^{(c)}(B_{1g}) &= \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z383}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 1) &= \mathbb{Q}_6^{(a)}(B_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z384}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 2) &= \mathbb{Q}_6^{(a)}(B_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z385}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}) &= \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z386}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}) &= \mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z387}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{1g}, 1) &= \mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z388}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{1g}, 2) &= \mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z389}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}) &= \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z390}} \quad \mathbb{Q}_4^{(1,1;c)}(B_{1g}) &= \mathbb{Q}_4^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z391}} \quad \mathbb{G}_3^{(1,0;c)}(B_{1g}) &= \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z392}} \quad \mathbb{G}_5^{(1,0;c)}(B_{1g}) &= \mathbb{G}_5^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z572}} \quad \mathbb{Q}_2^{(c)}(B_{2g}) &= \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z573}} \quad \mathbb{Q}_4^{(c)}(B_{2g}) &= \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z574}} \quad \mathbb{Q}_6^{(c)}(B_{2g}, 1) &= \mathbb{Q}_6^{(a)}(B_{2g}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z575}} \quad \mathbb{Q}_6^{(c)}(B_{2g}, 2) &= \mathbb{Q}_6^{(a)}(B_{2g}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z576}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{2g}) &= \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z577}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{2g}) &= \mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z578}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 1) &= \mathbb{Q}_6^{(1,-1;a)}(B_{2g}, 1)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z579}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 2) &= \mathbb{Q}_6^{(1,-1;a)}(B_{2g}, 2)\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z580}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{2g}) &= \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z581}} \quad \mathbb{Q}_4^{(1,1;c)}(B_{2g}) &= \mathbb{Q}_4^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z582}} \quad \mathbb{G}_3^{(1,0;c)}(B_{2g}) &= \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g}) \\
\boxed{\text{z583}} \quad \mathbb{G}_5^{(1,0;c)}(B_{2g}) &= \mathbb{G}_5^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})
\end{aligned}$$

$$\boxed{\text{z765}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z766}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z767}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z768}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z769}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z770}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z771}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z772}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z773}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z774}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z775}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z776}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z777}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z778}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z779}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z780}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z781}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z782}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z783}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z784}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z785}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z786}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z787}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z788}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z789}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z790}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z791}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z792}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z793}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z794}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z795}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z796}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z797}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z798}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z799}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z800}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z801}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z802}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z803}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z804}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z805}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z806}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

• U;U_001_1 : 'U'-'U' bond-cluster

* bra: $\langle \frac{3}{2}, \frac{3}{2}; d |, \langle \frac{3}{2}, \frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, \frac{5}{2}; d |, \langle \frac{5}{2}, \frac{3}{2}; d |, \langle \frac{5}{2}, \frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, -\frac{5}{2}; d |$

* ket: $|\frac{3}{2}, \frac{3}{2}; d\rangle, |\frac{3}{2}, \frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{5}{2}; d\rangle, |\frac{5}{2}, \frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, -\frac{5}{2}; d\rangle$

* wyckoff: 4b04c

$$\boxed{\text{z26}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z28}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z29}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) = -\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z30}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1) = \mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z31}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 2) = \mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z32}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z33}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_{1g}, a) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z34}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_{1g}, b) = -\mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z35}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z36}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z37}} \quad \mathbb{Q}_2^{(1,0;c)}(A_{1g}) = -\mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z38}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}, a) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z39}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}, b) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z40}} \quad \mathbb{Q}_2^{(1,1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z122}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{14} \left(2\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{14}$$

$$\begin{aligned}
\boxed{\text{z123}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{1u}) &= \frac{-\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z124}} \quad \mathbb{G}_0^{(c)}(A_{1u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z125}} \quad \mathbb{G}_2^{(c)}(A_{1u}) &= \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z126}} \quad \mathbb{G}_4^{(c)}(A_{1u}, 1) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z127}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z128}} \quad \mathbb{G}_2^{(1,-1;c)}(A_{1u}) &= \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z129}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 1a) &= \frac{\sqrt{34} \left(49\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{952} \\
\boxed{\text{z130}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 1b) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z131}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 2) &= \frac{\sqrt{34} \left(-3\sqrt{3}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{35}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{105}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{35}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{136} \\
\boxed{\text{z132}} \quad \mathbb{G}_2^{(1,0;c)}(A_{1u}) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z133}} \quad \mathbb{G}_4^{(1,0;c)}(A_{1u}, 1) &= \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z134}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z135}} \quad \mathbb{G}_2^{(1,1;c)}(A_{1u}) = \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z136}} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 1) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z213}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, a) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z214}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, b) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z215}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, c) = -\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z216}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}, a) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z217}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}, b) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z218}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}, c) = -\mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z219}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z220}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{2g}) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z221}} \quad \mathbb{G}_1^{(1,0;c)}(A_{2g}) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z222}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z297}} \quad \mathbb{Q}_1^{(c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z298}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z299}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z300}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z301}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 1) = \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16}$$

$$\boxed{\text{z302}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 2) = \frac{\sqrt{2} \left(21\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 21\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{112}$$

$$\boxed{\text{z303}} \quad \mathbb{Q}_1^{(1,0;c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z304}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}) = -\frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z305}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 2) = \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z306}} \quad \mathbb{Q}_1^{(1,1;c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z307}} \quad \mathbb{Q}_3^{(1,1;c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z308}} \quad \mathbb{G}_4^{(c)}(A_{2u}) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z309}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{2u}, a) = \frac{\sqrt{2} \left(-7\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{56}$$

$$\boxed{\text{z310}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{2u}, b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z311}} \quad \mathbb{G}_4^{(1,1;c)}(A_{2u}) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z393}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\begin{aligned}
\boxed{\text{z394}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, b) &= \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z395}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, c) &= -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z396}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, d) &= \frac{\sqrt{3} \left(\sqrt{7}\mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z397}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) &= \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z398}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) &= \frac{\sqrt{3} \left(\sqrt{5}\mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z399}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}, a) &= \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z400}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}, b) &= -\mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z401}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}, c) &= \frac{\sqrt{3} \left(\sqrt{7}\mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z402}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}, a) &= \mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z403}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}, b) &= \frac{\sqrt{3} \left(\sqrt{5}\mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z404}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}, a) &= \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z405}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}, b) &= \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z406}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}, c) &= -\mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z407}} \quad \mathbb{G}_3^{(1,0;c)}(B_{1g}) &= \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z488}} \quad \mathbb{Q}_3^{(c)}(B_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z489}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z490}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{1u}) &= -\frac{\sqrt{26} \left(2\sqrt{3}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26} \\
\boxed{\text{z491}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z492}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z493}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{1u}) &= \frac{\sqrt{2} \left(\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z494}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z495}} \quad \mathbb{G}_2^{(c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z496}} \quad \mathbb{G}_2^{(c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z497}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z498}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z499}} \quad \mathbb{G}_4^{(1,-1;c)}(B_{1u}) &= \frac{\sqrt{13} \left(13\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{208} \\
\boxed{\text{z500}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 1) &= \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} \\
\boxed{\text{z501}} \quad \mathbb{G}_2^{(1,1;c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z502}} \quad \mathbb{G}_2^{(1,1;c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z584}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, a) &= \mathbb{Q}_2^{(a)}(B_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z585}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, b) &= \mathbb{Q}_4^{(a)}(A_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z586}} \quad \mathbb{Q}_4^{(c)}(B_{2g}) &= \mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z587}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{2g}, a) &= \mathbb{Q}_2^{(1,-1;a)}(B_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z588}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{2g}, b) &= \mathbb{Q}_4^{(1,-1;a)}(A_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z589}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{2g}) &= \mathbb{Q}_4^{(1,-1;a)}(B_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z590}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{2g}, a) &= \mathbb{G}_1^{(1,0;a)}(A_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z591}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{2g}, b) &= -\mathbb{G}_3^{(1,0;a)}(A_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z592}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{2g}) &= \mathbb{Q}_2^{(1,1;a)}(B_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z593}} \quad \mathbb{G}_3^{(1,0;c)}(B_{2g}) &= \mathbb{G}_3^{(1,0;a)}(B_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z667}} \quad \mathbb{Q}_3^{(c)}(B_{2u}) &= \frac{\sqrt{2} \left(\sqrt{15} \mathbb{M}_{3,1}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,1}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15} \mathbb{M}_{3,2}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z668}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{2u}) &= \frac{\sqrt{2} \left(\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z669}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{2u}) &= \frac{\sqrt{26} \left(-\sqrt{105} \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 10 \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105} \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3} \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 10 \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{104} \\
\boxed{\text{z670}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z671}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{2u}, b) &= \frac{\sqrt{2} \left(-\sqrt{7} \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 3 \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7} \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 3 \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z672}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{2u}) &= \frac{\sqrt{2} \left(3\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z673}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{2u}) &= \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z674}} \quad \mathbb{G}_2^{(c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z675}} \quad \mathbb{G}_2^{(c)}(B_{2u}, b) &= \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z676}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z677}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{2u}, b) &= \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z678}} \quad \mathbb{G}_4^{(1,-1;c)}(B_{2u}) &= \frac{\sqrt{13} \left(\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{208} \\
\boxed{\text{z679}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 1) &= \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} \\
\boxed{\text{z680}} \quad \mathbb{G}_2^{(1,1;c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z681}} \quad \mathbb{G}_2^{(1,1;c)}(B_{2u}, b) &= \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{\text{z807}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z808}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}
\end{aligned}$$

$$\boxed{\text{z809}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z810}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z811}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\sqrt{2}\left(-\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\mathbb{Q}_{4,1}^{(a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z812}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\sqrt{2}\left(\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 3\mathbb{Q}_{4,2}^{(a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z813}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z814}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z815}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1b) = -\frac{\sqrt{2}\left(3\mathbb{Q}_{4,1}^{(a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z816}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1b) = \frac{\sqrt{2}\left(3\mathbb{Q}_{4,2}^{(a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z817}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z818}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z819}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z820}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z821}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\begin{aligned}
\boxed{\text{z822}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z823}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, c) &= \frac{\sqrt{2}\left(-\sqrt{7}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z824}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, c) &= \frac{\sqrt{2}\left(\sqrt{7}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 3\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z825}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z826}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z827}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1b) &= -\frac{\sqrt{2}\left(3\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z828}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{2}\left(3\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z829}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z830}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z831}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) &= -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z832}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z833}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) &= \frac{\sqrt{2}\left(\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{15}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z834}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) &= \frac{\sqrt{2}\left(-\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{15}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}
\end{aligned}$$

$$\boxed{\text{z835}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 1) = -\frac{\sqrt{2} \left(\sqrt{15} \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z836}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2} \left(\sqrt{15} \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z837}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z838}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z839}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, b) = -\frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,1;a)}(E_g) \mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z840}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,1;a)}(E_g) \mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z841}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2} \mathbb{G}_{1,1}^{(1,0;a)}(E_g) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z842}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2} \mathbb{G}_{1,2}^{(1,0;a)}(E_g) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z843}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2} \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z844}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2} \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z845}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2} \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z846}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2} \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1141}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u) = -\frac{\sqrt{2} \mathbb{M}_1^{(a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1142}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1143}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1) = \frac{\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1144}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1) = -\frac{\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1145}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2) = \frac{\sqrt{11} \left(\sqrt{15}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1146}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2) = \frac{\sqrt{11} \left(-\sqrt{15}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 8\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1147}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1148}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1149}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1150}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1151}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{11} \left(\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1152}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{11} \left(-\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 8\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1153}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2} \left(-\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{16}$$

$$\boxed{\text{z1154}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16}$$

$$\begin{aligned}
\boxed{\text{z1155}} \quad & \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 2) \\
= & \frac{\sqrt{46} \left(-16\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 11\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 16\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{368} \\
\boxed{\text{z1156}} \quad & \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 2) \\
= & \frac{\sqrt{46} \left(-16\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{35}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{368} \\
\boxed{\text{z1157}} \quad & \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 3) \\
= & \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 20\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{37352} \\
\boxed{\text{z1158}} \quad & \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 3) \\
= & \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 49\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{37352} \\
\boxed{\text{z1159}} \quad & \mathbb{Q}_{1,1}^{(1,0;c)}(E_u) = \frac{\sqrt{14} \left(-\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{14} \\
\boxed{\text{z1160}} \quad & \mathbb{Q}_{1,2}^{(1,0;c)}(E_u) = \frac{\sqrt{14} \left(-\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{14} \\
\boxed{\text{z1161}} \quad & \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1a) = \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 4\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z1162}} \quad & \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1a) = -\frac{\sqrt{14} \left(\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z1163}} \quad & \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1b) = \frac{\sqrt{13} \left(4\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{156} \\
\boxed{\text{z1164}} \quad & \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1b) = \frac{\sqrt{13} \left(4\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 6\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{156} \\
\boxed{\text{z1165}} \quad & \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2a) = -\frac{\sqrt{2} \left(\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{4}
\end{aligned}$$

$$\boxed{\text{z1166}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2a) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{4}$$

$$\boxed{\text{z1167}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2b) = \frac{\sqrt{377} \left(28\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 146\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 441\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 186\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 249\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{31668}$$

$$\boxed{\text{z1168}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2b) = \frac{\sqrt{377} \left(28\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 146\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 441\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 186\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 249\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{31668}$$

$$\boxed{\text{z1169}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{58} \left(1050\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 123\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 392\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 59\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 88\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{24360}$$

$$\boxed{\text{z1170}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{58} \left(1050\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 123\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 392\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 59\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 88\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{24360}$$

$$\boxed{\text{z1171}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{62} \left(150\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - 23\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{3720}$$

$$\boxed{\text{z1172}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{62} \left(150\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 16\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + 23\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 16\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{3720}$$

$$\boxed{\text{z1173}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 3) = \frac{\sqrt{62} \left(4\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 5\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{372}$$

$$\boxed{\text{z1174}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 3) = \frac{\sqrt{62} \left(4\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{372}$$

$$\boxed{\text{z1175}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1176}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1177}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1178}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1179}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{11} \left(\sqrt{15}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1180}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{11} \left(-\sqrt{15}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 8\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1181}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{22} \left(-\sqrt{5}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{22}$$

$$\boxed{\text{z1182}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\sqrt{22} \left(\sqrt{5}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{22}$$

$$\boxed{\text{z1183}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{22} \left(-\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{22}$$

$$\boxed{\text{z1184}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{22} \left(\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{22}$$

$$\boxed{\text{z1185}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 343\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 79\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{200564}$$

$$\boxed{\text{z1186}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 343\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 79\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{200564}$$

$$\boxed{\text{z1187}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{988}$$

$$\boxed{\text{z1188}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(Eu, 2) = \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 5\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{988}$$

$$\boxed{\text{z1189}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(Eu) = \frac{\sqrt{22} \left(-\sqrt{5}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{22}$$

$$\boxed{\text{z1190}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(Eu) = \frac{\sqrt{22} \left(\sqrt{5}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{22}$$

• **U;U_001_1** : 'U'-'U' bond-cluster

* bra: $\langle \frac{3}{2}, \frac{3}{2}; d |, \langle \frac{3}{2}, \frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, \frac{5}{2}; d |, \langle \frac{5}{2}, \frac{3}{2}; d |, \langle \frac{5}{2}, \frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, -\frac{5}{2}; d |$

* ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

* wyckoff: **4b@4c**

$$\boxed{\text{z41}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{T}_{1,2}^{(a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z42}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z43}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,1}^{(a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{3}\mathbb{T}_{3,2}^{(a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu)}{4}$$

$$\boxed{\text{z44}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1a) = \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{M}_{4,2}^{(a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z45}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1b) = \frac{\sqrt{22} \left(\sqrt{35}\mathbb{T}_{5,1}^{(a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 3\mathbb{T}_{5,1}^{(a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{35}\mathbb{T}_{5,2}^{(a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\mathbb{T}_{5,2}^{(a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{44}$$

$$\boxed{\text{z46}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{3}\mathbb{T}_{3,1}^{(a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{T}_{3,2}^{(a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu)}{4}$$

$$\boxed{\text{z47}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 2a) = \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{M}_{4,2}^{(a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z48}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 2b) = \frac{\sqrt{506} \left(-3\sqrt{15}\mathbb{T}_{5,1}^{(a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\sqrt{21}\mathbb{T}_{5,1}^{(a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 22\sqrt{7}\mathbb{T}_{5,1}^{(a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{15}\mathbb{T}_{5,2}^{(a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 5\sqrt{21}\mathbb{T}_{5,2}^{(a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 22\sqrt{7}\mathbb{T}_{5,2}^{(a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2024}$$

$$\boxed{\text{z49}} \quad \mathbb{Q}_6^{(c)}(A_{1g}, 1) = \frac{\sqrt{46} \left(3\sqrt{7}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 7\sqrt{5}\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,1}^{(a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5}\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,2}^{(a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{184}$$

$$\boxed{\text{z50}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_{1g}, a) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z51}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_{1g}, b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z52}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1a) = \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z53}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1b) = \frac{\sqrt{22} \left(\sqrt{35}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z54}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z55}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2a) = \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z56}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2b) = \frac{\sqrt{506} \left(-3\sqrt{15}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 5\sqrt{21}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 22\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{15}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{21}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 22\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2024}$$

$$\boxed{\text{z57}} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 1a) = -\frac{\sqrt{2} \left(\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z58}} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 1b) = \frac{\sqrt{46} \left(3\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 7\sqrt{5}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{184}$$

$$\boxed{\text{z59}} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 2) = -\frac{\sqrt{13} \left(3\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{52}$$

$$\begin{aligned}
\boxed{\text{z60}} \quad \mathbb{Q}_0^{(1,0;c)}(A_{1g}) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z61}} \quad \mathbb{Q}_2^{(1,0;c)}(A_{1g}, a) &= -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z62}} \quad \mathbb{Q}_2^{(1,0;c)}(A_{1g}, b) &= -\frac{\sqrt{3} \mathbb{T}_{3,1}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,1}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{T}_{3,2}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,2}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z63}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{1g}, 1a) &= \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z64}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{1g}, 1b) &= \frac{\sqrt{22} \left(\sqrt{35} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 3 \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{35} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 3 \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44} \\
\boxed{\text{z65}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{1g}, 1c) &= \frac{\sqrt{5} \mathbb{T}_{3,1}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3} \mathbb{T}_{3,1}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,2}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3} \mathbb{T}_{3,2}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z66}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{1g}, 2a) &= \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z67}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{1g}, 2b) &= \frac{\sqrt{506} \left(-3\sqrt{15} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 5\sqrt{21} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 22\sqrt{7} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{15} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{21} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 22\sqrt{7} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2024} \\
\boxed{\text{z68}} \quad \mathbb{Q}_6^{(1,0;c)}(A_{1g}, 1) &= \frac{\sqrt{46} \left(3\sqrt{7} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 7\sqrt{5} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15} \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{15} \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{184} \\
\boxed{\text{z69}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{1,2}^{(1,1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z70}} \quad \mathbb{Q}_2^{(1,1;c)}(A_{1g}, a) &= -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z71}} \quad \mathbb{Q}_2^{(1,1;c)}(A_{1g}, b) &= -\frac{\sqrt{3} \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4}
\end{aligned}$$

$$\boxed{\text{z72}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 1a) = \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z73}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 1b) = \frac{\sqrt{5} \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3} \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3} \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z74}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 2) = \frac{\sqrt{2} \left(\mathbb{M}_{4,1}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z75}} \quad \mathbb{G}_5^{(1,-1;c)}(A_{1g}) = \frac{\sqrt{13} \left(-\sqrt{5} \mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{11} \mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5} \mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{11} \mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{52}$$

$$\boxed{\text{z137}} \quad \mathbb{Q}_5^{(c)}(A_{1u}, a) = \mathbb{Q}_5^{(a)}(A_{1u}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z138}} \quad \mathbb{Q}_5^{(c)}(A_{1u}, b) = \mathbb{Q}_5^{(a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z139}} \quad \mathbb{Q}_5^{(c)}(A_{1u}, c) = \mathbb{Q}_3^{(a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z140}} \quad \mathbb{Q}_5^{(c)}(A_{1u}, d) = -\mathbb{G}_4^{(a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z141}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}, a) = \mathbb{Q}_5^{(1,-1;a)}(A_{1u}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z142}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}, b) = \mathbb{Q}_5^{(1,-1;a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z143}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}, c) = \mathbb{Q}_3^{(1,-1;a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z144}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}, d) = -\mathbb{G}_4^{(1,-1;a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z145}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}, e) = \frac{\sqrt{11} \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5} \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 2) \mathbb{Q}_2^{(b)}(B_{1g})}{4}$$

$$\boxed{\text{z146}} \quad \mathbb{Q}_7^{(1,-1;c)}(A_{1u}) = \frac{-\sqrt{5} \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{11} \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 2) \mathbb{Q}_2^{(b)}(B_{1g})}{4}$$

$$\boxed{\text{z147}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{1u}, a) = \mathbb{Q}_5^{(1,0;a)}(A_{1u}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z148}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{1u}, b) = \mathbb{Q}_5^{(1,0;a)}(B_{1u}) \mathbb{Q}_2^{(b)}(B_{1g})$$

$$\begin{aligned}
\boxed{\text{z149}} \quad & \mathbb{Q}_5^{(1,0;c)}(A_{1u}, c) = \mathbb{Q}_3^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z150}} \quad & \mathbb{Q}_5^{(1,0;c)}(A_{1u}, d) = -\mathbb{G}_4^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z151}} \quad & \mathbb{Q}_5^{(1,1;c)}(A_{1u}, a) = \mathbb{Q}_3^{(1,1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z152}} \quad & \mathbb{Q}_5^{(1,1;c)}(A_{1u}, b) = -\mathbb{G}_4^{(1,1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z153}} \quad & \mathbb{G}_0^{(c)}(A_{1u}) = \mathbb{G}_2^{(a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z154}} \quad & \mathbb{G}_2^{(c)}(A_{1u}) = \mathbb{G}_2^{(a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z155}} \quad & \mathbb{G}_4^{(c)}(A_{1u}, 1) = \mathbb{G}_4^{(a)}(A_{1u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z156}} \quad & \mathbb{G}_4^{(c)}(A_{1u}, 2) = \mathbb{G}_4^{(a)}(A_{1u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z157}} \quad & \mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \mathbb{G}_2^{(1,-1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z158}} \quad & \mathbb{G}_2^{(1,-1;c)}(A_{1u}) = \mathbb{G}_2^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z159}} \quad & \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 1) = \mathbb{G}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z160}} \quad & \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 2) = \mathbb{G}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z161}} \quad & \mathbb{G}_6^{(1,-1;c)}(A_{1u}, 1) = \mathbb{G}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z162}} \quad & \mathbb{G}_6^{(1,-1;c)}(A_{1u}, 2) = \mathbb{G}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z163}} \quad & \mathbb{G}_0^{(1,0;c)}(A_{1u}) = \mathbb{G}_2^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z164}} \quad & \mathbb{G}_2^{(1,0;c)}(A_{1u}) = \mathbb{G}_2^{(1,0;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z165}} \quad & \mathbb{G}_4^{(1,0;c)}(A_{1u}, 1) = \mathbb{G}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z166}} \quad & \mathbb{G}_4^{(1,0;c)}(A_{1u}, 2) = \mathbb{G}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z167}} \quad & \mathbb{G}_0^{(1,1;c)}(A_{1u}, a) = \mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})
\end{aligned}$$

$$\begin{aligned}
\boxed{z168} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}, b) &= \mathbb{G}_2^{(1,1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{z169} \quad \mathbb{G}_2^{(1,1;c)}(A_{1u}) &= \mathbb{G}_2^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{z170} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 1) &= \mathbb{G}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{z171} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 2) &= \mathbb{G}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{z223} \quad \mathbb{Q}_4^{(c)}(A_{2g}, a) &= \frac{\mathbb{M}_{4,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{z224} \quad \mathbb{Q}_4^{(c)}(A_{2g}, b) &= \frac{\sqrt{46} \left(-\sqrt{35}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_{5,1}^{(a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_{5,2}^{(a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{92} \\
\boxed{z225} \quad \mathbb{Q}_4^{(c)}(A_{2g}, c) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{z226} \quad \mathbb{Q}_6^{(c)}(A_{2g}) &= \frac{\sqrt{46} \left(-9\sqrt{7}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 13\sqrt{5}\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_{5,1}^{(a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{7}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 13\sqrt{5}\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,2}^{(a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{368} \\
\boxed{z227} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}, a) &= \frac{\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{z228} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}, b) &= \frac{\sqrt{46} \left(-\sqrt{35}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{92} \\
\boxed{z229} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{2g}, c) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{z230} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{2g}, a) &= \frac{\sqrt{13} \left(2\sqrt{2}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{33}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{15}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{2}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{33}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{15}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{104} \\
\boxed{z231} \quad \mathbb{Q}_6^{(1,-1;c)}(A_{2g}, b) &= \frac{\sqrt{46} \left(-9\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 13\sqrt{5}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 13\sqrt{5}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{368}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z232}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{2g}, a) &= \frac{\mathbb{M}_{4,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z233}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{2g}, b) &= \frac{\sqrt{46} \left(-\sqrt{35}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{92} \\
\boxed{\text{z234}} \quad \mathbb{Q}_4^{(1,0;c)}(A_{2g}, c) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z235}} \quad \mathbb{Q}_6^{(1,0;c)}(A_{2g}) &= \frac{\sqrt{46} \left(-9\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 13\sqrt{5}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 13\sqrt{5}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{368} \\
\boxed{\text{z236}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{2g}, a) &= \frac{\mathbb{M}_{4,1}^{(1,1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z237}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{2g}, b) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z238}} \quad \mathbb{G}_1^{(c)}(A_{2g}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z239}} \quad \mathbb{G}_1^{(c)}(A_{2g}, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z240}} \quad \mathbb{G}_3^{(c)}(A_{2g}, a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z241}} \quad \mathbb{G}_3^{(c)}(A_{2g}, b) &= \frac{-\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z242}} \quad \mathbb{G}_5^{(c)}(A_{2g}, 1) &= \frac{\sqrt{2} \left(\sqrt{15}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{T}_{5,1}^{(a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{7}\mathbb{T}_{5,2}^{(a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16} \\
\boxed{\text{z243}} \quad \mathbb{G}_1^{(1,-1;c)}(A_{2g}) &= \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z244}} \quad \mathbb{G}_3^{(1,-1;c)}(A_{2g}, a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z245}} \quad \mathbb{G}_3^{(1,-1;c)}(A_{2g}, b) &= \frac{-\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z246}} \quad \mathbb{G}_5^{(1,-1;c)}(A_{2g}, 1a) &= \frac{2\sqrt{6}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{6}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} \\
\boxed{\text{z247}} \quad \mathbb{G}_5^{(1,-1;c)}(A_{2g}, 1b) &= \frac{\sqrt{2}\left(\sqrt{15}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{16} \\
\boxed{\text{z248}} \quad \mathbb{G}_5^{(1,-1;c)}(A_{2g}, 2) &= \frac{\sqrt{13}\left(-2\sqrt{330}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 7\sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{330}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 7\sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{208} \\
\boxed{\text{z249}} \quad \mathbb{G}_1^{(1,0;c)}(A_{2g}, a) &= \frac{\sqrt{2}\left(\mathbb{M}_{2,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{2} \\
\boxed{\text{z250}} \quad \mathbb{G}_1^{(1,0;c)}(A_{2g}, b) &= \frac{\sqrt{2}\left(\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{2} \\
\boxed{\text{z251}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}, a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z252}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}, b) &= \frac{-\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z253}} \quad \mathbb{G}_5^{(1,0;c)}(A_{2g}, 1) &= \frac{\sqrt{2}\left(\sqrt{15}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{16} \\
\boxed{\text{z254}} \quad \mathbb{G}_1^{(1,1;c)}(A_{2g}, a) &= \frac{\sqrt{2}\left(\mathbb{M}_{2,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z255}} \quad \mathbb{G}_1^{(1,1;c)}(A_{2g}, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(1,1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z256}} \quad \mathbb{G}_3^{(1,1;c)}(A_{2g}, a) &= \frac{-\sqrt{7} \mathbb{M}_{4,1}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,1}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7} \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z257}} \quad \mathbb{G}_3^{(1,1;c)}(A_{2g}, b) &= \frac{-\sqrt{3} \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5} \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3} \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z312}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, a) &= \mathbb{Q}_1^{(a)}(A_{2u}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z313}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, b) &= \mathbb{Q}_3^{(a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z314}} \quad \mathbb{Q}_1^{(c)}(A_{2u}, c) &= -\mathbb{G}_2^{(a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z315}} \quad \mathbb{Q}_3^{(c)}(A_{2u}, a) &= \mathbb{Q}_5^{(a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z316}} \quad \mathbb{Q}_3^{(c)}(A_{2u}, b) &= \mathbb{Q}_3^{(a)}(A_{2u}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z317}} \quad \mathbb{Q}_3^{(c)}(A_{2u}, c) &= -\mathbb{G}_4^{(a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z318}} \quad \mathbb{Q}_5^{(c)}(A_{2u}, 1) &= \mathbb{Q}_5^{(a)}(A_{2u}, 1) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z319}} \quad \mathbb{Q}_5^{(c)}(A_{2u}, 2) &= \mathbb{Q}_5^{(a)}(A_{2u}, 2) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z320}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_{2u}, a) &= \mathbb{Q}_3^{(1,-1;a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z321}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_{2u}, b) &= -\mathbb{G}_2^{(1,-1;a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z322}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}, a) &= \mathbb{Q}_5^{(1,-1;a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z323}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}, b) &= \mathbb{Q}_3^{(1,-1;a)}(A_{2u}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z324}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}, c) &= -\mathbb{G}_4^{(1,-1;a)}(B_{2u}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z325}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 1a) &= \mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 1) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z326}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 1b) &= -\mathbb{G}_6^{(1,-1;a)}(B_{2u}, 2) \mathbb{Q}_2^{(b)}(B_{1g})
\end{aligned}$$

$$\boxed{\text{z327}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 2a) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z328}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 2b) = -\mathbb{G}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z329}} \quad \mathbb{Q}_1^{(1,0;c)}(A_{2u}, a) = \mathbb{Q}_1^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z330}} \quad \mathbb{Q}_1^{(1,0;c)}(A_{2u}, b) = \mathbb{Q}_3^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z331}} \quad \mathbb{Q}_1^{(1,0;c)}(A_{2u}, c) = -\mathbb{G}_2^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z332}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}, a) = \mathbb{Q}_5^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z333}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}, b) = \mathbb{Q}_3^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z334}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}, c) = -\mathbb{G}_4^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z335}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 1) = \mathbb{Q}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z336}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 2) = \mathbb{Q}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z337}} \quad \mathbb{Q}_1^{(1,1;c)}(A_{2u}, a) = \mathbb{Q}_1^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z338}} \quad \mathbb{Q}_1^{(1,1;c)}(A_{2u}, b) = \mathbb{Q}_3^{(1,1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z339}} \quad \mathbb{Q}_1^{(1,1;c)}(A_{2u}, c) = -\mathbb{G}_2^{(1,1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z340}} \quad \mathbb{Q}_3^{(1,1;c)}(A_{2u}, a) = \mathbb{Q}_3^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z341}} \quad \mathbb{Q}_3^{(1,1;c)}(A_{2u}, b) = -\mathbb{G}_4^{(1,1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z342}} \quad \mathbb{G}_4^{(c)}(A_{2u}) = \mathbb{G}_4^{(a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z343}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{2u}) = \mathbb{G}_4^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z344}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{2u}) = \mathbb{G}_6^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z345}} \quad \mathbb{G}_4^{(1,0;c)}(A_{2u}) = \mathbb{G}_4^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z346}} \quad \mathbb{G}_4^{(1,1;c)}(A_{2u}) &= \mathbb{G}_4^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z408}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, a) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z409}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z410}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, c) &= \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\
\boxed{\text{z411}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z412}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) &= \frac{\sqrt{34} \left(5\sqrt{5}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{21}\mathbb{T}_{5,1}^{(a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{5}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{7}\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{T}_{5,2}^{(a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{136} \\
\boxed{\text{z413}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, c) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\
\boxed{\text{z414}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 1) &= \frac{\sqrt{1598} \left(5\sqrt{105}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 21\sqrt{3}\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 94\mathbb{T}_{5,1}^{(a)}(E_u, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{105}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 21\sqrt{3}\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 94\mathbb{T}_{5,2}^{(a)}(E_u, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{6392} \\
\boxed{\text{z415}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 2) &= \frac{\sqrt{94} \left(3\sqrt{7}\mathbb{T}_{5,1}^{(a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 5\sqrt{5}\mathbb{T}_{5,1}^{(a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{T}_{5,2}^{(a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 5\sqrt{5}\mathbb{T}_{5,2}^{(a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{188} \\
\boxed{\text{z416}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}, a) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z417}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{1g}, b) &= \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\
\boxed{\text{z418}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}, a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z419}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}, b) &= \frac{\sqrt{34} \left(5\sqrt{5}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 2\sqrt{21}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) - 5\sqrt{5}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{21}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{136} \\
\boxed{\text{z420}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{1g}, c) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 3\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu)}{8} \\
\boxed{\text{z421}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{1g}, 1a) &= \frac{\sqrt{7} \left(3\sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{28} \\
\boxed{\text{z422}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{1g}, 1b) &= \frac{\sqrt{1598} \left(5\sqrt{105}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 21\sqrt{3}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 94\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) - 5\sqrt{105}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 21\sqrt{3}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 94\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{6392} \\
\boxed{\text{z423}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{1g}, 2a) &= \frac{\sqrt{133} \left(7\sqrt{2}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{33}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{15}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) - 7\sqrt{2}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{33}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{15}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{266} \\
\boxed{\text{z424}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{1g}, 2b) &= \frac{\sqrt{94} \left(3\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\sqrt{5}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 5\sqrt{5}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{188} \\
\boxed{\text{z425}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{1g}, a) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,0;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(1,0;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2} \\
\boxed{\text{z426}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{1g}, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,0;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) - \mathbb{T}_{1,2}^{(1,0;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2} \\
\boxed{\text{z427}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{1g}, c) &= \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu)}{8} \\
\boxed{\text{z428}} \quad \mathbb{Q}_4^{(1,0;c)}(B_{1g}, a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{M}_{4,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - \mathbb{M}_{4,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu)}{4} \\
\boxed{\text{z429}} \quad \mathbb{Q}_4^{(1,0;c)}(B_{1g}, b) &= \frac{\sqrt{34} \left(5\sqrt{5}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 2\sqrt{21}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) - 5\sqrt{5}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{21}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{136}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z430}} \quad \mathbb{Q}_4^{(1,0;c)}(B_{1g},c) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\
\boxed{\text{z431}} \quad \mathbb{Q}_6^{(1,0;c)}(B_{1g},1) &= \frac{\sqrt{1598}\left(5\sqrt{105}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) - 21\sqrt{3}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - 94\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{105}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) + 21\sqrt{3}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u) + 94\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{6392} \\
\boxed{\text{z432}} \quad \mathbb{Q}_6^{(1,0;c)}(B_{1g},2) &= \frac{\sqrt{94}\left(3\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + 5\sqrt{5}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - 5\sqrt{5}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{188} \\
\boxed{\text{z433}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g},a) &= \frac{\sqrt{2}\left(-\mathbb{M}_{2,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{2} \\
\boxed{\text{z434}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g},b) &= \frac{\sqrt{2}\left(\mathbb{T}_{1,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{2} \\
\boxed{\text{z435}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g},c) &= \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\
\boxed{\text{z436}} \quad \mathbb{Q}_4^{(1,1;c)}(B_{1g},a) &= \frac{-\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z437}} \quad \mathbb{Q}_4^{(1,1;c)}(B_{1g},b) &= \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\
\boxed{\text{z438}} \quad \mathbb{G}_3^{(c)}(B_{1g}) &= \frac{\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z439}} \quad \mathbb{G}_3^{(1,-1;c)}(B_{1g}) &= \frac{\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z440}} \quad \mathbb{G}_5^{(1,-1;c)}(B_{1g}) &= \frac{\sqrt{19}\left(4\sqrt{6}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{6}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{76} \\
\boxed{\text{z441}} \quad \mathbb{G}_3^{(1,0;c)}(B_{1g}) &= \frac{\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}
\end{aligned}$$

$$\boxed{\text{z442}} \quad \mathbb{G}_3^{(1,1;c)}(B_{1g}) = \frac{\mathbb{M}_{4,1}^{(1,1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z503}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, a) = \mathbb{Q}_5^{(a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z504}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, b) = \mathbb{Q}_3^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z505}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, c) = \mathbb{G}_2^{(a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z506}} \quad \mathbb{Q}_3^{(c)}(B_{1u}, d) = -\mathbb{G}_4^{(a)}(A_{1u}, 2)\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z507}} \quad \mathbb{Q}_5^{(c)}(B_{1u}, a) = \mathbb{Q}_5^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z508}} \quad \mathbb{Q}_5^{(c)}(B_{1u}, b) = -\mathbb{G}_4^{(a)}(A_{1u}, 1)\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z509}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}, a) = \mathbb{Q}_5^{(1,-1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z510}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}, b) = \mathbb{Q}_3^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z511}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}, c) = \mathbb{G}_2^{(1,-1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z512}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}, d) = -\mathbb{G}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z513}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{1u}, a) = \mathbb{Q}_5^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z514}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{1u}, b) = -\mathbb{G}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z515}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{1u}, c) = \frac{\sqrt{2} \left(\sqrt{7}\mathbb{G}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{G}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{4}$$

$$\boxed{\text{z516}} \quad \mathbb{Q}_7^{(1,-1;c)}(B_{1u}, 1) = \frac{\sqrt{2} \left(-\mathbb{G}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{G}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{4}$$

$$\boxed{\text{z517}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, a) = \mathbb{Q}_5^{(1,0;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z518}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, b) = \mathbb{Q}_3^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z519}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, c) &= \mathbb{G}_2^{(1,0;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z520}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, d) &= -\mathbb{G}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z521}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{1u}, a) &= \mathbb{Q}_5^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z522}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{1u}, b) &= -\mathbb{G}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z523}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{1u}, a) &= \mathbb{Q}_3^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z524}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{1u}, b) &= \mathbb{G}_2^{(1,1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z525}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{1u}, c) &= -\mathbb{G}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z526}} \quad \mathbb{Q}_5^{(1,1;c)}(B_{1u}) &= -\mathbb{G}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z527}} \quad \mathbb{G}_2^{(c)}(B_{1u}) &= \mathbb{G}_2^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z528}} \quad \mathbb{G}_4^{(c)}(B_{1u}) &= \mathbb{G}_4^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z529}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{1u}) &= \mathbb{G}_2^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z530}} \quad \mathbb{G}_4^{(1,-1;c)}(B_{1u}) &= \mathbb{G}_4^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z531}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 1) &= \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z532}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 2) &= \mathbb{G}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z533}} \quad \mathbb{G}_2^{(1,0;c)}(B_{1u}) &= \mathbb{G}_2^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z534}} \quad \mathbb{G}_4^{(1,0;c)}(B_{1u}) &= \mathbb{G}_4^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z535}} \quad \mathbb{G}_2^{(1,1;c)}(B_{1u}, a) &= \mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z536}} \quad \mathbb{G}_2^{(1,1;c)}(B_{1u}, b) &= \mathbb{G}_2^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z537}} \quad \mathbb{G}_4^{(1,1;c)}(B_{1u}) &= \mathbb{G}_4^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})
\end{aligned}$$

$$\boxed{\text{z594}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, a) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z595}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, b) = \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{T}_{1,2}^{(a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z596}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, c) = \frac{-\sqrt{3} \mathbb{T}_{3,1}^{(a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{3,1}^{(a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{3} \mathbb{T}_{3,2}^{(a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{3,2}^{(a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu)}{4}$$

$$\boxed{\text{z597}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, a) = \frac{\sqrt{7} \mathbb{M}_{4,1}^{(a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + 5 \mathbb{M}_{4,1}^{(a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7} \mathbb{M}_{4,2}^{(a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + 5 \mathbb{M}_{4,2}^{(a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu)}{8}$$

$$\boxed{\text{z598}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, b) = \frac{\sqrt{34} \left(\sqrt{5} \mathbb{T}_{5,1}^{(a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{7} \mathbb{T}_{5,1}^{(a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{5,2}^{(a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{7} \mathbb{T}_{5,2}^{(a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu) \right)}{68}$$

$$\boxed{\text{z599}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, c) = -\frac{\sqrt{5} \mathbb{T}_{3,1}^{(a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{3} \mathbb{T}_{3,1}^{(a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{3,2}^{(a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{3} \mathbb{T}_{3,2}^{(a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu)}{4}$$

$$\boxed{\text{z600}} \quad \mathbb{Q}_6^{(c)}(B_{2g}, 1) = \frac{3\sqrt{7} \mathbb{T}_{5,1}^{(a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{5,1}^{(a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{15} \mathbb{T}_{5,1}^{(a)}(Eu, 3) \mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{7} \mathbb{T}_{5,2}^{(a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{5,2}^{(a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu) - 2\sqrt{15} \mathbb{T}_{5,2}^{(a)}(Eu, 3) \mathbb{T}_{1,1}^{(b)}(Eu)}{16}$$

$$\boxed{\text{z601}} \quad \mathbb{Q}_6^{(c)}(B_{2g}, 2) = \frac{\sqrt{17} \left(3\sqrt{105} \mathbb{T}_{5,1}^{(a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + 5\sqrt{3} \mathbb{T}_{5,1}^{(a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + 34 \mathbb{T}_{5,1}^{(a)}(Eu, 3) \mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{105} \mathbb{T}_{5,2}^{(a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + 5\sqrt{3} \mathbb{T}_{5,2}^{(a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu) + 34 \mathbb{T}_{5,2}^{(a)}(Eu, 3) \mathbb{T}_{1,1}^{(b)}(Eu) \right)}{272}$$

$$\boxed{\text{z602}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{2g}, a) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z603}} \quad \mathbb{Q}_2^{(1,-1;c)}(B_{2g}, b) = \frac{-\sqrt{3} \mathbb{T}_{3,1}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{3,1}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{3} \mathbb{T}_{3,2}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{3,2}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu)}{4}$$

$$\boxed{\text{z604}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{2g}, a) = \frac{\sqrt{7} \mathbb{M}_{4,1}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + 5 \mathbb{M}_{4,1}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7} \mathbb{M}_{4,2}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + 5 \mathbb{M}_{4,2}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu)}{8}$$

$$\boxed{\text{z605}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{2g}, b) = \frac{\sqrt{34} \left(\sqrt{5} \mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{7} \mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{7} \mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu) \right)}{68}$$

$$\boxed{\text{z606}} \quad \mathbb{Q}_4^{(1,-1;c)}(B_{2g}, c) = -\frac{\sqrt{5} \mathbb{T}_{3,1}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{3} \mathbb{T}_{3,1}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5} \mathbb{T}_{3,2}^{(1,-1;a)}(Eu, 1) \mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{3} \mathbb{T}_{3,2}^{(1,-1;a)}(Eu, 2) \mathbb{T}_{1,1}^{(b)}(Eu)}{4}$$

$$\begin{aligned} & \boxed{\text{z607}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 1a) \\ &= \frac{4\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{6}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{330}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) + 4\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{6}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - \sqrt{330}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu)}{32} \end{aligned}$$

$$\begin{aligned} & \boxed{\text{z608}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 1b) \\ &= \frac{3\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{15}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 2\sqrt{15}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu)}{16} \end{aligned}$$

$$\begin{aligned} & \boxed{\text{z609}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 2a) = \\ &= \frac{\sqrt{19} \left(20\sqrt{15}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{110}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 37\sqrt{2}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) + 20\sqrt{15}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 3\sqrt{110}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 37\sqrt{2}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{608} \end{aligned}$$

$$\begin{aligned} & \boxed{\text{z610}} \quad \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 2b) \\ &= \frac{\sqrt{17} \left(3\sqrt{105}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 5\sqrt{3}\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 34\mathbb{T}_{5,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{105}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\sqrt{3}\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 34\mathbb{T}_{5,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{272} \end{aligned}$$

$$\boxed{\text{z611}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{2g}, a) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,0;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(1,0;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z612}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{2g}, b) = \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,0;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{T}_{1,2}^{(1,0;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z613}} \quad \mathbb{Q}_2^{(1,0;c)}(B_{2g}, c) = \frac{-\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{4}$$

$$\boxed{\text{z614}} \quad \mathbb{Q}_4^{(1,0;c)}(B_{2g}, a) = \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 5\mathbb{M}_{4,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\mathbb{M}_{4,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{8}$$

$$\boxed{\text{z615}} \quad \mathbb{Q}_4^{(1,0;c)}(B_{2g}, b) = \frac{\sqrt{34} \left(\sqrt{5}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{68}$$

$$\boxed{\text{z616}} \quad \mathbb{Q}_4^{(1,0;c)}(B_{2g}, c) = -\frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{4}$$

$$\begin{aligned} & \boxed{\text{z617}} \quad \mathbb{Q}_6^{(1,0;c)}(B_{2g}, 1) \\ &= \frac{3\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{15}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 2\sqrt{15}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu)}{16} \end{aligned}$$

$$\begin{aligned}
\boxed{\text{z618}} \quad & \mathbb{Q}_6^{(1,0;c)}(B_{2g}, 2) \\
= & \frac{\sqrt{17} \left(3\sqrt{105}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 5\sqrt{3}\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 34\mathbb{T}_{5,1}^{(1,0;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{105}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\sqrt{3}\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 34\mathbb{T}_{5,2}^{(1,0;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{272} \\
\boxed{\text{z619}} \quad & \mathbb{Q}_2^{(1,1;c)}(B_{2g}, a) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,1;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(1,1;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2} \\
\boxed{\text{z620}} \quad & \mathbb{Q}_2^{(1,1;c)}(B_{2g}, b) = \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,1;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{T}_{1,2}^{(1,1;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{2} \\
\boxed{\text{z621}} \quad & \mathbb{Q}_2^{(1,1;c)}(B_{2g}, c) = \frac{-\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{4} \\
\boxed{\text{z622}} \quad & \mathbb{Q}_4^{(1,1;c)}(B_{2g}, a) = \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 5\mathbb{M}_{4,1}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\mathbb{M}_{4,2}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{8} \\
\boxed{\text{z623}} \quad & \mathbb{Q}_4^{(1,1;c)}(B_{2g}, b) = -\frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{4} \\
\boxed{\text{z624}} \quad & \mathbb{G}_3^{(c)}(B_{2g}) = \frac{-5\mathbb{M}_{4,1}^{(a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,1}^{(a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 5\mathbb{M}_{4,2}^{(a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{8} \\
\boxed{\text{z625}} \quad & \mathbb{G}_3^{(1,-1;c)}(B_{2g}) = \frac{-5\mathbb{M}_{4,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 5\mathbb{M}_{4,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{8} \\
\boxed{\text{z626}} \quad & \mathbb{G}_5^{(1,-1;c)}(B_{2g}) \\
= & \frac{\sqrt{19} \left(-2\sqrt{6}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 7\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,2}^{(b)}(Eu) - 2\sqrt{6}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 7\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 3\sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(Eu, 3)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{152} \\
\boxed{\text{z627}} \quad & \mathbb{G}_3^{(1,0;c)}(B_{2g}) = \frac{-5\mathbb{M}_{4,1}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 5\mathbb{M}_{4,2}^{(1,0;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{8} \\
\boxed{\text{z628}} \quad & \mathbb{G}_3^{(1,1;c)}(B_{2g}) = \frac{-5\mathbb{M}_{4,1}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 5\mathbb{M}_{4,2}^{(1,1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(Eu, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{8} \\
\boxed{\text{z682}} \quad & \mathbb{Q}_3^{(c)}(B_{2u}, a) = \frac{\sqrt{17} \left(\sqrt{5}\mathbb{Q}_5^{(a)}(A_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{7}\mathbb{Q}_5^{(a)}(A_{2u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{34}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z683}} \quad & \mathbb{Q}_3^{(c)}(B_{2u}, b) = \mathbb{Q}_1^{(a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z684}} \quad & \mathbb{Q}_3^{(c)}(B_{2u}, c) = \mathbb{Q}_3^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z685}} \quad & \mathbb{Q}_3^{(c)}(B_{2u}, d) = -\mathbb{Q}_3^{(a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z686}} \quad & \mathbb{Q}_3^{(c)}(B_{2u}, e) = -\mathbb{G}_4^{(a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z687}} \quad & \mathbb{Q}_5^{(c)}(B_{2u}, a) = \mathbb{Q}_5^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z688}} \quad & \mathbb{Q}_5^{(c)}(B_{2u}, b) = \frac{\sqrt{17} \left(-3\sqrt{7}\mathbb{Q}_5^{(a)}(A_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_5^{(a)}(A_{2u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{34} \\
\boxed{\text{z689}} \quad & \mathbb{Q}_3^{(1,-1;c)}(B_{2u}, a) = \frac{\sqrt{17} \left(\sqrt{5}\mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{7}\mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{34} \\
\boxed{\text{z690}} \quad & \mathbb{Q}_3^{(1,-1;c)}(B_{2u}, b) = \mathbb{Q}_3^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z691}} \quad & \mathbb{Q}_3^{(1,-1;c)}(B_{2u}, c) = -\mathbb{Q}_3^{(1,-1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z692}} \quad & \mathbb{Q}_3^{(1,-1;c)}(B_{2u}, d) = -\mathbb{G}_4^{(1,-1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z693}} \quad & \mathbb{Q}_5^{(1,-1;c)}(B_{2u}, a) = \mathbb{Q}_5^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z694}} \quad & \mathbb{Q}_5^{(1,-1;c)}(B_{2u}, b) = \frac{\sqrt{17} \left(-3\sqrt{7}\mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{34} \\
\boxed{\text{z695}} \quad & \mathbb{Q}_5^{(1,-1;c)}(B_{2u}, c) = -\mathbb{G}_6^{(1,-1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z696}} \quad & \mathbb{Q}_3^{(1,0;c)}(B_{2u}, a) = \frac{\sqrt{17} \left(\sqrt{5}\mathbb{Q}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{7}\mathbb{Q}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{34} \\
\boxed{\text{z697}} \quad & \mathbb{Q}_3^{(1,0;c)}(B_{2u}, b) = \mathbb{Q}_1^{(1,0;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z698}} \quad & \mathbb{Q}_3^{(1,0;c)}(B_{2u}, c) = \mathbb{Q}_3^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z699}} \quad & \mathbb{Q}_3^{(1,0;c)}(B_{2u}, d) = -\mathbb{Q}_3^{(1,0;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z700}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{2u}, e) &= -\mathbb{G}_4^{(1,0;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z701}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{2u}, a) &= \mathbb{Q}_5^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z702}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{2u}, b) &= \frac{\sqrt{17} \left(-3\sqrt{7}\mathbb{Q}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{34} \\
\boxed{\text{z703}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{2u}, a) &= \mathbb{Q}_1^{(1,1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z704}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{2u}, b) &= \mathbb{Q}_3^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z705}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{2u}, c) &= -\mathbb{Q}_3^{(1,1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z706}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{2u}, d) &= -\mathbb{G}_4^{(1,1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z707}} \quad \mathbb{G}_2^{(c)}(B_{2u}) &= \mathbb{G}_2^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z708}} \quad \mathbb{G}_4^{(c)}(B_{2u}) &= \mathbb{G}_4^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z709}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{2u}) &= \mathbb{G}_2^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z710}} \quad \mathbb{G}_4^{(1,-1;c)}(B_{2u}) &= \mathbb{G}_4^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z711}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 1) &= \mathbb{G}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z712}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 2) &= \mathbb{G}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z713}} \quad \mathbb{G}_2^{(1,0;c)}(B_{2u}) &= \mathbb{G}_2^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z714}} \quad \mathbb{G}_4^{(1,0;c)}(B_{2u}) &= \mathbb{G}_4^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z715}} \quad \mathbb{G}_2^{(1,1;c)}(B_{2u}) &= \mathbb{G}_2^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z716}} \quad \mathbb{G}_4^{(1,1;c)}(B_{2u}) &= \mathbb{G}_4^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z847}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) &= \frac{\sqrt{10} \left(\sqrt{3}\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10}
\end{aligned}$$

$$\begin{aligned}
\text{z848} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10} \\
\text{z849} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) &= \frac{\sqrt{2}\mathbb{T}_1^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\text{z850} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) &= \frac{\sqrt{2}\mathbb{T}_1^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\text{z851} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) &= \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26} \\
\text{z852} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) &= \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{26} \\
\text{z853} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1a) &= \frac{-4\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{36} \\
\text{z854} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1a) &= \frac{4\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{36} \\
\text{z855} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1b) &= \frac{-3\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16} \\
\text{z856} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1b) &= \frac{3\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16} \\
\text{z857} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1c) &= -\frac{\sqrt{286} \left(7\sqrt{5}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 10\sqrt{3}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{572} \\
\text{z858} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1c) &= \frac{\sqrt{286} \left(-7\sqrt{5}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 10\sqrt{3}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{572} \\
\text{z859} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2a) &= \frac{\sqrt{209} \left(4\sqrt{105}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 94\sqrt{3}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) - 150\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 249\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{7524} \\
\text{z860} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2a) &= \frac{\sqrt{209} \left(-4\sqrt{105}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 94\sqrt{3}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 150\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 249\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{7524}
\end{aligned}$$

$$\begin{aligned}
\text{z861} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2b) &= \frac{\sqrt{403} \left(31\sqrt{21}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{15}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 31\sqrt{21}\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 35\sqrt{7}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{7}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{6448} \\
\text{z862} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2b) &= \frac{\sqrt{403} \left(-31\sqrt{21}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 13\sqrt{15}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 31\sqrt{21}\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 35\sqrt{7}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{7}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{6448} \\
\text{z863} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2c) &= \frac{\sqrt{22} \left(-\sqrt{15}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 5\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44} \\
\text{z864} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2c) &= \frac{\sqrt{22} \left(-\sqrt{15}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 2\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 5\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44} \\
\text{z865} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 1) &= \frac{\sqrt{87854} \left(-248\sqrt{5}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 39\sqrt{7}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 155\sqrt{5}\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 80\sqrt{15}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 106\sqrt{15}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{351416} \\
\text{z866} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 1) &= \frac{\sqrt{87854} \left(248\sqrt{5}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 39\sqrt{7}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 155\sqrt{5}\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 80\sqrt{15}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 106\sqrt{15}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{351416} \\
\text{z867} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 2) &= \frac{\sqrt{109} \left(-20\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 27\sqrt{35}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 67\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 92\sqrt{3}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{3488} \\
\text{z868} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 2) &= \frac{\sqrt{109} \left(20\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 27\sqrt{35}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 67\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 92\sqrt{3}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{3488} \\
\text{z869} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 3) &= \frac{4\sqrt{3}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 7\sqrt{3}\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 14\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{32} \\
\text{z870} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 3) &= \frac{-4\sqrt{3}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105}\mathbb{T}_5^{(a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{3}\mathbb{T}_5^{(a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 4\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 14\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{32} \\
\text{z871} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{10} \left(\sqrt{3}\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10} \\
\text{z872} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10} \\
\text{z873} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26}
\end{aligned}$$

$$\boxed{\text{z874}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{26}$$

$$\boxed{\text{z875}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1a) = \frac{-4\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{36}$$

$$\boxed{\text{z876}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1a) = \frac{4\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{36}$$

$$\boxed{\text{z877}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1b) = \frac{-3\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$$\boxed{\text{z878}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1b) = \frac{3\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$$\boxed{\text{z879}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1c) = -\frac{\sqrt{286} \left(7\sqrt{5}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 10\sqrt{3}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{572}$$

$$\boxed{\text{z880}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1c) = \frac{\sqrt{286} \left(-7\sqrt{5}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 10\sqrt{3}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{572}$$

$$\boxed{\text{z881}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2a) = \frac{\sqrt{209} \left(4\sqrt{105}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 94\sqrt{3}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) - 150\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 249\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{7524}$$

$$\boxed{\text{z882}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2a) = \frac{\sqrt{209} \left(-4\sqrt{105}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 94\sqrt{3}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 150\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 249\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{7524}$$

$$\boxed{\text{z883}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2b) = \frac{\sqrt{403} \left(31\sqrt{21}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{15}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 31\sqrt{21}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 35\sqrt{7}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{7}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{6448}$$

$$\boxed{\text{z884}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2b) = \frac{\sqrt{403} \left(-31\sqrt{21}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 13\sqrt{15}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 31\sqrt{21}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 35\sqrt{7}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{7}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{6448}$$

$$\begin{aligned}
\boxed{\text{z885}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2c) &= \frac{\sqrt{22} \left(-\sqrt{15} \mathbb{T}_3^{(1,-1;a)}(A_{2u}) \mathbb{T}_{1,2}^{(b)}(E_u) + 2 \mathbb{T}_3^{(1,-1;a)}(B_{1u}) \mathbb{T}_{1,1}^{(b)}(E_u) + 5 \mathbb{T}_3^{(1,-1;a)}(B_{2u}) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44} \\
\boxed{\text{z886}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2c) &= \frac{\sqrt{22} \left(-\sqrt{15} \mathbb{T}_3^{(1,-1;a)}(A_{2u}) \mathbb{T}_{1,1}^{(b)}(E_u) + 2 \mathbb{T}_3^{(1,-1;a)}(B_{1u}) \mathbb{T}_{1,2}^{(b)}(E_u) - 5 \mathbb{T}_3^{(1,-1;a)}(B_{2u}) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44} \\
\boxed{\text{z887}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{29} \left(8\sqrt{7} \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 4 \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{2} \mathbb{M}_6^{(1,-1;a)}(A_{2u}) \mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{3} \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{33} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{15} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{232} \\
\boxed{\text{z888}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{29} \left(-8\sqrt{7} \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 4 \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{2} \mathbb{M}_6^{(1,-1;a)}(A_{2u}) \mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{3} \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{33} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{15} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{232} \\
\boxed{\text{z889}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{87854} \left(-248\sqrt{5} \mathbb{T}_5^{(1,-1;a)}(A_{1u}) \mathbb{T}_{1,1}^{(b)}(E_u) + 39\sqrt{7} \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 155\sqrt{5} \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 80\sqrt{15} \mathbb{T}_5^{(1,-1;a)}(B_{1u}) \mathbb{T}_{1,1}^{(b)}(E_u) - 106\sqrt{15} \mathbb{T}_5^{(1,-1;a)}(B_{2u}) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{351416} \\
\boxed{\text{z890}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{87854} \left(248\sqrt{5} \mathbb{T}_5^{(1,-1;a)}(A_{1u}) \mathbb{T}_{1,2}^{(b)}(E_u) + 39\sqrt{7} \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 155\sqrt{5} \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 80\sqrt{15} \mathbb{T}_5^{(1,-1;a)}(B_{1u}) \mathbb{T}_{1,2}^{(b)}(E_u) + 106\sqrt{15} \mathbb{T}_5^{(1,-1;a)}(B_{2u}) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{351416} \\
\boxed{\text{z891}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{21547} \left(-16\sqrt{77} \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 340\sqrt{11} \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 62\sqrt{22} \mathbb{M}_6^{(1,-1;a)}(A_{2u}) \mathbb{T}_{1,2}^{(b)}(E_u) - 232\sqrt{15} \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 124\sqrt{33} \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 51\sqrt{3} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 51\sqrt{3} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{344752} \\
\boxed{\text{z892}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{21547} \left(16\sqrt{77} \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 340\sqrt{11} \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 62\sqrt{22} \mathbb{M}_6^{(1,-1;a)}(A_{2u}) \mathbb{T}_{1,1}^{(b)}(E_u) - 232\sqrt{15} \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 124\sqrt{33} \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 51\sqrt{3} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 51\sqrt{3} \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{344752} \\
\boxed{\text{z893}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{109} \left(-20 \mathbb{T}_5^{(1,-1;a)}(A_{1u}) \mathbb{T}_{1,1}^{(b)}(E_u) + 27\sqrt{35} \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 67 \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 92\sqrt{3} \mathbb{T}_5^{(1,-1;a)}(B_{1u}) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3} \mathbb{T}_5^{(1,-1;a)}(B_{2u}) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{3488}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z894}} \quad & \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 2b) \\
&= \frac{\sqrt{109} \left(20\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 27\sqrt{35}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 67\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 92\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{3488} \\
\boxed{\text{z895}} \quad & \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 3a) \\
&= \frac{\sqrt{517871} \left(-624\sqrt{1155}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 1372\sqrt{165}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3526\sqrt{330}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 30712\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 2620\sqrt{55}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 14883\sqrt{55}\mathbb{M}_6^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{91145296} \\
\boxed{\text{z896}} \quad & \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 3a) \\
&= \frac{\sqrt{517871} \left(624\sqrt{1155}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 1372\sqrt{165}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3526\sqrt{330}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 30712\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2620\sqrt{55}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 14883\sqrt{55}\mathbb{M}_6^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{91145296} \\
\boxed{\text{z897}} \quad & \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 3b) = \frac{4\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 7\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 14\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{32} \\
\boxed{\text{z898}} \quad & \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 3b) = \frac{-4\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 4\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 14\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{32} \\
\boxed{\text{z899}} \quad & \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) = \frac{\sqrt{10} \left(\sqrt{3}\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10} \\
\boxed{\text{z900}} \quad & \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10} \\
\boxed{\text{z901}} \quad & \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z902}} \quad & \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z903}} \quad & \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, c) = \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26} \\
\boxed{\text{z904}} \quad & \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, c) = \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{26} \\
\boxed{\text{z905}} \quad & \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 1a) = \frac{-4\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{36}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z906}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 1a) &= \frac{4\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{36} \\
\boxed{\text{z907}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 1b) &= \frac{-3\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16} \\
\boxed{\text{z908}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 1b) &= \frac{3\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{35}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16} \\
\boxed{\text{z909}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 1c) &= -\frac{\sqrt{286}\left(7\sqrt{5}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 10\sqrt{3}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{572} \\
\boxed{\text{z910}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 1c) &= \frac{\sqrt{286}\left(-7\sqrt{5}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 10\sqrt{3}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{572} \\
\boxed{\text{z911}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 2a) &= \frac{\sqrt{209}\left(4\sqrt{105}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 94\sqrt{3}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) - 150\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 249\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{7524} \\
\boxed{\text{z912}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 2a) &= \frac{\sqrt{209}\left(-4\sqrt{105}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 94\sqrt{3}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 150\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 249\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{7524} \\
\boxed{\text{z913}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 2b) &= \frac{\sqrt{403}\left(31\sqrt{21}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{15}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 31\sqrt{21}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 35\sqrt{7}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{7}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{6448} \\
\boxed{\text{z914}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 2b) &= \frac{\sqrt{403}\left(-31\sqrt{21}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 13\sqrt{15}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 31\sqrt{21}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 35\sqrt{7}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{7}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{6448} \\
\boxed{\text{z915}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 2c) &= \frac{\sqrt{22}\left(-\sqrt{15}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 5\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{44} \\
\boxed{\text{z916}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 2c) &= \frac{\sqrt{22}\left(-\sqrt{15}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 2\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 5\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{44}
\end{aligned}$$

$$\boxed{\text{z917}} \quad \mathbb{Q}_{6,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{87854} \left(-248\sqrt{5}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 39\sqrt{7}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 155\sqrt{5}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 80\sqrt{15}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 106\sqrt{15}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{351416}$$

$$\boxed{\text{z918}} \quad \mathbb{Q}_{6,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{87854} \left(248\sqrt{5}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 39\sqrt{7}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 155\sqrt{5}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 80\sqrt{15}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 106\sqrt{15}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{351416}$$

$$\boxed{\text{z919}} \quad \mathbb{Q}_{6,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{109} \left(-20\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 27\sqrt{35}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 67\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 92\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{3488}$$

$$\boxed{\text{z920}} \quad \mathbb{Q}_{6,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{109} \left(20\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 27\sqrt{35}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 67\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 92\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{3488}$$

$$\boxed{\text{z921}} \quad \mathbb{Q}_{6,1}^{(1,0;c)}(E_g, 3) = \frac{4\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 7\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 14\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{32}$$

$$\boxed{\text{z922}} \quad \mathbb{Q}_{6,2}^{(1,0;c)}(E_g, 3) = \frac{-4\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 4\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 14\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{32}$$

$$\boxed{\text{z923}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, a) = \frac{\sqrt{10} \left(\sqrt{3}\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z924}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, a) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z925}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z926}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z927}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, c) = \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26}$$

$$\boxed{\text{z928}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, c) = \frac{\sqrt{26} \left(-\sqrt{3}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{26}$$

$$\begin{aligned}
\boxed{\text{z929}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 1a) &= \frac{-4\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{36} \\
\boxed{\text{z930}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 1a) &= \frac{4\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{36} \\
\boxed{\text{z931}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 1b) &= -\frac{\sqrt{286}\left(7\sqrt{5}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 10\sqrt{3}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{572} \\
\boxed{\text{z932}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 1b) &= \frac{\sqrt{286}\left(-7\sqrt{5}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 10\sqrt{3}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{3}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{572} \\
\boxed{\text{z933}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 2a) &= \frac{\sqrt{209}\left(4\sqrt{105}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 94\sqrt{3}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) - 150\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 249\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{7524} \\
\boxed{\text{z934}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 2a) &= \frac{\sqrt{209}\left(-4\sqrt{105}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 94\sqrt{3}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{7}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 150\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 249\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{7524} \\
\boxed{\text{z935}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 2b) &= \frac{\sqrt{22}\left(-\sqrt{15}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 5\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{44} \\
\boxed{\text{z936}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 2b) &= \frac{\sqrt{22}\left(-\sqrt{15}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 2\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 5\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{44} \\
\boxed{\text{z937}} \quad \mathbb{G}_{1,1}^{(c)}(E_g) &= \frac{\sqrt{5}\left(-2\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{10} \\
\boxed{\text{z938}} \quad \mathbb{G}_{1,2}^{(c)}(E_g) &= \frac{\sqrt{5}\left(2\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{10} \\
\boxed{\text{z939}} \quad \mathbb{G}_{3,1}^{(c)}(E_g, 1a) &= \frac{\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z940}} \quad \mathbb{G}_{3,2}^{(c)}(E_g, 1a) &= \frac{\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}
\end{aligned}$$

$$\boxed{\text{z941}} \quad \mathbb{G}_{3,1}^{(c)}(E_g, 1b) = \frac{\sqrt{418} \left(97\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 17\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{5}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{21318}$$

$$\boxed{\text{z942}} \quad \mathbb{G}_{3,2}^{(c)}(E_g, 1b) = \frac{\sqrt{418} \left(-97\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 17\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{5}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{21318}$$

$$\boxed{\text{z943}} \quad \mathbb{G}_{3,1}^{(c)}(E_g, 2) = \frac{\sqrt{105}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 17\sqrt{3}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 32\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$$\boxed{\text{z944}} \quad \mathbb{G}_{3,2}^{(c)}(E_g, 2) = \frac{-\sqrt{105}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 17\sqrt{3}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 8\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 32\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$

$$\boxed{\text{z945}} \quad \mathbb{G}_{5,1}^{(c)}(E_g, 1) = \frac{\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z946}} \quad \mathbb{G}_{5,2}^{(c)}(E_g, 1) = \frac{-\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_4^{(a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z947}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{5} \left(-2\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z948}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{5} \left(2\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z949}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 1a) = \frac{\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z950}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 1a) = \frac{\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z951}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 1b) = \frac{\sqrt{418} \left(97\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 17\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{5}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{21318}$$

$$\boxed{\text{z952}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 1b) = \frac{\sqrt{418} \left(-97\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 17\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{5}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{21318}$$

$$\begin{aligned}
\boxed{\text{z953}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 2) &= \frac{\sqrt{105}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 17\sqrt{3}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 32\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{68} \\
\boxed{\text{z954}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 2) &= \frac{-\sqrt{105}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 17\sqrt{3}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 8\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 32\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{68} \\
\boxed{\text{z955}} \quad \mathbb{G}_{5,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12} \\
\boxed{\text{z956}} \quad \mathbb{G}_{5,2}^{(1,-1;c)}(E_g, 1a) &= \frac{-\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12} \\
\boxed{\text{z957}} \quad \mathbb{G}_{5,1}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{20852149} \left(20794\sqrt{33}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9382\sqrt{231}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 574\sqrt{462}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) - 7678\sqrt{35}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 19726\sqrt{77}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 41972\sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 962\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{1834989112} \\
\boxed{\text{z958}} \quad \mathbb{G}_{5,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{20852149} \left(-20794\sqrt{33}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 9382\sqrt{231}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 574\sqrt{462}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 7678\sqrt{35}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 19726\sqrt{77}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 41972\sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 962\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{1834989112} \\
\boxed{\text{z959}} \quad \mathbb{G}_{5,1}^{(1,-1;c)}(E_g, 2) &= \frac{\sqrt{89960419} \left(-2358\sqrt{105}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3418\sqrt{15}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 4954\sqrt{30}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 2402\sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 962\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 841\sqrt{33}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{3}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{719683352} \\
\boxed{\text{z960}} \quad \mathbb{G}_{5,2}^{(1,-1;c)}(E_g, 2) &= \frac{\sqrt{89960419} \left(2358\sqrt{105}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3418\sqrt{15}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4954\sqrt{30}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 2402\sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 962\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 841\sqrt{33}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 20\sqrt{3}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{719683352} \\
\boxed{\text{z961}} \quad \mathbb{G}_{5,1}^{(1,-1;c)}(E_g, 3) &= \frac{\sqrt{3007} \left(-32\sqrt{7}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 128\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 96\sqrt{2}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{165}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{3}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{33}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{24056} \\
\boxed{\text{z962}} \quad \mathbb{G}_{5,2}^{(1,-1;c)}(E_g, 3) &= \frac{\sqrt{3007} \left(32\sqrt{7}\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 128\mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 96\sqrt{2}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) - 15\sqrt{165}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 5\sqrt{3}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 20\sqrt{33}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{24056}
\end{aligned}$$

$$\boxed{\text{z963}} \quad \mathbb{G}_{7,1}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$$\boxed{\text{z964}} \quad \mathbb{G}_{7,2}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$$\boxed{\text{z965}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{5} \left(-2\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z966}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{5} \left(2\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z967}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 1a) = \frac{\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z968}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 1a) = \frac{\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z969}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 1b) = \frac{\sqrt{418} \left(97\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 17\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{5}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{21318}$$

$$\boxed{\text{z970}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 1b) = \frac{\sqrt{418} \left(-97\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 17\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{5}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{21318}$$

$$\boxed{\text{z971}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{105}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 17\sqrt{3}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 32\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$$\boxed{\text{z972}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 2) = \frac{-\sqrt{105}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 17\sqrt{3}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 8\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 32\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$

$$\boxed{\text{z973}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z974}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 1) = \frac{-\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z975}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z976}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(E_g, a) = -\frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z977}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(E_g, b) = \frac{\sqrt{5}\left(-2\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{10}$$

$$\boxed{\text{z978}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(E_g, b) = \frac{\sqrt{5}\left(2\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{10}$$

$$\boxed{\text{z979}} \quad \mathbb{G}_{3,1}^{(1,1;c)}(E_g, 1a) = \frac{\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z980}} \quad \mathbb{G}_{3,2}^{(1,1;c)}(E_g, 1a) = \frac{\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z981}} \quad \mathbb{G}_{3,1}^{(1,1;c)}(E_g, 1b) = \frac{\sqrt{418}\left(97\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 17\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{5}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{21318}$$

$$\boxed{\text{z982}} \quad \mathbb{G}_{3,2}^{(1,1;c)}(E_g, 1b) = \frac{\sqrt{418}\left(-97\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 17\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 57\sqrt{35}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 213\sqrt{5}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{5}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{21318}$$

$$\boxed{\text{z983}} \quad \mathbb{G}_{3,1}^{(1,1;c)}(E_g, 2) = \frac{\sqrt{105}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 17\sqrt{3}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u) - 32\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$$\boxed{\text{z984}} \quad \mathbb{G}_{3,2}^{(1,1;c)}(E_g, 2) = \frac{-\sqrt{105}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 17\sqrt{3}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{7}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u) + 8\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u) + 32\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$

$$\boxed{\text{z985}} \quad \mathbb{G}_{5,1}^{(1,1;c)}(E_g, 1) = \frac{\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 6\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z986}} \quad \mathbb{G}_{5,2}^{(1,1;c)}(E_g, 1) = \frac{-\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z1191}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1192}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1193}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, b) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1194}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, b) = -\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1195}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, c) = \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_{3,1}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1196}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, c) = \frac{-3\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_{3,2}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1197}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1198}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, d) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1199}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1a) = \frac{\sqrt{134} \left(2\sqrt{15}\mathbb{Q}_{5,1}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{7}\mathbb{Q}_{5,1}^{(a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{134}$$

$$\boxed{\text{z1200}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1a) = \frac{\sqrt{134} \left(-2\sqrt{15}\mathbb{Q}_{5,2}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{5,2}^{(a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{134}$$

$$\boxed{\text{z1201}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1202}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1203}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1c) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\begin{aligned}
\text{z1204} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1c) &= -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8} \\
\text{z1205} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1d) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\text{z1206} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1d) &= -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\text{z1207} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2a) &= \frac{\sqrt{67}\left(17\sqrt{35}\mathbb{Q}_{5,1}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 201\mathbb{Q}_{5,1}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 170\sqrt{3}\mathbb{Q}_{5,1}^{(a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{4288} \\
\text{z1208} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2a) &= \frac{\sqrt{67}\left(-17\sqrt{35}\mathbb{Q}_{5,2}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 201\mathbb{Q}_{5,2}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - 170\sqrt{3}\mathbb{Q}_{5,2}^{(a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{4288} \\
\text{z1209} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z1210} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z1211} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2c) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\text{z1212} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2c) &= -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\text{z1213} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z1214} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z1215} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 1b) &= \frac{3\sqrt{7}\mathbb{Q}_{5,1}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 17\sqrt{5}\mathbb{Q}_{5,1}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 6\sqrt{15}\mathbb{Q}_{5,1}^{(a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{64} \\
\text{z1216} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 1b) &= -\frac{3\sqrt{7}\mathbb{Q}_{5,2}^{(a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 17\sqrt{5}\mathbb{Q}_{5,2}^{(a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 6\sqrt{15}\mathbb{Q}_{5,2}^{(a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{64}
\end{aligned}$$

$$\boxed{\text{z1217}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1218}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1219}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1220}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1221}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u, a) = \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1222}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u, a) = \frac{-3\sqrt{3}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1223}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1224}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u, b) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1225}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1a) = \frac{\sqrt{134} \left(2\sqrt{15}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{7}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{134}$$

$$\boxed{\text{z1226}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1a) = \frac{\sqrt{134} \left(-2\sqrt{15}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{134}$$

$$\boxed{\text{z1227}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1228}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1229}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1c) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\begin{aligned}
\boxed{\text{z1230}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1c) &= -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8} \\
\boxed{\text{z1231}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1d) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1232}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1d) &= -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1233}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2a) &= \frac{\sqrt{67}\left(17\sqrt{35}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 201\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 170\sqrt{3}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{4288} \\
\boxed{\text{z1234}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2a) &= \frac{\sqrt{67}\left(-17\sqrt{35}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 201\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - 170\sqrt{3}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{4288} \\
\boxed{\text{z1235}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1236}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1237}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2c) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1238}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2c) &= -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1239}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1240}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1241}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 1b) &= \frac{3\sqrt{7}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 17\sqrt{5}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 6\sqrt{15}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{64} \\
\boxed{\text{z1242}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 1b) &= -\frac{3\sqrt{7}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 17\sqrt{5}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 6\sqrt{15}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{64}
\end{aligned}$$

$$\boxed{\text{z1243}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(Eu, 1c) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(Eu, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1244}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(Eu, 1c) = -\frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(Eu, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1245}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(Eu, 2a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(Eu, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1246}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(Eu, 2a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(Eu, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1247}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(Eu, 2b) = \frac{\sqrt{107} \left(8\sqrt{3}\mathbb{G}_{6,1}^{(1,-1;a)}(Eu, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{22}\mathbb{G}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{214}$$

$$\boxed{\text{z1248}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(Eu, 2b) = -\frac{\sqrt{107} \left(8\sqrt{3}\mathbb{G}_{6,2}^{(1,-1;a)}(Eu, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{22}\mathbb{G}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{214}$$

$$\boxed{\text{z1249}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(Eu, 3a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(Eu, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1250}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(Eu, 3a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(Eu, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1251}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(Eu, 3b) = \frac{\sqrt{107} \left(\sqrt{22}\mathbb{G}_{6,1}^{(1,-1;a)}(Eu, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 8\sqrt{3}\mathbb{G}_{6,1}^{(1,-1;a)}(Eu, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{214}$$

$$\boxed{\text{z1252}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(Eu, 3b) = \frac{\sqrt{107} \left(-\sqrt{22}\mathbb{G}_{6,2}^{(1,-1;a)}(Eu, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 8\sqrt{3}\mathbb{G}_{6,2}^{(1,-1;a)}(Eu, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{214}$$

$$\boxed{\text{z1253}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(Eu, a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(Eu)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1254}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(Eu, a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(Eu)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1255}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(Eu, b) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(Eu)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1256}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, b) = -\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1257}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, c) = \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1258}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, c) = \frac{-3\sqrt{3}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1259}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1260}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, d) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1261}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1a) = \frac{\sqrt{134}\left(2\sqrt{15}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{7}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{134}$$

$$\boxed{\text{z1262}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1a) = \frac{\sqrt{134}\left(-2\sqrt{15}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{134}$$

$$\boxed{\text{z1263}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1264}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1265}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1c) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1266}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1c) = -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$$\boxed{\text{z1267}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1d) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1268}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1d) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\begin{aligned}
\boxed{\text{z1269}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2a) &= \frac{\sqrt{67} \left(17\sqrt{35}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 201\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 170\sqrt{3}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{4288} \\
\boxed{\text{z1270}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2a) &= \frac{\sqrt{67} \left(-17\sqrt{35}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 201\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - 170\sqrt{3}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{4288} \\
\boxed{\text{z1271}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1272}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2b) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1273}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2c) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1274}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2c) &= -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1275}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1276}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1277}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 1b) &= \frac{3\sqrt{7}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 17\sqrt{5}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 6\sqrt{15}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{64} \\
\boxed{\text{z1278}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 1b) &= -\frac{3\sqrt{7}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 17\sqrt{5}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 6\sqrt{15}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{64} \\
\boxed{\text{z1279}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1280}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 2) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1281}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 3) &= \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z1282}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 3) &= \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1283}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1284}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1285}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u, b) &= \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1286}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u, b) &= -\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1287}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u, c) &= \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8} \\
\boxed{\text{z1288}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u, c) &= \frac{-3\sqrt{3}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8} \\
\boxed{\text{z1289}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u, d) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1290}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u, d) &= -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1291}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1292}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1293}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 1b) &= \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8} \\
\boxed{\text{z1294}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 1b) &= -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{3}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{8} \\
\boxed{\text{z1295}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 1c) &= \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2}
\end{aligned}$$

$$\boxed{\text{z1296}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 1c) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1297}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 2a) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1298}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 2a) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1299}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 2b) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1300}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 2b) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1301}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1302}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1303}} \quad \mathbb{G}_{4,1}^{(c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1304}} \quad \mathbb{G}_{4,2}^{(c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1305}} \quad \mathbb{G}_{4,1}^{(c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1306}} \quad \mathbb{G}_{4,2}^{(c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1307}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1308}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1309}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1310}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1311}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1312}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1313}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1314}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1315}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1316}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1317}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1318}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1319}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1320}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1321}} \quad \mathbb{G}_{4,1}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1322}} \quad \mathbb{G}_{4,2}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1323}} \quad \mathbb{G}_{4,1}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1324}} \quad \mathbb{G}_{4,2}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1325}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1326}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1327}} \quad \mathbb{G}_{4,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1328}} \quad \mathbb{G}_{4,2}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1329}} \quad \mathbb{G}_{4,1}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1330}} \quad \mathbb{G}_{4,2}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

• **U;U_001_1 : 'U'-'U' bond-cluster**

* bra: $\langle \frac{5}{2}, \frac{5}{2}; f |, \langle \frac{5}{2}, \frac{3}{2}; f |, \langle \frac{5}{2}, \frac{1}{2}; f |, \langle \frac{5}{2}, -\frac{1}{2}; f |, \langle \frac{5}{2}, -\frac{3}{2}; f |, \langle \frac{5}{2}, -\frac{5}{2}; f |, \langle \frac{7}{2}, \frac{7}{2}; f |, \langle \frac{7}{2}, \frac{5}{2}; f |, \langle \frac{7}{2}, \frac{3}{2}; f |, \langle \frac{7}{2}, \frac{1}{2}; f |, \langle \frac{7}{2}, -\frac{1}{2}; f |, \langle \frac{7}{2}, -\frac{3}{2}; f |, \langle \frac{7}{2}, -\frac{5}{2}; f |, \langle \frac{7}{2}, -\frac{7}{2}; f |$

* ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

* wyckoff: **4b04c**

$$\boxed{\text{z76}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z77}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z78}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z79}} \quad \mathbb{Q}_2^{(c)}(A_{1g}, b) = -\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z80}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1a) = \mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z81}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 1b) = \mathbb{Q}_6^{(a)}(B_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z82}} \quad \mathbb{Q}_4^{(c)}(A_{1g}, 2a) = \mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z83}} \quad & \mathbb{Q}_4^{(c)}(A_{1g}, 2b) = -\mathbb{Q}_6^{(a)}(B_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z84}} \quad & \mathbb{Q}_6^{(c)}(A_{1g}, 1) = \mathbb{Q}_6^{(a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z85}} \quad & \mathbb{Q}_6^{(c)}(A_{1g}, 2) = \mathbb{Q}_6^{(a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z86}} \quad & \mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z87}} \quad & \mathbb{Q}_2^{(1,-1;c)}(A_{1g}, a) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z88}} \quad & \mathbb{Q}_2^{(1,-1;c)}(A_{1g}, b) = -\mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z89}} \quad & \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1a) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z90}} \quad & \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1b) = \mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z91}} \quad & \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2a) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z92}} \quad & \mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2b) = -\mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z93}} \quad & \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 1) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z94}} \quad & \mathbb{Q}_6^{(1,-1;c)}(A_{1g}, 2) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z95}} \quad & \mathbb{Q}_2^{(1,0;c)}(A_{1g}) = -\mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z96}} \quad & \mathbb{Q}_4^{(1,0;c)}(A_{1g}, 1) = \mathbb{G}_5^{(1,0;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z97}} \quad & \mathbb{Q}_0^{(1,1;c)}(A_{1g}, a) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z98}} \quad & \mathbb{Q}_0^{(1,1;c)}(A_{1g}, b) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z99}} \quad & \mathbb{Q}_2^{(1,1;c)}(A_{1g}, a) = \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z100}} \quad & \mathbb{Q}_2^{(1,1;c)}(A_{1g}, b) = -\mathbb{Q}_4^{(1,1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z101}} \quad & \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 1) = \mathbb{Q}_4^{(1,1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z102}} \quad \mathbb{Q}_4^{(1,1;c)}(A_{1g}, 2) &= \mathbb{Q}_4^{(1,1;a)}(A_{1g}, 2) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z103}} \quad \mathbb{G}_5^{(1,0;c)}(A_{1g}) &= \mathbb{G}_5^{(1,0;a)}(A_{1g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z172}} \quad \mathbb{Q}_5^{(c)}(A_{1u}) &= \frac{\sqrt{14} \left(2\mathbb{M}_{5,1}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{5,1}^{(a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 2\mathbb{M}_{5,2}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{5,2}^{(a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{14} \\
\boxed{\text{z173}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{1u}) &= \frac{\sqrt{14} \left(2\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 2\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{14} \\
\boxed{\text{z174}} \quad \mathbb{Q}_7^{(1,-1;c)}(A_{1u}) &= \frac{\sqrt{5} \left(4\sqrt{2}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{13}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{11}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4) \mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{2}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{13}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{11}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{40} \\
\boxed{\text{z175}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{1u}, a) &= \frac{-\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z176}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{1u}, b) &= \frac{\sqrt{29} \left(-\sqrt{10}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 4\sqrt{3}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{10}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{58} \\
\boxed{\text{z177}} \quad \mathbb{Q}_7^{(1,0;c)}(A_{1u}) &= \frac{\sqrt{29} \left(-6\sqrt{66}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 29\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{55}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{66}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 29\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{55}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{464} \\
\boxed{\text{z178}} \quad \mathbb{Q}_5^{(1,1;c)}(A_{1u}) &= \frac{\sqrt{14} \left(2\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 2\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{14} \\
\boxed{\text{z179}} \quad \mathbb{G}_0^{(c)}(A_{1u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z180}} \quad \mathbb{G}_2^{(c)}(A_{1u}) &= \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z181}} \quad \mathbb{G}_4^{(c)}(A_{1u}, 1a) &= \frac{\sqrt{34} \left(49\sqrt{5}\mathbb{M}_{5,1}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,1}^{(a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{5}\mathbb{M}_{5,2}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{21}\mathbb{M}_{5,2}^{(a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{952}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z182}} \quad \mathbb{G}_4^{(c)}(A_{1u}, 1b) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z183}} \quad \mathbb{G}_4^{(c)}(A_{1u}, 2) &= \frac{\sqrt{34} \left(-3\sqrt{3}\mathbb{M}_{5,1}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{35}\mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{M}_{5,2}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{105}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{35}\mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{136} \\
\boxed{\text{z184}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z185}} \quad \mathbb{G}_2^{(1,-1;c)}(A_{1u}) &= \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z186}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 1a) &= \frac{\sqrt{34} \left(49\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{952} \\
\boxed{\text{z187}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 1b) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z188}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, 2) &= \frac{\sqrt{34} \left(-3\sqrt{3}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{35}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{105}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{35}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{136} \\
\boxed{\text{z189}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{1u}, 1) &= \frac{\sqrt{835} \left(20\sqrt{14}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{66}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{429}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 11\sqrt{3}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 16\sqrt{66}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{6680} \\
\boxed{\text{z190}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{1u}, 2) &= \frac{\sqrt{167} \left(-55\sqrt{42}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 35\sqrt{22}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 17\sqrt{143}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 383\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) + 55\sqrt{42}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 35\sqrt{22}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10688} \\
\boxed{\text{z191}} \quad \mathbb{G}_8^{(1,-1;c)}(A_{1u}, 1) &= \frac{\sqrt{858}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{1001}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{858}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{1001}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u)}{64}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z192}} \quad \mathbb{G}_2^{(1,0;c)}(A_{1u}) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z193}} \quad \mathbb{G}_4^{(1,0;c)}(A_{1u}, 1) &= \frac{\sqrt{7} \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7} \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z194}} \quad \mathbb{G}_6^{(1,0;c)}(A_{1u}, 1) &= \frac{-2\sqrt{6} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{11} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{6} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{11} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u)}{16} \\
\boxed{\text{z195}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z196}} \quad \mathbb{G}_2^{(1,1;c)}(A_{1u}) &= \frac{-\sqrt{3} \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5} \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5} \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z197}} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 1a) &= \frac{\sqrt{34} \left(49\sqrt{5} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{7} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{5} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{7} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 6\sqrt{21} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{952} \\
\boxed{\text{z198}} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 1b) &= \frac{\sqrt{5} \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3} \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5} \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u)}{4} \\
\boxed{\text{z199}} \quad \mathbb{G}_4^{(1,1;c)}(A_{1u}, 2) &= \frac{\sqrt{34} \left(-3\sqrt{3} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{105} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{35} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{3} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{105} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{35} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{136} \\
\boxed{\text{z258}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, a) &= \mathbb{Q}_2^{(a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z259}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, b) &= \mathbb{Q}_4^{(a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z260}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, c) &= -\mathbb{Q}_4^{(a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z261}} \quad \mathbb{Q}_4^{(c)}(A_{2g}, d) &= \frac{\sqrt{31} \left(3\sqrt{55} \mathbb{Q}_6^{(a)}(B_{2g}, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{Q}_6^{(a)}(B_{2g}, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{124} \\
\boxed{\text{z262}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, a) &= \mathbb{Q}_6^{(a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g})
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z263}} \quad \mathbb{Q}_6^{(c)}(A_{2g}, b) &= \frac{\sqrt{31} \left(\mathbb{Q}_6^{(a)}(B_{2g}, 1) \mathbb{Q}_2^{(b)}(B_{1g}) - 3\sqrt{55} \mathbb{Q}_6^{(a)}(B_{2g}, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{124} \\
\boxed{\text{z264}} \quad \mathbb{Q}_4^{(1, -1; c)}(A_{2g}, a) &= \mathbb{Q}_2^{(1, -1; a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z265}} \quad \mathbb{Q}_4^{(1, -1; c)}(A_{2g}, b) &= \mathbb{Q}_4^{(1, -1; a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z266}} \quad \mathbb{Q}_4^{(1, -1; c)}(A_{2g}, c) &= -\mathbb{Q}_4^{(1, -1; a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z267}} \quad \mathbb{Q}_4^{(1, -1; c)}(A_{2g}, d) &= \frac{\sqrt{31} \left(3\sqrt{55} \mathbb{Q}_6^{(1, -1; a)}(B_{2g}, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{Q}_6^{(1, -1; a)}(B_{2g}, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{124} \\
\boxed{\text{z268}} \quad \mathbb{Q}_6^{(1, -1; c)}(A_{2g}, a) &= \mathbb{Q}_6^{(1, -1; a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z269}} \quad \mathbb{Q}_6^{(1, -1; c)}(A_{2g}, b) &= \frac{\sqrt{31} \left(\mathbb{Q}_6^{(1, -1; a)}(B_{2g}, 1) \mathbb{Q}_2^{(b)}(B_{1g}) - 3\sqrt{55} \mathbb{Q}_6^{(1, -1; a)}(B_{2g}, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{124} \\
\boxed{\text{z270}} \quad \mathbb{Q}_4^{(1, 0; c)}(A_{2g}, a) &= -\mathbb{G}_5^{(1, 0; a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z271}} \quad \mathbb{Q}_4^{(1, 0; c)}(A_{2g}, b) &= \mathbb{G}_3^{(1, 0; a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z272}} \quad \mathbb{Q}_4^{(1, 1; c)}(A_{2g}, a) &= \mathbb{Q}_2^{(1, 1; a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z273}} \quad \mathbb{Q}_4^{(1, 1; c)}(A_{2g}, b) &= \mathbb{Q}_4^{(1, 1; a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z274}} \quad \mathbb{Q}_4^{(1, 1; c)}(A_{2g}, c) &= -\mathbb{Q}_4^{(1, 1; a)}(B_{2g}) \mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z275}} \quad \mathbb{G}_1^{(1, 0; c)}(A_{2g}) &= \mathbb{G}_1^{(1, 0; a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z276}} \quad \mathbb{G}_3^{(1, 0; c)}(A_{2g}) &= \mathbb{G}_3^{(1, 0; a)}(A_{2g}) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z277}} \quad \mathbb{G}_5^{(1, 0; c)}(A_{2g}, 1) &= \mathbb{G}_5^{(1, 0; a)}(A_{2g}, 1) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z278}} \quad \mathbb{G}_5^{(1, 0; c)}(A_{2g}, 2) &= \mathbb{G}_5^{(1, 0; a)}(A_{2g}, 2) \mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z347}} \quad \mathbb{Q}_1^{(c)}(A_{2u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z348}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) &= -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z349}} \quad \mathbb{Q}_5^{(c)}(A_{2u}, 1) &= \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_{5,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{5,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16} \\
\boxed{\text{z350}} \quad \mathbb{Q}_5^{(c)}(A_{2u}, 2) &= \frac{\sqrt{2} \left(21\sqrt{5}\mathbb{M}_{5,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 21\sqrt{5}\mathbb{M}_{5,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{112} \\
\boxed{\text{z351}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_{2u}) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z352}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) &= -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z353}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 1) &= \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16} \\
\boxed{\text{z354}} \quad \mathbb{Q}_5^{(1,-1;c)}(A_{2u}, 2) &= \frac{\sqrt{2} \left(21\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 21\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{112} \\
\boxed{\text{z355}} \quad \mathbb{Q}_7^{(1,-1;c)}(A_{2u}, 1) &= \frac{5\sqrt{14}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{429}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 15\sqrt{3}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{429}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{3}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} \\
\boxed{\text{z356}} \quad \mathbb{Q}_7^{(1,-1;c)}(A_{2u}, 2) &= \frac{\sqrt{5} \left(-3\sqrt{462}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 43\sqrt{2}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{13}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{11}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{462}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 43\sqrt{2}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{13}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{11}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{320} \\
\boxed{\text{z357}} \quad \mathbb{Q}_1^{(1,0;c)}(A_{2u}) &= \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z358}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}) &= -\frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z359}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 1) &= \frac{2\sqrt{6}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{11}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{6}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{11}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} \\
\boxed{\text{z360}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 2a) &= \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z361}} \quad \mathbb{Q}_5^{(1,0;c)}(A_{2u}, 2b) &= \frac{\sqrt{29}\left(-14\sqrt{10}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{165}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 23\sqrt{3}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 14\sqrt{10}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{165}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 23\sqrt{3}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{464} \\
\boxed{\text{z362}} \quad \mathbb{Q}_7^{(1,0;c)}(A_{2u}, 2) &= \frac{\sqrt{29}\left(2\sqrt{66}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 13\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{55}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{66}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 13\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{55}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{232} \\
\boxed{\text{z363}} \quad \mathbb{Q}_1^{(1,1;c)}(A_{2u}) &= \frac{\sqrt{2}\left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{2} \\
\boxed{\text{z364}} \quad \mathbb{Q}_3^{(1,1;c)}(A_{2u}) &= -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z365}} \quad \mathbb{Q}_5^{(1,1;c)}(A_{2u}, 1) &= \frac{\sqrt{2}\left(\sqrt{15}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{16} \\
\boxed{\text{z366}} \quad \mathbb{Q}_5^{(1,1;c)}(A_{2u}, 2) &= \frac{\sqrt{2}\left(21\sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 21\sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{112} \\
\boxed{\text{z367}} \quad \mathbb{G}_4^{(c)}(A_{2u}, a) &= \frac{\sqrt{2}\left(-7\mathbb{M}_{5,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\mathbb{M}_{5,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{56}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z368}} \quad \mathbb{G}_4^{(c)}(A_{2u}, b) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z369}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{2u}, a) &= \frac{\sqrt{2} \left(-7\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{56} \\
\boxed{\text{z370}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{2u}, b) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z371}} \quad \mathbb{G}_6^{(1,-1;c)}(A_{2u}) &= \frac{\sqrt{5} \left(\sqrt{42}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{22}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 7\sqrt{143}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 23\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{42}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{22}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{143}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 23\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{320} \\
\boxed{\text{z372}} \quad \mathbb{G}_8^{(1,-1;c)}(A_{2u}, 1) &= \frac{\sqrt{858}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{1001}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{858}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{1001}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} \\
\boxed{\text{z373}} \quad \mathbb{G}_4^{(1,1;c)}(A_{2u}, a) &= \frac{\sqrt{2} \left(-7\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{105}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{56} \\
\boxed{\text{z374}} \quad \mathbb{G}_4^{(1,1;c)}(A_{2u}, b) &= \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} \\
\boxed{\text{z443}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, a) &= \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z444}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, b) &= \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z445}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, c) &= -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z446}} \quad \mathbb{Q}_2^{(c)}(B_{1g}, d) &= \frac{\sqrt{3} \left(\sqrt{7}\mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z447}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, a) &= \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z448}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, b) &= \frac{\sqrt{3} \left(\sqrt{5}\mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z449}} \quad \mathbb{Q}_4^{(c)}(B_{1g}, c) = \frac{3\mathbb{Q}_6^{(a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{7}\mathbb{Q}_6^{(a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$$

$$\boxed{\text{z450}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 1a) = \mathbb{Q}_6^{(a)}(B_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z451}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 1b) = \frac{\sqrt{7}\mathbb{Q}_6^{(a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\mathbb{Q}_6^{(a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$$

$$\boxed{\text{z452}} \quad \mathbb{Q}_6^{(c)}(B_{1g}, 2) = \mathbb{Q}_6^{(a)}(B_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z453}} \quad \mathbb{Q}_2^{(1, -1; c)}(B_{1g}, a) = \mathbb{Q}_2^{(1, -1; a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z454}} \quad \mathbb{Q}_2^{(1, -1; c)}(B_{1g}, b) = -\mathbb{Q}_2^{(1, -1; a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z455}} \quad \mathbb{Q}_2^{(1, -1; c)}(B_{1g}, c) = \frac{\sqrt{3} \left(\sqrt{7}\mathbb{Q}_4^{(1, -1; a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_4^{(1, -1; a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6}$$

$$\boxed{\text{z456}} \quad \mathbb{Q}_4^{(1, -1; c)}(B_{1g}, a) = \mathbb{Q}_4^{(1, -1; a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z457}} \quad \mathbb{Q}_4^{(1, -1; c)}(B_{1g}, b) = \frac{\sqrt{3} \left(\sqrt{5}\mathbb{Q}_4^{(1, -1; a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_4^{(1, -1; a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6}$$

$$\boxed{\text{z458}} \quad \mathbb{Q}_4^{(1, -1; c)}(B_{1g}, c) = \frac{3\mathbb{Q}_6^{(1, -1; a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{7}\mathbb{Q}_6^{(1, -1; a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$$

$$\boxed{\text{z459}} \quad \mathbb{Q}_6^{(1, -1; c)}(B_{1g}, 1a) = \mathbb{Q}_6^{(1, -1; a)}(B_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z460}} \quad \mathbb{Q}_6^{(1, -1; c)}(B_{1g}, 1b) = \frac{\sqrt{7}\mathbb{Q}_6^{(1, -1; a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\mathbb{Q}_6^{(1, -1; a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$$

$$\boxed{\text{z461}} \quad \mathbb{Q}_6^{(1, -1; c)}(B_{1g}, 2) = \mathbb{Q}_6^{(1, -1; a)}(B_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z462}} \quad \mathbb{Q}_4^{(1, 0; c)}(B_{1g}) = \mathbb{G}_5^{(1, 0; a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z463}} \quad \mathbb{Q}_2^{(1, 1; c)}(B_{1g}, a) = \mathbb{Q}_0^{(1, 1; a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z464}} \quad \mathbb{Q}_2^{(1, 1; c)}(B_{1g}, b) = \mathbb{Q}_2^{(1, 1; a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z465}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}, c) &= -\mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z466}} \quad \mathbb{Q}_2^{(1,1;c)}(B_{1g}, d) &= \frac{\sqrt{3} \left(\sqrt{7}\mathbb{Q}_4^{(1,1;a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{5}\mathbb{Q}_4^{(1,1;a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z467}} \quad \mathbb{Q}_4^{(1,1;c)}(B_{1g}, a) &= \mathbb{Q}_4^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z468}} \quad \mathbb{Q}_4^{(1,1;c)}(B_{1g}, b) &= \frac{\sqrt{3} \left(\sqrt{5}\mathbb{Q}_4^{(1,1;a)}(A_{1g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_4^{(1,1;a)}(A_{1g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z469}} \quad \mathbb{G}_3^{(1,0;c)}(B_{1g}) &= \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z470}} \quad \mathbb{G}_5^{(1,0;c)}(B_{1g}) &= \mathbb{G}_5^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z538}} \quad \mathbb{Q}_3^{(c)}(B_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z539}} \quad \mathbb{Q}_5^{(c)}(B_{1u}) &= -\frac{\sqrt{26} \left(2\sqrt{3}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26} \\
\boxed{\text{z540}} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z541}} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{1u}) &= -\frac{\sqrt{26} \left(2\sqrt{3}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26} \\
\boxed{\text{z542}} \quad \mathbb{Q}_7^{(1,-1;c)}(B_{1u}, 1) &= -\frac{\sqrt{14} \left(3\sqrt{11}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{13}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{11}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{13}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{56} \\
\boxed{\text{z543}} \quad \mathbb{Q}_7^{(1,-1;c)}(B_{1u}, 2) &= \frac{\sqrt{35} \left(-112\sqrt{13}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 39\sqrt{2}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{286}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) - 112\sqrt{13}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 39\sqrt{2}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{286}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{3640} \\
\boxed{\text{z544}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z545}} \quad \mathbb{Q}_3^{(1,0;c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z546}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z547}} \quad \mathbb{Q}_5^{(1,0;c)}(B_{1u}, b) &= \frac{\sqrt{23} \left(\sqrt{30} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 4 \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{30} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 4 \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{46} \\
\boxed{\text{z548}} \quad \mathbb{Q}_7^{(1,0;c)}(B_{1u}, 1) &= \frac{\sqrt{667} \left(-30\sqrt{6} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 23\sqrt{11} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 45\sqrt{5} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) - 30\sqrt{6} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 23\sqrt{11} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 45\sqrt{5} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{5336} \\
\boxed{\text{z549}} \quad \mathbb{Q}_7^{(1,0;c)}(B_{1u}, 2) &= \frac{\sqrt{29} \left(2\sqrt{22} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{3} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{165} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{22} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 15\sqrt{3} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{165} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{232} \\
\boxed{\text{z550}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{1u}) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z551}} \quad \mathbb{Q}_5^{(1,1;c)}(B_{1u}) &= -\frac{\sqrt{26} \left(2\sqrt{3} \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{3} \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{26} \\
\boxed{\text{z552}} \quad \mathbb{G}_2^{(c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z553}} \quad \mathbb{G}_2^{(c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z554}} \quad \mathbb{G}_4^{(c)}(B_{1u}) &= \frac{\sqrt{13} \left(13\sqrt{5} \mathbb{M}_{5,1}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7} \mathbb{M}_{5,1}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21} \mathbb{M}_{5,1}^{(a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{5} \mathbb{M}_{5,2}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7} \mathbb{M}_{5,2}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{21} \mathbb{M}_{5,2}^{(a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{208} \\
\boxed{\text{z555}} \quad \mathbb{G}_6^{(c)}(B_{1u}, 1) &= \frac{3\sqrt{7} \mathbb{M}_{5,1}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5} \mathbb{M}_{5,1}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15} \mathbb{M}_{5,1}^{(a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7} \mathbb{M}_{5,2}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5} \mathbb{M}_{5,2}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15} \mathbb{M}_{5,2}^{(a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u)}{16} \\
\boxed{\text{z556}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{1u}, a) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z557}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{1u}, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z558}} \quad \mathbb{G}_4^{(1,-1;c)}(B_{1u}) \\ = \frac{\sqrt{13} \left(13\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{208}$$

$$\boxed{\text{z559}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 1a) \\ = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$$\boxed{\text{z560}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 1b) \\ = \frac{\sqrt{5} \left(-3\sqrt{26}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 13\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{143}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{26}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 13\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{143}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{130}$$

$$\boxed{\text{z561}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{1u}, 2) = \frac{\sqrt{2} \left(\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z562}} \quad \mathbb{G}_2^{(1,1;c)}(B_{1u}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z563}} \quad \mathbb{G}_2^{(1,1;c)}(B_{1u}, b) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z564}} \quad \mathbb{G}_4^{(1,1;c)}(B_{1u}) \\ = \frac{\sqrt{13} \left(13\sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{21}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{208}$$

$$\boxed{\text{z565}} \quad \mathbb{G}_6^{(1,1;c)}(B_{1u}, 1) \\ = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$$\boxed{\text{z629}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, a) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z630}} \quad \mathbb{Q}_2^{(c)}(B_{2g}, b) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\boxed{\text{z631}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, a) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z632}} \quad \mathbb{Q}_4^{(c)}(B_{2g}, b) = \mathbb{Q}_6^{(a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$$\begin{aligned}
\boxed{\text{z633}} \quad & \mathbb{Q}_6^{(c)}(B_{2g}, 1) = \mathbb{Q}_6^{(a)}(B_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z634}} \quad & \mathbb{Q}_6^{(c)}(B_{2g}, 2) = \mathbb{Q}_6^{(a)}(B_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z635}} \quad & \mathbb{Q}_2^{(1,-1;c)}(B_{2g}, a) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z636}} \quad & \mathbb{Q}_2^{(1,-1;c)}(B_{2g}, b) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z637}} \quad & \mathbb{Q}_4^{(1,-1;c)}(B_{2g}, a) = \mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z638}} \quad & \mathbb{Q}_4^{(1,-1;c)}(B_{2g}, b) = \mathbb{Q}_6^{(1,-1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z639}} \quad & \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 1) = \mathbb{Q}_6^{(1,-1;a)}(B_{2g}, 1)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z640}} \quad & \mathbb{Q}_6^{(1,-1;c)}(B_{2g}, 2) = \mathbb{Q}_6^{(1,-1;a)}(B_{2g}, 2)\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z641}} \quad & \mathbb{Q}_2^{(1,0;c)}(B_{2g}, a) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z642}} \quad & \mathbb{Q}_2^{(1,0;c)}(B_{2g}, b) = -\mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z643}} \quad & \mathbb{Q}_4^{(1,0;c)}(B_{2g}) = \frac{\sqrt{3} \left(-\sqrt{5}\mathbb{G}_5^{(1,0;a)}(A_{2g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{G}_5^{(1,0;a)}(A_{2g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z644}} \quad & \mathbb{Q}_6^{(1,0;c)}(B_{2g}, 1) = \frac{\sqrt{3} \left(\sqrt{7}\mathbb{G}_5^{(1,0;a)}(A_{2g}, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{5}\mathbb{G}_5^{(1,0;a)}(A_{2g}, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{6} \\
\boxed{\text{z645}} \quad & \mathbb{Q}_2^{(1,1;c)}(B_{2g}, a) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z646}} \quad & \mathbb{Q}_2^{(1,1;c)}(B_{2g}, b) = \mathbb{Q}_4^{(1,1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g}) \\
\boxed{\text{z647}} \quad & \mathbb{Q}_4^{(1,1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z648}} \quad & \mathbb{G}_3^{(1,0;c)}(B_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g}) \\
\boxed{\text{z649}} \quad & \mathbb{G}_5^{(1,0;c)}(B_{2g}) = \mathbb{G}_5^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})
\end{aligned}$$

$$\begin{aligned}
\boxed{z717} \quad \mathbb{Q}_3^{(c)}(B_{2u}) &= \frac{\sqrt{2} \left(\sqrt{15} \mathbb{M}_{3,1}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,1}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15} \mathbb{M}_{3,2}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{z718} \quad \mathbb{Q}_5^{(c)}(B_{2u}) &= \frac{\sqrt{26} \left(-\sqrt{105} \mathbb{M}_{5,1}^{(a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_{5,1}^{(a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 10 \mathbb{M}_{5,1}^{(a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105} \mathbb{M}_{5,2}^{(a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3} \mathbb{M}_{5,2}^{(a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 10 \mathbb{M}_{5,2}^{(a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{104} \\
\boxed{z719} \quad \mathbb{Q}_3^{(1,-1;c)}(B_{2u}) &= \frac{\sqrt{2} \left(\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{z720} \quad \mathbb{Q}_5^{(1,-1;c)}(B_{2u}) &= \frac{\sqrt{26} \left(-\sqrt{105} \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 10 \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105} \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3} \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 10 \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{104} \\
\boxed{z721} \quad \mathbb{Q}_7^{(1,-1;c)}(B_{2u}, 1) &= \frac{\sqrt{5} \left(\sqrt{6006} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{26} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 29 \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{143} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{6006} \mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{26} \mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \right)}{320} \\
\boxed{z722} \quad \mathbb{Q}_7^{(1,-1;c)}(B_{2u}, 2) &= \frac{15\sqrt{546} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 11\sqrt{286} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 39\sqrt{11} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) + 115\sqrt{13} \mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4) \mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{546} \mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{286} \mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \right)}{832} \\
\boxed{z723} \quad \mathbb{Q}_3^{(1,0;c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\boxed{z724} \quad \mathbb{Q}_3^{(1,0;c)}(B_{2u}, b) &= \frac{\sqrt{2} \left(-\sqrt{7} \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 3 \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7} \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 3 \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{z725} \quad \mathbb{Q}_5^{(1,0;c)}(B_{2u}, a) &= \frac{\sqrt{2} \left(3 \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{7} \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 3 \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{7} \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
\boxed{z726} \quad \mathbb{Q}_5^{(1,0;c)}(B_{2u}, b) &= \frac{\sqrt{23} \left(-2\sqrt{30} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{55} \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - 11 \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{30} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{55} \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 11 \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{184}
\end{aligned}$$

$$\begin{aligned}
& \boxed{\text{z727}} \quad \mathbb{Q}_7^{(1,0;c)}(B_{2u}, 1) \\
& = \frac{4\sqrt{11}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{6}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{330}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 4\sqrt{11}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{6}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{330}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} \\
& \boxed{\text{z728}} \quad \mathbb{Q}_7^{(1,0;c)}(B_{2u}, 2) \\
& = \frac{\sqrt{23} \left(-44\sqrt{3}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 13\sqrt{22}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{10}\mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 44\sqrt{3}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 13\sqrt{22}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{10}\mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{736} \\
& \boxed{\text{z729}} \quad \mathbb{Q}_3^{(1,1;c)}(B_{2u}) = \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
& \boxed{\text{z730}} \quad \mathbb{Q}_5^{(1,1;c)}(B_{2u}) \\
& = \frac{\sqrt{26} \left(-\sqrt{105}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 10\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{105}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 10\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{104} \\
& \boxed{\text{z731}} \quad \mathbb{G}_2^{(c)}(B_{2u}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
& \boxed{\text{z732}} \quad \mathbb{G}_2^{(c)}(B_{2u}, b) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
& \boxed{\text{z733}} \quad \mathbb{G}_4^{(c)}(B_{2u}) \\
& = \frac{\sqrt{13} \left(\sqrt{5}\mathbb{M}_{5,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{208} \\
& \boxed{\text{z734}} \quad \mathbb{G}_6^{(c)}(B_{2u}, 1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{M}_{5,2}^{(a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} \\
& \boxed{\text{z735}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{2u}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
& \boxed{\text{z736}} \quad \mathbb{G}_2^{(1,-1;c)}(B_{2u}, b) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8} \\
& \boxed{\text{z737}} \quad \mathbb{G}_4^{(1,-1;c)}(B_{2u}) \\
& = \frac{\sqrt{13} \left(\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{208}
\end{aligned}$$

$$\boxed{\text{z738}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 1a) \\ = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$$\boxed{\text{z739}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 1b) \\ = \frac{\sqrt{5} \left(-3\sqrt{66}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 19\sqrt{14}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{91}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 7\sqrt{77}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 19\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9\sqrt{14}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{91}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{77}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 19\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{91}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{77}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{320}$$

$$\boxed{\text{z740}} \quad \mathbb{G}_6^{(1,-1;c)}(B_{2u}, 2) \\ = \frac{-\sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{858}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 91\sqrt{33}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{39}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9\sqrt{858}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9\sqrt{14}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{91}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{77}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 19\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 9\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{91}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{77}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g, 4)\mathbb{T}_{1,1}^{(b)}(E_u)}{832}$$

$$\boxed{\text{z741}} \quad \mathbb{G}_2^{(1,1;c)}(B_{2u}, a) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z742}} \quad \mathbb{G}_2^{(1,1;c)}(B_{2u}, b) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{8}$$

$$\boxed{\text{z743}} \quad \mathbb{G}_4^{(1,1;c)}(B_{2u}) \\ = \frac{\sqrt{13} \left(\sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{21}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{21}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{208}$$

$$\boxed{\text{z744}} \quad \mathbb{G}_6^{(1,1;c)}(B_{2u}, 1) \\ = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{15}\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$$\boxed{\text{z987}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z988}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z989}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z990}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z991}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\sqrt{2} \left(-\sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + 3 \mathbb{Q}_{4,1}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z992}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\sqrt{2} \left(\sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) - 3 \mathbb{Q}_{4,2}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z993}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1a) = \frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_g, 1) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z994}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1a) = \frac{\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_g, 1) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z995}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1b) = -\frac{\sqrt{2} \left(3 \mathbb{Q}_{4,1}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7} \mathbb{Q}_{4,1}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z996}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1b) = \frac{\sqrt{2} \left(3 \mathbb{Q}_{4,2}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7} \mathbb{Q}_{4,2}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z997}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1c) = \frac{\sqrt{383} \left(3\sqrt{30} \mathbb{Q}_{6,1}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) - 3\sqrt{55} \mathbb{Q}_{6,1}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) - \mathbb{Q}_{6,1}^{(a)}(E_g, 3) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{766}$$

$$\boxed{\text{z998}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1c) = \frac{\sqrt{383} \left(-3\sqrt{30} \mathbb{Q}_{6,2}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{55} \mathbb{Q}_{6,2}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{Q}_{6,2}^{(a)}(E_g, 3) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{766}$$

$$\boxed{\text{z999}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2a) = \frac{\sqrt{2} \mathbb{Q}_{4,1}^{(a)}(E_g, 2) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1000}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2a) = \frac{\sqrt{2} \mathbb{Q}_{4,2}^{(a)}(E_g, 2) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1001}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 2b) = \frac{\sqrt{383} \left(-460 \mathbb{Q}_{6,1}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) - 51\sqrt{66} \mathbb{Q}_{6,1}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) + 303\sqrt{30} \mathbb{Q}_{6,1}^{(a)}(E_g, 3) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{49024}$$

$$\boxed{\text{z1002}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 2b) = \frac{\sqrt{383} \left(460 \mathbb{Q}_{6,2}^{(a)}(E_g, 1) \mathbb{Q}_2^{(b)}(B_{1g}) + 51\sqrt{66} \mathbb{Q}_{6,2}^{(a)}(E_g, 2) \mathbb{Q}_2^{(b)}(B_{1g}) - 303\sqrt{30} \mathbb{Q}_{6,2}^{(a)}(E_g, 3) \mathbb{Q}_2^{(b)}(B_{1g}) \right)}{49024}$$

$$\boxed{\text{z1003}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 1a) = \frac{\sqrt{2} \mathbb{Q}_{6,1}^{(a)}(E_g, 1) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1004}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 1a) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1005}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 1b) = \frac{12\sqrt{33}\mathbb{Q}_{6,1}^{(a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 35\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{110}\mathbb{Q}_{6,1}^{(a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{128}$$

$$\boxed{\text{z1006}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 1b) = -\frac{12\sqrt{33}\mathbb{Q}_{6,2}^{(a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 35\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{110}\mathbb{Q}_{6,2}^{(a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{128}$$

$$\boxed{\text{z1007}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1008}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1009}} \quad \mathbb{Q}_{6,1}^{(c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1010}} \quad \mathbb{Q}_{6,2}^{(c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1011}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1012}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1013}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1014}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1015}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, c) = \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z1016}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, c) = \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 3\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\begin{aligned}
\boxed{\text{z1017}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1018}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1019}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1b) &= -\frac{\sqrt{2}\left(3\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z1020}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{2}\left(3\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8} \\
\boxed{\text{z1021}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1c) &= \frac{\sqrt{383}\left(3\sqrt{30}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 3\sqrt{55}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - \mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{766} \\
\boxed{\text{z1022}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1c) &= \frac{\sqrt{383}\left(-3\sqrt{30}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{55}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{766} \\
\boxed{\text{z1023}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1024}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1025}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{383}\left(-460\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 51\sqrt{66}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 303\sqrt{30}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{49024} \\
\boxed{\text{z1026}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{383}\left(460\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 51\sqrt{66}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - 303\sqrt{30}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{49024} \\
\boxed{\text{z1027}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1028}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1029}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 1b) &= \frac{12\sqrt{33}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 35\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{110}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{128}
\end{aligned}$$

$$\boxed{\text{z1030}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 1b) = -\frac{12\sqrt{33}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 35\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\sqrt{110}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})}{128}$$

$$\boxed{\text{z1031}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1032}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1033}} \quad \mathbb{Q}_{6,1}^{(1,-1;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1034}} \quad \mathbb{Q}_{6,2}^{(1,-1;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1035}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) = -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1036}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$$\boxed{\text{z1037}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) = \frac{\sqrt{2}\left(\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - \sqrt{15}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z1038}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) = \frac{\sqrt{2}\left(-\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{15}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z1039}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 1a) = \frac{\sqrt{218}\left(-6\sqrt{3}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{G}_{5,1}^{(1,0;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{218}$$

$$\boxed{\text{z1040}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 1a) = \frac{\sqrt{218}\left(6\sqrt{3}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - \mathbb{G}_{5,2}^{(1,0;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{218}$$

$$\boxed{\text{z1041}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 1b) = -\frac{\sqrt{2}\left(\sqrt{15}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\boxed{\text{z1042}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 1b) = \frac{\sqrt{2}\left(\sqrt{15}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g})\right)}{8}$$

$$\begin{aligned}
\boxed{\text{z1043}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(E_g, 2) &= \frac{\sqrt{218} \left(109\sqrt{15}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 19\sqrt{21}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) - 342\sqrt{7}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{20928} \\
\boxed{\text{z1044}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(E_g, 2) &= \frac{\sqrt{218} \left(-109\sqrt{15}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 19\sqrt{21}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 342\sqrt{7}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{20928} \\
\boxed{\text{z1045}} \quad \mathbb{Q}_{6,1}^{(1,0;c)}(E_g, 1) &= -\frac{\sqrt{2} \left(19\sqrt{21}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{15}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 18\sqrt{5}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{192} \\
\boxed{\text{z1046}} \quad \mathbb{Q}_{6,2}^{(1,0;c)}(E_g, 1) &= \frac{\sqrt{2} \left(19\sqrt{21}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{15}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) + 18\sqrt{5}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 3)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{192} \\
\boxed{\text{z1047}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1048}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1049}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, b) &= -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1050}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, b) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2} \\
\boxed{\text{z1051}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, c) &= \frac{\sqrt{2} \left(-\sqrt{7}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + 3\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8} \\
\boxed{\text{z1052}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, c) &= \frac{\sqrt{2} \left(\sqrt{7}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) - 3\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8} \\
\boxed{\text{z1053}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1054}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\boxed{\text{z1055}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 1b) &= -\frac{\sqrt{2} \left(3\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}
\end{aligned}$$

$$\boxed{\text{z1056}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 1b) = \frac{\sqrt{2} \left(3\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 1)\mathbb{Q}_2^{(b)}(B_{1g}) + \sqrt{7}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 2)\mathbb{Q}_2^{(b)}(B_{1g}) \right)}{8}$$

$$\boxed{\text{z1057}} \quad \mathbb{Q}_{4,1}^{(1,1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1058}} \quad \mathbb{Q}_{4,2}^{(1,1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1059}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1060}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1061}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1062}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1063}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1064}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1065}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1066}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1067}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1068}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1069}} \quad \mathbb{G}_{5,1}^{(1,0;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1070}} \quad \mathbb{G}_{5,2}^{(1,0;c)}(E_g, 3) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g, 3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z1331}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1332}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1333}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 1) = \frac{\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1334}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 1) = -\frac{\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1335}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2) = \frac{\sqrt{11}\left(\sqrt{15}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{44}$$

$$\boxed{\text{z1336}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2) = \frac{\sqrt{11}\left(-\sqrt{15}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 8\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{44}$$

$$\boxed{\text{z1337}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 1) = \frac{\sqrt{2}\left(-\sqrt{15}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{16}$$

$$\boxed{\text{z1338}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 1) = \frac{\sqrt{2}\left(\sqrt{15}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{16}$$

$$\boxed{\text{z1339}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 2) = \frac{\sqrt{46}\left(-16\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 11\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 16\sqrt{3}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{368}$$

$$\boxed{\text{z1340}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 2) = \frac{\sqrt{46}\left(-16\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{35}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{3}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)\right)}{368}$$

$$\boxed{\text{z1341}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u, 3) = \frac{\sqrt{1334}\left(116\sqrt{21}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{15}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{21}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 20\sqrt{7}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{37352}$$

$$\boxed{\text{z1342}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u, 3) = \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 49\sqrt{15}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{21}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{7}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{37352}$$

$$\boxed{\text{z1343}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1344}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1345}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1346}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1347}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{11} \left(\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1348}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{11} \left(-\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 8\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1349}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2} \left(-\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{16}$$

$$\boxed{\text{z1350}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16}$$

$$\boxed{\text{z1351}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{46} \left(-16\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 11\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 16\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{368}$$

$$\boxed{\text{z1352}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 2) = \frac{\sqrt{46} \left(-16\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{35}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{368}$$

$$\boxed{\text{z1353}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 20\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{37352}$$

$$\begin{aligned}
\boxed{\text{z1354}} \quad & \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, 3) \\
= & \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 49\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{37352} \\
\boxed{\text{z1355}} \quad & \mathbb{Q}_{7,1}^{(1,-1;c)}(E_u, 1) = \frac{5\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{429}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 15\sqrt{3}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} \\
\boxed{\text{z1356}} \quad & \mathbb{Q}_{7,2}^{(1,-1;c)}(E_u, 1) = -\frac{5\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{66}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{429}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 15\sqrt{3}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} \\
\boxed{\text{z1357}} \quad & \mathbb{Q}_{7,1}^{(1,-1;c)}(E_u, 2) \\
= & \frac{-32\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{462}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 43\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 32\sqrt{6}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{13}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{192} \\
\boxed{\text{z1358}} \quad & \mathbb{Q}_{7,2}^{(1,-1;c)}(E_u, 2) \\
= & \frac{-32\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{462}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 43\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 32\sqrt{6}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 9\sqrt{13}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{192} \\
\boxed{\text{z1359}} \quad & \mathbb{Q}_{7,1}^{(1,-1;c)}(E_u, 3) \\
= & \frac{\sqrt{355} \left(320\sqrt{26}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 9\sqrt{6006}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 65\sqrt{26}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 192\sqrt{66}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 64\sqrt{78}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 465\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{68160} \\
\boxed{\text{z1360}} \quad & \mathbb{Q}_{7,2}^{(1,-1;c)}(E_u, 3) \\
= & \frac{\sqrt{355} \left(320\sqrt{26}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 9\sqrt{6006}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 65\sqrt{26}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 192\sqrt{66}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 64\sqrt{78}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 465\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{68160} \\
\boxed{\text{z1361}} \quad & \mathbb{Q}_{7,1}^{(1,-1;c)}(E_u, 4) \\
= & \frac{\sqrt{103447} \left(-27104\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 4371\sqrt{462}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 44165\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 7872\sqrt{858}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 95392\sqrt{6}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 85304\sqrt{13}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 11\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{218480064} \\
\boxed{\text{z1362}} \quad & \mathbb{Q}_{7,2}^{(1,-1;c)}(E_u, 4) \\
= & \frac{\sqrt{103447} \left(-27104\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 4371\sqrt{462}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 44165\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 7872\sqrt{858}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 95392\sqrt{6}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 85304\sqrt{13}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{218480064} \\
\boxed{\text{z1363}} \quad & \mathbb{Q}_{1,1}^{(1,0;c)}(E_u) = \frac{\sqrt{14} \left(-\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{14}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z1364}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u) &= \frac{\sqrt{14} \left(-\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{14} \\
\boxed{\text{z1365}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1a) &= \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 4\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z1366}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1a) &= -\frac{\sqrt{14} \left(\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z1367}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1b) &= \frac{\sqrt{13} \left(4\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 6\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{156} \\
\boxed{\text{z1368}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1b) &= \frac{\sqrt{13} \left(4\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 6\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{5}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{156} \\
\boxed{\text{z1369}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2a) &= -\frac{\sqrt{2} \left(\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{4} \\
\boxed{\text{z1370}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2a) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{4} \\
\boxed{\text{z1371}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2b) &= \frac{\sqrt{377} \left(28\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 146\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 441\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 186\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 249\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{31668} \\
\boxed{\text{z1372}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2b) &= \frac{\sqrt{377} \left(28\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 146\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 441\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 186\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 249\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{31668} \\
\boxed{\text{z1373}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u, 1a) &= \frac{\sqrt{58} \left(1050\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 123\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 392\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 59\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 88\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{24360} \\
\boxed{\text{z1374}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u, 1a) &= \frac{\sqrt{58} \left(1050\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 123\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 392\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 59\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 88\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{24360}
\end{aligned}$$

$$\begin{aligned} \boxed{\text{z1375}} \quad & \mathbb{Q}_{5,1}^{(1,0;c)}(Eu, 1b) \\ = & \frac{\sqrt{19} \left(8\sqrt{3}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 4\sqrt{21}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 2\sqrt{42}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + 12\sqrt{7}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 3\sqrt{77}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{35}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{304} \end{aligned}$$

$$\begin{aligned} \boxed{\text{z1376}} \quad & \mathbb{Q}_{5,2}^{(1,0;c)}(Eu, 1b) \\ = & \frac{\sqrt{19} \left(8\sqrt{3}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 4\sqrt{21}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 2\sqrt{42}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - 12\sqrt{7}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{77}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{35}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{304} \end{aligned}$$

$$\begin{aligned} \boxed{\text{z1377}} \quad & \mathbb{Q}_{5,1}^{(1,0;c)}(Eu, 2a) \\ = & \frac{\sqrt{62} \left(150\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 9\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 16\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) - 23\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(Eu) - 16\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{3720} \end{aligned}$$

$$\begin{aligned} \boxed{\text{z1378}} \quad & \mathbb{Q}_{5,2}^{(1,0;c)}(Eu, 2a) \\ = & \frac{\sqrt{62} \left(150\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 9\sqrt{35}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 16\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) + 23\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(Eu) - 16\sqrt{105}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{3720} \end{aligned}$$

$$\begin{aligned} \boxed{\text{z1379}} \quad & \mathbb{Q}_{5,1}^{(1,0;c)}(Eu, 2b) \\ = & \frac{-8\sqrt{35}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 4\sqrt{5}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 14\sqrt{10}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + 4\sqrt{15}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{165}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 23\sqrt{3}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu)}{112} \end{aligned}$$

$$\begin{aligned} \boxed{\text{z1380}} \quad & \mathbb{Q}_{5,2}^{(1,0;c)}(Eu, 2b) \\ = & \frac{-8\sqrt{35}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 4\sqrt{5}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 14\sqrt{10}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - 4\sqrt{15}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{165}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 23\sqrt{3}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu)}{112} \end{aligned}$$

$$\boxed{\text{z1381}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(Eu, 3a) = \frac{\sqrt{62} \left(4\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 5\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 6\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(Eu) + 6\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{372}$$

$$\boxed{\text{z1382}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(Eu, 3a) = \frac{\sqrt{62} \left(4\sqrt{15}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 5\sqrt{21}\mathbb{T}_4^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 6\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) + 3\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(Eu) + 6\sqrt{7}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{372}$$

$$\begin{aligned} \boxed{\text{z1383}} \quad & \mathbb{Q}_{5,1}^{(1,0;c)}(Eu, 3b) \\ = & \frac{\sqrt{930563} \left(-32\sqrt{105}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 4524\sqrt{15}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 1806\sqrt{30}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) - 7448\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 1324\sqrt{5}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 2193\sqrt{55}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 1324\sqrt{5}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{52111528} \end{aligned}$$

$$\begin{aligned}
& \boxed{\text{z1384}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(Eu, 3b) \\
& = \frac{\sqrt{930563} \left(-32\sqrt{105}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 4524\sqrt{15}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 1806\sqrt{30}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) + 7448\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 1324\sqrt{5}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 2193\sqrt{55}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{52111528} \\
& \boxed{\text{z1385}} \quad \mathbb{Q}_{7,1}^{(1,0;c)}(Eu, 1) \\
& = \frac{\sqrt{118671271} \left(5206\sqrt{7}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 10901\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 18318\sqrt{2}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + 784\sqrt{165}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 18741\sqrt{3}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 6579\sqrt{33}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{949370168} \\
& \boxed{\text{z1386}} \quad \mathbb{Q}_{7,2}^{(1,0;c)}(Eu, 1) \\
& = \frac{\sqrt{118671271} \left(5206\sqrt{7}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 10901\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 18318\sqrt{2}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - 784\sqrt{165}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 18741\sqrt{3}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 6579\sqrt{33}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{949370168} \\
& \boxed{\text{z1387}} \quad \mathbb{Q}_{7,1}^{(1,0;c)}(Eu, 2) \\
& = \frac{\sqrt{235031} \left(-1974\sqrt{33}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 99\sqrt{231}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 538\sqrt{462}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + 1056\sqrt{35}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + 663\sqrt{77}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 2087\sqrt{7}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{13161736} \\
& \boxed{\text{z1388}} \quad \mathbb{Q}_{7,2}^{(1,0;c)}(Eu, 2) \\
& = \frac{\sqrt{235031} \left(-1974\sqrt{33}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 99\sqrt{231}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 538\sqrt{462}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - 1056\sqrt{35}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 663\sqrt{77}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 2087\sqrt{7}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{13161736} \\
& \boxed{\text{z1389}} \quad \mathbb{Q}_{7,1}^{(1,0;c)}(Eu, 3) \\
& = \frac{\sqrt{117079} \left(896\sqrt{66}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 435\sqrt{462}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 128\sqrt{231}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + 2150\sqrt{70}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 691\sqrt{154}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) - 1056\sqrt{14}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{13112848} \\
& \boxed{\text{z1390}} \quad \mathbb{Q}_{7,2}^{(1,0;c)}(Eu, 3) \\
& = \frac{\sqrt{117079} \left(896\sqrt{66}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 435\sqrt{462}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 128\sqrt{231}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - 2150\sqrt{70}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 691\sqrt{154}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 1056\sqrt{14}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{13112848} \\
& \boxed{\text{z1391}} \quad \mathbb{Q}_{7,1}^{(1,0;c)}(Eu, 4) \\
& = \frac{\sqrt{1207} \left(-32\sqrt{14}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 193\sqrt{2}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 32\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(Eu) + 2\sqrt{330}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 43\sqrt{6}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 8\sqrt{66}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{19312} \\
& \boxed{\text{z1392}} \quad \mathbb{Q}_{7,2}^{(1,0;c)}(Eu, 4) \\
& = \frac{\sqrt{1207} \left(-32\sqrt{14}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) - 193\sqrt{2}\mathbb{T}_6^{(1,0;a)}(A_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 32\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(Eu) - 2\sqrt{330}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + 43\sqrt{6}\mathbb{T}_6^{(1,0;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 8\sqrt{66}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(Eu) - 40\sqrt{11}\mathbb{T}_6^{(1,0;a)}(B_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(Eu) \right)}{19312}
\end{aligned}$$

$$\boxed{\text{z1393}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1394}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z1395}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1396}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$$\boxed{\text{z1397}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{11} \left(\sqrt{15}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + 8\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1398}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{11} \left(-\sqrt{15}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - 8\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{44}$$

$$\boxed{\text{z1399}} \quad \mathbb{Q}_{5,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2} \left(-\sqrt{15}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{21}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{16}$$

$$\boxed{\text{z1400}} \quad \mathbb{Q}_{5,2}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{21}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 2\sqrt{7}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{16}$$

$$\boxed{\text{z1401}} \quad \mathbb{Q}_{5,1}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{46} \left(-16\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{35}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 11\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 16\sqrt{3}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{368}$$

$$\boxed{\text{z1402}} \quad \mathbb{Q}_{5,2}^{(1,1;c)}(E_u, 2) = \frac{\sqrt{46} \left(-16\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 3\sqrt{35}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 11\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 16\sqrt{3}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{3}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{368}$$

$$\boxed{\text{z1403}} \quad \mathbb{Q}_{5,1}^{(1,1;c)}(E_u, 3) = \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 49\sqrt{15}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{21}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 20\sqrt{7}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{37352}$$

$$\boxed{\text{z1404}} \quad \mathbb{Q}_{5,2}^{(1,1;c)}(E_u, 3) = \frac{\sqrt{1334} \left(116\sqrt{21}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 49\sqrt{15}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{21}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 20\sqrt{7}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 78\sqrt{7}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{37352}$$

$$\begin{aligned}
\boxed{\text{z1405}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) &= \frac{\sqrt{22} \left(-\sqrt{5}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{22} \\
\boxed{\text{z1406}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) &= \frac{\sqrt{22} \left(\sqrt{5}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{22} \\
\boxed{\text{z1407}} \quad \mathbb{G}_{4,1}^{(c)}(E_u, 1) &= \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 343\sqrt{5}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{7}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 79\sqrt{21}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{200564} \\
\boxed{\text{z1408}} \quad \mathbb{G}_{4,2}^{(c)}(E_u, 1) &= \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 343\sqrt{5}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{7}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 79\sqrt{21}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{200564} \\
\boxed{\text{z1409}} \quad \mathbb{G}_{4,1}^{(c)}(E_u, 2) &= \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{15}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{988} \\
\boxed{\text{z1410}} \quad \mathbb{G}_{4,2}^{(c)}(E_u, 2) &= \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_5^{(a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{5}\mathbb{M}_5^{(a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{15}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{988} \\
\boxed{\text{z1411}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{22} \left(-\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{22} \\
\boxed{\text{z1412}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{22} \left(\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{22} \\
\boxed{\text{z1413}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 343\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 79\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{200564} \\
\boxed{\text{z1414}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 1) &= \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 343\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 79\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{200564} \\
\boxed{\text{z1415}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 2) &= \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{988}
\end{aligned}$$

$$\begin{aligned}
& \boxed{\text{z1416}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 2) \\
& \quad \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{5}\mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{988} \\
& \boxed{\text{z1417}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 1) \\
& \quad \frac{\sqrt{7285} \left(-49610\sqrt{42}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 275373\sqrt{22}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 419265\sqrt{42}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 17416\sqrt{2002}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 703406\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{525044520} \\
& \boxed{\text{z1418}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 1) \\
& \quad \frac{\sqrt{7285} \left(-49610\sqrt{42}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 275373\sqrt{22}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 419265\sqrt{42}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 17416\sqrt{2002}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 703406\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{525044520} \\
& \boxed{\text{z1419}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 2) \\
& \quad \frac{\sqrt{36793} \left(145972\sqrt{21}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 662634\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 220878\sqrt{21}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 22792\sqrt{1001}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 146996\sqrt{7}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{964270944} \\
& \boxed{\text{z1420}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 2) \\
& \quad \frac{\sqrt{36793} \left(145972\sqrt{21}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 662634\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 220878\sqrt{21}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 22792\sqrt{1001}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 146996\sqrt{7}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{964270944} \\
& \boxed{\text{z1421}} \quad \mathbb{G}_{6,1}^{(1,-1;c)}(E_u, 3) \\
& \quad \frac{\sqrt{36793} \left(-900\sqrt{91}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 90\sqrt{429}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 90\sqrt{91}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 488\sqrt{231}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 44\sqrt{273}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 1485\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{3532128} \\
& \boxed{\text{z1422}} \quad \mathbb{G}_{6,2}^{(1,-1;c)}(E_u, 3) \\
& \quad \frac{\sqrt{36793} \left(-900\sqrt{91}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 90\sqrt{429}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 90\sqrt{91}\mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 488\sqrt{231}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 44\sqrt{273}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 1485\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(B_{1g}, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{3532128} \\
& \boxed{\text{z1423}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{22} \left(-\sqrt{5}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{22} \\
& \boxed{\text{z1424}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{22} \left(\sqrt{5}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{22}
\end{aligned}$$

$$\boxed{\text{z1425}} \quad \mathbb{G}_{4,1}^{(1,1;c)}(E_u, 1) \\ = \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 343\sqrt{5}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 87\sqrt{7}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 79\sqrt{21}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{200564}$$

$$\boxed{\text{z1426}} \quad \mathbb{G}_{4,2}^{(1,1;c)}(E_u, 1) \\ = \frac{\sqrt{14326} \left(-87\sqrt{7}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 343\sqrt{5}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 87\sqrt{7}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 79\sqrt{21}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) + 166\sqrt{21}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{200564}$$

$$\boxed{\text{z1427}} \quad \mathbb{G}_{4,1}^{(1,1;c)}(E_u, 2) \\ = \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{7}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 7\sqrt{5}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 5\sqrt{15}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{988}$$

$$\boxed{\text{z1428}} \quad \mathbb{G}_{4,2}^{(1,1;c)}(E_u, 2) \\ = \frac{\sqrt{494} \left(7\sqrt{5}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 3\sqrt{7}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + 7\sqrt{5}\mathbb{M}_5^{(1,1;a)}(A_{2g}, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 5\sqrt{15}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{15}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{988}$$

Atomic SAMB

- bra: $\langle \frac{3}{2}, \frac{3}{2}; d |, \langle \frac{3}{2}, \frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, \frac{5}{2}; d |, \langle \frac{5}{2}, \frac{3}{2}; d |, \langle \frac{5}{2}, \frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, -\frac{5}{2}; d |$
- ket: $|\frac{3}{2}, \frac{3}{2}; d \rangle, |\frac{3}{2}, \frac{1}{2}; d \rangle, |\frac{3}{2}, -\frac{1}{2}; d \rangle, |\frac{3}{2}, -\frac{3}{2}; d \rangle, |\frac{5}{2}, \frac{5}{2}; d \rangle, |\frac{5}{2}, \frac{3}{2}; d \rangle, |\frac{5}{2}, \frac{1}{2}; d \rangle, |\frac{5}{2}, -\frac{1}{2}; d \rangle, |\frac{5}{2}, -\frac{3}{2}; d \rangle, |\frac{5}{2}, -\frac{5}{2}; d \rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{Q}_2^{(a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{Q}_4^{(a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{\sqrt{15}}{60} \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{Q}_4^{(a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{30} & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} \\ 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} \end{bmatrix}$$

$$\boxed{\text{x5}} \quad \mathbb{Q}_4^{(a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_2^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 \\ -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} \\ 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{70} & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 \\ -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\ 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_4^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} \\ 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{Q}_2^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 \\ -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} \\ 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{70} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & -\frac{\sqrt{210}i}{70} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{70} & 0 \\ -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{70} & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} \\ 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{Q}_4^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & \frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} \\ 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{7}i}{70} & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & -\frac{\sqrt{105}i}{70} \\ -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 \\ -\frac{3\sqrt{42}i}{140} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & -\frac{\sqrt{105}i}{35} \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x11}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{7}}{70} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & \frac{\sqrt{105}}{70} \\ \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 \\ -\frac{3\sqrt{42}}{140} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 & \frac{\sqrt{105}}{35} \\ 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{Q}_{4,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{40} & 0 & \frac{\sqrt{2}i}{8} & 0 & \frac{i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & -\frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{15}i}{40} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{3}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{5}i}{40} \\ -\frac{\sqrt{5}i}{40} & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{8} & 0 & \frac{i}{8} & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} \\ 0 & \frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{2}i}{8} & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ -\frac{i}{8} & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{5}i}{20} \\ 0 & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{5}i}{40} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{5}i}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{Q}_{4,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{40} & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{1}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{15}}{40} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{3}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{5}}{40} \\ \frac{\sqrt{5}}{40} & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{8} & 0 & \frac{1}{8} & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} \\ 0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{2}}{8} & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 \\ \frac{1}{8} & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{5}}{20} \\ 0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{5}}{40} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{5}}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{Q}_{4,1}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{280} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{105}i}{40} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{40} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{8} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{280} \\ -\frac{\sqrt{35}i}{280} & 0 & \frac{\sqrt{105}i}{40} & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{7}i}{8} & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{70}i}{40} \\ 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{14}i}{56} & \frac{\sqrt{70}i}{40} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\ \frac{\sqrt{7}i}{8} & 0 & -\frac{\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{140} \\ 0 & -\frac{\sqrt{105}i}{40} & 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & -\frac{\sqrt{70}i}{40} & 0 & \frac{\sqrt{35}i}{140} & 0 \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{Q}_{4,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{280} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{105}}{40} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{40} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{21}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{8} & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{280} \\ \frac{\sqrt{35}}{280} & 0 & \frac{\sqrt{105}}{40} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{70}}{40} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{56} & 0 & -\frac{\sqrt{7}}{8} & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{70}}{40} \\ 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{14}}{56} & -\frac{\sqrt{70}}{40} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{7}}{8} & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{140} \\ 0 & \frac{\sqrt{105}}{40} & 0 & \frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{70}}{40} & 0 & \frac{\sqrt{35}}{140} & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{Q}_2^{(1,-1;a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \end{bmatrix}$$

$$\boxed{\text{x17}} \quad \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \end{bmatrix}$$

$$\boxed{\text{x18}} \quad \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{60} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 \\ \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & \frac{\sqrt{105}}{30} \\ 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \end{bmatrix}$$

$$\boxed{\text{x19}} \quad \mathbb{Q}_4^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x20}} \quad \mathbb{Q}_2^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} & 0 \\ -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} \\ 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & \frac{1}{5} & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 \\ -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & \frac{1}{5} \\ 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x21}} \quad \mathbb{Q}_4^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \\ 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x22}} \quad \mathbb{Q}_2^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{50} & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 \\ -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} \\ 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}i}{50} & \frac{i}{5} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 \\ -\frac{3\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & 0 & -\frac{i}{5} \\ 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x23}} \quad \mathbb{Q}_4^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\ \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x24}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{30}i}{100} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} & 0 & -\frac{3\sqrt{2}i}{20} \\ -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{100} & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & -\frac{2\sqrt{5}i}{25} & 0 & 0 & 0 \\ -\frac{9\sqrt{5}i}{100} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{\sqrt{2}i}{5} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 \end{bmatrix}$$

$$\boxed{\text{x25}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{30}}{100} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{100} & 0 & \frac{3\sqrt{2}}{20} \\ \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ -\frac{9\sqrt{5}}{100} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & -\frac{\sqrt{2}}{5} \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 \end{bmatrix}$$

$$\boxed{\text{x26}} \quad \mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{15}i}{80} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{3}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{80} \\ -\frac{\sqrt{5}i}{80} & 0 & -\frac{\sqrt{15}i}{80} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{16} & 0 & \frac{i}{16} & \frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{2}i}{16} & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{i}{16} & 0 & -\frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{5}i}{10} \\ 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{\sqrt{5}i}{80} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x27}} \quad \mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{80} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{1}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{15}}{80} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{3}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{80} \\ \frac{\sqrt{5}}{80} & 0 & -\frac{\sqrt{15}}{80} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{1}{16} & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & \frac{\sqrt{10}}{20} \\ 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{2}}{16} & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ \frac{1}{16} & 0 & -\frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{5}}{10} \\ 0 & -\frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{5}}{80} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x28}} \quad \mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{560} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{7}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{105}i}{80} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{21}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{16} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{35}i}{560} \\ -\frac{\sqrt{35}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{20} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{7}i}{16} & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{20} \\ 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{14}i}{112} & -\frac{\sqrt{70}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ \frac{\sqrt{7}i}{16} & 0 & -\frac{\sqrt{21}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{35}i}{70} \\ 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{70}i}{20} & 0 & -\frac{\sqrt{35}i}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x29}} \quad \mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{560} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{105}}{80} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{80} & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{21}}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{16} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{560} \\ \frac{\sqrt{35}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{7}}{16} & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{20} \\ 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{70}}{20} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{7}}{16} & 0 & -\frac{\sqrt{21}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{35}}{70} \\ 0 & \frac{\sqrt{105}}{80} & 0 & \frac{\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{70}}{20} & 0 & -\frac{\sqrt{35}}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x30}} \quad \mathbb{Q}_0^{(1,1;a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} \end{bmatrix}$$

$$\boxed{\text{x31}} \quad \mathbb{Q}_2^{(1,1;a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \\ -\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \end{bmatrix}$$

$$\boxed{\text{x32}} \quad \mathbb{Q}_2^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{25} & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & -\frac{8\sqrt{105}}{525} & 0 \\ \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{105}}{525} & 0 & 0 & 0 & -\frac{4\sqrt{21}}{105} \\ 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & 0 & 0 \\ 0 & 0 & \frac{8\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & -\frac{3\sqrt{14}}{140} & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 \\ -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} \\ 0 & -\frac{8\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x33}} \quad \mathbb{Q}_2^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}i}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{25} & \frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & \frac{8\sqrt{105}i}{525} & 0 \\ \frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & \frac{4\sqrt{21}i}{105} \\ 0 & \frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}i}{175} & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{700} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{70}i}{175} & -\frac{3\sqrt{14}i}{140} & 0 & 0 & 0 & \frac{9\sqrt{70}i}{700} & 0 \\ -\frac{2\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{700} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{140} \\ 0 & -\frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{700} & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{140} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x34}} \quad \mathbb{Q}_{2,1}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{525} & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{25} & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{2\sqrt{105}i}{525} & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{175} & 0 & -\frac{2\sqrt{7}i}{35} \\ -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{105}i}{525} & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{70}i}{350} & 0 & 0 & 0 \\ -\frac{3\sqrt{70}i}{175} & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{3\sqrt{70}i}{350} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{350} & 0 \\ 0 & 0 & -\frac{2\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{350} & 0 & -\frac{3\sqrt{7}i}{70} \\ 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x35}} \quad \mathbb{Q}_{2,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{105}}{25} & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{525} & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & -\frac{2\sqrt{105}}{525} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & \frac{2\sqrt{7}}{35} \\ \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{105}}{525} & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & -\frac{3\sqrt{70}}{350} & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{175} & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & -\frac{3\sqrt{70}}{350} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{105} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{350} & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{525} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{350} & 0 & \frac{3\sqrt{7}}{70} \\ 0 & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x36}} \quad \mathbb{G}_1^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x37}} \quad \mathbb{G}_3^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x38}} \quad \mathbb{G}_3^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x39}} \quad \mathbb{G}_3^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x40}} \quad \mathbb{G}_{1,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x41}} \quad \mathbb{G}_{1,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x42}} \quad \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{30}i}{80} & 0 & \frac{\sqrt{15}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}i}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & \frac{5i}{16} \\ 0 & 0 & 0 & 0 & -\frac{5i}{16} & 0 & -\frac{\sqrt{10}i}{80} & 0 & \frac{7\sqrt{5}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & \frac{3\sqrt{30}i}{80} & 0 & -\frac{\sqrt{3}i}{16} \\ -\frac{\sqrt{3}i}{16} & 0 & \frac{5i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{5}i}{80} & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{80} & 0 & -\frac{3\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{16} & 0 & -\frac{7\sqrt{5}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5i}{16} & 0 & \frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x43}} \quad \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{30}}{80} & 0 & \frac{\sqrt{15}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & \frac{5}{16} \\ 0 & 0 & 0 & 0 & \frac{5}{16} & 0 & -\frac{\sqrt{10}}{80} & 0 & -\frac{7\sqrt{5}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{16} & 0 & \frac{3\sqrt{30}}{80} & 0 & \frac{\sqrt{3}}{16} \\ \frac{\sqrt{3}}{16} & 0 & \frac{5}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{5}}{80} & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{80} & 0 & \frac{3\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{16} & 0 & -\frac{7\sqrt{5}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5}{16} & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x44}} \quad \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & -\frac{3i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}i}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{15}i}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{7\sqrt{3}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{16} \\ -\frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{3}i}{48} & 0 & -\frac{3i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{16} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{48} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3i}{16} & 0 & -\frac{7\sqrt{3}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{16} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x45}} \quad \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & -\frac{3}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}}{48} & 0 & -\frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{15}}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{6}}{48} & 0 & -\frac{7\sqrt{3}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{16} \\ \frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{3}}{48} & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{16} & 0 & -\frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{48} & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{16} & 0 & -\frac{7\sqrt{3}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{16} & 0 & \frac{\sqrt{5}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x46}} \quad \mathbb{M}_1^{(a)}(A_{2g}) = \begin{bmatrix} \frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$$

$$\boxed{\text{x47}} \quad \mathbb{M}_3^{(a)}(A_{2g}) = \begin{bmatrix} -\frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \end{bmatrix}$$

$$\boxed{\text{x48}} \quad \mathbb{M}_3^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{5} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & \frac{i}{10} & 0 \\ -\frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 \\ -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x49}} \quad \mathbb{M}_3^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{5} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & -\frac{1}{10} & 0 \\ -\frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 \\ -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x50}} \quad \mathbb{M}_{1,1}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & -\frac{1}{10} & 0 & \frac{\sqrt{10}}{100} & 0 & 0 & 0 \\ \frac{3\sqrt{15}}{50} & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & \frac{\sqrt{30}}{100} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{25} & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & \frac{\sqrt{15}}{50} & 0 \\ 0 & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & \frac{1}{10} \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & \frac{1}{5} & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{100} & 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{100} & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & \frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & \frac{1}{5} \\ 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 \end{bmatrix}$$

$$\boxed{\text{x51}} \quad \mathbb{M}_{1,2}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{3\sqrt{15}i}{50} & 0 & 0 & \frac{i}{10} & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & 0 \\ -\frac{3\sqrt{15}i}{50} & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & \frac{\sqrt{15}i}{50} & 0 & \frac{\sqrt{30}i}{100} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{25} & 0 & \frac{3\sqrt{15}i}{50} & 0 & 0 & \frac{\sqrt{30}i}{100} & 0 & \frac{\sqrt{15}i}{50} & 0 \\ 0 & 0 & -\frac{3\sqrt{15}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} & 0 & \frac{i}{10} \\ -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & -\frac{i}{5} & 0 & \frac{2\sqrt{10}i}{25} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{100} & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 & -\frac{2\sqrt{10}i}{25} & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{100} & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & \frac{2\sqrt{10}i}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{25} & 0 & \frac{i}{5} \\ 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 \end{bmatrix}$$

$$\boxed{\text{x52}} \quad \mathbb{M}_{3,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{15}}{50} & 0 & -\frac{\sqrt{5}}{10} & -\frac{3}{40} & 0 & \frac{9\sqrt{10}}{200} & 0 & -\frac{3\sqrt{5}}{40} & 0 \\ \frac{\sqrt{15}}{50} & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & \frac{7\sqrt{15}}{200} & 0 & -\frac{\sqrt{30}}{200} & 0 & -\frac{\sqrt{3}}{8} \\ 0 & -\frac{3\sqrt{5}}{50} & 0 & \frac{\sqrt{15}}{50} & \frac{\sqrt{3}}{8} & 0 & \frac{\sqrt{30}}{200} & 0 & -\frac{7\sqrt{15}}{200} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{15}}{50} & 0 & 0 & \frac{3\sqrt{5}}{40} & 0 & -\frac{9\sqrt{10}}{200} & 0 & \frac{3}{40} \\ -\frac{3}{40} & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & \frac{3}{20} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & \frac{7\sqrt{15}}{200} & 0 & \frac{3\sqrt{5}}{40} & \frac{3}{20} & 0 & -\frac{3\sqrt{10}}{200} & 0 & -\frac{\sqrt{5}}{10} & 0 \\ \frac{9\sqrt{10}}{200} & 0 & \frac{\sqrt{30}}{200} & 0 & 0 & -\frac{3\sqrt{10}}{200} & 0 & -\frac{3\sqrt{5}}{50} & 0 & -\frac{\sqrt{2}}{8} \\ 0 & -\frac{\sqrt{30}}{200} & 0 & -\frac{9\sqrt{10}}{200} & -\frac{\sqrt{2}}{8} & 0 & -\frac{3\sqrt{5}}{50} & 0 & -\frac{3\sqrt{10}}{200} & 0 \\ -\frac{3\sqrt{5}}{40} & 0 & -\frac{7\sqrt{15}}{200} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{3\sqrt{10}}{200} & 0 & \frac{3}{20} \\ 0 & -\frac{\sqrt{3}}{8} & 0 & \frac{3}{40} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{3}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x53}} \quad \mathbb{M}_{3,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{15}i}{50} & 0 & \frac{\sqrt{5}i}{10} & \frac{3i}{40} & 0 & \frac{9\sqrt{10}i}{200} & 0 & \frac{3\sqrt{5}i}{40} & 0 \\ -\frac{\sqrt{15}i}{50} & 0 & -\frac{3\sqrt{5}i}{50} & 0 & 0 & -\frac{7\sqrt{15}i}{200} & 0 & -\frac{\sqrt{30}i}{200} & 0 & \frac{\sqrt{3}i}{8} \\ 0 & \frac{3\sqrt{5}i}{50} & 0 & \frac{\sqrt{15}i}{50} & \frac{\sqrt{3}i}{8} & 0 & -\frac{\sqrt{30}i}{200} & 0 & -\frac{7\sqrt{15}i}{200} & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & \frac{3\sqrt{5}i}{40} & 0 & \frac{9\sqrt{10}i}{200} & 0 & \frac{3i}{40} \\ -\frac{3i}{40} & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & \frac{3i}{20} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & \frac{7\sqrt{15}i}{200} & 0 & -\frac{3\sqrt{5}i}{40} & -\frac{3i}{20} & 0 & -\frac{3\sqrt{10}i}{200} & 0 & \frac{\sqrt{5}i}{10} & 0 \\ -\frac{9\sqrt{10}i}{200} & 0 & \frac{\sqrt{30}i}{200} & 0 & 0 & \frac{3\sqrt{10}i}{200} & 0 & -\frac{3\sqrt{5}i}{50} & 0 & \frac{\sqrt{2}i}{8} \\ 0 & \frac{\sqrt{30}i}{200} & 0 & -\frac{9\sqrt{10}i}{200} & -\frac{\sqrt{2}i}{8} & 0 & \frac{3\sqrt{5}i}{50} & 0 & -\frac{3\sqrt{10}i}{200} & 0 \\ -\frac{3\sqrt{5}i}{40} & 0 & \frac{7\sqrt{15}i}{200} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{3\sqrt{10}i}{200} & 0 & \frac{3i}{20} \\ 0 & -\frac{\sqrt{3}i}{8} & 0 & -\frac{3i}{40} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{3i}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x54}} \quad \mathbb{M}_{3,1}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{1}{10} & 0 & \frac{\sqrt{3}}{10} & -\frac{\sqrt{15}}{40} & 0 & \frac{3\sqrt{6}}{40} & 0 & \frac{3\sqrt{3}}{40} & 0 \\ \frac{1}{10} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{7}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & \frac{3\sqrt{5}}{40} \\ 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{1}{10} & -\frac{3\sqrt{5}}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & -\frac{7}{40} & 0 \\ \frac{\sqrt{3}}{10} & 0 & \frac{1}{10} & 0 & 0 & -\frac{3\sqrt{3}}{40} & 0 & -\frac{3\sqrt{6}}{40} & 0 & \frac{\sqrt{15}}{40} \\ -\frac{\sqrt{15}}{40} & 0 & -\frac{3\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 \\ 0 & \frac{7}{40} & 0 & -\frac{3\sqrt{3}}{40} & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{3}}{10} & 0 \\ \frac{3\sqrt{6}}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{30}}{40} \\ 0 & -\frac{\sqrt{2}}{40} & 0 & -\frac{3\sqrt{6}}{40} & \frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{40} & 0 \\ \frac{3\sqrt{3}}{40} & 0 & -\frac{7}{40} & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{15}}{20} \\ 0 & \frac{3\sqrt{5}}{40} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{15}}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x55}} \quad \mathbb{M}_{3,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{i}{10} & 0 & -\frac{\sqrt{3}i}{10} & \frac{\sqrt{15}i}{40} & 0 & \frac{3\sqrt{6}i}{40} & 0 & -\frac{3\sqrt{3}i}{40} & 0 \\ -\frac{i}{10} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{7i}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & -\frac{3\sqrt{5}i}{40} \\ 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{i}{10} & -\frac{3\sqrt{5}i}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & -\frac{7i}{40} & 0 \\ \frac{\sqrt{3}i}{10} & 0 & -\frac{i}{10} & 0 & 0 & -\frac{3\sqrt{3}i}{40} & 0 & \frac{3\sqrt{6}i}{40} & 0 & \frac{\sqrt{15}i}{40} \\ -\frac{\sqrt{15}i}{40} & 0 & \frac{3\sqrt{5}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 \\ 0 & \frac{7i}{40} & 0 & \frac{3\sqrt{3}i}{40} & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{3}i}{10} & 0 \\ -\frac{3\sqrt{6}i}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{30}i}{40} \\ 0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{3\sqrt{6}i}{40} & \frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{40} & 0 \\ \frac{3\sqrt{3}i}{40} & 0 & \frac{7i}{40} & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{15}i}{20} \\ 0 & \frac{3\sqrt{5}i}{40} & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x56}} \quad \mathbb{M}_5^{(1,-1;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x57}} \quad \mathbb{M}_1^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} -\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} \end{bmatrix}$$

$$\boxed{\text{x58}} \quad \mathbb{M}_3^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} \frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} \end{bmatrix}$$

$$\boxed{\text{x59}} \quad \mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} \end{bmatrix}$$

$$\boxed{\text{x60}} \quad \mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x61}} \quad \mathbb{M}_3^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{70} & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 \\ \frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{105} \\ 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{70} & 0 \\ \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} \\ 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x62}} \quad \mathbb{M}_5^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x63}} \quad \mathbb{M}_3^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{70} & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 \\ \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{105} \\ 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 \\ \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \\ 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x64}} \quad \mathbb{M}_5^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x65}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{50} & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{25} & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{15}}{25} & 0 & -\frac{\sqrt{30}}{25} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & -\frac{\sqrt{2}}{5} & 0 \\ \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{25} & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{25} & 0 & \frac{\sqrt{15}}{25} & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{25} & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{\sqrt{2}}{10} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x66}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & -\frac{\sqrt{5}i}{25} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{50} & 0 & -\frac{\sqrt{10}i}{25} & 0 & 0 & -\frac{\sqrt{30}i}{25} & 0 & -\frac{\sqrt{15}i}{25} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{25} & 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{15}i}{25} & 0 & -\frac{\sqrt{30}i}{25} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} & 0 & -\frac{\sqrt{2}i}{5} \\ \frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & \frac{2\sqrt{5}i}{25} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{25} & 0 & \frac{\sqrt{15}i}{25} & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{3\sqrt{10}i}{50} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{25} & 0 & \frac{\sqrt{5}i}{25} & 0 & 0 & -\frac{3\sqrt{10}i}{50} & 0 & \frac{2\sqrt{5}i}{25} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{\sqrt{2}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x67}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{3\sqrt{35}}{700} & 0 & \frac{\sqrt{105}}{140} & \frac{\sqrt{21}}{70} & 0 & -\frac{3\sqrt{210}}{350} & 0 & \frac{\sqrt{105}}{70} & 0 \\ -\frac{3\sqrt{35}}{700} & 0 & \frac{3\sqrt{105}}{700} & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 & \frac{\sqrt{70}}{350} & 0 & \frac{\sqrt{7}}{14} \\ 0 & \frac{3\sqrt{105}}{700} & 0 & -\frac{3\sqrt{35}}{700} & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{70}}{350} & 0 & \frac{\sqrt{35}}{50} & 0 \\ \frac{\sqrt{105}}{140} & 0 & -\frac{3\sqrt{35}}{700} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{3\sqrt{210}}{350} & 0 & -\frac{\sqrt{21}}{70} \\ \frac{\sqrt{21}}{70} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{3\sqrt{21}}{70} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{50} & 0 & -\frac{\sqrt{105}}{70} & \frac{3\sqrt{21}}{70} & 0 & -\frac{3\sqrt{210}}{700} & 0 & -\frac{\sqrt{105}}{35} & 0 \\ -\frac{3\sqrt{210}}{350} & 0 & -\frac{\sqrt{70}}{350} & 0 & 0 & -\frac{3\sqrt{210}}{700} & 0 & -\frac{3\sqrt{105}}{175} & 0 & -\frac{\sqrt{42}}{28} \\ 0 & \frac{\sqrt{70}}{350} & 0 & \frac{3\sqrt{210}}{350} & -\frac{\sqrt{42}}{28} & 0 & -\frac{3\sqrt{105}}{175} & 0 & -\frac{3\sqrt{210}}{700} & 0 \\ \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{35}}{50} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & -\frac{3\sqrt{210}}{700} & 0 & \frac{3\sqrt{21}}{70} \\ 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{3\sqrt{21}}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x68}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{3\sqrt{35}i}{700} & 0 & -\frac{\sqrt{105}i}{140} & -\frac{\sqrt{21}i}{70} & 0 & -\frac{3\sqrt{210}i}{350} & 0 & -\frac{\sqrt{105}i}{70} & 0 \\ \frac{3\sqrt{35}i}{700} & 0 & \frac{3\sqrt{105}i}{700} & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & \frac{\sqrt{70}i}{350} & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & -\frac{3\sqrt{105}i}{700} & 0 & -\frac{3\sqrt{35}i}{700} & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{70}i}{350} & 0 & \frac{\sqrt{35}i}{50} & 0 \\ \frac{\sqrt{105}i}{140} & 0 & \frac{3\sqrt{35}i}{700} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & -\frac{3\sqrt{210}i}{350} & 0 & -\frac{\sqrt{21}i}{70} \\ \frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{3\sqrt{21}i}{70} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{50} & 0 & \frac{\sqrt{105}i}{70} & -\frac{3\sqrt{21}i}{70} & 0 & -\frac{3\sqrt{210}i}{700} & 0 & \frac{\sqrt{105}i}{35} & 0 \\ \frac{3\sqrt{210}i}{350} & 0 & -\frac{\sqrt{70}i}{350} & 0 & 0 & \frac{3\sqrt{210}i}{700} & 0 & -\frac{3\sqrt{105}i}{175} & 0 & \frac{\sqrt{42}i}{28} \\ 0 & -\frac{\sqrt{70}i}{350} & 0 & \frac{3\sqrt{210}i}{350} & -\frac{\sqrt{42}i}{28} & 0 & \frac{3\sqrt{105}i}{175} & 0 & -\frac{3\sqrt{210}i}{700} & 0 \\ \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & \frac{3\sqrt{210}i}{700} & 0 & \frac{3\sqrt{21}i}{70} \\ 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{21}i}{70} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{3\sqrt{21}i}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x69}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{21}}{140} & 0 & -\frac{3\sqrt{7}}{140} & \frac{\sqrt{35}}{70} & 0 & -\frac{3\sqrt{14}}{70} & 0 & -\frac{3\sqrt{7}}{70} & 0 \\ -\frac{\sqrt{21}}{140} & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & -\frac{\sqrt{21}}{30} & 0 & \frac{\sqrt{42}}{210} & 0 & -\frac{\sqrt{105}}{70} \\ 0 & \frac{3\sqrt{7}}{140} & 0 & -\frac{\sqrt{21}}{140} & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{42}}{210} & 0 & \frac{\sqrt{21}}{30} & 0 \\ -\frac{3\sqrt{7}}{140} & 0 & -\frac{\sqrt{21}}{140} & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & \frac{3\sqrt{14}}{70} & 0 & -\frac{\sqrt{35}}{70} \\ \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & \frac{3\sqrt{7}}{70} & \frac{3\sqrt{35}}{70} & 0 & -\frac{3\sqrt{14}}{140} & 0 & \frac{3\sqrt{7}}{35} & 0 \\ -\frac{3\sqrt{14}}{70} & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & -\frac{3\sqrt{7}}{35} & 0 & \frac{3\sqrt{70}}{140} \\ 0 & \frac{\sqrt{42}}{210} & 0 & \frac{3\sqrt{14}}{70} & \frac{3\sqrt{70}}{140} & 0 & -\frac{3\sqrt{7}}{35} & 0 & -\frac{3\sqrt{14}}{140} & 0 \\ -\frac{3\sqrt{7}}{70} & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & -\frac{3\sqrt{14}}{140} & 0 & \frac{3\sqrt{35}}{70} \\ 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & \frac{3\sqrt{35}}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x70}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{21}i}{140} & 0 & \frac{3\sqrt{7}i}{140} & -\frac{\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{14}i}{70} & 0 & \frac{3\sqrt{7}i}{70} & 0 \\ \frac{\sqrt{21}i}{140} & 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & \frac{\sqrt{42}i}{210} & 0 & \frac{\sqrt{105}i}{70} \\ 0 & -\frac{3\sqrt{7}i}{140} & 0 & -\frac{\sqrt{21}i}{140} & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{42}i}{210} & 0 & \frac{\sqrt{21}i}{30} & 0 \\ -\frac{3\sqrt{7}i}{140} & 0 & \frac{\sqrt{21}i}{140} & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & -\frac{3\sqrt{14}i}{70} & 0 & -\frac{\sqrt{35}i}{70} \\ \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & -\frac{3\sqrt{7}i}{70} & -\frac{3\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{14}i}{140} & 0 & -\frac{3\sqrt{7}i}{35} & 0 \\ \frac{3\sqrt{14}i}{70} & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & \frac{3\sqrt{14}i}{140} & 0 & -\frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{70}i}{140} \\ 0 & -\frac{\sqrt{42}i}{210} & 0 & \frac{3\sqrt{14}i}{70} & \frac{3\sqrt{70}i}{140} & 0 & \frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{14}i}{140} & 0 \\ -\frac{3\sqrt{7}i}{70} & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & \frac{3\sqrt{7}i}{35} & 0 & \frac{3\sqrt{14}i}{140} & 0 & \frac{3\sqrt{35}i}{70} \\ 0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & -\frac{3\sqrt{35}i}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x71}} \quad \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{70}}{48} & 0 & \frac{3\sqrt{7}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{7}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{7}}{56} & 0 & -\frac{\sqrt{70}}{48} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{48} & 0 & \frac{5\sqrt{7}}{56} & 0 & -\frac{5\sqrt{14}}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{48} & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{16} & 0 & -\frac{\sqrt{70}}{48} & 0 & \frac{\sqrt{35}}{112} & 0 \end{bmatrix}$$

$$\boxed{\text{x72}} \quad \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{70}i}{48} & 0 & \frac{3\sqrt{7}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{7}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & \frac{5\sqrt{7}i}{56} & 0 & \frac{\sqrt{70}i}{48} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{48} & 0 & -\frac{5\sqrt{7}i}{56} & 0 & -\frac{5\sqrt{14}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{48} & 0 & \frac{5\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{112} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{16} & 0 & -\frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{35}i}{112} & 0 \end{bmatrix}$$

$$\boxed{\text{x73}} \quad \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{16} \\ 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & -\frac{3\sqrt{5}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{5}}{8} & 0 & \frac{3\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{8} & 0 & -\frac{\sqrt{10}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{1}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{1}{16} & 0 \end{bmatrix}$$

$$\boxed{\text{x74}} \quad \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & \frac{\sqrt{5}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & -\frac{\sqrt{10}i}{16} & 0 & \frac{3\sqrt{5}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{5}i}{8} & 0 & -\frac{3\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{8} & 0 & -\frac{\sqrt{10}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & \frac{i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{i}{16} & 0 \end{bmatrix}$$

$$\boxed{\text{x75}} \quad \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{15}}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{15}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{15}}{12} & 0 & -\frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{15}}{12} & 0 & -\frac{\sqrt{30}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{3}}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{8} & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{3}}{24} & 0 \end{bmatrix}$$

$$\boxed{\text{x76}} \quad \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{15}i}{8} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{15}i}{12} & 0 & \frac{\sqrt{6}i}{24} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{15}i}{12} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{3}i}{24} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{8} & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{3}i}{24} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x77}} \quad \mathbb{M}_1^{(1,1;a)}(A_{2g}) = \begin{bmatrix} \frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{175} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x78}} \quad \mathbb{M}_3^{(1,1;a)}(A_{2g}) = \begin{bmatrix} -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 \\ 0 & \frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 \\ 0 & 0 & -\frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{525} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{525} & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} \end{bmatrix}$$

$$\boxed{\text{x79}} \quad \mathbb{M}_3^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{7}i}{35} & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 \\ -\frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{140} \\ 0 & \frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{210} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & \frac{\sqrt{210}i}{210} & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 \\ \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 & 0 & 0 & \frac{\sqrt{210}i}{210} \\ 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{210} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x80}} \quad \mathbb{M}_3^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{7}}{35} & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 \\ -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} \\ 0 & \frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{210} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & \frac{\sqrt{210}}{210} & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 \\ \frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 & 0 & -\frac{\sqrt{210}}{210} \\ 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{210} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x81}} \quad \mathbb{M}_{1,1}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{210}}{50} & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & -\frac{\sqrt{35}}{100} & 0 & 0 & 0 \\ \frac{\sqrt{210}}{50} & 0 & \frac{\sqrt{70}}{25} & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & -\frac{\sqrt{105}}{100} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{25} & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & \frac{\sqrt{105}}{100} & 0 & -\frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{100} & 0 & -\frac{\sqrt{14}}{20} \\ \frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{100} & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & -\frac{4\sqrt{35}}{175} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{100} & 0 & \frac{\sqrt{105}}{100} & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{100} & 0 & \frac{\sqrt{35}}{100} & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & -\frac{4\sqrt{35}}{175} & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & -\frac{\sqrt{14}}{35} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x82}} \quad \mathbb{M}_{1,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & -\frac{\sqrt{35}i}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{50} & 0 & \frac{\sqrt{70}i}{25} & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & -\frac{\sqrt{105}i}{100} & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{25} & 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & -\frac{\sqrt{105}i}{100} & 0 & -\frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{100} & 0 & -\frac{\sqrt{14}i}{20} \\ \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & \frac{\sqrt{14}i}{35} & 0 & -\frac{4\sqrt{35}i}{175} & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{100} & 0 & \frac{\sqrt{105}i}{100} & 0 & 0 & \frac{4\sqrt{35}i}{175} & 0 & -\frac{3\sqrt{70}i}{175} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{100} & 0 & \frac{\sqrt{35}i}{100} & 0 & 0 & \frac{3\sqrt{70}i}{175} & 0 & -\frac{4\sqrt{35}i}{175} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}i}{175} & 0 & -\frac{\sqrt{14}i}{35} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x83}} \quad \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{3\sqrt{105}}{175} & 0 & -\frac{3\sqrt{35}}{35} & \frac{9\sqrt{7}}{560} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & \frac{9\sqrt{35}}{560} & 0 \\ \frac{3\sqrt{105}}{175} & 0 & -\frac{9\sqrt{35}}{175} & 0 & 0 & -\frac{3\sqrt{105}}{400} & 0 & \frac{3\sqrt{210}}{2800} & 0 & \frac{3\sqrt{21}}{112} \\ 0 & -\frac{9\sqrt{35}}{175} & 0 & \frac{3\sqrt{105}}{175} & -\frac{3\sqrt{21}}{112} & 0 & -\frac{3\sqrt{210}}{2800} & 0 & \frac{3\sqrt{105}}{400} & 0 \\ -\frac{3\sqrt{35}}{35} & 0 & \frac{3\sqrt{105}}{175} & 0 & 0 & -\frac{9\sqrt{35}}{560} & 0 & \frac{27\sqrt{70}}{2800} & 0 & -\frac{9\sqrt{7}}{560} \\ \frac{9\sqrt{7}}{560} & 0 & -\frac{3\sqrt{21}}{112} & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{400} & 0 & -\frac{9\sqrt{35}}{560} & -\frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{70}}{700} & 0 & \frac{\sqrt{35}}{105} & 0 \\ -\frac{27\sqrt{70}}{2800} & 0 & -\frac{3\sqrt{210}}{2800} & 0 & 0 & \frac{\sqrt{70}}{700} & 0 & \frac{\sqrt{35}}{175} & 0 & \frac{\sqrt{14}}{84} \\ 0 & \frac{3\sqrt{210}}{2800} & 0 & \frac{27\sqrt{70}}{2800} & \frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{35}}{175} & 0 & \frac{\sqrt{70}}{700} & 0 \\ \frac{9\sqrt{35}}{560} & 0 & \frac{3\sqrt{105}}{400} & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & \frac{\sqrt{70}}{700} & 0 & -\frac{\sqrt{7}}{70} \\ 0 & \frac{3\sqrt{21}}{112} & 0 & -\frac{9\sqrt{7}}{560} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{7}}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x84}} \quad \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{3\sqrt{105}i}{175} & 0 & \frac{3\sqrt{35}i}{35} & -\frac{9\sqrt{7}i}{560} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{35}i}{560} & 0 \\ -\frac{3\sqrt{105}i}{175} & 0 & -\frac{9\sqrt{35}i}{175} & 0 & 0 & \frac{3\sqrt{105}i}{400} & 0 & \frac{3\sqrt{210}i}{2800} & 0 & -\frac{3\sqrt{21}i}{112} \\ 0 & \frac{9\sqrt{35}i}{175} & 0 & \frac{3\sqrt{105}i}{175} & -\frac{3\sqrt{21}i}{112} & 0 & \frac{3\sqrt{210}i}{2800} & 0 & \frac{3\sqrt{105}i}{400} & 0 \\ -\frac{3\sqrt{35}i}{35} & 0 & -\frac{3\sqrt{105}i}{175} & 0 & 0 & -\frac{9\sqrt{35}i}{560} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{7}i}{560} \\ \frac{9\sqrt{7}i}{560} & 0 & \frac{3\sqrt{21}i}{112} & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 \\ 0 & -\frac{3\sqrt{105}i}{400} & 0 & \frac{9\sqrt{35}i}{560} & \frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{70}i}{700} & 0 & -\frac{\sqrt{35}i}{105} & 0 \\ \frac{27\sqrt{70}i}{2800} & 0 & -\frac{3\sqrt{210}i}{2800} & 0 & 0 & -\frac{\sqrt{70}i}{700} & 0 & \frac{\sqrt{35}i}{175} & 0 & -\frac{\sqrt{14}i}{84} \\ 0 & -\frac{3\sqrt{210}i}{2800} & 0 & \frac{27\sqrt{70}i}{2800} & \frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{35}i}{175} & 0 & \frac{\sqrt{70}i}{700} & 0 \\ \frac{9\sqrt{35}i}{560} & 0 & -\frac{3\sqrt{105}i}{400} & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{70}i}{700} & 0 & -\frac{\sqrt{7}i}{70} \\ 0 & \frac{3\sqrt{21}i}{112} & 0 & \frac{9\sqrt{7}i}{560} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{7}i}{70} & 0 \end{bmatrix}$$

$$\boxed{\text{x85}} \quad \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{3\sqrt{7}}{35} & 0 & \frac{3\sqrt{21}}{35} & \frac{3\sqrt{105}}{560} & 0 & -\frac{9\sqrt{42}}{560} & 0 & -\frac{9\sqrt{21}}{560} & 0 \\ \frac{3\sqrt{7}}{35} & 0 & -\frac{3\sqrt{21}}{35} & 0 & 0 & -\frac{3\sqrt{7}}{80} & 0 & \frac{3\sqrt{14}}{560} & 0 & -\frac{9\sqrt{35}}{560} \\ 0 & -\frac{3\sqrt{21}}{35} & 0 & \frac{3\sqrt{7}}{35} & \frac{9\sqrt{35}}{560} & 0 & -\frac{3\sqrt{14}}{560} & 0 & \frac{3\sqrt{7}}{80} & 0 \\ \frac{3\sqrt{21}}{35} & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & \frac{9\sqrt{21}}{560} & 0 & \frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{105}}{560} \\ \frac{3\sqrt{105}}{560} & 0 & \frac{9\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{80} & 0 & \frac{9\sqrt{21}}{560} & -\frac{\sqrt{105}}{210} & 0 & \frac{\sqrt{42}}{420} & 0 & -\frac{\sqrt{21}}{105} & 0 \\ -\frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{14}}{560} & 0 & 0 & \frac{\sqrt{42}}{420} & 0 & \frac{\sqrt{21}}{105} & 0 & -\frac{\sqrt{210}}{420} \\ 0 & \frac{3\sqrt{14}}{560} & 0 & \frac{9\sqrt{42}}{560} & -\frac{\sqrt{210}}{420} & 0 & \frac{\sqrt{21}}{105} & 0 & \frac{\sqrt{42}}{420} & 0 \\ -\frac{9\sqrt{21}}{560} & 0 & \frac{3\sqrt{7}}{80} & 0 & 0 & -\frac{\sqrt{21}}{105} & 0 & \frac{\sqrt{42}}{420} & 0 & -\frac{\sqrt{105}}{210} \\ 0 & -\frac{9\sqrt{35}}{560} & 0 & -\frac{3\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{105}}{210} & 0 \end{bmatrix}$$

$$\boxed{\text{x86}} \quad \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{21}i}{35} & -\frac{3\sqrt{105}i}{560} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & \frac{9\sqrt{21}i}{560} & 0 \\ -\frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{21}i}{35} & 0 & 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{14}i}{560} & 0 & \frac{9\sqrt{35}i}{560} \\ 0 & \frac{3\sqrt{21}i}{35} & 0 & \frac{3\sqrt{7}i}{35} & \frac{9\sqrt{35}i}{560} & 0 & \frac{3\sqrt{14}i}{560} & 0 & \frac{3\sqrt{7}i}{80} & 0 \\ \frac{3\sqrt{21}i}{35} & 0 & -\frac{3\sqrt{7}i}{35} & 0 & 0 & \frac{9\sqrt{21}i}{560} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{105}i}{560} \\ \frac{3\sqrt{105}i}{560} & 0 & -\frac{9\sqrt{35}i}{560} & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 \\ 0 & -\frac{3\sqrt{7}i}{80} & 0 & -\frac{9\sqrt{21}i}{560} & \frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{42}i}{420} & 0 & \frac{\sqrt{21}i}{105} & 0 \\ \frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{14}i}{560} & 0 & 0 & -\frac{\sqrt{42}i}{420} & 0 & \frac{\sqrt{21}i}{105} & 0 & \frac{\sqrt{210}i}{420} \\ 0 & -\frac{3\sqrt{14}i}{560} & 0 & \frac{9\sqrt{42}i}{560} & -\frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{21}i}{105} & 0 & \frac{\sqrt{42}i}{420} & 0 \\ -\frac{9\sqrt{21}i}{560} & 0 & -\frac{3\sqrt{7}i}{80} & 0 & 0 & -\frac{\sqrt{21}i}{105} & 0 & -\frac{\sqrt{42}i}{420} & 0 & -\frac{\sqrt{105}i}{210} \\ 0 & -\frac{9\sqrt{35}i}{560} & 0 & \frac{3\sqrt{105}i}{560} & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & \frac{\sqrt{105}i}{210} & 0 \end{bmatrix}$$

$$\boxed{\text{x87}} \quad \mathbb{T}_2^{(1,0;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x88}} \quad \mathbb{T}_4^{(1,0;a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x89}} \quad \mathbb{T}_4^{(1,0;a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x90}} \quad \mathbb{T}_4^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x91}} \quad \mathbb{T}_2^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x92}} \quad \mathbb{T}_4^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x93}} \quad \mathbb{T}_2^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x94}} \quad \mathbb{T}_4^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x95}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & -\frac{\sqrt{70}}{28} \\ \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}}{28} & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{21}}{84} & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x96}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{28} \\ \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}i}{28} & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{84} & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x97}} \quad \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{5}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{30}}{16} & 0 & -\frac{\sqrt{3}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{30}}{16} & 0 & \frac{\sqrt{15}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & -\frac{1}{16} \\ \frac{1}{16} & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{5}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{16} & 0 & -\frac{1}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x98}} \quad \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{5}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{30}i}{16} & 0 & \frac{\sqrt{3}i}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{30}i}{16} & 0 & \frac{\sqrt{15}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{i}{16} \\ \frac{i}{16} & 0 & -\frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{16} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{16} & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{16} & 0 & \frac{i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x99}} \quad \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{35}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{21}}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{16} & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{16} & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{7}}{112} \\ \frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{21}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{35}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{16} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{16} & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x100}} \quad \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{21}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{16} & 0 & -\frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{16} & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{7}i}{112} \\ \frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{21}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{35}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{16} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{16} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- bra: $\langle \frac{3}{2}, \frac{3}{2}; d |, \langle \frac{3}{2}, \frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{1}{2}; d |, \langle \frac{3}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, \frac{5}{2}; d |, \langle \frac{5}{2}, \frac{3}{2}; d |, \langle \frac{5}{2}, \frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{1}{2}; d |, \langle \frac{5}{2}, -\frac{3}{2}; d |, \langle \frac{5}{2}, -\frac{5}{2}; d |$
- ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

$$\boxed{\text{x101}} \quad \mathbb{Q}_5^{(a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x102}} \quad \mathbb{Q}_1^{(a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x103}} \quad \mathbb{Q}_3^{(a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x104}} \quad \mathbb{Q}_5^{(a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\ -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x105}} \quad \mathbb{Q}_5^{(a)}(A_{2u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x106}} \quad \mathbb{Q}_3^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x107}} \quad \mathbb{Q}_5^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{2\sqrt{42}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x108}} \quad \mathbb{Q}_3^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 \\ \frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x109}} \quad \mathbb{Q}_5^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} \\ 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} \\ 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x110}} \quad \mathbb{Q}_{1,1}^{(a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{35} & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{35} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{42}}{28} \end{bmatrix}$$

$$\boxed{\text{x111}} \quad \mathbb{Q}_{1,2}^{(a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{35} & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{70} & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{30}i}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{42}i}{28} \end{bmatrix}$$

$$\boxed{\text{x112}} \quad \mathbb{Q}_{3,1}^{(a)}(E_u, 1) = \begin{bmatrix} -\frac{3\sqrt{105}}{560} & 0 & \frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{21}}{112} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{336} & 0 & 0 \\ 0 & \frac{3\sqrt{7}}{80} & 0 & -\frac{3\sqrt{14}}{560} & 0 & -\frac{3\sqrt{35}}{112} & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{210}}{168} & 0 \\ \frac{3\sqrt{35}}{112} & 0 & \frac{3\sqrt{14}}{560} & 0 & -\frac{3\sqrt{7}}{80} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} \\ 0 & \frac{3\sqrt{21}}{112} & 0 & -\frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\ 0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{42}}{280} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{280} & 0 & \frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & -\frac{3\sqrt{35}}{112} & 0 \\ \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & \frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{5}}{16} \\ 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{42}}{280} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{16} \end{bmatrix}$$

$$\boxed{\text{x113}} \quad \mathbb{Q}_{3,2}^{(a)}(E_u, 1) = \begin{bmatrix} -\frac{3\sqrt{105}i}{560} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 \\ 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{35}i}{112} & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 \\ -\frac{3\sqrt{35}i}{112} & 0 & \frac{3\sqrt{14}i}{560} & 0 & \frac{3\sqrt{7}i}{80} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & -\frac{3\sqrt{21}i}{112} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{105}i}{560} & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 \\ 0 & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{280} & 0 & -\frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{35}i}{112} & 0 \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{70} & 0 & -\frac{\sqrt{42}i}{280} & 0 & 0 & -\frac{3\sqrt{35}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{5}i}{16} \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{16} \end{bmatrix}$$

$$\boxed{\text{x114}} \quad \mathbb{Q}_{3,1}^{(a)}(E_u, 2) = \begin{bmatrix} -\frac{3\sqrt{7}}{112} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{9\sqrt{35}}{560} & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 \\ 0 & \frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{210}}{560} & 0 & \frac{3\sqrt{21}}{112} & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 \\ -\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & \frac{\sqrt{2}}{16} \\ 0 & -\frac{9\sqrt{35}}{560} & 0 & -\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{14}}{56} & -\frac{\sqrt{3}}{16} & 0 & \frac{5\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{336} & 0 & \frac{3\sqrt{21}}{112} & 0 \\ -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{105}}{336} & 0 & -\frac{5\sqrt{7}}{112} & 0 & \frac{\sqrt{3}}{16} \\ 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{48} \end{bmatrix}$$

$$\boxed{\text{x115}} \quad \mathbb{Q}_{3,2}^{(a)}(E_u, 2) = \begin{bmatrix} -\frac{3\sqrt{7}i}{112} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & \frac{9\sqrt{35}i}{560} & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{210}i}{560} & 0 & \frac{3\sqrt{21}i}{112} & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 \\ \frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} \\ 0 & \frac{9\sqrt{35}i}{560} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{112} & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{5\sqrt{42}i}{336} & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{14}i}{56} & \frac{\sqrt{3}i}{16} & 0 & \frac{5\sqrt{7}i}{112} & 0 & \frac{\sqrt{105}i}{336} & 0 & \frac{3\sqrt{21}i}{112} & 0 \\ \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{105}i}{336} & 0 & \frac{5\sqrt{7}i}{112} & 0 & \frac{\sqrt{3}i}{16} \\ 0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{280} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{5\sqrt{42}i}{336} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{30}i}{48} \end{bmatrix}$$

$$\boxed{\text{x116}} \quad \mathbb{Q}_{5,1}^{(a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{80} & 0 & \frac{1}{16} & 0 & -\frac{7\sqrt{15}}{240} & 0 & \frac{3\sqrt{35}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{240} & 0 & \frac{3\sqrt{5}}{80} & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{7\sqrt{15}}{240} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}}{240} & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{5}}{80} & 0 & -\frac{\sqrt{105}}{240} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{80} & 0 & -\frac{7\sqrt{15}}{240} & 0 & \frac{1}{16} & 0 & -\frac{\sqrt{5}}{80} & 0 \\ 0 & -\frac{\sqrt{30}}{224} & 0 & \frac{\sqrt{15}}{48} & 0 & -\frac{3\sqrt{6}}{32} & -\frac{\sqrt{7}}{224} & 0 & \frac{5\sqrt{3}}{224} & 0 & -\frac{\sqrt{5}}{32} & 0 & \frac{3}{32} & 0 \\ -\frac{\sqrt{30}}{224} & 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 & \frac{23\sqrt{5}}{1120} & 0 & -\frac{13}{224} & 0 & \frac{\sqrt{15}}{160} & 0 & \frac{3\sqrt{35}}{160} \\ 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{6}}{112} & 0 & \frac{\sqrt{15}}{48} & \frac{\sqrt{70}}{160} & 0 & -\frac{11\sqrt{30}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{3\sqrt{10}}{160} & 0 \\ \frac{\sqrt{15}}{48} & 0 & -\frac{5\sqrt{6}}{112} & 0 & \frac{5\sqrt{3}}{112} & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & -\frac{\sqrt{2}}{224} & 0 & \frac{11\sqrt{30}}{1120} & 0 & -\frac{\sqrt{70}}{160} \\ 0 & -\frac{5\sqrt{6}}{96} & 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{\sqrt{30}}{224} & -\frac{3\sqrt{35}}{160} & 0 & -\frac{\sqrt{15}}{160} & 0 & \frac{13}{224} & 0 & -\frac{23\sqrt{5}}{1120} & 0 \\ -\frac{3\sqrt{6}}{32} & 0 & \frac{\sqrt{15}}{48} & 0 & -\frac{\sqrt{30}}{224} & 0 & 0 & -\frac{3}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & -\frac{5\sqrt{3}}{224} & 0 & \frac{\sqrt{7}}{224} \end{bmatrix}$$

$$\boxed{\text{x117}} \quad \mathbb{Q}_{5,2}^{(a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{80} & 0 & -\frac{i}{16} & 0 & -\frac{7\sqrt{15}i}{240} & 0 & -\frac{3\sqrt{35}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{240} & 0 & \frac{3\sqrt{5}i}{80} & 0 & \frac{\sqrt{3}i}{16} & 0 & \frac{7\sqrt{15}i}{240} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{15}i}{240} & 0 & -\frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{105}i}{240} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{80} & 0 & \frac{7\sqrt{15}i}{240} & 0 & \frac{i}{16} & 0 & \frac{\sqrt{5}i}{80} & 0 \\ 0 & \frac{\sqrt{30}i}{224} & 0 & \frac{\sqrt{15}i}{48} & 0 & \frac{3\sqrt{6}i}{32} & -\frac{\sqrt{7}i}{224} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{5}i}{32} & 0 & -\frac{3i}{32} & 0 \\ -\frac{\sqrt{30}i}{224} & 0 & -\frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{23\sqrt{5}i}{1120} & 0 & \frac{13i}{224} & 0 & \frac{\sqrt{15}i}{160} & 0 & -\frac{3\sqrt{35}i}{160} \\ 0 & \frac{5\sqrt{3}i}{112} & 0 & \frac{5\sqrt{6}i}{112} & 0 & \frac{\sqrt{15}i}{48} & -\frac{\sqrt{70}i}{160} & 0 & -\frac{11\sqrt{30}i}{1120} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{3\sqrt{10}i}{160} & 0 \\ -\frac{\sqrt{15}i}{48} & 0 & -\frac{5\sqrt{6}i}{112} & 0 & -\frac{5\sqrt{3}i}{112} & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{11\sqrt{30}i}{1120} & 0 & -\frac{\sqrt{70}i}{160} \\ 0 & \frac{5\sqrt{6}i}{96} & 0 & \frac{5\sqrt{3}i}{112} & 0 & \frac{\sqrt{30}i}{224} & -\frac{3\sqrt{35}i}{160} & 0 & \frac{\sqrt{15}i}{160} & 0 & \frac{13i}{224} & 0 & \frac{23\sqrt{5}i}{1120} & 0 \\ -\frac{3\sqrt{6}i}{32} & 0 & -\frac{\sqrt{15}i}{48} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & -\frac{3i}{32} & 0 & -\frac{\sqrt{5}i}{32} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{7}i}{224} \end{bmatrix}$$

$$\boxed{\text{x118}} \quad \mathbb{Q}_{5,1}^{(a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{80} & 0 & \frac{\sqrt{35}}{80} & 0 & \frac{3\sqrt{21}}{80} & 0 & \frac{1}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{80} & 0 & \frac{3\sqrt{7}}{80} & 0 & -\frac{\sqrt{105}}{80} & 0 & -\frac{3\sqrt{21}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{80} & 0 & -\frac{\sqrt{105}}{80} & 0 & \frac{3\sqrt{7}}{80} & 0 & \frac{3\sqrt{3}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & \frac{3\sqrt{21}}{80} & 0 & \frac{\sqrt{35}}{80} & 0 & -\frac{\sqrt{7}}{80} & 0 \\ 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{224} & -\frac{\sqrt{5}}{160} & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{9\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 \\ -\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{105}}{112} & 0 & \frac{3\sqrt{210}}{224} & 0 & 0 & \frac{23\sqrt{7}}{1120} & 0 & -\frac{13\sqrt{35}}{1120} & 0 & -\frac{9\sqrt{21}}{1120} & 0 & \frac{1}{32} \\ 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{3\sqrt{21}}{112} & -\frac{9\sqrt{2}}{160} & 0 & -\frac{11\sqrt{42}}{1120} & 0 & \frac{\sqrt{70}}{1120} & 0 & -\frac{27\sqrt{14}}{1120} & 0 \\ -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{27\sqrt{14}}{1120} & 0 & -\frac{\sqrt{70}}{1120} & 0 & \frac{11\sqrt{42}}{1120} & 0 & \frac{9\sqrt{2}}{160} \\ 0 & \frac{3\sqrt{210}}{224} & 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{42}}{224} & -\frac{1}{32} & 0 & \frac{9\sqrt{21}}{1120} & 0 & \frac{13\sqrt{35}}{1120} & 0 & -\frac{23\sqrt{7}}{1120} & 0 \\ -\frac{\sqrt{210}}{224} & 0 & -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{\sqrt{35}}{224} & 0 & -\frac{9\sqrt{7}}{224} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{5}}{160} \end{bmatrix}$$

$$\boxed{\text{x119}} \quad \mathbb{Q}_{5,2}^{(a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{80} & 0 & -\frac{\sqrt{35}i}{80} & 0 & \frac{3\sqrt{21}i}{80} & 0 & -\frac{i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{80} & 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{\sqrt{105}i}{80} & 0 & -\frac{3\sqrt{21}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}i}{80} & 0 & -\frac{\sqrt{105}i}{80} & 0 & -\frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{3}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & -\frac{3\sqrt{21}i}{80} & 0 & \frac{\sqrt{35}i}{80} & 0 & \frac{\sqrt{7}i}{80} & 0 \\ 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{224} & -\frac{\sqrt{5}i}{160} & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{9\sqrt{7}i}{224} & 0 & -\frac{\sqrt{35}i}{224} & 0 \\ -\frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{105}i}{112} & 0 & \frac{3\sqrt{210}i}{224} & 0 & 0 & \frac{23\sqrt{7}i}{1120} & 0 & \frac{13\sqrt{35}i}{1120} & 0 & -\frac{9\sqrt{21}i}{1120} & 0 & -\frac{i}{32} \\ 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{21}i}{112} & \frac{9\sqrt{2}i}{160} & 0 & -\frac{11\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{27\sqrt{14}i}{1120} & 0 \\ \frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & -\frac{27\sqrt{14}i}{1120} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{11\sqrt{42}i}{1120} & 0 & \frac{9\sqrt{2}i}{160} \\ 0 & -\frac{3\sqrt{210}i}{224} & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{42}i}{224} & -\frac{i}{32} & 0 & -\frac{9\sqrt{21}i}{1120} & 0 & \frac{13\sqrt{35}i}{1120} & 0 & \frac{23\sqrt{7}i}{1120} & 0 \\ -\frac{\sqrt{210}i}{224} & 0 & \frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{\sqrt{35}i}{224} & 0 & \frac{9\sqrt{7}i}{224} & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{5}i}{160} \end{bmatrix}$$

$$\boxed{\text{x120}} \quad \mathbb{Q}_{5,1}^{(a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{120} & 0 & \frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{40} & 0 & \frac{\sqrt{21}}{40} & 0 & -\frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{7}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{21}}{40} & 0 & -\frac{1}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & -\frac{\sqrt{7}}{40} & 0 & \frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{21}}{120} & 0 \\ 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{3\sqrt{70}}{112} & -\frac{\sqrt{15}}{240} & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 \\ -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & \frac{23\sqrt{21}}{1680} & 0 & -\frac{13\sqrt{105}}{1680} & 0 & \frac{3\sqrt{7}}{560} & 0 & -\frac{\sqrt{3}}{16} \\ 0 & \frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{7}}{56} & \frac{\sqrt{6}}{80} & 0 & -\frac{11\sqrt{14}}{560} & 0 & \frac{\sqrt{210}}{1680} & 0 & \frac{3\sqrt{42}}{560} & 0 \\ \frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & -\frac{3\sqrt{42}}{560} & 0 & -\frac{\sqrt{210}}{1680} & 0 & \frac{11\sqrt{14}}{560} & 0 & -\frac{\sqrt{6}}{80} \\ 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{3}}{16} & 0 & -\frac{3\sqrt{7}}{560} & 0 & \frac{13\sqrt{105}}{1680} & 0 & -\frac{23\sqrt{21}}{1680} & 0 \\ \frac{3\sqrt{70}}{112} & 0 & \frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & \frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{15}}{240} \end{bmatrix}$$

$$\boxed{\text{x121}} \quad \mathbb{Q}_{5,2}^{(a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{120} & 0 & -\frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{40} & 0 & \frac{\sqrt{21}i}{40} & 0 & \frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{7}i}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{35}i}{40} & 0 & -\frac{\sqrt{21}i}{40} & 0 & -\frac{i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & \frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{105}i}{120} & 0 & \frac{\sqrt{21}i}{120} & 0 \\ 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{3\sqrt{70}i}{112} & -\frac{\sqrt{15}i}{240} & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & \frac{23\sqrt{21}i}{1680} & 0 & \frac{13\sqrt{105}i}{1680} & 0 & \frac{3\sqrt{7}i}{560} & 0 & \frac{\sqrt{3}i}{16} \\ 0 & \frac{\sqrt{35}i}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{7}i}{56} & -\frac{\sqrt{6}i}{80} & 0 & -\frac{11\sqrt{14}i}{560} & 0 & -\frac{\sqrt{210}i}{1680} & 0 & \frac{3\sqrt{42}i}{560} & 0 \\ -\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{35}i}{56} & 0 & 0 & \frac{3\sqrt{42}i}{560} & 0 & -\frac{\sqrt{210}i}{1680} & 0 & -\frac{11\sqrt{14}i}{560} & 0 & -\frac{\sqrt{6}i}{80} \\ 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{56} & 0 & \frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{16} & 0 & \frac{3\sqrt{7}i}{560} & 0 & \frac{13\sqrt{105}i}{1680} & 0 & \frac{23\sqrt{21}i}{1680} & 0 \\ \frac{3\sqrt{70}i}{112} & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{15}i}{240} \end{bmatrix}$$

$$\boxed{\text{x122}} \quad \mathbb{Q}_5^{(1,-1;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x123}} \quad \mathbb{Q}_3^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 \end{bmatrix}$$

$$\boxed{\text{x124}} \quad \mathbb{Q}_5^{(1,-1;a)}(A_{2u},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 \\ -\frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x125}} \quad \mathbb{Q}_5^{(1,-1;a)}(A_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x126}} \quad \mathbb{Q}_3^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\ \frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 \\ 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x127}} \quad \mathbb{Q}_5^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} \\ 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{\sqrt{10}i}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{8\sqrt{35}i}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & \frac{\sqrt{10}i}{25} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x128}} \quad \mathbb{Q}_3^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ \frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x129}} \quad \mathbb{Q}_5^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{200} & 0 & 0 & 0 & \frac{\sqrt{14}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}}{200} \\ 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{10}}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & \frac{\sqrt{10}}{25} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x130}} \quad \mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{140} & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & -\frac{3\sqrt{5}}{56} & 0 & \frac{3}{28} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{40} & 0 & -\frac{1}{140} & 0 & -\frac{\sqrt{10}}{56} & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & \frac{3\sqrt{3}}{56} & 0 & -\frac{\sqrt{15}}{28} & 0 \\ \frac{\sqrt{10}}{56} & 0 & \frac{1}{140} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{3\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} \\ 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{3\sqrt{3}}{140} & 0 & \frac{\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & \frac{3}{28} & 0 & -\frac{3\sqrt{5}}{56} & 0 \\ 0 & -\frac{3\sqrt{30}}{280} & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ -\frac{3\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{280} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{3}}{280} & 0 & \frac{3\sqrt{6}}{140} & 0 & \frac{\sqrt{15}}{56} & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{3\sqrt{10}}{56} & 0 \\ \frac{\sqrt{15}}{56} & 0 & \frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{3}}{280} & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & -\frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{70}}{56} \\ 0 & \frac{\sqrt{6}}{28} & 0 & \frac{3\sqrt{3}}{280} & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{1}{14} & 0 & \frac{\sqrt{5}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{56} & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{7}}{28} \end{bmatrix}$$

$$\boxed{\text{x131}} \quad \mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{30}i}{280} & 0 & -\frac{3\sqrt{3}i}{140} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & -\frac{3i}{28} & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{40} & 0 & \frac{i}{140} & 0 & -\frac{\sqrt{10}i}{56} & \frac{\sqrt{105}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\ -\frac{\sqrt{10}i}{56} & 0 & \frac{i}{140} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{3\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} \\ 0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{3\sqrt{3}i}{140} & 0 & -\frac{\sqrt{30}i}{280} & 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & \frac{3i}{28} & 0 & \frac{3\sqrt{5}i}{56} & 0 \\ 0 & \frac{3\sqrt{30}i}{280} & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ -\frac{3\sqrt{30}i}{280} & 0 & -\frac{3\sqrt{3}i}{280} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{i}{14} & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{3}i}{280} & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{\sqrt{15}i}{56} & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{3\sqrt{10}i}{56} & 0 \\ -\frac{\sqrt{15}i}{56} & 0 & \frac{3\sqrt{6}i}{140} & 0 & -\frac{3\sqrt{3}i}{280} & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{70}i}{56} \\ 0 & -\frac{\sqrt{6}i}{28} & 0 & \frac{3\sqrt{3}i}{280} & 0 & \frac{3\sqrt{30}i}{280} & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{5}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{7}i}{28} \end{bmatrix}$$

$$\boxed{\text{x132}} \quad \mathbb{Q}_{3,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{10}}{280} & 0 & 0 & -\frac{5\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & \frac{3}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{15}}{420} & 0 & \frac{\sqrt{6}}{56} & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{3}{28} & 0 \\ -\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{15}}{420} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{3}{28} & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} \\ 0 & -\frac{3\sqrt{10}}{280} & 0 & -\frac{3\sqrt{5}}{140} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & -\frac{5\sqrt{3}}{56} & 0 \\ 0 & -\frac{3\sqrt{2}}{56} & 0 & -\frac{3}{56} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{280} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{3}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{280} & 0 & \frac{3\sqrt{10}}{140} & 0 & -\frac{3}{56} & \frac{\sqrt{42}}{56} & 0 & -\frac{5\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{168} & 0 & -\frac{3\sqrt{6}}{56} & 0 \\ -\frac{3}{56} & 0 & \frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{5}}{280} & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{168} & 0 & \frac{5\sqrt{2}}{56} & 0 & -\frac{\sqrt{42}}{56} \\ 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{5}}{280} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{3}{28} & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{5\sqrt{3}}{84} & 0 \\ 0 & 0 & -\frac{3}{56} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{84} \end{bmatrix}$$

$$\boxed{\text{x133}} \quad \mathbb{Q}_{3,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 & -\frac{5\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{3i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{15}i}{420} & 0 & \frac{\sqrt{6}i}{56} & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{3i}{28} & 0 \\ \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{420} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{3i}{28} & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & \frac{3\sqrt{10}i}{280} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{3i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{5\sqrt{3}i}{56} & 0 \\ 0 & \frac{3\sqrt{2}i}{56} & 0 & -\frac{3i}{56} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & -\frac{5\sqrt{3}i}{84} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{3i}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{3i}{56} & -\frac{\sqrt{42}i}{56} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & -\frac{\sqrt{30}i}{168} & 0 & -\frac{3\sqrt{6}i}{56} & 0 \\ \frac{3i}{56} & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{3\sqrt{5}i}{280} & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{168} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56} \\ 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{5}i}{280} & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & -\frac{3i}{28} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{5\sqrt{3}i}{84} & 0 \\ 0 & 0 & \frac{3i}{56} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{105}i}{84} \end{bmatrix}$$

$$\boxed{\text{x134}} \quad \mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{160} & 0 & \frac{\sqrt{30}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 & \frac{3\sqrt{42}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{160} & 0 & \frac{3\sqrt{6}}{160} & 0 & -\frac{3\sqrt{10}}{160} & 0 & \frac{7\sqrt{2}}{160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}}{160} & 0 & -\frac{3\sqrt{10}}{160} & 0 & \frac{3\sqrt{6}}{160} & 0 & -\frac{\sqrt{14}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 & \frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{6}}{160} & 0 \\ 0 & -\frac{1}{112} & 0 & \frac{\sqrt{2}}{48} & 0 & -\frac{3\sqrt{5}}{80} & \frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{3\sqrt{30}}{80} & 0 \\ -\frac{1}{112} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{48} & 0 & 0 & -\frac{23\sqrt{6}}{560} & 0 & \frac{13\sqrt{30}}{560} & 0 & -\frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{80} \\ 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{2}}{48} & -\frac{\sqrt{21}}{40} & 0 & \frac{33}{280} & 0 & -\frac{\sqrt{15}}{280} & 0 & -\frac{3\sqrt{3}}{40} & 0 \\ \frac{\sqrt{2}}{48} & 0 & -\frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & \frac{3\sqrt{3}}{40} & 0 & \frac{\sqrt{15}}{280} & 0 & -\frac{33}{280} & 0 & \frac{\sqrt{21}}{40} \\ 0 & -\frac{\sqrt{5}}{48} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{1}{112} & \frac{3\sqrt{42}}{80} & 0 & \frac{3\sqrt{2}}{80} & 0 & -\frac{13\sqrt{30}}{560} & 0 & \frac{23\sqrt{6}}{560} & 0 \\ -\frac{3\sqrt{5}}{80} & 0 & \frac{\sqrt{2}}{48} & 0 & -\frac{1}{112} & 0 & 0 & \frac{3\sqrt{30}}{80} & 0 & -\frac{\sqrt{6}}{16} & 0 & \frac{3\sqrt{10}}{112} & 0 & -\frac{\sqrt{210}}{560} \end{bmatrix}$$

$$\boxed{\text{x135}} \quad \mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & -\frac{\sqrt{30}i}{160} & 0 & -\frac{7\sqrt{2}i}{160} & 0 & -\frac{3\sqrt{42}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{160} & 0 & \frac{3\sqrt{6}i}{160} & 0 & \frac{3\sqrt{10}i}{160} & 0 & \frac{7\sqrt{2}i}{160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}i}{160} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & -\frac{3\sqrt{6}i}{160} & 0 & -\frac{\sqrt{14}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{160} & 0 & \frac{7\sqrt{2}i}{160} & 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{6}i}{160} & 0 \\ 0 & \frac{i}{112} & 0 & \frac{\sqrt{2}i}{48} & 0 & \frac{3\sqrt{5}i}{80} & \frac{\sqrt{210}i}{560} & 0 & \frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{3\sqrt{30}i}{80} & 0 \\ -\frac{i}{112} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{48} & 0 & 0 & -\frac{23\sqrt{6}i}{560} & 0 & -\frac{13\sqrt{30}i}{560} & 0 & -\frac{3\sqrt{2}i}{80} & 0 & \frac{3\sqrt{42}i}{80} \\ 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{48} & \frac{\sqrt{21}i}{40} & 0 & \frac{33i}{280} & 0 & \frac{\sqrt{15}i}{280} & 0 & -\frac{3\sqrt{3}i}{40} & 0 \\ -\frac{\sqrt{2}i}{48} & 0 & -\frac{\sqrt{5}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 & -\frac{3\sqrt{3}i}{40} & 0 & \frac{\sqrt{15}i}{280} & 0 & \frac{33i}{280} & 0 & \frac{\sqrt{21}i}{40} \\ 0 & \frac{\sqrt{5}i}{48} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{i}{112} & \frac{3\sqrt{42}i}{80} & 0 & -\frac{3\sqrt{2}i}{80} & 0 & -\frac{13\sqrt{30}i}{560} & 0 & -\frac{23\sqrt{6}i}{560} & 0 \\ -\frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{2}i}{48} & 0 & -\frac{i}{112} & 0 & 0 & \frac{3\sqrt{30}i}{80} & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{210}i}{560} \end{bmatrix}$$

$$\boxed{\text{x136}} \quad \mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{800} & 0 & \frac{\sqrt{42}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 & \frac{\sqrt{30}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{800} & 0 & \frac{3\sqrt{210}}{800} & 0 & -\frac{3\sqrt{14}}{160} & 0 & -\frac{9\sqrt{70}}{800} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{800} & 0 & -\frac{3\sqrt{14}}{160} & 0 & \frac{3\sqrt{210}}{800} & 0 & \frac{9\sqrt{10}}{800} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 & \frac{\sqrt{42}}{160} & 0 & -\frac{\sqrt{210}}{800} & 0 \\ 0 & -\frac{\sqrt{35}}{560} & 0 & -\frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{7}}{112} & \frac{\sqrt{6}}{80} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{9\sqrt{210}}{560} & 0 & -\frac{\sqrt{42}}{112} & 0 \\ -\frac{\sqrt{35}}{560} & 0 & \frac{\sqrt{14}}{112} & 0 & \frac{3\sqrt{7}}{112} & 0 & 0 & -\frac{23\sqrt{210}}{2800} & 0 & \frac{13\sqrt{42}}{560} & 0 & \frac{27\sqrt{70}}{2800} & 0 & -\frac{\sqrt{30}}{80} \\ 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{56} & 0 & -\frac{3\sqrt{70}}{560} & \frac{9\sqrt{15}}{200} & 0 & \frac{33\sqrt{35}}{1400} & 0 & -\frac{\sqrt{21}}{280} & 0 & \frac{27\sqrt{105}}{1400} & 0 \\ -\frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{27\sqrt{105}}{1400} & 0 & \frac{\sqrt{21}}{280} & 0 & -\frac{33\sqrt{35}}{1400} & 0 & -\frac{9\sqrt{15}}{200} \\ 0 & \frac{3\sqrt{7}}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{35}}{560} & \frac{\sqrt{30}}{80} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & -\frac{13\sqrt{42}}{560} & 0 & \frac{23\sqrt{210}}{2800} & 0 \\ -\frac{\sqrt{7}}{112} & 0 & -\frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{35}}{560} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{9\sqrt{210}}{560} & 0 & \frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{6}}{80} \end{bmatrix}$$

$$\boxed{\text{x137}} \quad \mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{800} & 0 & -\frac{\sqrt{42}i}{160} & 0 & \frac{9\sqrt{70}i}{800} & 0 & -\frac{\sqrt{30}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{800} & 0 & \frac{3\sqrt{210}i}{800} & 0 & \frac{3\sqrt{14}i}{160} & 0 & -\frac{9\sqrt{70}i}{800} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{800} & 0 & -\frac{3\sqrt{14}i}{160} & 0 & -\frac{3\sqrt{210}i}{800} & 0 & \frac{9\sqrt{10}i}{800} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{9\sqrt{70}i}{800} & 0 & \frac{\sqrt{42}i}{160} & 0 & \frac{\sqrt{210}i}{800} & 0 \\ 0 & \frac{\sqrt{35}i}{560} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & \frac{\sqrt{7}i}{112} & \frac{\sqrt{6}i}{80} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{9\sqrt{210}i}{560} & 0 & \frac{\sqrt{42}i}{112} & 0 \\ -\frac{\sqrt{35}i}{560} & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{3\sqrt{7}i}{112} & 0 & 0 & -\frac{23\sqrt{210}i}{2800} & 0 & -\frac{13\sqrt{42}i}{560} & 0 & \frac{27\sqrt{70}i}{2800} & 0 & \frac{\sqrt{30}i}{80} \\ 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{3\sqrt{70}i}{560} & -\frac{9\sqrt{15}i}{200} & 0 & \frac{33\sqrt{35}i}{1400} & 0 & \frac{\sqrt{21}i}{280} & 0 & \frac{27\sqrt{105}i}{1400} & 0 \\ \frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{27\sqrt{105}i}{1400} & 0 & \frac{\sqrt{21}i}{280} & 0 & \frac{33\sqrt{35}i}{1400} & 0 & -\frac{9\sqrt{15}i}{200} \\ 0 & -\frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{560} & \frac{\sqrt{30}i}{80} & 0 & \frac{27\sqrt{70}i}{2800} & 0 & -\frac{13\sqrt{42}i}{560} & 0 & -\frac{23\sqrt{210}i}{2800} & 0 \\ -\frac{\sqrt{7}i}{112} & 0 & \frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{9\sqrt{210}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{6}i}{80} \end{bmatrix}$$

$$\boxed{\text{x138}} \quad \mathbb{Q}_{5,1}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{400} & 0 & \frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 & -\frac{3\sqrt{10}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{400} & 0 & \frac{3\sqrt{70}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & \frac{\sqrt{210}}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & \frac{3\sqrt{70}}{400} & 0 & -\frac{\sqrt{30}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 & \frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{70}}{400} & 0 \\ 0 & -\frac{\sqrt{105}}{840} & 0 & \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{21}}{56} & \frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{105}}{840} & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & -\frac{23\sqrt{70}}{1400} & 0 & \frac{13\sqrt{14}}{280} & 0 & -\frac{3\sqrt{210}}{1400} & 0 & \frac{3\sqrt{10}}{40} \\ 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{210}}{840} & -\frac{3\sqrt{5}}{100} & 0 & \frac{11\sqrt{105}}{700} & 0 & -\frac{\sqrt{7}}{140} & 0 & -\frac{9\sqrt{35}}{700} & 0 \\ \frac{\sqrt{210}}{840} & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{9\sqrt{35}}{700} & 0 & \frac{\sqrt{7}}{140} & 0 & -\frac{11\sqrt{105}}{700} & 0 & \frac{3\sqrt{5}}{100} \\ 0 & -\frac{\sqrt{21}}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{840} & -\frac{3\sqrt{10}}{40} & 0 & \frac{3\sqrt{210}}{1400} & 0 & -\frac{13\sqrt{14}}{280} & 0 & \frac{23\sqrt{70}}{1400} & 0 \\ \frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{840} & 0 & -\frac{\sqrt{105}}{840} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{2}}{40} \end{bmatrix}$$

$$\boxed{\text{x139}} \quad \mathbb{Q}_{5,2}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{400} & 0 & -\frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{210}i}{400} & 0 & \frac{3\sqrt{10}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{400} & 0 & \frac{3\sqrt{70}i}{400} & 0 & \frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & -\frac{3\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 & \frac{\sqrt{14}i}{80} & 0 & \frac{\sqrt{70}i}{400} & 0 \\ 0 & \frac{\sqrt{105}i}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{21}i}{56} & \frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{14}i}{56} & 0 \\ -\frac{\sqrt{105}i}{840} & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & -\frac{23\sqrt{70}i}{1400} & 0 & -\frac{13\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & -\frac{3\sqrt{10}i}{40} \\ 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{840} & \frac{3\sqrt{5}i}{100} & 0 & \frac{11\sqrt{105}i}{700} & 0 & \frac{\sqrt{7}i}{140} & 0 & -\frac{9\sqrt{35}i}{700} & 0 \\ -\frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{21}i}{84} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{9\sqrt{35}i}{700} & 0 & \frac{\sqrt{7}i}{140} & 0 & \frac{11\sqrt{105}i}{700} & 0 & \frac{3\sqrt{5}i}{100} \\ 0 & \frac{\sqrt{21}i}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{105}i}{840} & -\frac{3\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & -\frac{13\sqrt{14}i}{280} & 0 & -\frac{23\sqrt{70}i}{1400} & 0 \\ \frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{105}i}{840} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{40} \end{bmatrix}$$

$$\boxed{\text{x140}} \quad \mathbb{Q}_5^{(1,0;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2100} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x141}} \quad \mathbb{Q}_1^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x142}} \quad \mathbb{Q}_3^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 \end{bmatrix}$$

$$\boxed{\text{x143}} \quad \mathbb{Q}_5^{(1,0;a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 \end{bmatrix}$$

$$\boxed{\text{x144}} \quad \mathbb{Q}_5^{(1,0;a)}(A_{2u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x145}} \quad \mathbb{Q}_3^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x146}} \quad \mathbb{Q}_5^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} \\ 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{10}i}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{210} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}i}{525} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{300} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x147}} \quad \mathbb{Q}_3^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{56} & 0 & 0 & 0 & \frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & \frac{\sqrt{7}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & \frac{\sqrt{2}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x148}} \quad \mathbb{Q}_5^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{100} & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{100} & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} \\ 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{10}}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 \\ \frac{\sqrt{210}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}}{300} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x149}} \quad \mathbb{Q}_{1,1}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{10}}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{70} & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{6}{35} & 0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2}}{70} & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{6}{35} & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & \frac{\sqrt{21}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x150}} \quad \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{10}i}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{6i}{35} & 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{6i}{35} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{21}i}{28} \end{bmatrix}$$

$$\boxed{\text{x151}} \quad \mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} \frac{3\sqrt{35}}{1120} & 0 & -\frac{9\sqrt{14}}{1120} & 0 & \frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{105}}{224} & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 & \frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 \\ -\frac{\sqrt{105}}{224} & 0 & -\frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{21}}{160} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} \\ 0 & -\frac{3\sqrt{7}}{224} & 0 & \frac{9\sqrt{14}}{1120} & 0 & -\frac{3\sqrt{35}}{1120} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{210}}{112} & 0 \\ 0 & \frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{672} & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{140} & 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{\sqrt{42}}{336} & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{7}}{70} & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{105}}{224} & 0 \\ -\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{7}}{70} & 0 & -\frac{3\sqrt{14}}{280} & 0 & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{15}}{96} \\ 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{14}}{280} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{42}}{336} & 0 & -\frac{\sqrt{210}}{672} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{6}}{96} \end{bmatrix}$$

$$\boxed{\text{x152}} \quad \mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} \frac{3\sqrt{35}i}{1120} & 0 & \frac{9\sqrt{14}i}{1120} & 0 & \frac{3\sqrt{7}i}{224} & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{160} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & \frac{\sqrt{105}i}{224} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 \\ \frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{21}i}{160} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} \\ 0 & \frac{3\sqrt{7}i}{224} & 0 & \frac{9\sqrt{14}i}{1120} & 0 & \frac{3\sqrt{35}i}{1120} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{210}i}{112} & 0 \\ 0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{672} & 0 & 0 & 0 \\ \frac{3\sqrt{35}i}{140} & 0 & \frac{3\sqrt{14}i}{280} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{42}i}{336} & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{280} & 0 & \frac{3\sqrt{7}i}{70} & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{15}i}{96} & 0 & \frac{\sqrt{35}i}{224} & 0 & \frac{\sqrt{21}i}{672} & 0 & -\frac{\sqrt{105}i}{224} & 0 \\ \frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{14}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{224} & 0 & -\frac{\sqrt{15}i}{96} \\ 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{672} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{96} \end{bmatrix}$$

$$\boxed{\text{x153}} \quad \mathbb{Q}_{3,1}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{21}}{224} & 0 & -\frac{3\sqrt{210}}{1120} & 0 & -\frac{3\sqrt{105}}{1120} & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{160} & 0 & \frac{\sqrt{70}}{1120} & 0 & -\frac{3\sqrt{7}}{224} & \frac{\sqrt{6}}{16} & 0 & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 \\ \frac{3\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{1120} & 0 & \frac{\sqrt{35}}{160} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & \frac{\sqrt{6}}{16} \\ 0 & \frac{3\sqrt{105}}{1120} & 0 & \frac{3\sqrt{210}}{1120} & 0 & -\frac{\sqrt{21}}{224} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{5\sqrt{14}}{112} & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{224} & 0 & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{5\sqrt{14}}{672} & 0 & \frac{\sqrt{70}}{336} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{42}}{56} & -\frac{1}{32} & 0 & \frac{5\sqrt{21}}{672} & 0 & -\frac{\sqrt{35}}{672} & 0 & \frac{3\sqrt{7}}{224} & 0 \\ \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{672} & 0 & -\frac{5\sqrt{21}}{672} & 0 & \frac{1}{32} \\ 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{70}}{336} & 0 & -\frac{5\sqrt{14}}{672} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{10}}{96} \end{bmatrix}$$

$$\boxed{\text{x154}} \quad \mathbb{Q}_{3,2}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{21}i}{224} & 0 & \frac{3\sqrt{210}i}{1120} & 0 & -\frac{3\sqrt{105}i}{1120} & 0 & 0 & -\frac{5\sqrt{14}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{160} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{3\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{42}i}{56} & 0 \\ -\frac{3\sqrt{7}i}{224} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{35}i}{160} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{210}i}{112} & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} \\ 0 & -\frac{3\sqrt{105}i}{1120} & 0 & \frac{3\sqrt{210}i}{1120} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{5\sqrt{14}i}{112} & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{14}i}{224} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{5\sqrt{14}i}{672} & 0 & -\frac{\sqrt{70}i}{336} & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{42}i}{56} & \frac{i}{32} & 0 & \frac{5\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{672} & 0 & \frac{3\sqrt{7}i}{224} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & \frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{672} & 0 & \frac{5\sqrt{21}i}{672} & 0 & \frac{i}{32} \\ 0 & -\frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{70}i}{336} & 0 & \frac{5\sqrt{14}i}{672} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{10}i}{96} \end{bmatrix}$$

$$\boxed{\text{x155}} \quad \mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{7\sqrt{2}}{80} & 0 & \frac{3\sqrt{42}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{80} & 0 & \frac{3\sqrt{6}}{80} & 0 & -\frac{3\sqrt{10}}{80} & 0 & \frac{7\sqrt{2}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}}{80} & 0 & -\frac{3\sqrt{10}}{80} & 0 & \frac{3\sqrt{6}}{80} & 0 & -\frac{\sqrt{14}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{80} & 0 & -\frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{\sqrt{6}}{80} & 0 \\ 0 & \frac{3}{112} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{9\sqrt{5}}{80} & -\frac{\sqrt{210}}{6720} & 0 & \frac{\sqrt{10}}{448} & 0 & -\frac{\sqrt{6}}{192} & 0 & \frac{\sqrt{30}}{320} & 0 \\ \frac{3}{112} & 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & 0 & \frac{23\sqrt{6}}{6720} & 0 & -\frac{13\sqrt{30}}{6720} & 0 & \frac{\sqrt{2}}{320} & 0 & \frac{\sqrt{42}}{320} \\ 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{3\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{16} & \frac{\sqrt{21}}{480} & 0 & -\frac{11}{1120} & 0 & \frac{\sqrt{15}}{3360} & 0 & \frac{\sqrt{3}}{160} & 0 \\ -\frac{\sqrt{2}}{16} & 0 & \frac{3\sqrt{5}}{56} & 0 & -\frac{3\sqrt{10}}{112} & 0 & 0 & -\frac{\sqrt{3}}{160} & 0 & -\frac{\sqrt{15}}{3360} & 0 & \frac{11}{1120} & 0 & -\frac{\sqrt{21}}{480} \\ 0 & \frac{\sqrt{5}}{16} & 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{3}{112} & -\frac{\sqrt{42}}{320} & 0 & -\frac{\sqrt{2}}{320} & 0 & \frac{13\sqrt{30}}{6720} & 0 & -\frac{23\sqrt{6}}{6720} & 0 \\ \frac{9\sqrt{5}}{80} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{3}{112} & 0 & 0 & -\frac{\sqrt{30}}{320} & 0 & \frac{\sqrt{6}}{192} & 0 & -\frac{\sqrt{10}}{448} & 0 & \frac{\sqrt{210}}{6720} \end{bmatrix}$$

$$\boxed{\text{x156}} \quad \mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{7\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & \frac{3\sqrt{6}i}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{10}i}{80} & 0 & -\frac{3\sqrt{6}i}{80} & 0 & -\frac{\sqrt{14}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 & \frac{\sqrt{30}i}{80} & 0 & \frac{\sqrt{6}i}{80} & 0 \\ 0 & -\frac{3i}{112} & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{9\sqrt{5}i}{80} & -\frac{\sqrt{210}i}{6720} & 0 & -\frac{\sqrt{10}i}{448} & 0 & -\frac{\sqrt{6}i}{192} & 0 & -\frac{\sqrt{30}i}{320} & 0 \\ \frac{3i}{112} & 0 & \frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & \frac{23\sqrt{6}i}{6720} & 0 & \frac{13\sqrt{30}i}{6720} & 0 & \frac{\sqrt{2}i}{320} & 0 & -\frac{\sqrt{42}i}{320} \\ 0 & -\frac{3\sqrt{10}i}{112} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & -\frac{\sqrt{2}i}{16} & -\frac{\sqrt{21}i}{480} & 0 & -\frac{11i}{1120} & 0 & -\frac{\sqrt{15}i}{3360} & 0 & \frac{\sqrt{3}i}{160} & 0 \\ \frac{\sqrt{2}i}{16} & 0 & \frac{3\sqrt{5}i}{56} & 0 & \frac{3\sqrt{10}i}{112} & 0 & 0 & \frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{15}i}{3360} & 0 & -\frac{11i}{1120} & 0 & -\frac{\sqrt{21}i}{480} \\ 0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{3\sqrt{10}i}{112} & 0 & -\frac{3i}{112} & -\frac{\sqrt{42}i}{320} & 0 & \frac{\sqrt{2}i}{320} & 0 & \frac{13\sqrt{30}i}{6720} & 0 & \frac{23\sqrt{6}i}{6720} & 0 \\ \frac{9\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{3i}{112} & 0 & 0 & -\frac{\sqrt{30}i}{320} & 0 & -\frac{\sqrt{6}i}{192} & 0 & -\frac{\sqrt{10}i}{448} & 0 & -\frac{\sqrt{210}i}{6720} \end{bmatrix}$$

$$\boxed{\text{x157}} \quad \mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & \frac{9\sqrt{70}}{400} & 0 & \frac{\sqrt{30}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{400} & 0 & \frac{3\sqrt{210}}{400} & 0 & -\frac{3\sqrt{14}}{80} & 0 & -\frac{9\sqrt{70}}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{400} & 0 & -\frac{3\sqrt{14}}{80} & 0 & \frac{3\sqrt{210}}{400} & 0 & \frac{9\sqrt{10}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & \frac{9\sqrt{70}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 \\ 0 & \frac{3\sqrt{35}}{560} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{112} & -\frac{\sqrt{6}}{960} & 0 & \frac{\sqrt{14}}{448} & 0 & \frac{3\sqrt{210}}{2240} & 0 & \frac{\sqrt{42}}{1344} & 0 \\ \frac{3\sqrt{35}}{560} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{9\sqrt{7}}{112} & 0 & 0 & \frac{23\sqrt{210}}{33600} & 0 & -\frac{13\sqrt{42}}{6720} & 0 & -\frac{9\sqrt{70}}{11200} & 0 & \frac{\sqrt{30}}{960} \\ 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{9\sqrt{70}}{560} & -\frac{3\sqrt{15}}{800} & 0 & -\frac{11\sqrt{35}}{5600} & 0 & \frac{\sqrt{21}}{3360} & 0 & -\frac{9\sqrt{105}}{5600} & 0 \\ \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{56} & 0 & -\frac{3\sqrt{14}}{112} & 0 & 0 & \frac{9\sqrt{105}}{5600} & 0 & -\frac{\sqrt{21}}{3360} & 0 & \frac{11\sqrt{35}}{5600} & 0 & \frac{3\sqrt{15}}{800} \\ 0 & -\frac{9\sqrt{7}}{112} & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{35}}{560} & -\frac{\sqrt{30}}{960} & 0 & \frac{9\sqrt{70}}{11200} & 0 & \frac{13\sqrt{42}}{6720} & 0 & -\frac{23\sqrt{210}}{33600} & 0 \\ \frac{3\sqrt{7}}{112} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{42}}{1344} & 0 & -\frac{3\sqrt{210}}{2240} & 0 & -\frac{\sqrt{14}}{448} & 0 & \frac{\sqrt{6}}{960} \end{bmatrix}$$

$$\boxed{\text{x158}} \quad \mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{9\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{400} & 0 & \frac{3\sqrt{210}i}{400} & 0 & \frac{3\sqrt{14}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{400} & 0 & -\frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{210}i}{400} & 0 & \frac{9\sqrt{10}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 & \frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\ 0 & -\frac{3\sqrt{35}i}{560} & 0 & \frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{112} & -\frac{\sqrt{6}i}{960} & 0 & -\frac{\sqrt{14}i}{448} & 0 & \frac{3\sqrt{210}i}{2240} & 0 & -\frac{\sqrt{42}i}{1344} & 0 \\ \frac{3\sqrt{35}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{9\sqrt{7}i}{112} & 0 & 0 & \frac{23\sqrt{210}i}{33600} & 0 & \frac{13\sqrt{42}i}{6720} & 0 & -\frac{9\sqrt{70}i}{11200} & 0 & -\frac{\sqrt{30}i}{960} \\ 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{9\sqrt{70}i}{560} & \frac{3\sqrt{15}i}{800} & 0 & -\frac{11\sqrt{35}i}{5600} & 0 & -\frac{\sqrt{21}i}{3360} & 0 & -\frac{9\sqrt{105}i}{5600} & 0 \\ -\frac{9\sqrt{70}i}{560} & 0 & \frac{3\sqrt{7}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & -\frac{9\sqrt{105}i}{5600} & 0 & -\frac{\sqrt{21}i}{3360} & 0 & -\frac{11\sqrt{35}i}{5600} & 0 & \frac{3\sqrt{15}i}{800} \\ 0 & \frac{9\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{35}i}{560} & -\frac{\sqrt{30}i}{960} & 0 & -\frac{9\sqrt{70}i}{11200} & 0 & \frac{13\sqrt{42}i}{6720} & 0 & \frac{23\sqrt{210}i}{33600} & 0 \\ \frac{3\sqrt{7}i}{112} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & \frac{3\sqrt{35}i}{560} & 0 & 0 & -\frac{\sqrt{42}i}{1344} & 0 & \frac{3\sqrt{210}i}{2240} & 0 & -\frac{\sqrt{14}i}{448} & 0 & -\frac{\sqrt{6}i}{960} \end{bmatrix}$$

$$\boxed{\text{x159}} \quad \mathbb{Q}_{5,1}^{(1,0;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{200} & 0 & \frac{\sqrt{14}}{40} & 0 & -\frac{\sqrt{210}}{200} & 0 & -\frac{3\sqrt{10}}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{200} & 0 & \frac{3\sqrt{70}}{200} & 0 & -\frac{\sqrt{42}}{40} & 0 & \frac{\sqrt{210}}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 & -\frac{\sqrt{42}}{40} & 0 & \frac{3\sqrt{70}}{200} & 0 & -\frac{\sqrt{30}}{200} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{40} & 0 & -\frac{\sqrt{210}}{200} & 0 & \frac{\sqrt{14}}{40} & 0 & -\frac{\sqrt{70}}{200} & 0 \\ 0 & \frac{\sqrt{105}}{280} & 0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{3\sqrt{21}}{56} & -\frac{\sqrt{2}}{480} & 0 & \frac{\sqrt{42}}{672} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{14}}{224} & 0 \\ \frac{\sqrt{105}}{280} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{56} & 0 & 0 & \frac{23\sqrt{70}}{16800} & 0 & -\frac{13\sqrt{14}}{3360} & 0 & \frac{\sqrt{210}}{5600} & 0 & -\frac{\sqrt{10}}{160} \\ 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{280} & \frac{\sqrt{5}}{400} & 0 & -\frac{11\sqrt{105}}{8400} & 0 & \frac{\sqrt{7}}{1680} & 0 & \frac{3\sqrt{35}}{2800} & 0 \\ -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{3\sqrt{35}}{2800} & 0 & -\frac{\sqrt{7}}{1680} & 0 & \frac{11\sqrt{105}}{8400} & 0 & -\frac{\sqrt{5}}{400} \\ 0 & \frac{\sqrt{21}}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{105}}{280} & \frac{\sqrt{10}}{160} & 0 & -\frac{\sqrt{210}}{5600} & 0 & \frac{13\sqrt{14}}{3360} & 0 & -\frac{23\sqrt{70}}{16800} & 0 \\ -\frac{3\sqrt{21}}{56} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & \frac{\sqrt{14}}{224} & 0 & \frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{42}}{672} & 0 & \frac{\sqrt{2}}{480} \end{bmatrix}$$

$$\boxed{\text{x160}} \quad \mathbb{Q}_{5,2}^{(1,0;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{200} & 0 & -\frac{\sqrt{14}i}{40} & 0 & -\frac{\sqrt{210}i}{200} & 0 & \frac{3\sqrt{10}i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{200} & 0 & \frac{3\sqrt{70}i}{200} & 0 & \frac{\sqrt{42}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & -\frac{\sqrt{42}i}{40} & 0 & -\frac{3\sqrt{70}i}{200} & 0 & -\frac{\sqrt{30}i}{200} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 & \frac{\sqrt{14}i}{40} & 0 & \frac{\sqrt{70}i}{200} & 0 \\ 0 & -\frac{\sqrt{105}i}{280} & 0 & -\frac{\sqrt{210}i}{280} & 0 & \frac{3\sqrt{21}i}{56} & -\frac{\sqrt{2}i}{480} & 0 & -\frac{\sqrt{42}i}{672} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{\sqrt{14}i}{224} & 0 \\ \frac{\sqrt{105}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{56} & 0 & 0 & \frac{23\sqrt{70}i}{16800} & 0 & \frac{13\sqrt{14}i}{3360} & 0 & \frac{\sqrt{210}i}{5600} & 0 & \frac{\sqrt{10}i}{160} \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{210}i}{280} & -\frac{\sqrt{5}i}{400} & 0 & -\frac{11\sqrt{105}i}{8400} & 0 & -\frac{\sqrt{7}i}{1680} & 0 & \frac{3\sqrt{35}i}{2800} & 0 \\ \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{3\sqrt{35}i}{2800} & 0 & -\frac{\sqrt{7}i}{1680} & 0 & -\frac{11\sqrt{105}i}{8400} & 0 & -\frac{\sqrt{5}i}{400} \\ 0 & -\frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{105}i}{280} & \frac{\sqrt{10}i}{160} & 0 & \frac{\sqrt{210}i}{5600} & 0 & \frac{13\sqrt{14}i}{3360} & 0 & \frac{23\sqrt{70}i}{16800} & 0 \\ -\frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & \frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{42}i}{672} & 0 & -\frac{\sqrt{2}i}{480} \end{bmatrix}$$

$$\boxed{\text{x161}} \quad \mathbb{Q}_1^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x162}} \quad \mathbb{Q}_3^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x163}} \quad \mathbb{Q}_3^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{15i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\ 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\ 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x164}} \quad \mathbb{Q}_3^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ -\frac{15}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x165}} \quad \mathbb{Q}_{1,1}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{2}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{6}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{2}}{35} & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56} \end{bmatrix}$$

$$\boxed{\text{x166}} \quad \mathbb{Q}_{1,2}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}i}{35} & 0 & \frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{2}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{6i}{35} & 0 & \frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{2}i}{35} & 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56} \end{bmatrix}$$

$$\boxed{\text{x167}} \quad \mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} \frac{9\sqrt{5}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & \frac{45}{224} & 0 & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{10}}{112} & 0 & 0 \\ 0 & -\frac{3\sqrt{3}}{32} & 0 & \frac{3\sqrt{6}}{224} & 0 & \frac{15\sqrt{15}}{224} & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & \frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{10}}{56} & 0 \\ -\frac{15\sqrt{15}}{224} & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{3\sqrt{3}}{32} & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{3\sqrt{2}}{112} & 0 & 0 & 0 & -\frac{\sqrt{70}}{112} \\ 0 & -\frac{45}{224} & 0 & \frac{27\sqrt{2}}{224} & 0 & -\frac{9\sqrt{5}}{224} & 0 & 0 & -\frac{\sqrt{10}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 \\ 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{5\sqrt{10}}{168} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{5}{42} & 0 & 0 & \frac{\sqrt{30}}{224} & 0 & \frac{\sqrt{6}}{112} & 0 & -\frac{3\sqrt{10}}{224} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{56} & 0 & \frac{1}{14} & 0 & \frac{5\sqrt{10}}{168} & \frac{\sqrt{105}}{224} & 0 & \frac{3\sqrt{5}}{224} & 0 & -\frac{\sqrt{3}}{224} & 0 & -\frac{3\sqrt{15}}{224} & 0 \\ \frac{5\sqrt{10}}{168} & 0 & \frac{1}{14} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3\sqrt{15}}{224} & 0 & \frac{\sqrt{3}}{224} & 0 & -\frac{3\sqrt{5}}{224} & 0 & -\frac{\sqrt{105}}{224} \\ 0 & \frac{5}{42} & 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & \frac{3\sqrt{10}}{224} & 0 & -\frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{30}}{224} & 0 \\ 0 & 0 & \frac{5\sqrt{10}}{168} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{224} & 0 & -\frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{224} \end{bmatrix}$$

$$\boxed{\text{x168}} \quad \mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} \frac{9\sqrt{5}i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 & \frac{45i}{224} & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 \\ 0 & -\frac{3\sqrt{3}i}{32} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & \frac{15\sqrt{15}i}{224} & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 \\ \frac{15\sqrt{15}i}{224} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & -\frac{3\sqrt{3}i}{32} & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} \\ 0 & \frac{45i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 & \frac{9\sqrt{5}i}{224} & 0 & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 \\ 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{5\sqrt{10}i}{168} & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{5i}{42} & 0 & 0 & \frac{\sqrt{30}i}{224} & 0 & -\frac{\sqrt{6}i}{112} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{56} & 0 & -\frac{i}{14} & 0 & \frac{5\sqrt{10}i}{168} & -\frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & -\frac{3\sqrt{15}i}{224} & 0 \\ -\frac{5\sqrt{10}i}{168} & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{3\sqrt{15}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & \frac{3\sqrt{5}i}{224} & 0 & -\frac{\sqrt{105}i}{224} \\ 0 & -\frac{5i}{42} & 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & -\frac{3\sqrt{10}i}{224} & 0 & -\frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{30}i}{224} & 0 \\ 0 & 0 & -\frac{5\sqrt{10}i}{168} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{224} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{42}i}{224} \end{bmatrix}$$

$$\boxed{\text{x169}} \quad \mathbb{Q}_{3,1}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} \frac{15\sqrt{3}}{224} & 0 & -\frac{9\sqrt{30}}{224} & 0 & -\frac{9\sqrt{15}}{224} & 0 & 0 & -\frac{5\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{112} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{32} & 0 & \frac{3\sqrt{10}}{224} & 0 & -\frac{45}{224} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 \\ \frac{45}{224} & 0 & -\frac{3\sqrt{10}}{224} & 0 & \frac{3\sqrt{5}}{32} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{112} & 0 & 0 & 0 & \frac{\sqrt{42}}{112} \\ 0 & \frac{9\sqrt{15}}{224} & 0 & \frac{9\sqrt{30}}{224} & 0 & -\frac{15\sqrt{3}}{224} & 0 & 0 & \frac{\sqrt{6}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{5\sqrt{2}}{112} & 0 \\ 0 & -\frac{5\sqrt{3}}{84} & 0 & -\frac{5\sqrt{6}}{168} & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{3\sqrt{2}}{224} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{30}}{168} & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & \frac{5\sqrt{2}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & \frac{3\sqrt{6}}{224} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{5\sqrt{6}}{168} & -\frac{3\sqrt{7}}{224} & 0 & \frac{5\sqrt{3}}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & \frac{9}{224} & 0 \\ -\frac{5\sqrt{6}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{9}{224} & 0 & \frac{\sqrt{5}}{224} & 0 & -\frac{5\sqrt{3}}{224} & 0 & \frac{3\sqrt{7}}{224} \\ 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{30}}{168} & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & -\frac{3\sqrt{6}}{224} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{5\sqrt{2}}{224} & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{168} & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{224} & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{70}}{224} \end{bmatrix}$$

$$\boxed{\text{x170}} \quad \mathbb{Q}_{3,2}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} \frac{15\sqrt{3}i}{224} & 0 & \frac{9\sqrt{30}i}{224} & 0 & -\frac{9\sqrt{15}i}{224} & 0 & 0 & -\frac{5\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{32} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & -\frac{45i}{224} & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 \\ -\frac{45i}{224} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & -\frac{3\sqrt{5}i}{32} & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\ 0 & -\frac{9\sqrt{15}i}{224} & 0 & \frac{9\sqrt{30}i}{224} & 0 & \frac{15\sqrt{3}i}{224} & 0 & 0 & -\frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{5\sqrt{2}i}{112} & 0 \\ 0 & \frac{5\sqrt{3}i}{84} & 0 & -\frac{5\sqrt{6}i}{168} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{2}i}{224} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & \frac{5\sqrt{2}i}{224} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{3\sqrt{6}i}{224} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{5\sqrt{6}i}{168} & \frac{3\sqrt{7}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & \frac{9i}{224} & 0 \\ \frac{5\sqrt{6}i}{168} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{9i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & \frac{3\sqrt{7}i}{224} \\ 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{30}i}{168} & 0 & \frac{5\sqrt{3}i}{84} & 0 & 0 & \frac{3\sqrt{6}i}{224} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{5\sqrt{2}i}{224} & 0 \\ 0 & 0 & \frac{5\sqrt{6}i}{168} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{224} & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{70}i}{224} \end{bmatrix}$$

$$\boxed{\text{x171}} \quad \mathbb{G}_2^{(a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\ -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x172}} \quad \mathbb{G}_4^{(a)}(A_{1u}, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 \\ \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x173}} \quad \mathbb{G}_4^{(a)}(A_{1u}, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{15}i}{84} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{40} & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 \\ \frac{\sqrt{15}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x174}} \quad \mathbb{G}_4^{(a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x175}} \quad \mathbb{G}_2^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}i}{35} & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 \\ 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 \\ 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x176}} \quad \mathbb{G}_4^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{9\sqrt{3}i}{140} & 0 & 0 & 0 & -\frac{17i}{140} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{17i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x177}} \quad \mathbb{G}_2^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 \\ 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & \frac{3\sqrt{15}}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 \\ 0 & 0 & \frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x178}} \quad \mathbb{G}_4^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{6}}{280} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 & 0 & \frac{33\sqrt{2}}{280} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{280} \\ 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 & 0 & 0 & \frac{17}{140} & 0 & 0 \\ 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & \frac{17}{140} & 0 & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x179}} \quad \mathbb{G}_{2,1}^{(a)}(E_u) = \begin{bmatrix} \frac{\sqrt{6}}{14} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{70} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & \frac{1}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & \frac{1}{14} & 0 \\ 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & \frac{11\sqrt{5}}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}}{140} & 0 & -\frac{1}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{35}}{28} \end{bmatrix}$$

$$\boxed{\text{x180}} \quad \mathbb{G}_{2,2}^{(a)}(E_u) = \begin{bmatrix} \frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{70} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & -\frac{i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & -\frac{i}{14} & 0 \\ 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{11\sqrt{5}i}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}i}{140} & 0 & \frac{i}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{35}i}{28} \end{bmatrix}$$

$$\boxed{\text{x181}} \quad \mathbb{G}_{4,1}^{(a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{7}}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & 0 & \frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{210}}{280} & 0 & \frac{9\sqrt{14}}{560} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{21}}{112} & -\frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{140} & 0 & -\frac{3\sqrt{70}}{560} & 0 & \frac{3\sqrt{14}}{280} & 0 \\ \frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{70}}{560} & 0 & -\frac{3\sqrt{42}}{140} & 0 & -\frac{3\sqrt{2}}{80} \\ 0 & -\frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & \frac{9\sqrt{14}}{560} & 0 & \frac{3\sqrt{210}}{280} & 0 & \frac{9\sqrt{42}}{560} & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\ \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{42}}{560} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{14}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & \frac{3\sqrt{3}}{80} & 0 & \frac{\sqrt{7}}{560} & 0 & -\frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{21}}{560} & 0 \\ \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{21}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{7}}{560} & 0 & -\frac{3\sqrt{3}}{80} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{13\sqrt{42}}{560} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{30}}{80} \end{bmatrix}$$

$$\boxed{\text{x182}} \quad \mathbb{G}_{4,2}^{(a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{7}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{112} & 0 & 0 & \frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & \frac{9\sqrt{14}i}{560} & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & \frac{3\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{140} & 0 & \frac{3\sqrt{70}i}{560} & 0 & \frac{3\sqrt{14}i}{280} & 0 \\ -\frac{\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & -\frac{3\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & \frac{3\sqrt{42}i}{140} & 0 & -\frac{3\sqrt{2}i}{80} \\ 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & -\frac{9\sqrt{14}i}{560} & 0 & \frac{3\sqrt{210}i}{280} & 0 & -\frac{9\sqrt{42}i}{560} & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{42}i}{560} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{14}i}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & -\frac{3\sqrt{3}i}{80} & 0 & \frac{\sqrt{7}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & -\frac{\sqrt{21}i}{560} & 0 \\ -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{7}i}{560} & 0 & -\frac{3\sqrt{3}i}{80} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{13\sqrt{42}i}{560} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{30}i}{80} \end{bmatrix}$$

$$\boxed{\text{x183}} \quad \mathbb{G}_{4,1}^{(a)}(E_u, 2) = \begin{bmatrix} \frac{1}{112} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{16} & 0 & 0 & \frac{9\sqrt{6}}{560} & 0 & \frac{3\sqrt{30}}{280} & 0 & -\frac{9\sqrt{2}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{112} & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{3}}{16} & \frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{6}}{140} & 0 & -\frac{3\sqrt{10}}{560} & 0 & -\frac{3\sqrt{2}}{40} & 0 \\ -\frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{15}}{112} & 0 & 0 & -\frac{3\sqrt{2}}{40} & 0 & -\frac{3\sqrt{10}}{560} & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{14}}{80} \\ 0 & \frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{1}{112} & 0 & 0 & -\frac{9\sqrt{2}}{80} & 0 & \frac{3\sqrt{30}}{280} & 0 & \frac{9\sqrt{6}}{560} & 0 \\ 0 & -\frac{1}{28} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{210}}{560} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ \frac{1}{28} & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}}{560} & 0 & -\frac{\sqrt{30}}{280} & 0 & -\frac{7\sqrt{2}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & -\frac{3\sqrt{21}}{80} & 0 & \frac{1}{560} & 0 & -\frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{3}}{80} & 0 \\ -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & -\frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{15}}{80} & 0 & -\frac{1}{560} & 0 & \frac{3\sqrt{21}}{80} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{1}{28} & 0 & 0 & \frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{280} & 0 & \frac{13\sqrt{6}}{560} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{8} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{560} \end{bmatrix}$$

$$\boxed{\text{x184}} \quad \mathbb{G}_{4,2}^{(a)}(E_u, 2) = \begin{bmatrix} \frac{i}{112} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & \frac{9\sqrt{6}i}{560} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & -\frac{9\sqrt{2}i}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{112} & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{3}i}{16} & -\frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{560} & 0 & -\frac{3\sqrt{2}i}{40} & 0 \\ \frac{\sqrt{3}i}{16} & 0 & \frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{15}i}{112} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & -\frac{3\sqrt{10}i}{560} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{14}i}{80} \\ 0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{i}{112} & 0 & 0 & \frac{9\sqrt{2}i}{80} & 0 & \frac{3\sqrt{30}i}{280} & 0 & -\frac{9\sqrt{6}i}{560} & 0 \\ 0 & \frac{i}{28} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{210}i}{560} & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ \frac{i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}i}{560} & 0 & \frac{\sqrt{30}i}{280} & 0 & -\frac{7\sqrt{2}i}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & \frac{3\sqrt{21}i}{80} & 0 & \frac{i}{560} & 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{\sqrt{3}i}{80} & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{3}i}{80} & 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{i}{560} & 0 & \frac{3\sqrt{21}i}{80} \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{i}{28} & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & \frac{\sqrt{30}i}{280} & 0 & -\frac{13\sqrt{6}i}{560} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{560} \end{bmatrix}$$

$$\boxed{\text{x185}} \quad \mathbb{G}_2^{(1,-1;a)}(A_{1u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 \\ \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x186}} \quad \mathbb{G}_4^{(1,-1;a)}(A_{1u},1) = \begin{bmatrix} 0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{35}i}{168} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\ -\frac{\sqrt{7}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{70}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 \end{bmatrix}$$

$$\boxed{\text{x187}} \quad \mathbb{G}_4^{(1,-1;a)}(A_{1u},2) = \begin{bmatrix} 0 & \frac{\sqrt{5}i}{168} & 0 & 0 & 0 & -\frac{i}{24} & 0 & 0 & \frac{5\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{10}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{10}i}{168} & 0 & 0 & 0 \\ \frac{i}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{168} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{5\sqrt{2}i}{56} & 0 & 0 \\ -\frac{\sqrt{5}i}{84} & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & -\frac{5\sqrt{30}i}{168} & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{15}i}{168} & 0 & 0 & 0 \\ \frac{i}{12} & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{42} & 0 & 0 \\ 0 & \frac{i}{12} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{84} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & \frac{5\sqrt{30}i}{168} & 0 \end{bmatrix}$$

$$\boxed{\text{x188}} \quad \mathbb{G}_6^{(1,-1;a)}(A_{1u},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{264} & 0 & 0 & 0 & -\frac{7\sqrt{11}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 & \frac{7\sqrt{165}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{264} & 0 \end{bmatrix}$$

$$\boxed{\text{x189}} \quad \mathbb{G}_6^{(1,-1;a)}(A_{1u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{264} & 0 & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & \frac{\sqrt{330}i}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & \frac{\sqrt{385}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{264} & 0 \end{bmatrix}$$

$$\boxed{\text{x190}} \quad \mathbb{G}_4^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x191}} \quad \mathbb{G}_6^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x192}} \quad \mathbb{G}_2^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 \\ 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} \\ 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 \\ 0 & \frac{3i}{35} & 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & \frac{\sqrt{30}i}{35} & 0 \\ 0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x193}} \quad \mathbb{G}_4^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\ 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{9i}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & \frac{17\sqrt{3}i}{168} & 0 & 0 & 0 & -\frac{9i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x194}} \quad \mathbb{G}_6^{(1,-1;a)}(B_{1u},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & 0 & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{24} & 0 & 0 & 0 & -\frac{i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x195}} \quad \mathbb{G}_6^{(1,-1;a)}(B_{1u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{264} & 0 & 0 & 0 & \frac{\sqrt{11}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{264} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x196}} \quad \mathbb{G}_2^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & \frac{\sqrt{30}}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 \\ 0 & -\frac{3}{35} & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0 \\ 0 & 0 & -\frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x197}} \quad \mathbb{G}_4^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{11\sqrt{6}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & \frac{9}{56} & 0 & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 & 0 & \frac{9}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$$

x198

x199

$$\boxed{\text{x200}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} -\frac{3\sqrt{10}}{140} & 0 & -\frac{9}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{140} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 & -\frac{6}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{9}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 \\ 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{35} & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & \frac{11\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{140} & 0 & -\frac{\sqrt{15}}{140} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & -\frac{\sqrt{21}}{28} \end{bmatrix}$$

$$\boxed{\text{x201}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} -\frac{3\sqrt{10}i}{140} & 0 & \frac{9i}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{140} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & -\frac{6i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{9i}{140} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 \\ 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{35} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{140} & 0 & -\frac{11\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}i}{140} & 0 & \frac{\sqrt{15}i}{140} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{28} & 0 & \frac{\sqrt{21}i}{28} \end{bmatrix}$$

$$\boxed{\text{x202}} \quad \mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{105}}{672} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{224} & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{7}}{224} & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{42}}{168} & 0 \\ -\frac{\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{35}}{224} & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{6}}{48} \\ 0 & \frac{\sqrt{105}}{672} & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{\sqrt{21}}{672} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{14}}{112} & 0 \\ 0 & \frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & \frac{5\sqrt{210}}{336} & 0 & \frac{5\sqrt{14}}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{224} & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{96} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & \frac{3}{32} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{35}}{32} & 0 & -\frac{\sqrt{7}}{224} & 0 \\ -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{32} & 0 & -\frac{\sqrt{21}}{672} & 0 & -\frac{3}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{42}}{96} & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{13\sqrt{14}}{224} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{224} & 0 & -\frac{5\sqrt{210}}{336} & 0 & -\frac{\sqrt{10}}{32} \end{bmatrix}$$

$$\boxed{\text{x203}} \quad \mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{105}i}{672} & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{224} & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{42}i}{168} & 0 \\ \frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{35}i}{224} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{6}i}{48} \\ 0 & -\frac{\sqrt{105}i}{672} & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{21}i}{672} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 \\ 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}i}{224} & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{42}i}{96} & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & -\frac{3i}{32} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{32} & 0 & -\frac{\sqrt{7}i}{224} & 0 \\ \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{32} & 0 & \frac{\sqrt{21}i}{672} & 0 & -\frac{3i}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{96} & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{13\sqrt{14}i}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & \frac{\sqrt{10}i}{32} \end{bmatrix}$$

$$\boxed{\text{x204}} \quad \mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{3}}{672} & 0 & -\frac{\sqrt{30}}{672} & 0 & \frac{\sqrt{15}}{96} & 0 & 0 & -\frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\ 0 & \frac{\sqrt{5}}{224} & 0 & \frac{\sqrt{10}}{224} & 0 & -\frac{1}{32} & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{\sqrt{6}}{24} & 0 \\ \frac{1}{32} & 0 & -\frac{\sqrt{10}}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{42}}{48} \\ 0 & -\frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{30}}{672} & 0 & \frac{\sqrt{3}}{672} & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{3\sqrt{2}}{112} & 0 \\ 0 & \frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{5\sqrt{30}}{336} & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{224} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{7\sqrt{6}}{96} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & -\frac{3\sqrt{7}}{32} & 0 & \frac{\sqrt{3}}{672} & 0 & -\frac{\sqrt{5}}{32} & 0 & \frac{1}{32} & 0 \\ \frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{1}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & -\frac{\sqrt{3}}{672} & 0 & \frac{3\sqrt{7}}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & \frac{7\sqrt{6}}{96} & 0 & \frac{\sqrt{10}}{112} & 0 & \frac{13\sqrt{2}}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{32} & 0 & -\frac{5\sqrt{30}}{336} & 0 & -\frac{\sqrt{70}}{224} \end{bmatrix}$$

$$\boxed{\text{x205}} \quad \mathbb{G}_{4,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{30}i}{672} & 0 & \frac{\sqrt{15}i}{96} & 0 & 0 & -\frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & -\frac{i}{32} & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{30}i}{336} & 0 & \frac{\sqrt{6}i}{24} & 0 \\ -\frac{i}{32} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{30}i}{336} & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{42}i}{48} \\ 0 & \frac{\sqrt{15}i}{96} & 0 & \frac{\sqrt{30}i}{672} & 0 & -\frac{\sqrt{3}i}{672} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 \\ 0 & -\frac{\sqrt{3}i}{84} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & -\frac{5\sqrt{30}i}{336} & 0 & -\frac{5\sqrt{2}i}{32} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{7\sqrt{6}i}{96} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & \frac{3\sqrt{7}i}{32} & 0 & \frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{5}i}{32} & 0 & \frac{i}{32} & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{i}{32} & 0 & \frac{\sqrt{5}i}{32} & 0 & \frac{\sqrt{3}i}{672} & 0 & \frac{3\sqrt{7}i}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & -\frac{7\sqrt{6}i}{96} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{13\sqrt{2}i}{224} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & -\frac{5\sqrt{30}i}{336} & 0 & \frac{\sqrt{70}i}{224} \end{bmatrix}$$

$$\boxed{\text{x206}} \quad \mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{1056} & 0 & \frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{462}}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1056} & 0 & -\frac{5\sqrt{462}}{1056} & 0 & \frac{3\sqrt{770}}{352} & 0 & \frac{\sqrt{330}}{96} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{385}}{176} & 0 & \frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{1155}}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{385}}{176} & 0 & \frac{\sqrt{165}}{176} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{96} & 0 & -\frac{3\sqrt{770}}{352} & 0 & \frac{5\sqrt{462}}{1056} & 0 & \frac{\sqrt{2310}}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{96} & 0 & \frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{66}}{1056} \end{bmatrix}$$

$$\boxed{\text{x207}} \quad \mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{1056} & 0 & -\frac{\sqrt{154}i}{352} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{462}i}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1056} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & \frac{3\sqrt{770}i}{352} & 0 & -\frac{\sqrt{330}i}{96} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{176} & 0 & \frac{\sqrt{385}i}{176} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & -\frac{\sqrt{1155}i}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{176} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{385}i}{176} & 0 & \frac{\sqrt{165}i}{176} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{96} & 0 & \frac{3\sqrt{770}i}{352} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & -\frac{\sqrt{2310}i}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{96} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & -\frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{66}i}{1056} \end{bmatrix}$$

$$\boxed{\text{x208}} \quad \mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & \frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & \frac{\sqrt{7}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{5}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & -\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{10}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & -\frac{1}{64} & 0 \end{bmatrix}$$

$$\boxed{\text{x209}} \quad \mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & -\frac{\sqrt{7}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{70}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & \frac{i}{64} & 0 \end{bmatrix}$$

$$\boxed{\text{x210}} \quad \mathbb{G}_{6,1}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{2112} & 0 & \frac{\sqrt{1155}}{2112} & 0 & -\frac{9\sqrt{77}}{704} & 0 & \frac{\sqrt{385}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{2112} & 0 & -\frac{5\sqrt{385}}{2112} & 0 & \frac{9\sqrt{231}}{704} & 0 & -\frac{5\sqrt{11}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{2112} & 0 & \frac{5\sqrt{770}}{2112} & 0 & -\frac{9\sqrt{154}}{704} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{704} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & -\frac{5\sqrt{462}}{2112} & 0 & \frac{9\sqrt{22}}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{64} & 0 & -\frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{385}}{2112} & 0 & \frac{5\sqrt{77}}{2112} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{64} & 0 & \frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & -\frac{\sqrt{55}}{2112} & 0 \end{bmatrix}$$

$$\boxed{\text{x211}} \quad \mathbb{G}_{6,2}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{2112} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & -\frac{\sqrt{385}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{2112} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & \frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{11}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}i}{704} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & -\frac{9\sqrt{154}i}{704} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}i}{704} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & \frac{9\sqrt{22}i}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}i}{64} & 0 & \frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & -\frac{5\sqrt{77}i}{2112} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{64} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & \frac{\sqrt{55}i}{2112} & 0 \end{bmatrix}$$

$$\boxed{\text{x212}} \quad \mathbb{G}_2^{(1,0;a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\ -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x213}} \quad \mathbb{G}_4^{(1,0;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{105}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & \frac{9\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{14}i}{280} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{6}i}{40} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{168} & 0 & 0 & 0 & \frac{\sqrt{105}i}{840} & 0 & 0 & \frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{42}i}{280} & 0 & 0 \\ \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{\sqrt{210}i}{840} & 0 & 0 \\ 0 & -\frac{3\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{i}{40} \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{i}{40} & 0 & 0 & 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{210}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 \end{bmatrix}$$

$$\boxed{\text{x214}} \quad \mathbb{G}_4^{(1,0;a)}(A_{1u}, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{3}i}{168} & 0 & 0 & 0 & \frac{\sqrt{15}i}{120} & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{200} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{200} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{3\sqrt{210}i}{200} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{120} & 0 & 0 & 0 & \frac{\sqrt{3}i}{168} & 0 & 0 & -\frac{9\sqrt{10}i}{200} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 \\ \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{120} & 0 & 0 \\ 0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{200} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & \frac{\sqrt{6}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 \end{bmatrix}$$

$$\boxed{\text{x215}} \quad \mathbb{G}_4^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}}{700} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x216}} \quad \mathbb{G}_2^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & -\frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \\ 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\ 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x217}} \quad \mathbb{G}_4^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{9\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{\sqrt{15}i}{280} & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{1400} \\ 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{3\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{280} & 0 & 0 & 0 \\ -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 & 0 & 0 & \frac{17\sqrt{5}i}{1400} & 0 & 0 \\ 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 & 0 & \frac{9\sqrt{15}i}{1400} & 0 \\ 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{1400} \\ 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x218}} \quad \mathbb{G}_2^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \\ 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 \\ 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x219}} \quad \mathbb{G}_4^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & -\frac{9\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{280} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & \frac{99\sqrt{10}}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}}{280} & 0 & 0 & -\frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}}{1400} \\ 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{3\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 \\ -\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}}{1400} \\ 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x220}} \quad \mathbb{G}_{2,1}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{1}{14} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 \\ 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{168} & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & -\frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{168} \end{bmatrix}$$

$$\boxed{\text{x221}} \quad \mathbb{G}_{2,2}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{i}{14} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{210} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 \\ 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 \\ \frac{3i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & \frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & \frac{i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & -\frac{\sqrt{6}i}{168} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{168} \end{bmatrix}$$

$$\boxed{\text{x222}} \quad \mathbb{G}_{4,1}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{35}}{1120} & 0 & \frac{\sqrt{14}}{224} & 0 & \frac{\sqrt{7}}{224} & 0 & 0 & -\frac{27\sqrt{210}}{2800} & 0 & -\frac{9\sqrt{42}}{280} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{224} & 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{105}}{1120} & \frac{9\sqrt{10}}{400} & 0 & \frac{9\sqrt{210}}{700} & 0 & \frac{9\sqrt{14}}{560} & 0 & -\frac{9\sqrt{70}}{1400} & 0 \\ \frac{\sqrt{105}}{1120} & 0 & \frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{21}}{224} & 0 & 0 & -\frac{9\sqrt{70}}{1400} & 0 & \frac{9\sqrt{14}}{560} & 0 & \frac{9\sqrt{210}}{700} & 0 & \frac{9\sqrt{10}}{400} \\ 0 & -\frac{\sqrt{7}}{224} & 0 & -\frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{35}}{1120} & 0 & 0 & -\frac{27\sqrt{70}}{2800} & 0 & -\frac{9\sqrt{42}}{280} & 0 & -\frac{27\sqrt{210}}{2800} & 0 \\ 0 & -\frac{3\sqrt{35}}{140} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{6}}{160} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{1120} & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}}{5600} & 0 & \frac{\sqrt{42}}{560} & 0 & -\frac{\sqrt{70}}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & -\frac{3\sqrt{15}}{800} & 0 & -\frac{\sqrt{35}}{5600} & 0 & \frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{105}}{5600} & 0 \\ \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{105}}{5600} & 0 & -\frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{35}}{5600} & 0 & \frac{3\sqrt{15}}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{70}}{800} & 0 & -\frac{\sqrt{42}}{560} & 0 & -\frac{13\sqrt{210}}{5600} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{1120} & 0 & \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{6}}{160} \end{bmatrix}$$

$$\boxed{\text{x223}} \quad \mathbb{G}_{4,2}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{35}i}{1120} & 0 & -\frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{7}i}{224} & 0 & 0 & -\frac{27\sqrt{210}i}{2800} & 0 & \frac{9\sqrt{42}i}{280} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{224} & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{105}i}{1120} & -\frac{9\sqrt{10}i}{400} & 0 & \frac{9\sqrt{210}i}{700} & 0 & -\frac{9\sqrt{14}i}{560} & 0 & -\frac{9\sqrt{70}i}{1400} & 0 \\ -\frac{\sqrt{105}i}{1120} & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{21}i}{224} & 0 & 0 & \frac{9\sqrt{70}i}{1400} & 0 & \frac{9\sqrt{14}i}{560} & 0 & -\frac{9\sqrt{210}i}{700} & 0 & \frac{9\sqrt{10}i}{400} \\ 0 & \frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{35}i}{1120} & 0 & 0 & \frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{42}i}{280} & 0 & \frac{27\sqrt{210}i}{2800} & 0 \\ 0 & \frac{3\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{1120} & 0 & 0 & 0 \\ \frac{3\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}i}{5600} & 0 & -\frac{\sqrt{42}i}{560} & 0 & -\frac{\sqrt{70}i}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & \frac{3\sqrt{15}i}{800} & 0 & -\frac{\sqrt{35}i}{5600} & 0 & -\frac{\sqrt{21}i}{160} & 0 & \frac{\sqrt{105}i}{5600} & 0 \\ -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{105}i}{5600} & 0 & -\frac{\sqrt{21}i}{160} & 0 & -\frac{\sqrt{35}i}{5600} & 0 & \frac{3\sqrt{15}i}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & -\frac{\sqrt{70}i}{800} & 0 & -\frac{\sqrt{42}i}{560} & 0 & \frac{13\sqrt{210}i}{5600} & 0 \\ 0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{1120} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{160} \end{bmatrix}$$

$$\boxed{x224} \quad \mathbb{G}_{4,1}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{5}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{1}{32} & 0 & 0 & -\frac{27\sqrt{30}}{2800} & 0 & -\frac{9\sqrt{6}}{280} & 0 & \frac{27\sqrt{10}}{400} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{224} & 0 & -\frac{\sqrt{6}}{224} & 0 & \frac{\sqrt{15}}{160} & -\frac{9\sqrt{70}}{400} & 0 & \frac{9\sqrt{30}}{700} & 0 & \frac{9\sqrt{2}}{560} & 0 & \frac{9\sqrt{10}}{200} & 0 \\ -\frac{\sqrt{15}}{160} & 0 & \frac{\sqrt{6}}{224} & 0 & \frac{\sqrt{3}}{224} & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 & \frac{9\sqrt{2}}{560} & 0 & \frac{9\sqrt{30}}{700} & 0 & -\frac{9\sqrt{70}}{400} \\ 0 & \frac{1}{32} & 0 & -\frac{\sqrt{2}}{224} & 0 & -\frac{\sqrt{5}}{1120} & 0 & 0 & \frac{27\sqrt{10}}{400} & 0 & -\frac{9\sqrt{6}}{280} & 0 & -\frac{27\sqrt{30}}{2800} & 0 \\ 0 & -\frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{10}}{40} & 0 & 0 & -\frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{160} & 0 & 0 & 0 \\ \frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}}{5600} & 0 & \frac{\sqrt{6}}{560} & 0 & \frac{7\sqrt{10}}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{40} & \frac{3\sqrt{105}}{800} & 0 & -\frac{\sqrt{5}}{5600} & 0 & \frac{\sqrt{3}}{160} & 0 & -\frac{\sqrt{15}}{800} & 0 \\ -\frac{3\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{\sqrt{15}}{800} & 0 & -\frac{\sqrt{3}}{160} & 0 & \frac{\sqrt{5}}{5600} & 0 & -\frac{3\sqrt{105}}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{7\sqrt{10}}{800} & 0 & -\frac{\sqrt{6}}{560} & 0 & -\frac{13\sqrt{30}}{5600} & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{40} & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & \frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{1120} \end{bmatrix}$$

$$\boxed{x225} \quad \mathbb{G}_{4,2}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{5}i}{1120} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{i}{32} & 0 & 0 & -\frac{27\sqrt{30}i}{2800} & 0 & \frac{9\sqrt{6}i}{280} & 0 & \frac{27\sqrt{10}i}{400} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{224} & 0 & \frac{\sqrt{6}i}{224} & 0 & \frac{\sqrt{15}i}{160} & \frac{9\sqrt{70}i}{400} & 0 & \frac{9\sqrt{30}i}{700} & 0 & -\frac{9\sqrt{2}i}{560} & 0 & \frac{9\sqrt{10}i}{200} & 0 \\ \frac{\sqrt{15}i}{160} & 0 & \frac{\sqrt{6}i}{224} & 0 & -\frac{\sqrt{3}i}{224} & 0 & 0 & -\frac{9\sqrt{10}i}{200} & 0 & \frac{9\sqrt{2}i}{560} & 0 & -\frac{9\sqrt{30}i}{700} & 0 & -\frac{9\sqrt{70}i}{400} \\ 0 & -\frac{i}{32} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{\sqrt{5}i}{1120} & 0 & 0 & -\frac{27\sqrt{10}i}{400} & 0 & -\frac{9\sqrt{6}i}{280} & 0 & \frac{27\sqrt{30}i}{2800} & 0 \\ 0 & \frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{40} & 0 & 0 & -\frac{\sqrt{42}i}{1120} & 0 & \frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 & 0 \\ \frac{3\sqrt{5}i}{140} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{5600} & 0 & -\frac{\sqrt{6}i}{560} & 0 & \frac{7\sqrt{10}i}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{40} & -\frac{3\sqrt{105}i}{800} & 0 & -\frac{\sqrt{5}i}{5600} & 0 & -\frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{15}i}{800} & 0 \\ \frac{3\sqrt{10}i}{40} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & -\frac{\sqrt{15}i}{800} & 0 & -\frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{5}i}{5600} & 0 & -\frac{3\sqrt{105}i}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & \frac{7\sqrt{10}i}{800} & 0 & -\frac{\sqrt{6}i}{560} & 0 & \frac{13\sqrt{30}i}{5600} & 0 \\ 0 & 0 & -\frac{3\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{42}i}{1120} \end{bmatrix}$$

$$\boxed{\text{x226}} \quad \mathbb{G}_0^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x227}} \quad \mathbb{G}_2^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\ \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 \end{bmatrix}$$

$$\boxed{\text{x228}} \quad \mathbb{G}_4^{(1,1;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{770}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{42} & 0 & 0 & \frac{\sqrt{77}i}{140} & 0 & 0 & 0 & \frac{\sqrt{231}i}{420} & 0 \\ 0 & 0 & \frac{\sqrt{1155}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{420} & 0 & 0 & 0 & \frac{\sqrt{11}i}{60} \\ 0 & 0 & 0 & -\frac{\sqrt{1155}i}{105} & 0 & 0 & \frac{\sqrt{11}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{420} & 0 & 0 & 0 \\ \frac{\sqrt{154}i}{42} & 0 & 0 & 0 & \frac{\sqrt{770}i}{210} & 0 & 0 & \frac{\sqrt{231}i}{420} & 0 & 0 & 0 & \frac{\sqrt{77}i}{140} & 0 & 0 \\ -\frac{\sqrt{770}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{168} & 0 & 0 & \frac{\sqrt{1155}i}{1155} & 0 & 0 & 0 & \frac{\sqrt{385}i}{1155} & 0 & 0 \\ 0 & \frac{\sqrt{770}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{168} & 0 & 0 & -\frac{4\sqrt{77}i}{1155} & 0 & 0 & 0 & \frac{2\sqrt{231}i}{1155} & 0 \\ 0 & 0 & -\frac{\sqrt{770}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{2310} & 0 & 0 & 0 & \frac{\sqrt{66}i}{330} \\ 0 & 0 & 0 & -\frac{\sqrt{770}i}{420} & 0 & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{2310} & 0 & 0 & 0 \\ -\frac{\sqrt{154}i}{168} & 0 & 0 & 0 & \frac{\sqrt{770}i}{280} & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 & 0 & \frac{4\sqrt{77}i}{1155} & 0 & 0 \\ 0 & -\frac{\sqrt{154}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{840} & 0 & 0 & -\frac{\sqrt{385}i}{1155} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{1155} & 0 \end{bmatrix}$$

$$\boxed{\text{x229}} \quad \mathbb{G}_4^{(1,1;a)}(A_{1u}, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{22}i}{42} & 0 & 0 & 0 & \frac{\sqrt{110}i}{30} & 0 & 0 & \frac{\sqrt{55}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{300} \\ 0 & 0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & -\frac{\sqrt{385}i}{300} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{110}i}{30} & 0 & 0 & 0 & \frac{\sqrt{22}i}{42} & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 & 0 & 0 & \frac{\sqrt{55}i}{140} & 0 & 0 \\ -\frac{\sqrt{22}i}{168} & 0 & 0 & 0 & \frac{\sqrt{110}i}{120} & 0 & 0 & \frac{\sqrt{33}i}{231} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{165} & 0 & 0 \\ 0 & \frac{\sqrt{22}i}{56} & 0 & 0 & 0 & \frac{\sqrt{110}i}{120} & 0 & 0 & -\frac{4\sqrt{55}i}{1155} & 0 & 0 & 0 & -\frac{2\sqrt{165}i}{825} & 0 \\ 0 & 0 & -\frac{\sqrt{22}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1650} \\ 0 & 0 & 0 & -\frac{\sqrt{22}i}{84} & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 & 0 & \frac{\sqrt{66}i}{462} & 0 & 0 & 0 \\ \frac{\sqrt{110}i}{120} & 0 & 0 & 0 & \frac{\sqrt{22}i}{56} & 0 & 0 & \frac{2\sqrt{165}i}{825} & 0 & 0 & 0 & \frac{4\sqrt{55}i}{1155} & 0 & 0 \\ 0 & \frac{\sqrt{110}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{168} & 0 & 0 & \frac{\sqrt{11}i}{165} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{231} & 0 \end{bmatrix}$$

$$\boxed{\text{x230}} \quad \mathbb{G}_4^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{275} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x231}} \quad \mathbb{G}_2^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 \\ -\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{8i}{35} & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 \\ 0 & -\frac{8i}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 \\ 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} \\ 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}i}{105} & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{105} & 0 & 0 \\ 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}i}{105} & 0 \\ 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{105} \\ 0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{x232} \quad \mathbb{G}_4^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & -\frac{\sqrt{385}i}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{110}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 \\ 0 & \frac{\sqrt{22}i}{14} & 0 & 0 & 0 & \frac{\sqrt{110}i}{70} & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 \\ 0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{700} \\ 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & \frac{3\sqrt{385}i}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 \\ \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & -\frac{9\sqrt{110}i}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 \\ 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & -\frac{17\sqrt{330}i}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 \\ 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1925} \\ 0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{x233} \quad \mathbb{G}_2^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 \\ \frac{4\sqrt{5}}{35} & 0 & 0 & 0 & \frac{8}{35} & 0 & 0 & \frac{3\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 \\ 0 & \frac{8}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & \frac{3\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 \\ 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\ 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & -\frac{\sqrt{70}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 \\ -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 \\ 0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 \\ 0 & 0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{105} \\ 0 & 0 & 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{x234} \quad \mathbb{G}_4^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{700} & 0 & 0 & 0 & \frac{3\sqrt{11}}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{70} & 0 & 0 & 0 & \frac{\sqrt{22}}{14} & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 & 0 & 0 & \frac{\sqrt{55}}{700} & 0 & 0 \\ 0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & -\frac{\sqrt{110}}{70} & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 & 0 & -\frac{11\sqrt{165}}{2100} & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 & \frac{\sqrt{385}}{700} \\ 0 & 0 & -\frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & \frac{3\sqrt{385}}{1925} & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 \\ \frac{\sqrt{165}}{140} & 0 & 0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & -\frac{\sqrt{165}}{140} & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 & \frac{3\sqrt{385}}{1925} \\ 0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{x235} \quad \mathbb{G}_{2,1}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{2\sqrt{15}}{35} & 0 & \frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{6}}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 \\ 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{15}}{70} & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{210} & 0 & -\frac{11\sqrt{2}}{210} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{105} & 0 & -\frac{1}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & \frac{1}{15} & 0 & -\frac{\sqrt{15}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{210} & 0 & \frac{\sqrt{10}}{210} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & \frac{\sqrt{14}}{42} \end{bmatrix}$$

$$\boxed{\text{x236}} \quad \mathbb{G}_{2,2}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{2\sqrt{15}i}{35} & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2i}{35} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{6}i}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 \\ 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{15}i}{70} & 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{210} & 0 & \frac{11\sqrt{2}i}{210} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{105} & 0 & \frac{i}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{15} & 0 & \frac{\sqrt{15}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}i}{210} & 0 & -\frac{\sqrt{10}i}{210} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{14}i}{42} \end{bmatrix}$$

$$\boxed{\text{x237}} \quad \mathbb{G}_{4,1}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{3\sqrt{385}}{1400} & 0 & -\frac{\sqrt{77}}{140} & 0 & -\frac{\sqrt{1155}}{1400} & 0 & 0 \\ 0 & -\frac{\sqrt{154}}{56} & 0 & -\frac{\sqrt{77}}{28} & 0 & -\frac{\sqrt{770}}{280} & \frac{\sqrt{165}}{600} & 0 & \frac{\sqrt{385}}{350} & 0 & \frac{\sqrt{231}}{840} & 0 & -\frac{\sqrt{1155}}{2100} & 0 \\ \frac{\sqrt{770}}{280} & 0 & \frac{\sqrt{77}}{28} & 0 & \frac{\sqrt{154}}{56} & 0 & 0 & -\frac{\sqrt{1155}}{2100} & 0 & \frac{\sqrt{231}}{840} & 0 & \frac{\sqrt{385}}{350} & 0 & \frac{\sqrt{165}}{600} \\ 0 & -\frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{231}}{84} & 0 & -\frac{\sqrt{2310}}{840} & 0 & 0 & -\frac{\sqrt{1155}}{1400} & 0 & -\frac{\sqrt{77}}{140} & 0 & -\frac{3\sqrt{385}}{1400} & 0 \\ 0 & \frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{1155}}{840} & 0 & 0 & -\frac{\sqrt{11}}{220} & 0 & -\frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{385}}{1540} & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{840} & 0 & -\frac{\sqrt{231}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}}{7700} & 0 & \frac{\sqrt{77}}{770} & 0 & -\frac{\sqrt{1155}}{3300} & 0 & 0 \\ 0 & \frac{\sqrt{231}}{168} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{840} & -\frac{3\sqrt{110}}{2200} & 0 & -\frac{\sqrt{2310}}{46200} & 0 & \frac{\sqrt{154}}{440} & 0 & \frac{\sqrt{770}}{15400} & 0 \\ -\frac{\sqrt{1155}}{840} & 0 & 0 & 0 & \frac{\sqrt{231}}{168} & 0 & 0 & -\frac{\sqrt{770}}{15400} & 0 & -\frac{\sqrt{154}}{440} & 0 & \frac{\sqrt{2310}}{46200} & 0 & \frac{3\sqrt{110}}{2200} \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{168} & 0 & -\frac{\sqrt{2310}}{840} & 0 & 0 & \frac{\sqrt{1155}}{3300} & 0 & -\frac{\sqrt{77}}{770} & 0 & -\frac{13\sqrt{385}}{7700} & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{840} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{1540} & 0 & \frac{\sqrt{231}}{462} & 0 & \frac{\sqrt{11}}{220} \end{bmatrix}$$

$$\boxed{\text{x238}} \quad \mathbb{G}_{4,2}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{2310}i}{840} & 0 & -\frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{462}i}{168} & 0 & 0 & -\frac{3\sqrt{385}i}{1400} & 0 & \frac{\sqrt{77}i}{140} & 0 & -\frac{\sqrt{1155}i}{1400} & 0 & 0 \\ 0 & -\frac{\sqrt{154}i}{56} & 0 & \frac{\sqrt{77}i}{28} & 0 & -\frac{\sqrt{770}i}{280} & -\frac{\sqrt{165}i}{600} & 0 & \frac{\sqrt{385}i}{350} & 0 & -\frac{\sqrt{231}i}{840} & 0 & -\frac{\sqrt{1155}i}{2100} & 0 \\ -\frac{\sqrt{770}i}{280} & 0 & \frac{\sqrt{77}i}{28} & 0 & -\frac{\sqrt{154}i}{56} & 0 & 0 & \frac{\sqrt{1155}i}{2100} & 0 & \frac{\sqrt{231}i}{840} & 0 & -\frac{\sqrt{385}i}{350} & 0 & \frac{\sqrt{165}i}{600} \\ 0 & \frac{\sqrt{462}i}{168} & 0 & -\frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & \frac{\sqrt{1155}i}{1400} & 0 & -\frac{\sqrt{77}i}{140} & 0 & \frac{3\sqrt{385}i}{1400} & 0 \\ 0 & -\frac{\sqrt{2310}i}{840} & 0 & \frac{\sqrt{1155}i}{840} & 0 & 0 & -\frac{\sqrt{11}i}{220} & 0 & \frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{385}i}{1540} & 0 & 0 & 0 \\ -\frac{\sqrt{2310}i}{840} & 0 & \frac{\sqrt{231}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}i}{7700} & 0 & -\frac{\sqrt{77}i}{770} & 0 & -\frac{\sqrt{1155}i}{3300} & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{840} & \frac{3\sqrt{110}i}{2200} & 0 & -\frac{\sqrt{2310}i}{46200} & 0 & -\frac{\sqrt{154}i}{440} & 0 & \frac{\sqrt{770}i}{15400} & 0 \\ \frac{\sqrt{1155}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{168} & 0 & 0 & \frac{\sqrt{770}i}{15400} & 0 & -\frac{\sqrt{154}i}{440} & 0 & -\frac{\sqrt{2310}i}{46200} & 0 & \frac{3\sqrt{110}i}{2200} \\ 0 & 0 & 0 & -\frac{\sqrt{231}i}{168} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & -\frac{\sqrt{1155}i}{3300} & 0 & -\frac{\sqrt{77}i}{770} & 0 & \frac{13\sqrt{385}i}{7700} & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{840} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{1540} & 0 & \frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{11}i}{220} \end{bmatrix}$$

$$\boxed{\text{x239}} \quad \mathbb{G}_{4,1}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{330}}{840} & 0 & \frac{\sqrt{33}}{84} & 0 & -\frac{\sqrt{66}}{24} & 0 & 0 & -\frac{3\sqrt{55}}{1400} & 0 & -\frac{\sqrt{11}}{140} & 0 & \frac{\sqrt{165}}{200} & 0 & 0 \\ 0 & -\frac{\sqrt{22}}{56} & 0 & -\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{110}}{40} & -\frac{\sqrt{1155}}{600} & 0 & \frac{\sqrt{55}}{350} & 0 & \frac{\sqrt{33}}{840} & 0 & \frac{\sqrt{165}}{300} & 0 \\ -\frac{\sqrt{110}}{40} & 0 & \frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{22}}{56} & 0 & 0 & \frac{\sqrt{165}}{300} & 0 & \frac{\sqrt{33}}{840} & 0 & \frac{\sqrt{55}}{350} & 0 & -\frac{\sqrt{1155}}{600} \\ 0 & \frac{\sqrt{66}}{24} & 0 & -\frac{\sqrt{33}}{84} & 0 & -\frac{\sqrt{330}}{840} & 0 & 0 & \frac{\sqrt{165}}{200} & 0 & -\frac{\sqrt{11}}{140} & 0 & -\frac{3\sqrt{55}}{1400} & 0 \\ 0 & \frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{165}}{120} & 0 & 0 & -\frac{\sqrt{77}}{1540} & 0 & -\frac{\sqrt{33}}{462} & 0 & \frac{\sqrt{55}}{220} & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{33}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}}{7700} & 0 & \frac{\sqrt{11}}{770} & 0 & \frac{7\sqrt{165}}{3300} & 0 & 0 \\ 0 & \frac{\sqrt{33}}{168} & 0 & 0 & 0 & \frac{\sqrt{165}}{120} & \frac{3\sqrt{770}}{2200} & 0 & -\frac{\sqrt{330}}{46200} & 0 & \frac{\sqrt{22}}{440} & 0 & -\frac{\sqrt{110}}{2200} & 0 \\ \frac{\sqrt{165}}{120} & 0 & 0 & 0 & \frac{\sqrt{33}}{168} & 0 & 0 & \frac{\sqrt{110}}{2200} & 0 & -\frac{\sqrt{22}}{440} & 0 & \frac{\sqrt{330}}{46200} & 0 & -\frac{3\sqrt{770}}{2200} \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{168} & 0 & -\frac{\sqrt{330}}{840} & 0 & 0 & -\frac{7\sqrt{165}}{3300} & 0 & -\frac{\sqrt{11}}{770} & 0 & -\frac{13\sqrt{55}}{7700} & 0 \\ 0 & 0 & -\frac{\sqrt{165}}{120} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{220} & 0 & \frac{\sqrt{33}}{462} & 0 & \frac{\sqrt{77}}{1540} \end{bmatrix}$$

$$\boxed{\text{x240}} \quad \mathbb{G}_{4,2}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{330}i}{840} & 0 & -\frac{\sqrt{33}i}{84} & 0 & -\frac{\sqrt{66}i}{24} & 0 & 0 & -\frac{3\sqrt{55}i}{1400} & 0 & \frac{\sqrt{11}i}{140} & 0 & \frac{\sqrt{165}i}{200} & 0 & 0 \\ 0 & -\frac{\sqrt{22}i}{56} & 0 & \frac{\sqrt{11}i}{28} & 0 & \frac{\sqrt{110}i}{40} & \frac{\sqrt{1155}i}{600} & 0 & \frac{\sqrt{55}i}{350} & 0 & -\frac{\sqrt{33}i}{840} & 0 & \frac{\sqrt{165}i}{300} & 0 \\ \frac{\sqrt{110}i}{40} & 0 & \frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{22}i}{56} & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 & \frac{\sqrt{33}i}{840} & 0 & -\frac{\sqrt{55}i}{350} & 0 & -\frac{\sqrt{1155}i}{600} \\ 0 & -\frac{\sqrt{66}i}{24} & 0 & -\frac{\sqrt{33}i}{84} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & -\frac{\sqrt{165}i}{200} & 0 & -\frac{\sqrt{11}i}{140} & 0 & \frac{3\sqrt{55}i}{1400} & 0 \\ 0 & -\frac{\sqrt{330}i}{840} & 0 & -\frac{\sqrt{165}i}{120} & 0 & 0 & -\frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{33}i}{462} & 0 & \frac{\sqrt{55}i}{220} & 0 & 0 & 0 \\ -\frac{\sqrt{330}i}{840} & 0 & \frac{\sqrt{33}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}i}{7700} & 0 & -\frac{\sqrt{11}i}{770} & 0 & \frac{7\sqrt{165}i}{3300} & 0 & 0 \\ 0 & \frac{\sqrt{33}i}{168} & 0 & 0 & 0 & \frac{\sqrt{165}i}{120} & -\frac{3\sqrt{770}i}{2200} & 0 & -\frac{\sqrt{330}i}{46200} & 0 & -\frac{\sqrt{22}i}{440} & 0 & -\frac{\sqrt{110}i}{2200} & 0 \\ -\frac{\sqrt{165}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{168} & 0 & 0 & -\frac{\sqrt{110}i}{2200} & 0 & -\frac{\sqrt{22}i}{440} & 0 & -\frac{\sqrt{330}i}{46200} & 0 & -\frac{3\sqrt{770}i}{2200} \\ 0 & 0 & 0 & -\frac{\sqrt{33}i}{168} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & \frac{7\sqrt{165}i}{3300} & 0 & -\frac{\sqrt{11}i}{770} & 0 & \frac{13\sqrt{55}i}{7700} & 0 \\ 0 & 0 & \frac{\sqrt{165}i}{120} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{220} & 0 & \frac{\sqrt{33}i}{462} & 0 & -\frac{\sqrt{77}i}{1540} \end{bmatrix}$$

$$\boxed{\text{x241}} \quad \mathbb{M}_2^{(a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x242}} \quad \mathbb{M}_4^{(a)}(A_{1u}, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 \\ \frac{\sqrt{21}}{84} & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 \\ 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 \\ 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x243}} \quad \mathbb{M}_4^{(a)}(A_{1u}, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{15}}{84} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{3\sqrt{6}}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}}{40} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{42}}{40} & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & 0 & -\frac{3\sqrt{6}}{56} & 0 & 0 \\ \frac{\sqrt{15}}{84} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & \frac{\sqrt{7}}{20} \\ 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x244}} \quad \mathbb{M}_4^{(a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x245}} \quad \mathbb{M}_2^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 \\ 0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 \\ 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x246}} \quad \mathbb{M}_4^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}}{280} \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 & 0 & 0 & -\frac{17}{140} & 0 & 0 \\ 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{17}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x247}} \quad \mathbb{M}_2^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \\ 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \\ 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 \\ 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x248}} \quad \mathbb{M}_4^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & \frac{33\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{3\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 & 0 & 0 & -\frac{17i}{140} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{17i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x249}} \quad \mathbb{M}_{2,1}^{(a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{70} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & -\frac{i}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & -\frac{i}{14} & 0 \\ 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{11\sqrt{5}i}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{11\sqrt{5}i}{140} & 0 & \frac{i}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{35}i}{28} \end{bmatrix}$$

$$\boxed{\text{x250}} \quad \mathbb{M}_{2,2}^{(a)}(E_u) = \begin{bmatrix} \frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{70} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & -\frac{1}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & -\frac{1}{14} & 0 \\ 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{11\sqrt{5}}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}}{140} & 0 & \frac{1}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{35}}{28} \end{bmatrix}$$

$$\boxed{\text{x251}} \quad \mathbb{M}_{4,1}^{(a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{7}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{35}i}{112} & 0 & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & -\frac{9\sqrt{14}i}{560} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{21}i}{112} & \frac{3\sqrt{2}i}{80} & 0 & \frac{3\sqrt{42}i}{140} & 0 & \frac{3\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{14}i}{280} & 0 \\ -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & -\frac{3\sqrt{14}i}{280} & 0 & \frac{3\sqrt{70}i}{560} & 0 & \frac{3\sqrt{42}i}{140} & 0 & \frac{3\sqrt{2}i}{80} \\ 0 & \frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & -\frac{9\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & -\frac{9\sqrt{42}i}{560} & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{42}i}{560} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{14}i}{80} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & -\frac{3\sqrt{3}i}{80} & 0 & -\frac{\sqrt{7}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{21}i}{560} & 0 \\ -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{560} & 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{7}i}{560} & 0 & \frac{3\sqrt{3}i}{80} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{210}i}{280} & 0 & -\frac{13\sqrt{42}i}{560} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{30}i}{80} \end{bmatrix}$$

$$\begin{aligned}
\boxed{\text{x252}} \quad \mathbb{M}_{4,2}^{(a)}(E_u, 1) &= \begin{bmatrix} \frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & 0 & \frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{210}}{280} & 0 & \frac{9\sqrt{14}}{560} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{21}}{112} & \frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{140} & 0 & \frac{3\sqrt{70}}{560} & 0 & \frac{3\sqrt{14}}{280} & 0 \\ -\frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 & 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{70}}{560} & 0 & \frac{3\sqrt{42}}{140} & 0 & -\frac{3\sqrt{2}}{80} \\ 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & -\frac{9\sqrt{14}}{560} & 0 & \frac{3\sqrt{210}}{280} & 0 & -\frac{9\sqrt{42}}{560} & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\ \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{42}}{560} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{14}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & -\frac{3\sqrt{3}}{80} & 0 & \frac{\sqrt{7}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{21}}{560} & 0 \\ -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{21}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & \frac{\sqrt{7}}{560} & 0 & -\frac{3\sqrt{3}}{80} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{210}}{280} & 0 & -\frac{13\sqrt{42}}{560} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{80} \end{bmatrix} \\
\boxed{\text{x253}} \quad \mathbb{M}_{4,1}^{(a)}(E_u, 2) &= \begin{bmatrix} -\frac{i}{112} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & -\frac{9\sqrt{6}i}{560} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & \frac{9\sqrt{2}i}{80} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{112} & 0 & \frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{3}i}{16} & -\frac{3\sqrt{14}i}{80} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{560} & 0 & \frac{3\sqrt{2}i}{40} & 0 \\ \frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{15}i}{112} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & \frac{3\sqrt{10}i}{560} & 0 & \frac{3\sqrt{6}i}{140} & 0 & -\frac{3\sqrt{14}i}{80} \\ 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{i}{112} & 0 & 0 & \frac{9\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & -\frac{9\sqrt{6}i}{560} & 0 \\ 0 & \frac{i}{28} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{210}i}{560} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ -\frac{i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{6}i}{560} & 0 & \frac{\sqrt{30}i}{280} & 0 & \frac{7\sqrt{2}i}{80} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{3\sqrt{21}i}{80} & 0 & -\frac{i}{560} & 0 & \frac{\sqrt{15}i}{80} & 0 & -\frac{\sqrt{3}i}{80} & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{3}i}{80} & 0 & -\frac{\sqrt{15}i}{80} & 0 & \frac{i}{560} & 0 & -\frac{3\sqrt{21}i}{80} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{i}{28} & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & -\frac{\sqrt{30}i}{280} & 0 & -\frac{13\sqrt{6}i}{560} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{560} \end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x254}} \quad \mathbb{M}_{4,2}^{(a)}(E_u, 2) = \begin{bmatrix} \frac{1}{112} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{16} & 0 & 0 & \frac{9\sqrt{6}}{560} & 0 & -\frac{3\sqrt{30}}{280} & 0 & -\frac{9\sqrt{2}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{112} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{3}}{16} & -\frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{10}}{560} & 0 & -\frac{3\sqrt{2}}{40} & 0 \\ \frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{\sqrt{15}}{112} & 0 & 0 & \frac{3\sqrt{2}}{40} & 0 & -\frac{3\sqrt{10}}{560} & 0 & \frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{14}}{80} \\ 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{112} & 0 & \frac{1}{112} & 0 & 0 & \frac{9\sqrt{2}}{80} & 0 & \frac{3\sqrt{30}}{280} & 0 & -\frac{9\sqrt{6}}{560} & 0 \\ 0 & \frac{1}{28} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ \frac{1}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}}{560} & 0 & \frac{\sqrt{30}}{280} & 0 & -\frac{7\sqrt{2}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & \frac{3\sqrt{21}}{80} & 0 & \frac{1}{560} & 0 & \frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{3}}{80} & 0 \\ \frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & \frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{15}}{80} & 0 & \frac{1}{560} & 0 & \frac{3\sqrt{21}}{80} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{1}{28} & 0 & 0 & -\frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{280} & 0 & -\frac{13\sqrt{6}}{560} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{560} \end{bmatrix}$$

$$\boxed{\text{x255}} \quad \mathbb{M}_2^{(1,-1;a)}(A_{1u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x256}} \quad \mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{35}}{168} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{\sqrt{7}}{84} & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 \end{bmatrix}$$

$$\boxed{\text{x257}} \quad \mathbb{M}_4^{(1,-1;a)}(A_{1u}, 2) = \begin{bmatrix} 0 & \frac{\sqrt{5}}{168} & 0 & 0 & 0 & -\frac{1}{24} & 0 & 0 & \frac{5\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{10}}{168} & 0 & 0 & 0 & -\frac{\sqrt{14}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{10}}{168} & 0 & 0 & 0 \\ \frac{1}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}}{168} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{5\sqrt{2}}{56} & 0 & 0 \\ -\frac{\sqrt{5}}{84} & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & -\frac{5\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 \\ 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}}{168} & 0 & 0 & 0 & \frac{\sqrt{21}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{15}}{168} & 0 & 0 & 0 \\ \frac{1}{12} & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{5\sqrt{2}}{42} & 0 & 0 \\ 0 & \frac{1}{12} & 0 & 0 & 0 & -\frac{\sqrt{5}}{84} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & \frac{5\sqrt{30}}{168} & 0 \end{bmatrix}$$

$$\boxed{\text{x258}} \quad \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{7\sqrt{11}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 & \frac{7\sqrt{165}}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{11}}{88} & 0 & 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x259}} \quad \mathbb{M}_6^{(1,-1;a)}(A_{1u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{264} & 0 & 0 & 0 & \frac{\sqrt{77}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{88} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{264} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & 0 & \frac{\sqrt{385}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}}{264} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x260}} \quad \mathbb{M}_4^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x261}} \quad \mathbb{M}_6^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x262}} \quad \mathbb{M}_2^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 \\ 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{35} & 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 \\ 0 & \frac{3}{35} & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0 \\ 0 & 0 & \frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x263}} \quad \mathbb{M}_4^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \\ 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{9}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 & 0 & -\frac{9}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x264}} \quad \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{24} & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & 0 & 0 & \frac{\sqrt{35}}{24} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & \frac{\sqrt{14}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{24} & 0 & 0 & 0 & -\frac{1}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x265}} \quad \mathbb{M}_6^{(1,-1;a)}(B_{1u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{264} & 0 & 0 & 0 & \frac{\sqrt{11}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{264} & 0 & 0 & 0 & \frac{\sqrt{2310}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{8} & 0 & 0 & 0 & -\frac{\sqrt{385}}{264} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x266}} \quad \mathbb{M}_2^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & \frac{\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 \\ 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} \\ 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{3i}{35} & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 \\ 0 & \frac{3i}{35} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 \\ 0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x267}} \quad \mathbb{M}_4^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & -\frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\ 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{9i}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 & 0 & -\frac{9i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x268}} \quad \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x269}} \quad \mathbb{M}_6^{(1,-1;a)}(B_{2u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & 0 & 0 & \frac{\sqrt{385}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{462}i}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{66} & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{66} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x270}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} \frac{3\sqrt{10}i}{140} & 0 & \frac{9i}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{140} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9i}{140} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 \\ 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{35} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{140} & 0 & -\frac{11\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2i}{35} & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}i}{140} & 0 & \frac{\sqrt{15}i}{140} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & \frac{\sqrt{21}i}{28} \end{bmatrix}$$

$$\boxed{\text{x271}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} -\frac{3\sqrt{10}}{140} & 0 & \frac{9}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{140} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & \frac{6}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 & -\frac{6}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{9}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 \\ 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{35} & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & -\frac{11\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{140} & 0 & \frac{\sqrt{15}}{140} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & \frac{\sqrt{21}}{28} \end{bmatrix}$$

$$\begin{aligned}
\boxed{\text{x272}} \quad \mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 1) = & \begin{bmatrix} \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{210}i}{672} & 0 & \frac{\sqrt{105}i}{672} & 0 & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{224} & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{42}i}{168} & 0 \\ \frac{\sqrt{7}i}{224} & 0 & \frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{35}i}{224} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{6}i}{48} \\ 0 & -\frac{\sqrt{105}i}{672} & 0 & -\frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{21}i}{672} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 \\ 0 & -\frac{\sqrt{21}i}{84} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{14}i}{224} & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{96} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & -\frac{3i}{32} & 0 & -\frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{32} & 0 & \frac{\sqrt{7}i}{224} & 0 \\ \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{35}i}{32} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{3i}{32} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{96} & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{13\sqrt{14}i}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & \frac{5\sqrt{210}i}{336} & 0 & \frac{\sqrt{10}i}{32} \end{bmatrix} \\
\boxed{\text{x273}} \quad \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 1) = & \begin{bmatrix} -\frac{\sqrt{21}}{672} & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{105}}{672} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{7}}{224} & -\frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{42}}{168} & 0 \\ \frac{\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{6}}{48} \\ 0 & -\frac{\sqrt{105}}{672} & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{21}}{672} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 \\ 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & -\frac{5\sqrt{210}}{336} & 0 & \frac{5\sqrt{14}}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{224} & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{96} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & -\frac{3}{32} & 0 & \frac{\sqrt{21}}{672} & 0 & \frac{\sqrt{35}}{32} & 0 & -\frac{\sqrt{7}}{224} & 0 \\ \frac{\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{32} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{3}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{42}}{96} & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{13\sqrt{14}}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{224} & 0 & -\frac{5\sqrt{210}}{336} & 0 & \frac{\sqrt{10}}{32} \end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x274}} \quad \mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{30}i}{672} & 0 & -\frac{\sqrt{15}i}{96} & 0 & 0 & \frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{i}{32} & \frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{30}i}{336} & 0 & -\frac{\sqrt{6}i}{24} & 0 \\ -\frac{i}{32} & 0 & \frac{\sqrt{10}i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{30}i}{336} & 0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{42}i}{48} \\ 0 & \frac{\sqrt{15}i}{96} & 0 & -\frac{\sqrt{30}i}{672} & 0 & -\frac{\sqrt{3}i}{672} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 \\ 0 & -\frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{5\sqrt{30}i}{336} & 0 & \frac{5\sqrt{2}i}{32} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{7\sqrt{6}i}{96} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & \frac{3\sqrt{7}i}{32} & 0 & -\frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{5}i}{32} & 0 & -\frac{i}{32} & 0 \\ -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{i}{32} & 0 & -\frac{\sqrt{5}i}{32} & 0 & \frac{\sqrt{3}i}{672} & 0 & -\frac{3\sqrt{7}i}{32} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & -\frac{7\sqrt{6}i}{96} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{13\sqrt{2}i}{224} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & \frac{5\sqrt{30}i}{336} & 0 & \frac{\sqrt{70}i}{224} \end{bmatrix}$$

$$\boxed{\text{x275}} \quad \mathbb{M}_{4,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{3}}{672} & 0 & \frac{\sqrt{30}}{672} & 0 & \frac{\sqrt{15}}{96} & 0 & 0 & -\frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\ 0 & \frac{\sqrt{5}}{224} & 0 & -\frac{\sqrt{10}}{224} & 0 & -\frac{1}{32} & \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{30}}{336} & 0 & \frac{\sqrt{6}}{24} & 0 \\ -\frac{1}{32} & 0 & -\frac{\sqrt{10}}{224} & 0 & \frac{\sqrt{5}}{224} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{30}}{336} & 0 & -\frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{42}}{48} \\ 0 & \frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{30}}{672} & 0 & -\frac{\sqrt{3}}{672} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{3\sqrt{2}}{112} & 0 \\ 0 & -\frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{5\sqrt{30}}{336} & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{84} & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{7\sqrt{6}}{96} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & \frac{3\sqrt{7}}{32} & 0 & \frac{\sqrt{3}}{672} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{1}{32} & 0 \\ -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & \frac{1}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{\sqrt{3}}{672} & 0 & \frac{3\sqrt{7}}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & -\frac{7\sqrt{6}}{96} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{13\sqrt{2}}{224} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{32} & 0 & -\frac{5\sqrt{30}}{336} & 0 & \frac{\sqrt{70}}{224} \end{bmatrix}$$

$$\boxed{\text{x276}} \quad \mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{1056} & 0 & -\frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{462}i}{96} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1056} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & -\frac{3\sqrt{770}i}{352} & 0 & -\frac{\sqrt{330}i}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{385}i}{176} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{1155}i}{176} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{176} & 0 & \frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{385}i}{176} & 0 & -\frac{\sqrt{165}i}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{96} & 0 & \frac{3\sqrt{770}i}{352} & 0 & -\frac{5\sqrt{462}i}{1056} & 0 & -\frac{\sqrt{2310}i}{1056} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{96} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{66}i}{1056} & 0 \end{bmatrix}$$

$$\boxed{\text{x277}} \quad \mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{1056} & 0 & -\frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{2310}}{352} & 0 & \frac{\sqrt{462}}{96} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1056} & 0 & \frac{5\sqrt{462}}{1056} & 0 & \frac{3\sqrt{770}}{352} & 0 & -\frac{\sqrt{330}}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{385}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{1155}}{176} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & \frac{\sqrt{385}}{176} & 0 & \frac{\sqrt{165}}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{96} & 0 & \frac{3\sqrt{770}}{352} & 0 & \frac{5\sqrt{462}}{1056} & 0 & -\frac{\sqrt{2310}}{1056} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{96} & 0 & -\frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{154}}{352} & 0 & \frac{\sqrt{66}}{1056} & 0 \end{bmatrix}$$

$$\boxed{\text{x278}} \quad \mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & -\frac{\sqrt{7}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{64} & 0 & -\frac{\sqrt{210}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & -\frac{\sqrt{70}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{64} & 0 & \frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & \frac{\sqrt{10}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & \frac{i}{64} & 0 \end{bmatrix}$$

$$\boxed{\text{x279}} \quad \mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{7}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & \frac{\sqrt{5}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{10}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & \frac{1}{64} & 0 \end{bmatrix}$$

$$\boxed{\text{x280}} \quad \mathbb{M}_{6,1}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{2112} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & \frac{9\sqrt{77}i}{704} & 0 & -\frac{\sqrt{385}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{2112} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & -\frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{11}i}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{462}i}{2112} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & \frac{9\sqrt{154}i}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}i}{704} & 0 & \frac{5\sqrt{770}i}{2112} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{64} & 0 & \frac{9\sqrt{231}i}{704} & 0 & -\frac{5\sqrt{385}i}{2112} & 0 & -\frac{5\sqrt{77}i}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{64} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & \frac{\sqrt{1155}i}{2112} & 0 & \frac{\sqrt{55}i}{2112} \end{bmatrix}$$

$$\boxed{\text{x281}} \quad \mathbb{M}_{6,2}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{2112} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & -\frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{385}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{2112} & 0 & \frac{5\sqrt{385}}{2112} & 0 & \frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{11}}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{2112} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & -\frac{9\sqrt{154}}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}}{704} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & \frac{5\sqrt{462}}{2112} & 0 & \frac{9\sqrt{22}}{704} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{64} & 0 & \frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{385}}{2112} & 0 & -\frac{5\sqrt{77}}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{64} & 0 & -\frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & \frac{\sqrt{55}}{2112} \end{bmatrix}$$

$$\boxed{\text{x282}} \quad \mathbb{M}_2^{(1,0;a)}(A_{1u}) = \begin{bmatrix} 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x283}} \quad \mathbb{M}_4^{(1,0;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & \frac{\sqrt{105}}{840} & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & -\frac{9\sqrt{42}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{6}}{40} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{6}}{40} & 0 & 0 & 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{105}}{840} & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{42}}{280} & 0 & 0 \\ -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{\sqrt{210}}{840} & 0 & 0 \\ 0 & \frac{3\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & -\frac{1}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{1}{40} & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{3\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{210}}{840} & 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 \end{bmatrix}$$

$$\boxed{\text{x284}} \quad \mathbb{M}_4^{(1,0;a)}(A_{1u}, 2) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{168} & 0 & 0 & 0 & -\frac{\sqrt{15}}{120} & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & 0 & 0 & \frac{3\sqrt{210}}{200} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3\sqrt{210}}{200} & 0 & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{120} & 0 & 0 & 0 & -\frac{\sqrt{3}}{168} & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 \\ -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{\sqrt{6}}{120} & 0 & 0 \\ 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{10}}{100} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{35}}{200} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{20} & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{6}}{120} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 \end{bmatrix}$$

$$\boxed{\text{x285}} \quad \mathbb{M}_4^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}i}{700} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x286}} \quad \mathbb{M}_2^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{30}}{105} & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\ 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 \\ 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x287}} \quad \mathbb{M}_4^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & \frac{9\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{280} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}}{280} & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}}{1400} \\ 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{3\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 \\ \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{9\sqrt{15}}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 \\ 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & \frac{17\sqrt{5}}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}}{1400} \\ 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x288}} \quad \mathbb{M}_2^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \\ 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\ 0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x289}} \quad \mathbb{M}_4^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{9\sqrt{210}i}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}i}{280} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{280} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{1400} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{280} & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}i}{1400} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}i}{1400} \\ 0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{3\sqrt{210}i}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 \\ -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 \\ 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 \\ 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}i}{1400} \\ 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x290}} \quad \mathbb{M}_{2,1}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{i}{14} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{210} & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & \frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 \\ 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 \\ \frac{3i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & -\frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & -\frac{i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & \frac{\sqrt{6}i}{168} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{168} \end{bmatrix}$$

$$\boxed{\text{x291}} \quad \mathbb{M}_{2,2}^{(1,0;a)}(E_u) = \begin{bmatrix} -\frac{1}{14} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{140} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 \\ 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{168} & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & -\frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{168} \end{bmatrix}$$

$$\begin{aligned}
\boxed{\text{x292}} \quad \mathbb{M}_{4,1}^{(1,0;a)}(E_u, 1) = & \begin{bmatrix} \frac{\sqrt{35}i}{1120} & 0 & \frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{7}i}{224} & 0 & 0 & -\frac{27\sqrt{210}i}{2800} & 0 & -\frac{9\sqrt{42}i}{280} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{224} & 0 & -\frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{105}i}{1120} & \frac{9\sqrt{10}i}{400} & 0 & \frac{9\sqrt{210}i}{700} & 0 & \frac{9\sqrt{14}i}{560} & 0 & -\frac{9\sqrt{70}i}{1400} & 0 \\ \frac{\sqrt{105}i}{1120} & 0 & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 & -\frac{9\sqrt{70}i}{1400} & 0 & \frac{9\sqrt{14}i}{560} & 0 & \frac{9\sqrt{210}i}{700} & 0 & \frac{9\sqrt{10}i}{400} \\ 0 & -\frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{35}i}{1120} & 0 & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{42}i}{280} & 0 & -\frac{27\sqrt{210}i}{2800} & 0 \\ 0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{1120} & 0 & 0 & 0 \\ \frac{3\sqrt{35}i}{140} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}i}{5600} & 0 & \frac{\sqrt{42}i}{560} & 0 & -\frac{\sqrt{70}i}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & -\frac{3\sqrt{15}i}{800} & 0 & -\frac{\sqrt{35}i}{5600} & 0 & \frac{\sqrt{21}i}{160} & 0 & \frac{\sqrt{105}i}{5600} & 0 \\ \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{5600} & 0 & -\frac{\sqrt{21}i}{160} & 0 & \frac{\sqrt{35}i}{5600} & 0 & \frac{3\sqrt{15}i}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{800} & 0 & -\frac{\sqrt{42}i}{560} & 0 & -\frac{13\sqrt{210}i}{5600} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{1120} & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{6}i}{160} \end{bmatrix} \\
\boxed{\text{x293}} \quad \mathbb{M}_{4,2}^{(1,0;a)}(E_u, 1) = & \begin{bmatrix} -\frac{\sqrt{35}}{1120} & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{7}}{224} & 0 & 0 & \frac{27\sqrt{210}}{2800} & 0 & -\frac{9\sqrt{42}}{280} & 0 & \frac{27\sqrt{70}}{2800} & 0 & 0 \\ 0 & \frac{\sqrt{21}}{224} & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{105}}{1120} & \frac{9\sqrt{10}}{400} & 0 & -\frac{9\sqrt{210}}{700} & 0 & \frac{9\sqrt{14}}{560} & 0 & \frac{9\sqrt{70}}{1400} & 0 \\ \frac{\sqrt{105}}{1120} & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{21}}{224} & 0 & 0 & -\frac{9\sqrt{70}}{1400} & 0 & -\frac{9\sqrt{14}}{560} & 0 & \frac{9\sqrt{210}}{700} & 0 & -\frac{9\sqrt{10}}{400} \\ 0 & -\frac{\sqrt{7}}{224} & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{35}}{1120} & 0 & 0 & -\frac{27\sqrt{70}}{2800} & 0 & \frac{9\sqrt{42}}{280} & 0 & -\frac{27\sqrt{210}}{2800} & 0 \\ 0 & -\frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & \frac{\sqrt{6}}{160} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{210}}{1120} & 0 & 0 & 0 \\ -\frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{210}}{5600} & 0 & \frac{\sqrt{42}}{560} & 0 & \frac{\sqrt{70}}{800} & 0 & 0 \\ 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & -\frac{3\sqrt{15}}{800} & 0 & \frac{\sqrt{35}}{5600} & 0 & \frac{\sqrt{21}}{160} & 0 & -\frac{\sqrt{105}}{5600} & 0 \\ \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{105}}{5600} & 0 & \frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{35}}{5600} & 0 & -\frac{3\sqrt{15}}{800} \\ 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{70}}{800} & 0 & \frac{\sqrt{42}}{560} & 0 & -\frac{13\sqrt{210}}{5600} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{1120} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{6}}{160} \end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x294}} \quad \mathbb{M}_{4,1}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{5}i}{1120} & 0 & \frac{\sqrt{2}i}{224} & 0 & -\frac{i}{32} & 0 & 0 & -\frac{27\sqrt{30}i}{2800} & 0 & -\frac{9\sqrt{6}i}{280} & 0 & \frac{27\sqrt{10}i}{400} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{224} & 0 & -\frac{\sqrt{6}i}{224} & 0 & \frac{\sqrt{15}i}{160} & -\frac{9\sqrt{70}i}{400} & 0 & \frac{9\sqrt{30}i}{700} & 0 & \frac{9\sqrt{2}i}{560} & 0 & \frac{9\sqrt{10}i}{200} & 0 \\ -\frac{\sqrt{15}i}{160} & 0 & \frac{\sqrt{6}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & 0 & \frac{9\sqrt{10}i}{200} & 0 & \frac{9\sqrt{2}i}{560} & 0 & \frac{9\sqrt{30}i}{700} & 0 & -\frac{9\sqrt{70}i}{400} \\ 0 & \frac{i}{32} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{\sqrt{5}i}{1120} & 0 & 0 & \frac{27\sqrt{10}i}{400} & 0 & -\frac{9\sqrt{6}i}{280} & 0 & -\frac{27\sqrt{30}i}{2800} & 0 \\ 0 & -\frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{40} & 0 & 0 & -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 & 0 \\ \frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{5600} & 0 & \frac{\sqrt{6}i}{560} & 0 & \frac{7\sqrt{10}i}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{40} & \frac{3\sqrt{105}i}{800} & 0 & -\frac{\sqrt{5}i}{5600} & 0 & \frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{15}i}{800} & 0 \\ -\frac{3\sqrt{10}i}{40} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & \frac{\sqrt{15}i}{800} & 0 & -\frac{\sqrt{3}i}{160} & 0 & \frac{\sqrt{5}i}{5600} & 0 & -\frac{3\sqrt{105}i}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{7\sqrt{10}i}{800} & 0 & -\frac{\sqrt{6}i}{560} & 0 & -\frac{13\sqrt{30}i}{5600} & 0 \\ 0 & 0 & \frac{3\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{42}i}{1120} \end{bmatrix}$$

$$\boxed{\text{x295}} \quad \mathbb{M}_{4,2}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{5}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{1}{32} & 0 & 0 & \frac{27\sqrt{30}}{2800} & 0 & -\frac{9\sqrt{6}}{280} & 0 & -\frac{27\sqrt{10}}{400} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{224} & 0 & -\frac{\sqrt{6}}{224} & 0 & -\frac{\sqrt{15}}{160} & -\frac{9\sqrt{70}}{400} & 0 & -\frac{9\sqrt{30}}{700} & 0 & \frac{9\sqrt{2}}{560} & 0 & -\frac{9\sqrt{10}}{200} & 0 \\ -\frac{\sqrt{15}}{160} & 0 & -\frac{\sqrt{6}}{224} & 0 & \frac{\sqrt{3}}{224} & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 & -\frac{9\sqrt{2}}{560} & 0 & \frac{9\sqrt{30}}{700} & 0 & \frac{9\sqrt{70}}{400} \\ 0 & \frac{1}{32} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{\sqrt{5}}{1120} & 0 & 0 & \frac{27\sqrt{10}}{400} & 0 & \frac{9\sqrt{6}}{280} & 0 & -\frac{27\sqrt{30}}{2800} & 0 \\ 0 & -\frac{3\sqrt{5}}{140} & 0 & -\frac{3\sqrt{10}}{40} & 0 & 0 & \frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{30}}{5600} & 0 & \frac{\sqrt{6}}{560} & 0 & -\frac{7\sqrt{10}}{800} & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}}{40} & \frac{3\sqrt{105}}{800} & 0 & \frac{\sqrt{5}}{5600} & 0 & \frac{\sqrt{3}}{160} & 0 & \frac{\sqrt{15}}{800} & 0 \\ -\frac{3\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{\sqrt{15}}{800} & 0 & \frac{\sqrt{3}}{160} & 0 & \frac{\sqrt{5}}{5600} & 0 & \frac{3\sqrt{105}}{800} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{7\sqrt{10}}{800} & 0 & \frac{\sqrt{6}}{560} & 0 & -\frac{13\sqrt{30}}{5600} & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{40} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{1120} \end{bmatrix}$$

$$\boxed{\text{x296}} \quad \mathbb{M}_0^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x297}} \quad \mathbb{M}_2^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\ \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 \end{bmatrix}$$

$$\boxed{\text{x298}} \quad \mathbb{M}_4^{(1,1;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{770}}{210} & 0 & 0 & 0 & -\frac{\sqrt{154}}{42} & 0 & 0 & \frac{\sqrt{77}}{140} & 0 & 0 & 0 & \frac{\sqrt{231}}{420} & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{420} & 0 & 0 & 0 & \frac{\sqrt{11}}{60} \\ 0 & 0 & 0 & -\frac{\sqrt{1155}}{105} & 0 & 0 & \frac{\sqrt{11}}{60} & 0 & 0 & 0 & -\frac{\sqrt{385}}{420} & 0 & 0 & 0 \\ \frac{\sqrt{154}}{42} & 0 & 0 & 0 & \frac{\sqrt{770}}{210} & 0 & 0 & \frac{\sqrt{231}}{420} & 0 & 0 & 0 & \frac{\sqrt{77}}{140} & 0 & 0 \\ -\frac{\sqrt{770}}{840} & 0 & 0 & 0 & -\frac{\sqrt{154}}{168} & 0 & 0 & \frac{\sqrt{1155}}{1155} & 0 & 0 & 0 & \frac{\sqrt{385}}{1155} & 0 & 0 \\ 0 & \frac{\sqrt{770}}{280} & 0 & 0 & 0 & -\frac{\sqrt{154}}{168} & 0 & 0 & -\frac{4\sqrt{77}}{1155} & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 \\ 0 & 0 & -\frac{\sqrt{770}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{2310} & 0 & 0 & 0 & \frac{\sqrt{66}}{330} \\ 0 & 0 & 0 & -\frac{\sqrt{770}}{420} & 0 & 0 & -\frac{\sqrt{66}}{330} & 0 & 0 & 0 & \frac{\sqrt{2310}}{2310} & 0 & 0 & 0 \\ -\frac{\sqrt{154}}{168} & 0 & 0 & 0 & \frac{\sqrt{770}}{280} & 0 & 0 & -\frac{2\sqrt{231}}{1155} & 0 & 0 & 0 & \frac{4\sqrt{77}}{1155} & 0 & 0 \\ 0 & -\frac{\sqrt{154}}{168} & 0 & 0 & 0 & -\frac{\sqrt{770}}{840} & 0 & 0 & -\frac{\sqrt{385}}{1155} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{1155} & 0 \end{bmatrix}$$

$$\boxed{\text{x299}} \quad \mathbb{M}_4^{(1,1;a)}(A_{1u}, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{22}}{42} & 0 & 0 & 0 & \frac{\sqrt{110}}{30} & 0 & 0 & \frac{\sqrt{55}}{140} & 0 & 0 & 0 & -\frac{\sqrt{165}}{300} & 0 \\ 0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{84} & 0 & 0 & 0 & -\frac{\sqrt{385}}{300} \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{300} & 0 & 0 & 0 & -\frac{\sqrt{11}}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{30} & 0 & 0 & 0 & \frac{\sqrt{22}}{42} & 0 & 0 & -\frac{\sqrt{165}}{300} & 0 & 0 & 0 & \frac{\sqrt{55}}{140} & 0 & 0 \\ -\frac{\sqrt{22}}{168} & 0 & 0 & 0 & \frac{\sqrt{110}}{120} & 0 & 0 & \frac{\sqrt{33}}{231} & 0 & 0 & 0 & -\frac{\sqrt{11}}{165} & 0 & 0 \\ 0 & \frac{\sqrt{22}}{56} & 0 & 0 & 0 & \frac{\sqrt{110}}{120} & 0 & 0 & -\frac{4\sqrt{55}}{1155} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{825} & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{462} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1650} \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{84} & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 & 0 & \frac{\sqrt{66}}{462} & 0 & 0 & 0 \\ \frac{\sqrt{110}}{120} & 0 & 0 & 0 & \frac{\sqrt{22}}{56} & 0 & 0 & \frac{2\sqrt{165}}{825} & 0 & 0 & 0 & \frac{4\sqrt{55}}{1155} & 0 & 0 \\ 0 & \frac{\sqrt{110}}{120} & 0 & 0 & 0 & -\frac{\sqrt{22}}{168} & 0 & 0 & \frac{\sqrt{11}}{165} & 0 & 0 & 0 & -\frac{\sqrt{33}}{231} & 0 \end{bmatrix}$$

$$\boxed{\text{x300}} \quad \mathbb{M}_4^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{275} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x301}} \quad \mathbb{M}_2^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 \\ -\frac{4\sqrt{5}}{35} & 0 & 0 & 0 & \frac{8}{35} & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 \\ 0 & -\frac{8}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 \\ 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\ 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & \frac{\sqrt{70}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & \frac{4\sqrt{5}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 \\ 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 \\ 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{105} \\ 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{105} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x302}} \quad \mathbb{M}_4^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 \\ 0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & \frac{\sqrt{110}}{70} & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}}{700} \\ 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & \frac{3\sqrt{385}}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}}{385} & 0 & 0 & 0 \\ \frac{\sqrt{165}}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}}{11550} & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}}{3850} & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{1925} \\ 0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x303}} \quad \mathbb{M}_2^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{280} & 0 & 0 & 0 \\ -\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{8i}{35} & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 \\ 0 & -\frac{8i}{35} & 0 & 0 & 0 & -\frac{4\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{280} & 0 \\ 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} \\ 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 \\ 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 \\ 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 & \frac{\sqrt{70}i}{105} \\ 0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x304}} \quad \mathbb{M}_4^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & \frac{\sqrt{385}i}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 \\ \frac{\sqrt{110}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & -\frac{11\sqrt{165}i}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 \\ 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & 0 & \frac{\sqrt{110}i}{70} & 0 & 0 & \frac{\sqrt{55}i}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{700} \\ 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & -\frac{3\sqrt{385}i}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 \\ -\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 \\ 0 & \frac{\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1925} \\ 0 & 0 & 0 & -\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x305}} \quad \mathbb{M}_{2,1}^{(1,1;a)}(E_u) = \begin{bmatrix} -\frac{2\sqrt{15}i}{35} & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{2i}{35} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{6}i}{140} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 \\ 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}i}{70} & 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{210} & 0 & \frac{11\sqrt{2}i}{210} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{105} & 0 & \frac{i}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{15} & 0 & \frac{\sqrt{15}i}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}i}{70} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{2}i}{210} & 0 & -\frac{\sqrt{10}i}{210} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{14}i}{42} \end{bmatrix}$$

$$\boxed{\text{x306}} \quad \mathbb{M}_{2,2}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{2\sqrt{15}}{35} & 0 & -\frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{6}}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 \\ 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{15}}{70} & 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{210} & 0 & \frac{11\sqrt{2}}{210} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{105} & 0 & \frac{1}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & \frac{1}{15} & 0 & \frac{\sqrt{15}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{210} & 0 & -\frac{\sqrt{10}}{210} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{14}}{42} \end{bmatrix}$$

$$\boxed{\text{x307}} \quad \mathbb{M}_{4,1}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{2310}i}{840} & 0 & -\frac{\sqrt{231}i}{84} & 0 & -\frac{\sqrt{462}i}{168} & 0 & 0 & \frac{3\sqrt{385}i}{1400} & 0 & \frac{\sqrt{77}i}{140} & 0 & \frac{\sqrt{1155}i}{1400} & 0 & 0 \\ 0 & \frac{\sqrt{154}i}{56} & 0 & \frac{\sqrt{77}i}{28} & 0 & \frac{\sqrt{770}i}{280} & -\frac{\sqrt{165}i}{600} & 0 & -\frac{\sqrt{385}i}{350} & 0 & -\frac{\sqrt{231}i}{840} & 0 & \frac{\sqrt{1155}i}{2100} & 0 \\ -\frac{\sqrt{770}i}{280} & 0 & -\frac{\sqrt{77}i}{28} & 0 & -\frac{\sqrt{154}i}{56} & 0 & 0 & \frac{\sqrt{1155}i}{2100} & 0 & -\frac{\sqrt{231}i}{840} & 0 & -\frac{\sqrt{385}i}{350} & 0 & -\frac{\sqrt{165}i}{600} \\ 0 & \frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & \frac{\sqrt{1155}i}{1400} & 0 & \frac{\sqrt{77}i}{140} & 0 & \frac{3\sqrt{385}i}{1400} & 0 \\ 0 & -\frac{\sqrt{2310}i}{840} & 0 & -\frac{\sqrt{1155}i}{840} & 0 & 0 & \frac{\sqrt{11}i}{220} & 0 & \frac{\sqrt{231}i}{462} & 0 & \frac{\sqrt{385}i}{1540} & 0 & 0 & 0 \\ \frac{\sqrt{2310}i}{840} & 0 & \frac{\sqrt{231}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{385}i}{7700} & 0 & -\frac{\sqrt{77}i}{770} & 0 & \frac{\sqrt{1155}i}{3300} & 0 & 0 \\ 0 & -\frac{\sqrt{231}i}{168} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{840} & \frac{3\sqrt{110}i}{2200} & 0 & \frac{\sqrt{2310}i}{46200} & 0 & -\frac{\sqrt{154}i}{440} & 0 & -\frac{\sqrt{770}i}{15400} & 0 \\ \frac{\sqrt{1155}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{168} & 0 & 0 & \frac{\sqrt{770}i}{15400} & 0 & \frac{\sqrt{154}i}{440} & 0 & -\frac{\sqrt{2310}i}{46200} & 0 & -\frac{3\sqrt{110}i}{2200} \\ 0 & 0 & 0 & \frac{\sqrt{231}i}{168} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & -\frac{\sqrt{1155}i}{3300} & 0 & \frac{\sqrt{77}i}{770} & 0 & \frac{13\sqrt{385}i}{7700} & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{840} & 0 & -\frac{\sqrt{2310}i}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{1540} & 0 & -\frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{11}i}{220} \end{bmatrix}$$

$$\begin{aligned}
\boxed{\text{x308}} \quad \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 1) &= \begin{bmatrix} \frac{\sqrt{2310}}{840} & 0 & -\frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{3\sqrt{385}}{1400} & 0 & \frac{\sqrt{77}}{140} & 0 & -\frac{\sqrt{1155}}{1400} & 0 & 0 \\ 0 & -\frac{\sqrt{154}}{56} & 0 & \frac{\sqrt{77}}{28} & 0 & -\frac{\sqrt{770}}{280} & -\frac{\sqrt{165}}{600} & 0 & \frac{\sqrt{385}}{350} & 0 & -\frac{\sqrt{231}}{840} & 0 & -\frac{\sqrt{1155}}{2100} & 0 \\ -\frac{\sqrt{770}}{280} & 0 & \frac{\sqrt{77}}{28} & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & \frac{\sqrt{1155}}{2100} & 0 & \frac{\sqrt{231}}{840} & 0 & -\frac{\sqrt{385}}{350} & 0 & \frac{\sqrt{165}}{600} \\ 0 & \frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & \frac{\sqrt{1155}}{1400} & 0 & -\frac{\sqrt{77}}{140} & 0 & \frac{3\sqrt{385}}{1400} & 0 \\ 0 & -\frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{1155}}{840} & 0 & 0 & -\frac{\sqrt{11}}{220} & 0 & \frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{385}}{1540} & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{231}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}}{7700} & 0 & -\frac{\sqrt{77}}{770} & 0 & -\frac{\sqrt{1155}}{3300} & 0 & 0 \\ 0 & \frac{\sqrt{231}}{168} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{840} & \frac{3\sqrt{110}}{2200} & 0 & -\frac{\sqrt{2310}}{46200} & 0 & -\frac{\sqrt{154}}{440} & 0 & \frac{\sqrt{770}}{15400} & 0 \\ \frac{\sqrt{1155}}{840} & 0 & 0 & 0 & -\frac{\sqrt{231}}{168} & 0 & 0 & \frac{\sqrt{770}}{15400} & 0 & -\frac{\sqrt{154}}{440} & 0 & -\frac{\sqrt{2310}}{46200} & 0 & \frac{3\sqrt{110}}{2200} \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{168} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & -\frac{\sqrt{1155}}{3300} & 0 & -\frac{\sqrt{77}}{770} & 0 & \frac{13\sqrt{385}}{7700} & 0 \\ 0 & 0 & -\frac{\sqrt{1155}}{840} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{1540} & 0 & \frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{11}}{220} \end{bmatrix} \\
\boxed{\text{x309}} \quad \mathbb{M}_{4,1}^{(1,1;a)}(E_u, 2) &= \begin{bmatrix} -\frac{\sqrt{330i}}{840} & 0 & -\frac{\sqrt{33i}}{84} & 0 & \frac{\sqrt{66i}}{24} & 0 & 0 & \frac{3\sqrt{55i}}{1400} & 0 & \frac{\sqrt{11i}}{140} & 0 & -\frac{\sqrt{165i}}{200} & 0 & 0 \\ 0 & \frac{\sqrt{22i}}{56} & 0 & \frac{\sqrt{11i}}{28} & 0 & -\frac{\sqrt{110i}}{40} & \frac{\sqrt{1155i}}{600} & 0 & -\frac{\sqrt{55i}}{350} & 0 & -\frac{\sqrt{33i}}{840} & 0 & -\frac{\sqrt{165i}}{300} & 0 \\ \frac{\sqrt{110i}}{40} & 0 & -\frac{\sqrt{11i}}{28} & 0 & -\frac{\sqrt{22i}}{56} & 0 & 0 & -\frac{\sqrt{165i}}{300} & 0 & -\frac{\sqrt{33i}}{840} & 0 & -\frac{\sqrt{55i}}{350} & 0 & \frac{\sqrt{1155i}}{600} \\ 0 & -\frac{\sqrt{66i}}{24} & 0 & \frac{\sqrt{33i}}{84} & 0 & \frac{\sqrt{330i}}{840} & 0 & 0 & -\frac{\sqrt{165i}}{200} & 0 & \frac{\sqrt{11i}}{140} & 0 & \frac{3\sqrt{55i}}{1400} & 0 \\ 0 & -\frac{\sqrt{330i}}{840} & 0 & \frac{\sqrt{165i}}{120} & 0 & 0 & \frac{\sqrt{77i}}{1540} & 0 & \frac{\sqrt{33i}}{462} & 0 & -\frac{\sqrt{55i}}{220} & 0 & 0 & 0 \\ \frac{\sqrt{330i}}{840} & 0 & \frac{\sqrt{33i}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{55i}}{7700} & 0 & -\frac{\sqrt{11i}}{770} & 0 & -\frac{7\sqrt{165i}}{3300} & 0 & 0 \\ 0 & -\frac{\sqrt{33i}}{168} & 0 & 0 & 0 & -\frac{\sqrt{165i}}{120} & -\frac{3\sqrt{770i}}{2200} & 0 & \frac{\sqrt{330i}}{46200} & 0 & -\frac{\sqrt{22i}}{440} & 0 & \frac{\sqrt{110i}}{2200} & 0 \\ -\frac{\sqrt{165i}}{120} & 0 & 0 & 0 & -\frac{\sqrt{33i}}{168} & 0 & 0 & -\frac{\sqrt{110i}}{2200} & 0 & \frac{\sqrt{22i}}{440} & 0 & -\frac{\sqrt{330i}}{46200} & 0 & \frac{3\sqrt{770i}}{2200} \\ 0 & 0 & 0 & \frac{\sqrt{33i}}{168} & 0 & \frac{\sqrt{330i}}{840} & 0 & 0 & \frac{7\sqrt{165i}}{3300} & 0 & \frac{\sqrt{11i}}{770} & 0 & \frac{13\sqrt{55i}}{7700} & 0 \\ 0 & 0 & \frac{\sqrt{165i}}{120} & 0 & -\frac{\sqrt{330i}}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{55i}}{220} & 0 & -\frac{\sqrt{33i}}{462} & 0 & -\frac{\sqrt{77i}}{1540} \end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x310}} \quad \mathbb{M}_{4,2}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{33}}{84} & 0 & -\frac{\sqrt{66}}{24} & 0 & 0 & -\frac{3\sqrt{55}}{1400} & 0 & \frac{\sqrt{11}}{140} & 0 & \frac{\sqrt{165}}{200} & 0 & 0 \\ 0 & -\frac{\sqrt{22}}{56} & 0 & \frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{110}}{40} & \frac{\sqrt{1155}}{600} & 0 & \frac{\sqrt{55}}{350} & 0 & -\frac{\sqrt{33}}{840} & 0 & \frac{\sqrt{165}}{300} & 0 \\ \frac{\sqrt{110}}{40} & 0 & \frac{\sqrt{11}}{28} & 0 & -\frac{\sqrt{22}}{56} & 0 & 0 & -\frac{\sqrt{165}}{300} & 0 & \frac{\sqrt{33}}{840} & 0 & -\frac{\sqrt{55}}{350} & 0 & -\frac{\sqrt{1155}}{600} \\ 0 & -\frac{\sqrt{66}}{24} & 0 & -\frac{\sqrt{33}}{84} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & -\frac{\sqrt{165}}{200} & 0 & -\frac{\sqrt{11}}{140} & 0 & \frac{3\sqrt{55}}{1400} & 0 \\ 0 & -\frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{165}}{120} & 0 & 0 & -\frac{\sqrt{77}}{1540} & 0 & \frac{\sqrt{33}}{462} & 0 & \frac{\sqrt{55}}{220} & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{840} & 0 & \frac{\sqrt{33}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}}{7700} & 0 & -\frac{\sqrt{11}}{770} & 0 & \frac{7\sqrt{165}}{3300} & 0 & 0 \\ 0 & \frac{\sqrt{33}}{168} & 0 & 0 & 0 & \frac{\sqrt{165}}{120} & -\frac{3\sqrt{770}}{2200} & 0 & -\frac{\sqrt{330}}{46200} & 0 & -\frac{\sqrt{22}}{440} & 0 & -\frac{\sqrt{110}}{2200} & 0 \\ -\frac{\sqrt{165}}{120} & 0 & 0 & 0 & -\frac{\sqrt{33}}{168} & 0 & 0 & -\frac{\sqrt{110}}{2200} & 0 & -\frac{\sqrt{22}}{440} & 0 & -\frac{\sqrt{330}}{46200} & 0 & -\frac{3\sqrt{770}}{2200} \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{168} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & \frac{7\sqrt{165}}{3300} & 0 & -\frac{\sqrt{11}}{770} & 0 & \frac{13\sqrt{55}}{7700} & 0 \\ 0 & 0 & \frac{\sqrt{165}}{120} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{220} & 0 & \frac{\sqrt{33}}{462} & 0 & -\frac{\sqrt{77}}{1540} \end{bmatrix}$$

$$\boxed{\text{x311}} \quad \mathbb{T}_5^{(a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x312}} \quad \mathbb{T}_1^{(a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x313}} \quad \mathbb{T}_3^{(a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x314}} \quad \mathbb{T}_5^{(a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\ -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x315}} \quad \mathbb{T}_5^{(a)}(A_{2u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x316}} \quad \mathbb{T}_3^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x317}} \quad \mathbb{T}_5^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{60} \\ 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & -\frac{2\sqrt{42}}{105} & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 & \frac{\sqrt{3}}{30} \\ 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x318}} \quad \mathbb{T}_3^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} \\ 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x319}} \quad \mathbb{T}_5^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{20} & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{3}i}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x320}} \quad \mathbb{T}_{1,1}^{(a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{35} & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{70} & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{30}i}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{42}i}{28} \end{bmatrix}$$

$$\boxed{\text{x321}} \quad \mathbb{T}_{1,2}^{(a)}(E_u) = \begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{35} & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{35} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{42}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x322}} \quad \mathbb{T}_{3,1}^{(a)}(E_u, 1) = \begin{bmatrix} -\frac{3\sqrt{105}i}{560} & 0 & \frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 \\ 0 & \frac{3\sqrt{7}i}{80} & 0 & -\frac{3\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{35}i}{112} & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 \\ \frac{3\sqrt{35}i}{112} & 0 & \frac{3\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{7}i}{80} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & \frac{3\sqrt{21}i}{112} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & \frac{3\sqrt{105}i}{560} & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 \\ 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{210}i}{168} & \frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{35}i}{112} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{42}i}{280} & 0 & 0 & \frac{3\sqrt{35}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{5}i}{16} \\ 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{42}i}{280} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{16} \end{bmatrix}$$

$$\boxed{\text{x323}} \quad \mathbb{T}_{3,2}^{(a)}(E_u, 1) = \begin{bmatrix} \frac{3\sqrt{105}}{560} & 0 & \frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{21}}{112} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{80} & 0 & -\frac{3\sqrt{14}}{560} & 0 & \frac{3\sqrt{35}}{112} & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{210}}{168} & 0 \\ \frac{3\sqrt{35}}{112} & 0 & -\frac{3\sqrt{14}}{560} & 0 & -\frac{3\sqrt{7}}{80} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & \frac{\sqrt{30}}{48} \\ 0 & \frac{3\sqrt{21}}{112} & 0 & \frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\ 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{42}}{280} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{280} & 0 & \frac{\sqrt{21}}{70} & 0 & -\frac{\sqrt{210}}{168} & \frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & \frac{3\sqrt{35}}{112} & 0 \\ \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{5}}{16} \\ 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{42}}{280} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{16} \end{bmatrix}$$

$$\boxed{\text{x324}} \quad \mathbb{T}_{3,1}^{(a)}(E_u, 2) = \begin{bmatrix} -\frac{3\sqrt{7}i}{112} & 0 & \frac{9\sqrt{70}i}{560} & 0 & \frac{9\sqrt{35}i}{560} & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{80} & 0 & -\frac{\sqrt{210}i}{560} & 0 & \frac{3\sqrt{21}i}{112} & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 \\ -\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{560} & 0 & -\frac{\sqrt{105}i}{80} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} \\ 0 & -\frac{9\sqrt{35}i}{560} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & \frac{3\sqrt{7}i}{112} & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{280} & 0 & \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{16} & 0 & \frac{5\sqrt{7}i}{112} & 0 & -\frac{\sqrt{105}i}{336} & 0 & \frac{3\sqrt{21}i}{112} & 0 \\ -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & -\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{105}i}{336} & 0 & -\frac{5\sqrt{7}i}{112} & 0 & \frac{\sqrt{3}i}{16} \\ 0 & -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{30}i}{48} \end{bmatrix}$$

$$\boxed{\text{x325}} \quad \mathbb{T}_{3,2}^{(a)}(E_u, 2) = \begin{bmatrix} \frac{3\sqrt{7}}{112} & 0 & \frac{9\sqrt{70}}{560} & 0 & -\frac{9\sqrt{35}}{560} & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{21}}{112} & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 \\ -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} \\ 0 & -\frac{9\sqrt{35}}{560} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\ 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 \\ \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{3\sqrt{14}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{3}}{16} & 0 & -\frac{5\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{336} & 0 & -\frac{3\sqrt{21}}{112} & 0 \\ -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{105}}{336} & 0 & -\frac{5\sqrt{7}}{112} & 0 & -\frac{\sqrt{3}}{16} \\ 0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{48} \end{bmatrix}$$

$$\boxed{\text{x326}} \quad \mathbb{T}_{5,1}^{(a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{80} & 0 & \frac{i}{16} & 0 & -\frac{7\sqrt{15}i}{240} & 0 & \frac{3\sqrt{35}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{240} & 0 & \frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{3}i}{16} & 0 & \frac{7\sqrt{15}i}{240} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}i}{240} & 0 & -\frac{\sqrt{3}i}{16} & 0 & \frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{105}i}{240} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{80} & 0 & -\frac{7\sqrt{15}i}{240} & 0 & \frac{i}{16} & 0 & -\frac{\sqrt{5}i}{80} & 0 \\ 0 & -\frac{\sqrt{30}i}{224} & 0 & \frac{\sqrt{15}i}{48} & 0 & -\frac{3\sqrt{6}i}{32} & -\frac{\sqrt{7}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{5}i}{32} & 0 & \frac{3i}{32} & 0 \\ -\frac{\sqrt{30}i}{224} & 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{23\sqrt{5}i}{1120} & 0 & -\frac{13i}{224} & 0 & \frac{\sqrt{15}i}{160} & 0 & \frac{3\sqrt{35}i}{160} \\ 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{6}i}{112} & 0 & \frac{\sqrt{15}i}{48} & \frac{\sqrt{70}i}{160} & 0 & -\frac{11\sqrt{30}i}{1120} & 0 & \frac{\sqrt{2}i}{224} & 0 & \frac{3\sqrt{10}i}{160} & 0 \\ \frac{\sqrt{15}i}{48} & 0 & -\frac{5\sqrt{6}i}{112} & 0 & \frac{5\sqrt{3}i}{112} & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{11\sqrt{30}i}{1120} & 0 & -\frac{\sqrt{70}i}{160} \\ 0 & -\frac{5\sqrt{6}i}{96} & 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & -\frac{3\sqrt{35}i}{160} & 0 & -\frac{\sqrt{15}i}{160} & 0 & \frac{13i}{224} & 0 & -\frac{23\sqrt{5}i}{1120} & 0 \\ -\frac{3\sqrt{6}i}{32} & 0 & \frac{\sqrt{15}i}{48} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & -\frac{3i}{32} & 0 & \frac{\sqrt{5}i}{32} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & \frac{\sqrt{7}i}{224} \end{bmatrix}$$

$$\boxed{\text{x327}} \quad \mathbb{T}_{5,2}^{(a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{80} & 0 & \frac{1}{16} & 0 & \frac{7\sqrt{15}}{240} & 0 & \frac{3\sqrt{35}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{240} & 0 & -\frac{3\sqrt{5}}{80} & 0 & -\frac{\sqrt{3}}{16} & 0 & -\frac{7\sqrt{15}}{240} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}}{240} & 0 & \frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{5}}{80} & 0 & \frac{\sqrt{105}}{240} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{80} & 0 & -\frac{7\sqrt{15}}{240} & 0 & -\frac{1}{16} & 0 & -\frac{\sqrt{5}}{80} & 0 \\ 0 & -\frac{\sqrt{30}}{224} & 0 & -\frac{\sqrt{15}}{48} & 0 & -\frac{3\sqrt{6}}{32} & \frac{\sqrt{7}}{224} & 0 & \frac{5\sqrt{3}}{224} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{3}{32} & 0 \\ \frac{\sqrt{30}}{224} & 0 & \frac{5\sqrt{3}}{112} & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & -\frac{23\sqrt{5}}{1120} & 0 & -\frac{13}{224} & 0 & -\frac{\sqrt{15}}{160} & 0 & \frac{3\sqrt{35}}{160} \\ 0 & -\frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{6}}{112} & 0 & -\frac{\sqrt{15}}{48} & \frac{\sqrt{70}}{160} & 0 & \frac{11\sqrt{30}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{3\sqrt{10}}{160} & 0 \\ \frac{\sqrt{15}}{48} & 0 & \frac{5\sqrt{6}}{112} & 0 & \frac{5\sqrt{3}}{112} & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{11\sqrt{30}}{1120} & 0 & \frac{\sqrt{70}}{160} \\ 0 & -\frac{5\sqrt{6}}{96} & 0 & -\frac{5\sqrt{3}}{112} & 0 & -\frac{\sqrt{30}}{224} & \frac{3\sqrt{35}}{160} & 0 & -\frac{\sqrt{15}}{160} & 0 & -\frac{13}{224} & 0 & -\frac{23\sqrt{5}}{1120} & 0 \\ \frac{3\sqrt{6}}{32} & 0 & \frac{\sqrt{15}}{48} & 0 & \frac{\sqrt{30}}{224} & 0 & 0 & \frac{3}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{5\sqrt{3}}{224} & 0 & \frac{\sqrt{7}}{224} \end{bmatrix}$$

$$\boxed{\text{x328}} \quad \mathbb{T}_{5,1}^{(a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{80} & 0 & \frac{\sqrt{35}i}{80} & 0 & \frac{3\sqrt{21}i}{80} & 0 & \frac{i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{80} & 0 & \frac{3\sqrt{7}i}{80} & 0 & -\frac{\sqrt{105}i}{80} & 0 & -\frac{3\sqrt{21}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{80} & 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{3}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & \frac{3\sqrt{21}i}{80} & 0 & \frac{\sqrt{35}i}{80} & 0 & -\frac{\sqrt{7}i}{80} & 0 \\ 0 & -\frac{\sqrt{42}i}{224} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{224} & -\frac{\sqrt{5}i}{160} & 0 & \frac{\sqrt{105}i}{224} & 0 & \frac{9\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{224} & 0 \\ -\frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{3\sqrt{210}i}{224} & 0 & 0 & \frac{23\sqrt{7}i}{1120} & 0 & -\frac{13\sqrt{35}i}{1120} & 0 & -\frac{9\sqrt{21}i}{1120} & 0 & \frac{i}{32} \\ 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{21}i}{112} & -\frac{9\sqrt{2}i}{160} & 0 & -\frac{11\sqrt{42}i}{1120} & 0 & \frac{\sqrt{70}i}{1120} & 0 & -\frac{27\sqrt{14}i}{1120} & 0 \\ -\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 & 0 & \frac{27\sqrt{14}i}{1120} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{11\sqrt{42}i}{1120} & 0 & \frac{9\sqrt{2}i}{160} \\ 0 & \frac{3\sqrt{210}i}{224} & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{42}i}{224} & -\frac{i}{32} & 0 & \frac{9\sqrt{21}i}{1120} & 0 & \frac{13\sqrt{35}i}{1120} & 0 & -\frac{23\sqrt{7}i}{1120} & 0 \\ -\frac{\sqrt{210}i}{224} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{\sqrt{35}i}{224} & 0 & -\frac{9\sqrt{7}i}{224} & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{5}i}{160} \end{bmatrix}$$

$$\boxed{\text{x329}} \quad \mathbb{T}_{5,2}^{(a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{80} & 0 & \frac{\sqrt{35}}{80} & 0 & -\frac{3\sqrt{21}}{80} & 0 & \frac{1}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{80} & 0 & -\frac{3\sqrt{7}}{80} & 0 & -\frac{\sqrt{105}}{80} & 0 & \frac{3\sqrt{21}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{80} & 0 & \frac{\sqrt{105}}{80} & 0 & \frac{3\sqrt{7}}{80} & 0 & -\frac{3\sqrt{3}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{16} & 0 & \frac{3\sqrt{21}}{80} & 0 & -\frac{\sqrt{35}}{80} & 0 & -\frac{\sqrt{7}}{80} & 0 \\ 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{224} & \frac{\sqrt{5}}{160} & 0 & \frac{\sqrt{105}}{224} & 0 & -\frac{9\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 \\ \frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{3\sqrt{210}}{224} & 0 & 0 & -\frac{23\sqrt{7}}{1120} & 0 & -\frac{13\sqrt{35}}{1120} & 0 & \frac{9\sqrt{21}}{1120} & 0 & \frac{1}{32} \\ 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{3\sqrt{21}}{112} & -\frac{9\sqrt{2}}{160} & 0 & \frac{11\sqrt{42}}{1120} & 0 & \frac{\sqrt{70}}{1120} & 0 & \frac{27\sqrt{14}}{1120} & 0 \\ -\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{27\sqrt{14}}{1120} & 0 & \frac{\sqrt{70}}{1120} & 0 & \frac{11\sqrt{42}}{1120} & 0 & -\frac{9\sqrt{2}}{160} \\ 0 & \frac{3\sqrt{210}}{224} & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{42}}{224} & \frac{1}{32} & 0 & \frac{9\sqrt{21}}{1120} & 0 & -\frac{13\sqrt{35}}{1120} & 0 & -\frac{23\sqrt{7}}{1120} & 0 \\ \frac{\sqrt{210}}{224} & 0 & -\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{9\sqrt{7}}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{5}}{160} \end{bmatrix}$$

$$\boxed{\text{x330}} \quad \mathbb{T}_{5,1}^{(a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{120} & 0 & \frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{3}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{40} & 0 & \frac{\sqrt{21}i}{40} & 0 & -\frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{7}i}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{21}i}{40} & 0 & -\frac{i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{21}i}{120} & 0 \\ 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & \frac{3\sqrt{70}i}{112} & -\frac{\sqrt{15}i}{240} & 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & \frac{23\sqrt{21}i}{1680} & 0 & -\frac{13\sqrt{105}i}{1680} & 0 & \frac{3\sqrt{7}i}{560} & 0 & -\frac{\sqrt{3}i}{16} \\ 0 & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{7}i}{56} & \frac{\sqrt{6}i}{80} & 0 & -\frac{11\sqrt{14}i}{560} & 0 & \frac{\sqrt{210}i}{1680} & 0 & \frac{3\sqrt{42}i}{560} & 0 \\ \frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{35}i}{56} & 0 & 0 & -\frac{3\sqrt{42}i}{560} & 0 & -\frac{\sqrt{210}i}{1680} & 0 & \frac{11\sqrt{14}i}{560} & 0 & -\frac{\sqrt{6}i}{80} \\ 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{7}i}{560} & 0 & \frac{13\sqrt{105}i}{1680} & 0 & -\frac{23\sqrt{21}i}{1680} & 0 \\ \frac{3\sqrt{70}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{15}i}{240} \end{bmatrix}$$

$$\boxed{\text{x331}} \quad \mathbb{T}_{5,2}^{(a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{120} & 0 & \frac{\sqrt{105}}{120} & 0 & \frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{40} & 0 & -\frac{\sqrt{21}}{40} & 0 & -\frac{\sqrt{35}}{40} & 0 & -\frac{\sqrt{7}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & \frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{21}}{40} & 0 & \frac{1}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & -\frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{21}}{120} & 0 \\ 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{56} & 0 & \frac{3\sqrt{70}}{112} & \frac{\sqrt{15}}{240} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 \\ \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & -\frac{23\sqrt{21}}{1680} & 0 & -\frac{13\sqrt{105}}{1680} & 0 & -\frac{3\sqrt{7}}{560} & 0 & -\frac{\sqrt{3}}{16} \\ 0 & -\frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{7}}{56} & \frac{\sqrt{6}}{80} & 0 & \frac{11\sqrt{14}}{560} & 0 & \frac{\sqrt{210}}{1680} & 0 & -\frac{3\sqrt{42}}{560} & 0 \\ \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & -\frac{3\sqrt{42}}{560} & 0 & \frac{\sqrt{210}}{1680} & 0 & \frac{11\sqrt{14}}{560} & 0 & \frac{\sqrt{6}}{80} \\ 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{14}}{112} & -\frac{\sqrt{3}}{16} & 0 & -\frac{3\sqrt{7}}{560} & 0 & -\frac{13\sqrt{105}}{1680} & 0 & -\frac{23\sqrt{21}}{1680} & 0 \\ -\frac{3\sqrt{70}}{112} & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{15}}{240} \end{bmatrix}$$

$$\boxed{\text{x332}} \quad \mathbb{T}_5^{(1,-1;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x333}} \quad \mathbb{T}_3^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 \end{bmatrix}$$

$$\boxed{\text{x334}} \quad \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 \\ -\frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 \end{bmatrix}$$

$$\boxed{\text{x335}} \quad \mathbb{T}_5^{(1,-1;a)}(A_{2u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x336}} \quad \mathbb{T}_3^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ -\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x337}} \quad \mathbb{T}_5^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{200} & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 & \frac{\sqrt{10}}{200} \\ 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{10}}{25} & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 & \frac{8\sqrt{35}}{175} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & -\frac{\sqrt{10}}{25} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x338}} \quad \mathbb{T}_3^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{3i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\ \frac{3i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 \\ 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x339}} \quad \mathbb{T}_5^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{200} & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} \\ 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{10}i}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & \frac{\sqrt{10}i}{25} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x340}} \quad \mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{30}i}{280} & 0 & \frac{3\sqrt{3}i}{140} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{3i}{28} & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{i}{140} & 0 & -\frac{\sqrt{10}i}{56} & -\frac{\sqrt{105}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\ \frac{\sqrt{10}i}{56} & 0 & \frac{i}{140} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{3\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} \\ 0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{3\sqrt{3}i}{140} & 0 & \frac{\sqrt{30}i}{280} & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & \frac{3i}{28} & 0 & -\frac{3\sqrt{5}i}{56} & 0 \\ 0 & -\frac{3\sqrt{30}i}{280} & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ -\frac{3\sqrt{30}i}{280} & 0 & \frac{3\sqrt{3}i}{280} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{i}{14} & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{3}i}{280} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{\sqrt{15}i}{56} & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{3\sqrt{10}i}{56} & 0 \\ \frac{\sqrt{15}i}{56} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{3}i}{280} & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{70}i}{56} \\ 0 & \frac{\sqrt{6}i}{28} & 0 & \frac{3\sqrt{3}i}{280} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{7}i}{28} \end{bmatrix}$$

$$\boxed{\text{x341}} \quad \mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{140} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{3}{28} & 0 & \frac{\sqrt{15}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{40} & 0 & -\frac{1}{140} & 0 & \frac{\sqrt{10}}{56} & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 \\ \frac{\sqrt{10}}{56} & 0 & -\frac{1}{140} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{3\sqrt{3}}{56} & 0 & 0 & 0 & \frac{\sqrt{105}}{56} \\ 0 & \frac{\sqrt{6}}{56} & 0 & \frac{3\sqrt{3}}{140} & 0 & \frac{\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{3}{28} & 0 & -\frac{3\sqrt{5}}{56} & 0 \\ 0 & -\frac{3\sqrt{30}}{280} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{280} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & -\frac{1}{14} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 \\ 0 & -\frac{3\sqrt{3}}{280} & 0 & \frac{3\sqrt{6}}{140} & 0 & -\frac{\sqrt{15}}{56} & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{3\sqrt{10}}{56} & 0 \\ \frac{\sqrt{15}}{56} & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{3}}{280} & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{70}}{56} \\ 0 & \frac{\sqrt{6}}{28} & 0 & -\frac{3\sqrt{3}}{280} & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{5}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{56} & 0 & \frac{3\sqrt{30}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{7}}{28} \end{bmatrix}$$

$$\boxed{\text{x342}} \quad \mathbb{T}_{3,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{2}i}{56} & 0 & \frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 & -\frac{5\sqrt{3}i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{3i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{15}i}{420} & 0 & \frac{\sqrt{6}i}{56} & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & \frac{3i}{28} & 0 \\ -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{420} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & \frac{3i}{28} & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\ 0 & -\frac{3\sqrt{10}i}{280} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & \frac{3i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & -\frac{5\sqrt{3}i}{56} & 0 \\ 0 & -\frac{3\sqrt{2}i}{56} & 0 & -\frac{3i}{56} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{56} & 0 & \frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{3i}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{280} & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{3i}{56} & \frac{\sqrt{42}i}{56} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{3\sqrt{6}i}{56} & 0 \\ -\frac{3i}{56} & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{5}i}{280} & 0 & 0 & \frac{3\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{168} & 0 & \frac{5\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56} \\ 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & \frac{3i}{28} & 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{5\sqrt{3}i}{84} & 0 \\ 0 & 0 & -\frac{3i}{56} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{105}i}{84} \end{bmatrix}$$

$$\boxed{\text{x343}} \quad \mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 & \frac{5\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & -\frac{3}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{15}}{420} & 0 & -\frac{\sqrt{6}}{56} & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & -\frac{3}{28} & 0 \\ -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{15}}{420} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{3}{28} & 0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} \\ 0 & -\frac{3\sqrt{10}}{280} & 0 & \frac{3\sqrt{5}}{140} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3}{56} & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{5\sqrt{3}}{56} & 0 \\ 0 & -\frac{3\sqrt{2}}{56} & 0 & \frac{3}{56} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{280} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{3}{28} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{280} & 0 & \frac{3\sqrt{10}}{140} & 0 & \frac{3}{56} & \frac{\sqrt{42}}{56} & 0 & \frac{5\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{168} & 0 & \frac{3\sqrt{6}}{56} & 0 \\ -\frac{3}{56} & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{5}}{280} & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{168} & 0 & \frac{5\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56} \\ 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{5}}{280} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{3}{28} & 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{5\sqrt{3}}{84} & 0 \\ 0 & 0 & -\frac{3}{56} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{84} \end{bmatrix}$$

$$\boxed{\text{x344}} \quad \mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{7\sqrt{2}i}{160} & 0 & \frac{3\sqrt{42}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{160} & 0 & \frac{3\sqrt{6}i}{160} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & \frac{7\sqrt{2}i}{160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}i}{160} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & \frac{3\sqrt{6}i}{160} & 0 & -\frac{\sqrt{14}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{160} & 0 & -\frac{7\sqrt{2}i}{160} & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{\sqrt{6}i}{160} & 0 \\ 0 & -\frac{i}{112} & 0 & \frac{\sqrt{2}i}{48} & 0 & -\frac{3\sqrt{5}i}{80} & \frac{\sqrt{210}i}{560} & 0 & -\frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{3\sqrt{30}i}{80} & 0 \\ -\frac{i}{112} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{48} & 0 & 0 & -\frac{23\sqrt{6}i}{560} & 0 & \frac{13\sqrt{30}i}{560} & 0 & -\frac{3\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{80} \\ 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{48} & -\frac{\sqrt{21}i}{40} & 0 & \frac{33i}{280} & 0 & -\frac{\sqrt{15}i}{280} & 0 & -\frac{3\sqrt{3}i}{40} & 0 \\ \frac{\sqrt{2}i}{48} & 0 & -\frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{10}i}{112} & 0 & 0 & \frac{3\sqrt{3}i}{40} & 0 & \frac{\sqrt{15}i}{280} & 0 & -\frac{33i}{280} & 0 & \frac{\sqrt{21}i}{40} \\ 0 & -\frac{\sqrt{5}i}{48} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{i}{112} & \frac{3\sqrt{42}i}{80} & 0 & \frac{3\sqrt{2}i}{80} & 0 & -\frac{13\sqrt{30}i}{560} & 0 & \frac{23\sqrt{6}i}{560} & 0 \\ -\frac{3\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{48} & 0 & -\frac{i}{112} & 0 & 0 & \frac{3\sqrt{30}i}{80} & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{\sqrt{210}i}{560} \end{bmatrix}$$

$$\boxed{\text{x345}} \quad \mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{160} & 0 & \frac{\sqrt{30}}{160} & 0 & \frac{7\sqrt{2}}{160} & 0 & \frac{3\sqrt{42}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{160} & 0 & -\frac{3\sqrt{6}}{160} & 0 & -\frac{3\sqrt{10}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}}{160} & 0 & \frac{3\sqrt{10}}{160} & 0 & \frac{3\sqrt{6}}{160} & 0 & \frac{\sqrt{14}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{6}}{160} & 0 \\ 0 & -\frac{1}{112} & 0 & -\frac{\sqrt{2}}{48} & 0 & -\frac{3\sqrt{5}}{80} & -\frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{3\sqrt{30}}{80} & 0 \\ \frac{1}{112} & 0 & \frac{\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{48} & 0 & 0 & \frac{23\sqrt{6}}{560} & 0 & \frac{13\sqrt{30}}{560} & 0 & \frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{80} \\ 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{48} & -\frac{\sqrt{21}}{40} & 0 & -\frac{33}{280} & 0 & -\frac{\sqrt{15}}{280} & 0 & \frac{3\sqrt{3}}{40} & 0 \\ \frac{\sqrt{2}}{48} & 0 & \frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & \frac{3\sqrt{3}}{40} & 0 & -\frac{\sqrt{15}}{280} & 0 & -\frac{33}{280} & 0 & -\frac{\sqrt{21}}{40} \\ 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{1}{112} & -\frac{3\sqrt{42}}{80} & 0 & \frac{3\sqrt{2}}{80} & 0 & \frac{13\sqrt{30}}{560} & 0 & \frac{23\sqrt{6}}{560} & 0 \\ \frac{3\sqrt{5}}{80} & 0 & \frac{\sqrt{2}}{48} & 0 & \frac{1}{112} & 0 & 0 & -\frac{3\sqrt{30}}{80} & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{\sqrt{210}}{560} \end{bmatrix}$$

$$\boxed{\text{x346}} \quad \mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{800} & 0 & \frac{\sqrt{42}i}{160} & 0 & \frac{9\sqrt{70}i}{800} & 0 & \frac{\sqrt{30}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{800} & 0 & \frac{3\sqrt{210}i}{800} & 0 & -\frac{3\sqrt{14}i}{160} & 0 & -\frac{9\sqrt{70}i}{800} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{800} & 0 & -\frac{3\sqrt{14}i}{160} & 0 & \frac{3\sqrt{210}i}{800} & 0 & \frac{9\sqrt{10}i}{800} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{9\sqrt{70}i}{800} & 0 & \frac{\sqrt{42}i}{160} & 0 & -\frac{\sqrt{210}i}{800} & 0 \\ 0 & -\frac{\sqrt{35}i}{560} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{7}i}{112} & \frac{\sqrt{6}i}{80} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{9\sqrt{210}i}{560} & 0 & -\frac{\sqrt{42}i}{112} & 0 \\ -\frac{\sqrt{35}i}{560} & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{3\sqrt{7}i}{112} & 0 & 0 & -\frac{23\sqrt{210}i}{2800} & 0 & \frac{13\sqrt{42}i}{560} & 0 & \frac{27\sqrt{70}i}{2800} & 0 & -\frac{\sqrt{30}i}{80} \\ 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{3\sqrt{70}i}{560} & \frac{9\sqrt{15}i}{200} & 0 & \frac{33\sqrt{35}i}{1400} & 0 & -\frac{\sqrt{21}i}{280} & 0 & \frac{27\sqrt{105}i}{1400} & 0 \\ -\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{7}i}{56} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & -\frac{27\sqrt{105}i}{1400} & 0 & \frac{\sqrt{21}i}{280} & 0 & -\frac{33\sqrt{35}i}{1400} & 0 & -\frac{9\sqrt{15}i}{200} \\ 0 & \frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{35}i}{560} & \frac{\sqrt{30}i}{80} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{13\sqrt{42}i}{560} & 0 & \frac{23\sqrt{210}i}{2800} & 0 \\ -\frac{\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{9\sqrt{210}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{80} \end{bmatrix}$$

$$\boxed{\text{x347}} \quad \mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{800} & 0 & \frac{\sqrt{42}}{160} & 0 & -\frac{9\sqrt{70}}{800} & 0 & \frac{\sqrt{30}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{800} & 0 & -\frac{3\sqrt{210}}{800} & 0 & -\frac{3\sqrt{14}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{800} & 0 & \frac{3\sqrt{14}}{160} & 0 & \frac{3\sqrt{210}}{800} & 0 & -\frac{9\sqrt{10}}{800} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 & -\frac{\sqrt{42}}{160} & 0 & -\frac{\sqrt{210}}{800} & 0 \\ 0 & -\frac{\sqrt{35}}{560} & 0 & \frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{7}}{112} & -\frac{\sqrt{6}}{80} & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{9\sqrt{210}}{560} & 0 & -\frac{\sqrt{42}}{112} & 0 \\ \frac{\sqrt{35}}{560} & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{3\sqrt{7}}{112} & 0 & 0 & \frac{23\sqrt{210}}{2800} & 0 & \frac{13\sqrt{42}}{560} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & -\frac{\sqrt{30}}{80} \\ 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{56} & 0 & \frac{3\sqrt{70}}{560} & \frac{9\sqrt{15}}{200} & 0 & -\frac{33\sqrt{35}}{1400} & 0 & -\frac{\sqrt{21}}{280} & 0 & -\frac{27\sqrt{105}}{1400} & 0 \\ -\frac{3\sqrt{70}}{560} & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{27\sqrt{105}}{1400} & 0 & -\frac{\sqrt{21}}{280} & 0 & -\frac{33\sqrt{35}}{1400} & 0 & \frac{9\sqrt{15}}{200} \\ 0 & \frac{3\sqrt{7}}{112} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{35}}{560} & -\frac{\sqrt{30}}{80} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & \frac{13\sqrt{42}}{560} & 0 & \frac{23\sqrt{210}}{2800} & 0 \\ \frac{\sqrt{7}}{112} & 0 & -\frac{3\sqrt{70}}{560} & 0 & \frac{\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{9\sqrt{210}}{560} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{6}}{80} \end{bmatrix}$$

$$\boxed{\text{x348}} \quad \mathbb{T}_{5,1}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{400} & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{210}i}{400} & 0 & -\frac{3\sqrt{10}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{400} & 0 & \frac{3\sqrt{70}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{3\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{210}i}{400} & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{70}i}{400} & 0 \\ 0 & -\frac{\sqrt{105}i}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & \frac{\sqrt{21}i}{56} & \frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{3\sqrt{14}i}{56} & 0 \\ -\frac{\sqrt{105}i}{840} & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & -\frac{23\sqrt{70}i}{1400} & 0 & \frac{13\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & \frac{3\sqrt{10}i}{40} \\ 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{840} & -\frac{3\sqrt{5}i}{100} & 0 & \frac{11\sqrt{105}i}{700} & 0 & -\frac{\sqrt{7}i}{140} & 0 & -\frac{9\sqrt{35}i}{700} & 0 \\ \frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{9\sqrt{35}i}{700} & 0 & \frac{\sqrt{7}i}{140} & 0 & -\frac{11\sqrt{105}i}{700} & 0 & \frac{3\sqrt{5}i}{100} \\ 0 & -\frac{\sqrt{21}i}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{105}i}{840} & -\frac{3\sqrt{10}i}{40} & 0 & \frac{3\sqrt{210}i}{1400} & 0 & -\frac{13\sqrt{14}i}{280} & 0 & \frac{23\sqrt{70}i}{1400} & 0 \\ \frac{\sqrt{21}i}{56} & 0 & \frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{105}i}{840} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{40} \end{bmatrix}$$

$$\boxed{\text{x349}} \quad \mathbb{T}_{5,2}^{(1,-1;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{400} & 0 & \frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{210}}{400} & 0 & -\frac{3\sqrt{10}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{400} & 0 & -\frac{3\sqrt{70}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & \frac{3\sqrt{70}}{400} & 0 & \frac{\sqrt{30}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 & -\frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{70}}{400} & 0 \\ 0 & -\frac{\sqrt{105}}{840} & 0 & -\frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{21}}{56} & -\frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{14}}{56} & 0 \\ \frac{\sqrt{105}}{840} & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & \frac{23\sqrt{70}}{1400} & 0 & \frac{13\sqrt{14}}{280} & 0 & \frac{3\sqrt{210}}{1400} & 0 & \frac{3\sqrt{10}}{40} \\ 0 & -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{210}}{840} & -\frac{3\sqrt{5}}{100} & 0 & -\frac{11\sqrt{105}}{700} & 0 & -\frac{\sqrt{7}}{140} & 0 & \frac{9\sqrt{35}}{700} & 0 \\ \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{9\sqrt{35}}{700} & 0 & -\frac{\sqrt{7}}{140} & 0 & -\frac{11\sqrt{105}}{700} & 0 & -\frac{3\sqrt{5}}{100} \\ 0 & -\frac{\sqrt{21}}{168} & 0 & -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{840} & \frac{3\sqrt{10}}{40} & 0 & \frac{3\sqrt{210}}{1400} & 0 & \frac{13\sqrt{14}}{280} & 0 & \frac{23\sqrt{70}}{1400} & 0 \\ -\frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{105}}{840} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{2}}{40} \end{bmatrix}$$

$$\boxed{\text{x350}} \quad \mathbb{T}_5^{(1,0;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x351}} \quad \mathbb{T}_1^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x352}} \quad \mathbb{T}_3^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 \end{bmatrix}$$

$$\boxed{\text{x353}} \quad \mathbb{T}_5^{(1,0;a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 \end{bmatrix}$$

$$\boxed{\text{x354}} \quad \mathbb{T}_5^{(1,0;a)}(A_{2u}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x355}} \quad \mathbb{T}_3^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\ 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x356}} \quad \mathbb{T}_5^{(1,0;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} \\ 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{10}}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{2\sqrt{35}}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}}{300} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{420} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x357}} \quad \mathbb{T}_3^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x358}} \quad \mathbb{T}_5^{(1,0;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} \\ 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{10}i}{300} & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 & \frac{\sqrt{10}i}{300} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x359}} \quad \mathbb{T}_{1,1}^{(1,0;a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{10}i}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{6i}{35} & 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{2}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{6i}{35} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{21}i}{28} \end{bmatrix}$$

$$\boxed{\text{x360}} \quad \mathbb{T}_{1,2}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{10}}{20} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{70} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{6}{35} & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2}}{70} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{6}{35} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{15}}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{21}}{28} \end{bmatrix}$$

$$\boxed{\text{x361}} \quad \mathbb{T}_{3,1}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} -\frac{3\sqrt{35}i}{1120} & 0 & \frac{9\sqrt{14}i}{1120} & 0 & -\frac{3\sqrt{7}i}{224} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{160} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{105}i}{224} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 \\ \frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{21}i}{160} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} \\ 0 & \frac{3\sqrt{7}i}{224} & 0 & -\frac{9\sqrt{14}i}{1120} & 0 & \frac{3\sqrt{35}i}{1120} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{210}i}{112} & 0 \\ 0 & -\frac{3\sqrt{35}i}{140} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{210}i}{672} & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{140} & 0 & \frac{3\sqrt{14}i}{280} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{42}i}{336} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 \\ 0 & \frac{3\sqrt{14}i}{280} & 0 & \frac{3\sqrt{7}i}{70} & 0 & \frac{\sqrt{70}i}{56} & -\frac{\sqrt{15}i}{96} & 0 & -\frac{\sqrt{35}i}{224} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{105}i}{224} & 0 \\ \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{14}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{224} & 0 & \frac{\sqrt{15}i}{96} \\ 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{672} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{96} \end{bmatrix}$$

$$\boxed{\text{x362}} \quad \mathbb{T}_{3,2}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} \frac{3\sqrt{35}}{1120} & 0 & \frac{9\sqrt{14}}{1120} & 0 & \frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{160} & 0 & -\frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{105}}{224} & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 \\ \frac{\sqrt{105}}{224} & 0 & -\frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{21}}{160} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} \\ 0 & \frac{3\sqrt{7}}{224} & 0 & \frac{9\sqrt{14}}{1120} & 0 & \frac{3\sqrt{35}}{1120} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 \\ 0 & -\frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{672} & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{14}}{280} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{42}}{336} & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}}{280} & 0 & \frac{3\sqrt{7}}{70} & 0 & -\frac{\sqrt{70}}{56} & -\frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{35}}{224} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{105}}{224} & 0 \\ \frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{7}}{70} & 0 & \frac{3\sqrt{14}}{280} & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{21}}{672} & 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{15}}{96} \\ 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{672} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{6}}{96} \end{bmatrix}$$

$$\boxed{\text{x363}} \quad \mathbb{T}_{3,1}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} -\frac{\sqrt{21}i}{224} & 0 & \frac{3\sqrt{210}i}{1120} & 0 & \frac{3\sqrt{105}i}{1120} & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{160} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{3\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 \\ -\frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{35}i}{160} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} \\ 0 & -\frac{3\sqrt{105}i}{1120} & 0 & -\frac{3\sqrt{210}i}{1120} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{5\sqrt{14}i}{112} & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{5\sqrt{14}i}{672} & 0 & -\frac{\sqrt{70}i}{336} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{42}i}{56} & \frac{i}{32} & 0 & -\frac{5\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{672} & 0 & -\frac{3\sqrt{7}i}{224} & 0 \\ -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & \frac{3\sqrt{7}i}{224} & 0 & -\frac{\sqrt{35}i}{672} & 0 & \frac{5\sqrt{21}i}{672} & 0 & -\frac{i}{32} \\ 0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{70}i}{336} & 0 & \frac{5\sqrt{14}i}{672} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{10}i}{96} \end{bmatrix}$$

$$\boxed{\text{x364}} \quad \mathbb{T}_{3,2}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} \frac{\sqrt{21}}{224} & 0 & \frac{3\sqrt{210}}{1120} & 0 & -\frac{3\sqrt{105}}{1120} & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{160} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{3\sqrt{7}}{224} & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 \\ -\frac{3\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{35}}{160} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & \frac{\sqrt{6}}{16} \\ 0 & -\frac{3\sqrt{105}}{1120} & 0 & \frac{3\sqrt{210}}{1120} & 0 & \frac{\sqrt{21}}{224} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{5\sqrt{14}}{112} & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{224} & 0 & 0 & 0 \\ \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{5\sqrt{14}}{672} & 0 & -\frac{\sqrt{70}}{336} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{42}}{56} & \frac{1}{32} & 0 & \frac{5\sqrt{21}}{672} & 0 & \frac{\sqrt{35}}{672} & 0 & \frac{3\sqrt{7}}{224} & 0 \\ -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & \frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{672} & 0 & \frac{5\sqrt{21}}{672} & 0 & \frac{1}{32} \\ 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{70}}{336} & 0 & \frac{5\sqrt{14}}{672} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{10}}{96} \end{bmatrix}$$

$$\boxed{\text{x365}} \quad \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{6}i}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{7\sqrt{2}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{3\sqrt{6}i}{80} & 0 & \frac{\sqrt{14}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & \frac{\sqrt{6}i}{80} & 0 \\ 0 & -\frac{3i}{112} & 0 & \frac{\sqrt{2}i}{16} & 0 & -\frac{9\sqrt{5}i}{80} & \frac{\sqrt{210}i}{6720} & 0 & -\frac{\sqrt{10}i}{448} & 0 & \frac{\sqrt{6}i}{192} & 0 & -\frac{\sqrt{30}i}{320} & 0 \\ -\frac{3i}{112} & 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & -\frac{23\sqrt{6}i}{6720} & 0 & \frac{13\sqrt{30}i}{6720} & 0 & -\frac{\sqrt{2}i}{320} & 0 & -\frac{\sqrt{42}i}{320} \\ 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{16} & -\frac{\sqrt{21}i}{480} & 0 & \frac{11i}{1120} & 0 & -\frac{\sqrt{15}i}{3360} & 0 & -\frac{\sqrt{3}i}{160} & 0 \\ \frac{\sqrt{2}i}{16} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{3\sqrt{10}i}{112} & 0 & 0 & \frac{\sqrt{3}i}{160} & 0 & \frac{\sqrt{15}i}{3360} & 0 & -\frac{11i}{1120} & 0 & \frac{\sqrt{21}i}{480} \\ 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{3i}{112} & \frac{\sqrt{42}i}{320} & 0 & \frac{\sqrt{2}i}{320} & 0 & -\frac{13\sqrt{30}i}{6720} & 0 & \frac{23\sqrt{6}i}{6720} & 0 \\ -\frac{9\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{16} & 0 & -\frac{3i}{112} & 0 & 0 & \frac{\sqrt{30}i}{320} & 0 & -\frac{\sqrt{6}i}{192} & 0 & \frac{\sqrt{10}i}{448} & 0 & -\frac{\sqrt{210}i}{6720} \end{bmatrix}$$

$$\boxed{\text{x366}} \quad \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{30}}{80} & 0 & -\frac{7\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{80} & 0 & \frac{3\sqrt{6}}{80} & 0 & \frac{3\sqrt{10}}{80} & 0 & \frac{7\sqrt{2}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}}{80} & 0 & -\frac{3\sqrt{10}}{80} & 0 & -\frac{3\sqrt{6}}{80} & 0 & -\frac{\sqrt{14}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{80} & 0 & \frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & \frac{\sqrt{6}}{80} & 0 \\ 0 & -\frac{3}{112} & 0 & -\frac{\sqrt{2}}{16} & 0 & -\frac{9\sqrt{5}}{80} & -\frac{\sqrt{210}}{6720} & 0 & -\frac{\sqrt{10}}{448} & 0 & -\frac{\sqrt{6}}{192} & 0 & -\frac{\sqrt{30}}{320} & 0 \\ \frac{3}{112} & 0 & \frac{3\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & 0 & \frac{23\sqrt{6}}{6720} & 0 & \frac{13\sqrt{30}}{6720} & 0 & \frac{\sqrt{2}}{320} & 0 & -\frac{\sqrt{42}}{320} \\ 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{3\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{16} & -\frac{\sqrt{21}}{480} & 0 & -\frac{11}{1120} & 0 & -\frac{\sqrt{15}}{3360} & 0 & \frac{\sqrt{3}}{160} & 0 \\ \frac{\sqrt{2}}{16} & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{3\sqrt{10}}{112} & 0 & 0 & \frac{\sqrt{3}}{160} & 0 & -\frac{\sqrt{15}}{3360} & 0 & -\frac{11}{1120} & 0 & -\frac{\sqrt{21}}{480} \\ 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{3}{112} & -\frac{\sqrt{42}}{320} & 0 & \frac{\sqrt{2}}{320} & 0 & \frac{13\sqrt{30}}{6720} & 0 & \frac{23\sqrt{6}}{6720} & 0 \\ \frac{9\sqrt{5}}{80} & 0 & \frac{\sqrt{2}}{16} & 0 & \frac{3}{112} & 0 & 0 & -\frac{\sqrt{30}}{320} & 0 & -\frac{\sqrt{6}}{192} & 0 & -\frac{\sqrt{10}}{448} & 0 & -\frac{\sqrt{210}}{6720} \end{bmatrix}$$

$$\boxed{\text{x367}} \quad \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{400} & 0 & -\frac{3\sqrt{210}i}{400} & 0 & \frac{3\sqrt{14}i}{80} & 0 & \frac{9\sqrt{70}i}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{400} & 0 & \frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{210}i}{400} & 0 & -\frac{9\sqrt{10}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\ 0 & -\frac{3\sqrt{35}i}{560} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{112} & \frac{\sqrt{6}i}{960} & 0 & -\frac{\sqrt{14}i}{448} & 0 & -\frac{3\sqrt{210}i}{2240} & 0 & -\frac{\sqrt{42}i}{1344} & 0 \\ -\frac{3\sqrt{35}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{9\sqrt{7}i}{112} & 0 & 0 & -\frac{23\sqrt{210}i}{33600} & 0 & \frac{13\sqrt{42}i}{6720} & 0 & \frac{9\sqrt{70}i}{11200} & 0 & -\frac{\sqrt{30}i}{960} \\ 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & -\frac{9\sqrt{70}i}{560} & \frac{3\sqrt{15}i}{800} & 0 & \frac{11\sqrt{35}i}{5600} & 0 & -\frac{\sqrt{21}i}{3360} & 0 & \frac{9\sqrt{105}i}{5600} & 0 \\ -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & -\frac{9\sqrt{105}i}{5600} & 0 & \frac{\sqrt{21}i}{3360} & 0 & -\frac{11\sqrt{35}i}{5600} & 0 & -\frac{3\sqrt{15}i}{800} \\ 0 & \frac{9\sqrt{7}i}{112} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{35}i}{560} & \frac{\sqrt{30}i}{960} & 0 & -\frac{9\sqrt{70}i}{11200} & 0 & -\frac{13\sqrt{42}i}{6720} & 0 & \frac{23\sqrt{210}i}{33600} & 0 \\ -\frac{3\sqrt{7}i}{112} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{42}i}{1344} & 0 & \frac{3\sqrt{210}i}{2240} & 0 & \frac{\sqrt{14}i}{448} & 0 & -\frac{\sqrt{6}i}{960} \end{bmatrix}$$

$$\boxed{\text{x368}} \quad \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & \frac{9\sqrt{70}}{400} & 0 & -\frac{\sqrt{30}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{400} & 0 & \frac{3\sqrt{210}}{400} & 0 & \frac{3\sqrt{14}}{80} & 0 & -\frac{9\sqrt{70}}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}}{400} & 0 & -\frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{210}}{400} & 0 & \frac{9\sqrt{10}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{9\sqrt{70}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & \frac{\sqrt{210}}{400} & 0 \\ 0 & -\frac{3\sqrt{35}}{560} & 0 & \frac{9\sqrt{70}}{560} & 0 & -\frac{3\sqrt{7}}{112} & -\frac{\sqrt{6}}{960} & 0 & -\frac{\sqrt{14}}{448} & 0 & \frac{3\sqrt{210}}{2240} & 0 & -\frac{\sqrt{42}}{1344} & 0 \\ \frac{3\sqrt{35}}{560} & 0 & \frac{3\sqrt{14}}{112} & 0 & -\frac{9\sqrt{7}}{112} & 0 & 0 & \frac{23\sqrt{210}}{33600} & 0 & \frac{13\sqrt{42}}{6720} & 0 & -\frac{9\sqrt{70}}{11200} & 0 & -\frac{\sqrt{30}}{960} \\ 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{3\sqrt{7}}{56} & 0 & \frac{9\sqrt{70}}{560} & \frac{3\sqrt{15}}{800} & 0 & -\frac{11\sqrt{35}}{5600} & 0 & -\frac{\sqrt{21}}{3360} & 0 & -\frac{9\sqrt{105}}{5600} & 0 \\ -\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & -\frac{9\sqrt{105}}{5600} & 0 & -\frac{\sqrt{21}}{3360} & 0 & -\frac{11\sqrt{35}}{5600} & 0 & \frac{3\sqrt{15}}{800} \\ 0 & \frac{9\sqrt{7}}{112} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{3\sqrt{35}}{560} & -\frac{\sqrt{30}}{960} & 0 & -\frac{9\sqrt{70}}{11200} & 0 & \frac{13\sqrt{42}}{6720} & 0 & \frac{23\sqrt{210}}{33600} & 0 \\ \frac{3\sqrt{7}}{112} & 0 & -\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{42}}{1344} & 0 & \frac{3\sqrt{210}}{2240} & 0 & -\frac{\sqrt{14}}{448} & 0 & -\frac{\sqrt{6}}{960} \end{bmatrix}$$

$$\boxed{\text{x369}} \quad \mathbb{T}_{5,1}^{(1,0;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{200} & 0 & -\frac{\sqrt{14}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 & \frac{3\sqrt{10}i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{200} & 0 & -\frac{3\sqrt{70}i}{200} & 0 & \frac{\sqrt{42}i}{40} & 0 & -\frac{\sqrt{210}i}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & \frac{\sqrt{42}i}{40} & 0 & -\frac{3\sqrt{70}i}{200} & 0 & \frac{\sqrt{30}i}{200} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 & -\frac{\sqrt{14}i}{40} & 0 & \frac{\sqrt{70}i}{200} & 0 \\ 0 & -\frac{\sqrt{105}i}{280} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{3\sqrt{21}i}{56} & \frac{\sqrt{2}i}{480} & 0 & -\frac{\sqrt{42}i}{672} & 0 & \frac{\sqrt{70}i}{1120} & 0 & \frac{\sqrt{14}i}{224} & 0 \\ -\frac{\sqrt{105}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{56} & 0 & 0 & -\frac{23\sqrt{70}i}{16800} & 0 & \frac{13\sqrt{14}i}{3360} & 0 & -\frac{\sqrt{210}i}{5600} & 0 & \frac{\sqrt{10}i}{160} \\ 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{280} & -\frac{\sqrt{5}i}{400} & 0 & \frac{11\sqrt{105}i}{8400} & 0 & -\frac{\sqrt{7}i}{1680} & 0 & -\frac{3\sqrt{35}i}{2800} & 0 \\ \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{3\sqrt{35}i}{2800} & 0 & \frac{\sqrt{7}i}{1680} & 0 & -\frac{11\sqrt{105}i}{8400} & 0 & \frac{\sqrt{5}i}{400} \\ 0 & -\frac{\sqrt{21}i}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{105}i}{280} & -\frac{\sqrt{10}i}{160} & 0 & \frac{\sqrt{210}i}{5600} & 0 & -\frac{13\sqrt{14}i}{3360} & 0 & \frac{23\sqrt{70}i}{16800} & 0 \\ \frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 & -\frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{\sqrt{42}i}{672} & 0 & -\frac{\sqrt{2}i}{480} \end{bmatrix}$$

$$\boxed{\text{x370}} \quad \mathbb{T}_{5,2}^{(1,0;a)}(E_u, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{200} & 0 & -\frac{\sqrt{14}}{40} & 0 & -\frac{\sqrt{210}}{200} & 0 & \frac{3\sqrt{10}}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{200} & 0 & \frac{3\sqrt{70}}{200} & 0 & \frac{\sqrt{42}}{40} & 0 & \frac{\sqrt{210}}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 & -\frac{\sqrt{42}}{40} & 0 & -\frac{3\sqrt{70}}{200} & 0 & -\frac{\sqrt{30}}{200} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{40} & 0 & \frac{\sqrt{210}}{200} & 0 & \frac{\sqrt{14}}{40} & 0 & \frac{\sqrt{70}}{200} & 0 \\ 0 & -\frac{\sqrt{105}}{280} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{3\sqrt{21}}{56} & -\frac{\sqrt{2}}{480} & 0 & -\frac{\sqrt{42}}{672} & 0 & -\frac{\sqrt{70}}{1120} & 0 & \frac{\sqrt{14}}{224} & 0 \\ \frac{\sqrt{105}}{280} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{56} & 0 & 0 & \frac{23\sqrt{70}}{16800} & 0 & \frac{13\sqrt{14}}{3360} & 0 & \frac{\sqrt{210}}{5600} & 0 & \frac{\sqrt{10}}{160} \\ 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{280} & -\frac{\sqrt{5}}{400} & 0 & -\frac{11\sqrt{105}}{8400} & 0 & -\frac{\sqrt{7}}{1680} & 0 & \frac{3\sqrt{35}}{2800} & 0 \\ \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{35}}{2800} & 0 & -\frac{\sqrt{7}}{1680} & 0 & -\frac{11\sqrt{105}}{8400} & 0 & -\frac{\sqrt{5}}{400} \\ 0 & -\frac{\sqrt{21}}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{105}}{280} & \frac{\sqrt{10}}{160} & 0 & \frac{\sqrt{210}}{5600} & 0 & \frac{13\sqrt{14}}{3360} & 0 & \frac{23\sqrt{70}}{16800} & 0 \\ -\frac{3\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{42}}{672} & 0 & -\frac{\sqrt{2}}{480} \end{bmatrix}$$

$$\boxed{\text{x371}} \quad \mathbb{T}_1^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x372}} \quad \mathbb{T}_3^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x373}} \quad \mathbb{T}_3^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{15}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x374}} \quad \mathbb{T}_3^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{15i}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{15i}{56} & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} \\ 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} \\ 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x375}} \quad \mathbb{T}_{1,1}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}i}{35} & 0 & -\frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{2}i}{35} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{6i}{35} & 0 & -\frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{2}i}{35} & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{42}i}{56} \end{bmatrix}$$

$$\boxed{\text{x376}} \quad \mathbb{T}_{1,2}^{(1,1;a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{5}}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{2}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{6}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{2}}{35} & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56} \end{bmatrix}$$

$$\boxed{\text{x377}} \quad \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} \frac{9\sqrt{5}i}{224} & 0 & -\frac{27\sqrt{2}i}{224} & 0 & \frac{45i}{224} & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 \\ 0 & -\frac{3\sqrt{3}i}{32} & 0 & \frac{3\sqrt{6}i}{224} & 0 & \frac{15\sqrt{15}i}{224} & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 \\ -\frac{15\sqrt{15}i}{224} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & \frac{3\sqrt{3}i}{32} & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} \\ 0 & -\frac{45i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 & -\frac{9\sqrt{5}i}{224} & 0 & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{112} & 0 \\ 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{5\sqrt{10}i}{168} & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & \frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{5i}{42} & 0 & 0 & \frac{\sqrt{30}i}{224} & 0 & \frac{\sqrt{6}i}{112} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{i}{14} & 0 & \frac{5\sqrt{10}i}{168} & \frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{224} & 0 & -\frac{\sqrt{3}i}{224} & 0 & -\frac{3\sqrt{15}i}{224} & 0 \\ \frac{5\sqrt{10}i}{168} & 0 & \frac{i}{14} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & \frac{3\sqrt{15}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & -\frac{3\sqrt{5}i}{224} & 0 & -\frac{\sqrt{105}i}{224} \\ 0 & \frac{5i}{42} & 0 & \frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & \frac{3\sqrt{10}i}{224} & 0 & -\frac{\sqrt{6}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & 0 \\ 0 & 0 & \frac{5\sqrt{10}i}{168} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{224} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{42}i}{224} \end{bmatrix}$$

$$\boxed{\text{x378}} \quad \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 1) = \begin{bmatrix} -\frac{9\sqrt{5}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & -\frac{45}{224} & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 \\ 0 & \frac{3\sqrt{3}}{32} & 0 & \frac{3\sqrt{6}}{224} & 0 & -\frac{15\sqrt{15}}{224} & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & \frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 \\ -\frac{15\sqrt{15}}{224} & 0 & \frac{3\sqrt{6}}{224} & 0 & \frac{3\sqrt{3}}{32} & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{3\sqrt{2}}{112} & 0 & 0 & 0 & \frac{\sqrt{70}}{112} \\ 0 & -\frac{45}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & -\frac{9\sqrt{5}}{224} & 0 & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 \\ 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{5\sqrt{10}}{168} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & \frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{224} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{5}{42} & 0 & 0 & -\frac{\sqrt{30}}{224} & 0 & \frac{\sqrt{6}}{112} & 0 & \frac{3\sqrt{10}}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{56} & 0 & \frac{1}{14} & 0 & -\frac{5\sqrt{10}}{168} & \frac{\sqrt{105}}{224} & 0 & -\frac{3\sqrt{5}}{224} & 0 & -\frac{\sqrt{3}}{224} & 0 & \frac{3\sqrt{15}}{224} & 0 \\ \frac{5\sqrt{10}}{168} & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3\sqrt{15}}{224} & 0 & -\frac{\sqrt{3}}{224} & 0 & -\frac{3\sqrt{5}}{224} & 0 & \frac{\sqrt{105}}{224} \\ 0 & \frac{5}{42} & 0 & -\frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & \frac{3\sqrt{10}}{224} & 0 & \frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{30}}{224} & 0 \\ 0 & 0 & \frac{5\sqrt{10}}{168} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{224} & 0 & \frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{224} \end{bmatrix}$$

$$\boxed{\text{x379}} \quad \mathbb{T}_{3,1}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} \frac{15\sqrt{3}i}{224} & 0 & -\frac{9\sqrt{30}i}{224} & 0 & -\frac{9\sqrt{15}i}{224} & 0 & 0 & -\frac{5\sqrt{2}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{32} & 0 & \frac{3\sqrt{10}i}{224} & 0 & -\frac{45i}{224} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 \\ \frac{45i}{224} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & \frac{3\sqrt{5}i}{32} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\ 0 & \frac{9\sqrt{15}i}{224} & 0 & \frac{9\sqrt{30}i}{224} & 0 & -\frac{15\sqrt{3}i}{224} & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{5\sqrt{2}i}{112} & 0 \\ 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{5\sqrt{6}i}{168} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{2}i}{224} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}i}{84} & 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & \frac{5\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{3\sqrt{6}i}{224} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{168} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{5\sqrt{6}i}{168} & -\frac{3\sqrt{7}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{5}i}{224} & 0 & \frac{9i}{224} & 0 \\ -\frac{5\sqrt{6}i}{168} & 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{9i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & \frac{3\sqrt{7}i}{224} \\ 0 & -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & 0 & -\frac{3\sqrt{6}i}{224} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{5\sqrt{2}i}{224} & 0 \\ 0 & 0 & -\frac{5\sqrt{6}i}{168} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{224} & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{70}i}{224} \end{bmatrix}$$

$$\boxed{\text{x380}} \quad \mathbb{T}_{3,2}^{(1,1;a)}(E_u, 2) = \begin{bmatrix} -\frac{15\sqrt{3}}{224} & 0 & -\frac{9\sqrt{30}}{224} & 0 & \frac{9\sqrt{15}}{224} & 0 & 0 & \frac{5\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{112} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{32} & 0 & \frac{3\sqrt{10}}{224} & 0 & \frac{45}{224} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{\sqrt{6}}{56} & 0 \\ \frac{45}{224} & 0 & \frac{3\sqrt{10}}{224} & 0 & \frac{3\sqrt{5}}{32} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & \frac{9\sqrt{15}}{224} & 0 & -\frac{9\sqrt{30}}{224} & 0 & -\frac{15\sqrt{3}}{224} & 0 & 0 & \frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{5\sqrt{2}}{112} & 0 \\ 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{5\sqrt{6}}{168} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{3\sqrt{2}}{224} & 0 & 0 & 0 \\ \frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & -\frac{5\sqrt{2}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{3\sqrt{6}}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{5\sqrt{6}}{168} & -\frac{3\sqrt{7}}{224} & 0 & -\frac{5\sqrt{3}}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & -\frac{9}{224} & 0 \\ -\frac{5\sqrt{6}}{168} & 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{9}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & -\frac{5\sqrt{3}}{224} & 0 & -\frac{3\sqrt{7}}{224} \\ 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{30}}{168} & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{5\sqrt{2}}{224} & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{168} & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{70}}{224} \end{bmatrix}$$

- bra: $\langle \frac{5}{2}, \frac{5}{2}; f |, \langle \frac{5}{2}, \frac{3}{2}; f |, \langle \frac{5}{2}, \frac{1}{2}; f |, \langle \frac{5}{2}, -\frac{1}{2}; f |, \langle \frac{5}{2}, -\frac{3}{2}; f |, \langle \frac{5}{2}, -\frac{5}{2}; f |, \langle \frac{7}{2}, \frac{7}{2}; f |, \langle \frac{7}{2}, \frac{5}{2}; f |, \langle \frac{7}{2}, \frac{3}{2}; f |, \langle \frac{7}{2}, \frac{1}{2}; f |, \langle \frac{7}{2}, -\frac{1}{2}; f |, \langle \frac{7}{2}, -\frac{3}{2}; f |, \langle \frac{7}{2}, -\frac{5}{2}; f |, \langle \frac{7}{2}, -\frac{7}{2}; f |$
- ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

$$\boxed{\text{x381}} \quad \mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \end{bmatrix}$$

$$\boxed{\text{x382}} \quad \mathbb{Q}_2^{(a)}(A_{1g}) = \begin{bmatrix} -\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} \end{bmatrix}$$

$$\boxed{\text{x383}} \quad \mathbb{Q}_4^{(a)}(A_{1g}, 1) = \begin{bmatrix} \frac{\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{84} & 0 & 0 & \frac{5\sqrt{22}}{154} & 0 & 0 & 0 & \frac{5\sqrt{66}}{462} & 0 & 0 \\ 0 & -\frac{\sqrt{33}}{28} & 0 & 0 & 0 & \frac{\sqrt{165}}{84} & 0 & 0 & -\frac{2\sqrt{330}}{231} & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 \\ 0 & 0 & \frac{\sqrt{33}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{154} & 0 & 0 & 0 & \frac{\sqrt{385}}{154} \\ 0 & 0 & 0 & \frac{\sqrt{33}}{42} & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & 0 \\ \frac{\sqrt{165}}{84} & 0 & 0 & 0 & -\frac{\sqrt{33}}{28} & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & \frac{2\sqrt{330}}{231} & 0 & 0 \\ 0 & \frac{\sqrt{165}}{84} & 0 & 0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & -\frac{5\sqrt{66}}{462} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{154} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 \\ \frac{5\sqrt{22}}{154} & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & -\frac{13\sqrt{33}}{308} & 0 & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 & 0 \\ 0 & -\frac{2\sqrt{330}}{231} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{462} & 0 & 0 & -\frac{3\sqrt{33}}{308} & 0 & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 \\ 0 & 0 & -\frac{5\sqrt{11}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} \\ 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 \\ \frac{5\sqrt{66}}{462} & 0 & 0 & 0 & \frac{2\sqrt{330}}{231} & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{33}}{308} & 0 & 0 \\ 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{154} & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 & 0 & 0 & -\frac{13\sqrt{33}}{308} & 0 \\ 0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{\sqrt{33}}{44} \end{bmatrix}$$

$$\boxed{\text{x384}} \quad \mathbb{Q}_4^{(a)}(A_{1g}, 2) = \begin{bmatrix} \frac{\sqrt{1155}}{588} & 0 & 0 & 0 & -\frac{\sqrt{231}}{84} & 0 & 0 & \frac{5\sqrt{770}}{1078} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 \\ 0 & -\frac{\sqrt{1155}}{196} & 0 & 0 & 0 & -\frac{\sqrt{231}}{84} & 0 & 0 & -\frac{10\sqrt{462}}{1617} & 0 & 0 & 0 & -\frac{\sqrt{154}}{77} & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{1078} & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{294} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & \frac{5\sqrt{385}}{1078} & 0 & 0 & 0 \\ -\frac{\sqrt{231}}{84} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{196} & 0 & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & 0 & \frac{10\sqrt{462}}{1617} & 0 & 0 \\ 0 & -\frac{\sqrt{231}}{84} & 0 & 0 & 0 & \frac{\sqrt{1155}}{588} & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1078} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 \\ \frac{5\sqrt{770}}{1078} & 0 & 0 & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & -\frac{13\sqrt{1155}}{2156} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & 0 \\ 0 & -\frac{10\sqrt{462}}{1617} & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & -\frac{3\sqrt{1155}}{2156} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 \\ 0 & 0 & -\frac{5\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{1155}}{2156} & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} \\ 0 & 0 & 0 & \frac{5\sqrt{385}}{1078} & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & \frac{9\sqrt{1155}}{2156} & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{462} & 0 & 0 & 0 & \frac{10\sqrt{462}}{1617} & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{1155}}{2156} & 0 & 0 \\ 0 & -\frac{\sqrt{154}}{77} & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1078} & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & 0 & 0 & -\frac{13\sqrt{1155}}{2156} & 0 \\ 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} \end{bmatrix}$$

$$\boxed{\text{x385}} \quad \mathbb{Q}_6^{(a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & -\frac{\sqrt{770}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{308} & 0 & 0 & 0 & \frac{\sqrt{55}}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{88} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & 0 & \frac{\sqrt{154}}{616} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{770}}{88} & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & -\frac{3\sqrt{231}}{616} & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 \\ 0 & 0 & -\frac{5\sqrt{77}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{165}}{88} \\ 0 & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & 0 & 0 \\ \frac{\sqrt{462}}{88} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & -\frac{3\sqrt{231}}{616} & 0 & 0 \\ 0 & -\frac{\sqrt{770}}{88} & 0 & 0 & 0 & \frac{\sqrt{154}}{616} & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 \\ 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}}{1848} \end{bmatrix}$$

$$\boxed{\text{x386}} \quad \mathbb{Q}_6^{(a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 & 0 & -\frac{\sqrt{385}}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{308} & 0 & 0 & 0 & \frac{5\sqrt{11}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & 0 & \frac{\sqrt{22}}{88} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{385}}{308} & 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & -\frac{3\sqrt{33}}{88} & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 \\ 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} \\ 0 & 0 & 0 & \frac{5\sqrt{11}}{44} & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 \\ -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & -\frac{3\sqrt{33}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 \\ 0 & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & 0 & 0 & -\frac{\sqrt{33}}{264} \end{bmatrix}$$

$$\boxed{\text{x387}} \quad \mathbb{Q}_4^{(a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{66}i}{77} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x388}} \quad \mathbb{Q}_6^{(a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{308} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x389}} \quad \mathbb{Q}_2^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 \\ -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 \\ 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 \\ 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} \\ 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 \\ 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x390}} \quad \mathbb{Q}_4^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & \frac{3\sqrt{66}}{154} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1078} & 0 & 0 & 0 \\ -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & -\frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & \frac{17\sqrt{77}}{1078} & 0 & 0 \\ 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & -\frac{17\sqrt{77}}{1078} & 0 & 0 & 0 & \frac{9\sqrt{231}}{1078} & 0 \\ 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1078} & 0 & 0 & 0 & -\frac{3\sqrt{66}}{154} \\ 0 & 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{77}}{1078} & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 \\ -\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & \frac{\sqrt{2310}}{1078} & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{1078} & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 \\ 0 & 0 & \frac{17\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} \\ 0 & 0 & 0 & \frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x391}} \quad \mathbb{Q}_6^{(a)}(B_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & \frac{\sqrt{42}}{168} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & -\frac{1}{8} & 0 \\ 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & \frac{\sqrt{105}}{168} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x392}} \quad \mathbb{Q}_6^{(a)}(B_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{\sqrt{462}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{77}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & -\frac{\sqrt{77}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 \\ 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{1848} \\ 0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & -\frac{\sqrt{77}}{56} & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{462}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} & 0 & 0 & -\frac{\sqrt{77}}{56} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x393}} \quad \mathbb{Q}_2^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{7}i}{98} & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & \frac{2\sqrt{21}i}{147} & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & \frac{9\sqrt{7}i}{98} & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 \\ 0 & -\frac{9\sqrt{7}i}{98} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 \\ 0 & 0 & -\frac{9\sqrt{7}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{147} & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} \\ 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{294} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}i}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{294} & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{294} & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{147} & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{294} & 0 & 0 \\ 0 & -\frac{2\sqrt{21}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{294} & 0 & 0 & -\frac{5\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{5\sqrt{70}i}{196} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x394}} \quad \mathbb{Q}_4^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & -\frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 \\ \frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & \frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & \frac{17\sqrt{77}i}{1078} & 0 & 0 \\ 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{196} & 0 & 0 & \frac{17\sqrt{77}i}{1078} & 0 & 0 & 0 & \frac{9\sqrt{231}i}{1078} & 0 \\ 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 & -\frac{3\sqrt{66}i}{154} \\ 0 & 0 & 0 & \frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{539} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{77}i}{1078} & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & \frac{6\sqrt{231}i}{539} & 0 & 0 & 0 \\ \frac{5\sqrt{462}i}{539} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1078} & 0 & 0 & \frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 & \frac{6\sqrt{231}i}{539} & 0 & 0 \\ 0 & \frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 & \frac{5\sqrt{462}i}{539} & 0 & 0 & -\frac{6\sqrt{231}i}{539} & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{1078} & 0 \\ 0 & 0 & -\frac{17\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{231}i}{539} & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} \\ 0 & 0 & 0 & -\frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x395}} \quad \mathbb{Q}_6^{(a)}(B_{2g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x396}} \quad \mathbb{Q}_6^{(a)}(B_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 & \frac{\sqrt{462}i}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 & 0 \\ -\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & \frac{\sqrt{330}i}{66} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 \\ 0 & \frac{\sqrt{330}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 \\ 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} \\ 0 & 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x397}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{84} & 0 & \frac{5\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}i}{98} & 0 & \frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{588} & 0 & \frac{11\sqrt{21}i}{588} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}i}{49} & 0 & -\frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}i}{588} & 0 & -\frac{\sqrt{105}i}{588} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{196} & 0 & -\frac{5\sqrt{3}i}{84} \\ -\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & \frac{5\sqrt{42}i}{147} & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{7}i}{196} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{147} & 0 & \frac{5\sqrt{210}i}{588} & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{21}i}{588} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{11\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{588} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & \frac{5\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}i}{588} & 0 & -\frac{5\sqrt{42}i}{147} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{147} & 0 & -\frac{5\sqrt{2}i}{28} \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x398}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & -\frac{5\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{98} & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{588} & 0 & -\frac{11\sqrt{21}}{588} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{49} & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}}{588} & 0 & \frac{\sqrt{105}}{588} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{196} & 0 & \frac{5\sqrt{3}}{84} \\ \frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{588} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{7}}{196} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{21}}{588} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{11\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & -\frac{5\sqrt{7}}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & \frac{5\sqrt{42}}{147} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & \frac{5\sqrt{2}}{28} \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x399}} \quad \mathbb{Q}_{4,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{11}i}{28} & 0 & \frac{\sqrt{22}i}{56} & 0 & 0 & \frac{\sqrt{2310}i}{616} & 0 & \frac{5\sqrt{110}i}{308} & 0 & \frac{5\sqrt{66}i}{616} & 0 & 0 & 0 \\ -\frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{110}i}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{66}i}{616} & 0 & -\frac{\sqrt{330}i}{308} & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 \\ 0 & \frac{\sqrt{110}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{56} & \frac{3\sqrt{231}i}{616} & 0 & \frac{\sqrt{11}i}{616} & 0 & -\frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{33}i}{616} & 0 \\ -\frac{\sqrt{22}i}{56} & 0 & 0 & 0 & \frac{\sqrt{110}i}{56} & 0 & 0 & \frac{\sqrt{33}i}{616} & 0 & \frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{11}i}{616} & 0 & -\frac{3\sqrt{231}i}{616} \\ 0 & 0 & 0 & -\frac{\sqrt{110}i}{56} & 0 & -\frac{\sqrt{11}i}{28} & 0 & 0 & -\frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{330}i}{308} & 0 & \frac{13\sqrt{66}i}{616} & 0 \\ 0 & 0 & \frac{\sqrt{22}i}{56} & 0 & \frac{\sqrt{11}i}{28} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{616} & 0 & -\frac{5\sqrt{110}i}{308} & 0 & -\frac{\sqrt{2310}i}{616} \\ -\frac{\sqrt{2310}i}{616} & 0 & -\frac{3\sqrt{231}i}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{308} & 0 & \frac{3\sqrt{77}i}{308} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{66}i}{616} & 0 & -\frac{\sqrt{33}i}{616} & 0 & 0 & -\frac{3\sqrt{385}i}{308} & 0 & -\frac{3\sqrt{165}i}{308} & 0 & \frac{3\sqrt{11}i}{154} & 0 & 0 & 0 \\ -\frac{5\sqrt{110}i}{308} & 0 & -\frac{\sqrt{11}i}{616} & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 & \frac{3\sqrt{165}i}{308} & 0 & -\frac{9\sqrt{33}i}{308} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}i}{308} & 0 & -\frac{\sqrt{165}i}{88} & 0 & \frac{5\sqrt{66}i}{616} & -\frac{3\sqrt{77}i}{308} & 0 & \frac{9\sqrt{33}i}{308} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{154} & 0 \\ -\frac{5\sqrt{66}i}{616} & 0 & \frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{330}i}{308} & 0 & 0 & -\frac{3\sqrt{11}i}{154} & 0 & 0 & 0 & \frac{9\sqrt{33}i}{308} & 0 & -\frac{3\sqrt{77}i}{308} \\ 0 & -\frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{11}i}{616} & 0 & \frac{5\sqrt{110}i}{308} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{33}i}{308} & 0 & \frac{3\sqrt{165}i}{308} & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{616} & 0 & -\frac{13\sqrt{66}i}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{154} & 0 & -\frac{3\sqrt{165}i}{308} & 0 & -\frac{3\sqrt{385}i}{308} \\ 0 & 0 & 0 & \frac{3\sqrt{231}i}{616} & 0 & \frac{\sqrt{2310}i}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{308} & 0 & \frac{3\sqrt{385}i}{308} & 0 \end{bmatrix}$$

$$\boxed{x400} \quad \mathbb{Q}_{4,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{22}}{56} & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & -\frac{5\sqrt{110}}{308} & 0 & \frac{5\sqrt{66}}{616} & 0 & 0 & 0 \\ -\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{110}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{66}}{616} & 0 & \frac{\sqrt{330}}{308} & 0 & \frac{\sqrt{22}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{110}}{56} & 0 & 0 & 0 & -\frac{\sqrt{22}}{56} & -\frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{11}}{616} & 0 & \frac{\sqrt{165}}{88} & 0 & -\frac{\sqrt{33}}{616} & 0 \\ \frac{\sqrt{22}}{56} & 0 & 0 & 0 & -\frac{\sqrt{110}}{56} & 0 & 0 & -\frac{\sqrt{33}}{616} & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{11}}{616} & 0 & -\frac{3\sqrt{231}}{616} \\ 0 & 0 & 0 & -\frac{\sqrt{110}}{56} & 0 & \frac{\sqrt{11}}{28} & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{330}}{308} & 0 & -\frac{13\sqrt{66}}{616} & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{56} & 0 & \frac{\sqrt{11}}{28} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{616} & 0 & -\frac{5\sqrt{110}}{308} & 0 & \frac{\sqrt{2310}}{616} \\ \frac{\sqrt{2310}}{616} & 0 & -\frac{3\sqrt{231}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & \frac{3\sqrt{77}}{308} & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{66}}{616} & 0 & -\frac{\sqrt{33}}{616} & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & \frac{3\sqrt{165}}{308} & 0 & \frac{3\sqrt{11}}{154} & 0 & 0 & 0 \\ -\frac{5\sqrt{110}}{308} & 0 & \frac{\sqrt{11}}{616} & 0 & \frac{\sqrt{22}}{88} & 0 & 0 & \frac{3\sqrt{165}}{308} & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{308} & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{5\sqrt{66}}{616} & \frac{3\sqrt{77}}{308} & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 \\ \frac{5\sqrt{66}}{616} & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{330}}{308} & 0 & 0 & \frac{3\sqrt{11}}{154} & 0 & 0 & 0 & -\frac{9\sqrt{33}}{308} & 0 & -\frac{3\sqrt{77}}{308} \\ 0 & \frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{11}}{616} & 0 & -\frac{5\sqrt{110}}{308} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{33}}{308} & 0 & -\frac{3\sqrt{165}}{308} & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{616} & 0 & -\frac{13\sqrt{66}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 & -\frac{3\sqrt{165}}{308} & 0 & \frac{3\sqrt{385}}{308} \\ 0 & 0 & 0 & -\frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{308} & 0 & \frac{3\sqrt{385}}{308} & 0 \end{bmatrix}$$

$$\boxed{x401} \quad \mathbb{Q}_{4,1}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{\sqrt{77}i}{196} & 0 & -\frac{\sqrt{154}i}{56} & 0 & 0 & \frac{\sqrt{330}i}{616} & 0 & \frac{5\sqrt{770}i}{2156} & 0 & -\frac{5\sqrt{462}i}{616} & 0 & 0 & 0 \\ -\frac{\sqrt{77}i}{196} & 0 & -\frac{\sqrt{770}i}{392} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}i}{4312} & 0 & -\frac{\sqrt{2310}i}{2156} & 0 & -\frac{\sqrt{154}i}{88} & 0 & 0 \\ 0 & \frac{\sqrt{770}i}{392} & 0 & 0 & 0 & \frac{\sqrt{154}i}{56} & -\frac{3\sqrt{33}i}{88} & 0 & \frac{\sqrt{77}i}{4312} & 0 & -\frac{\sqrt{1155}i}{616} & 0 & \frac{\sqrt{231}i}{616} & 0 \\ \frac{\sqrt{154}i}{56} & 0 & 0 & 0 & \frac{\sqrt{770}i}{392} & 0 & 0 & -\frac{\sqrt{231}i}{616} & 0 & \frac{\sqrt{1155}i}{616} & 0 & -\frac{\sqrt{77}i}{4312} & 0 & \frac{3\sqrt{33}i}{88} \\ 0 & 0 & 0 & -\frac{\sqrt{770}i}{392} & 0 & -\frac{\sqrt{77}i}{196} & 0 & 0 & \frac{\sqrt{154}i}{88} & 0 & \frac{\sqrt{2310}i}{2156} & 0 & \frac{13\sqrt{462}i}{4312} & 0 \\ 0 & 0 & -\frac{\sqrt{154}i}{56} & 0 & \frac{\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{616} & 0 & -\frac{5\sqrt{770}i}{2156} & 0 & -\frac{\sqrt{330}i}{616} \\ -\frac{\sqrt{330}i}{616} & 0 & \frac{3\sqrt{33}i}{88} & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{308} & 0 & -\frac{3\sqrt{11}i}{44} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{462}i}{4312} & 0 & \frac{\sqrt{231}i}{616} & 0 & 0 & -\frac{3\sqrt{55}i}{308} & 0 & -\frac{3\sqrt{1155}i}{2156} & 0 & -\frac{3\sqrt{77}i}{154} & 0 & 0 & 0 \\ -\frac{5\sqrt{770}i}{2156} & 0 & -\frac{\sqrt{77}i}{4312} & 0 & -\frac{\sqrt{154}i}{88} & 0 & 0 & \frac{3\sqrt{1155}i}{2156} & 0 & -\frac{9\sqrt{231}i}{2156} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}i}{2156} & 0 & -\frac{\sqrt{1155}i}{616} & 0 & -\frac{5\sqrt{462}i}{616} & \frac{3\sqrt{11}i}{44} & 0 & \frac{9\sqrt{231}i}{2156} & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 \\ \frac{5\sqrt{462}i}{616} & 0 & \frac{\sqrt{1155}i}{616} & 0 & -\frac{\sqrt{2310}i}{2156} & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & \frac{9\sqrt{231}i}{2156} & 0 & \frac{3\sqrt{11}i}{44} \\ 0 & \frac{\sqrt{154}i}{88} & 0 & \frac{\sqrt{77}i}{4312} & 0 & \frac{5\sqrt{770}i}{2156} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}i}{2156} & 0 & \frac{3\sqrt{1155}i}{2156} & 0 \\ 0 & 0 & -\frac{\sqrt{231}i}{616} & 0 & -\frac{13\sqrt{462}i}{4312} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & -\frac{3\sqrt{1155}i}{2156} & 0 & -\frac{3\sqrt{55}i}{308} \\ 0 & 0 & 0 & -\frac{3\sqrt{33}i}{88} & 0 & \frac{\sqrt{330}i}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{55}i}{308} & 0 \end{bmatrix}$$

$$\boxed{\text{x402}} \quad \mathbb{Q}_{4,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{77}}{196} & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & \frac{\sqrt{330}}{616} & 0 & -\frac{5\sqrt{770}}{2156} & 0 & -\frac{5\sqrt{462}}{616} & 0 & 0 & 0 \\ -\frac{\sqrt{77}}{196} & 0 & \frac{\sqrt{770}}{392} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}}{4312} & 0 & \frac{\sqrt{2310}}{2156} & 0 & -\frac{\sqrt{154}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{770}}{392} & 0 & 0 & 0 & \frac{\sqrt{154}}{56} & \frac{3\sqrt{33}}{88} & 0 & \frac{\sqrt{77}}{4312} & 0 & \frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{231}}{616} & 0 \\ -\frac{\sqrt{154}}{56} & 0 & 0 & 0 & -\frac{\sqrt{770}}{392} & 0 & 0 & \frac{\sqrt{231}}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{77}}{4312} & 0 & \frac{3\sqrt{33}}{88} \\ 0 & 0 & 0 & -\frac{\sqrt{770}}{392} & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & -\frac{\sqrt{154}}{88} & 0 & \frac{\sqrt{2310}}{2156} & 0 & -\frac{13\sqrt{462}}{4312} & 0 \\ 0 & 0 & \frac{\sqrt{154}}{56} & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{616} & 0 & -\frac{5\sqrt{770}}{2156} & 0 & \frac{\sqrt{330}}{616} \\ \frac{\sqrt{330}}{616} & 0 & \frac{3\sqrt{33}}{88} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{308} & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{462}}{4312} & 0 & \frac{\sqrt{231}}{616} & 0 & 0 & -\frac{3\sqrt{55}}{308} & 0 & \frac{3\sqrt{1155}}{2156} & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & 0 \\ -\frac{5\sqrt{770}}{2156} & 0 & \frac{\sqrt{77}}{4312} & 0 & -\frac{\sqrt{154}}{88} & 0 & 0 & \frac{3\sqrt{1155}}{2156} & 0 & \frac{9\sqrt{231}}{2156} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{2156} & 0 & \frac{\sqrt{1155}}{616} & 0 & -\frac{5\sqrt{462}}{616} & -\frac{3\sqrt{11}}{44} & 0 & \frac{9\sqrt{231}}{2156} & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 \\ -\frac{5\sqrt{462}}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{2310}}{2156} & 0 & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & 0 & -\frac{9\sqrt{231}}{2156} & 0 & \frac{3\sqrt{11}}{44} \\ 0 & -\frac{\sqrt{154}}{88} & 0 & \frac{\sqrt{77}}{4312} & 0 & -\frac{5\sqrt{770}}{2156} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}}{2156} & 0 & -\frac{3\sqrt{1155}}{2156} & 0 \\ 0 & 0 & \frac{\sqrt{231}}{616} & 0 & -\frac{13\sqrt{462}}{4312} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & -\frac{3\sqrt{1155}}{2156} & 0 & \frac{3\sqrt{55}}{308} \\ 0 & 0 & 0 & \frac{3\sqrt{33}}{88} & 0 & \frac{\sqrt{330}}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{55}}{308} & 0 \end{bmatrix}$$

$$\boxed{\text{x403}} \quad \mathbb{Q}_{6,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{1232} & 0 & \frac{\sqrt{33}i}{176} & 0 & -\frac{3\sqrt{55}i}{176} & 0 & -\frac{\sqrt{11}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{176} & 0 & -\frac{5\sqrt{11}i}{176} & 0 & \frac{3\sqrt{165}i}{176} & 0 & \frac{\sqrt{385}i}{112} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}i}{1232} & 0 & \frac{\sqrt{330}i}{176} & 0 & \frac{5\sqrt{22}i}{176} & 0 & -\frac{3\sqrt{110}i}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{176} & 0 & -\frac{5\sqrt{22}i}{176} & 0 & -\frac{\sqrt{330}i}{176} & 0 & \frac{3\sqrt{770}i}{1232} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{112} & 0 & -\frac{3\sqrt{165}i}{176} & 0 & \frac{5\sqrt{11}i}{176} & 0 & \frac{\sqrt{55}i}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{16} & 0 & \frac{3\sqrt{55}i}{176} & 0 & -\frac{\sqrt{33}i}{176} & 0 & -\frac{\sqrt{77}i}{1232} \\ -\frac{\sqrt{77}i}{1232} & 0 & \frac{3\sqrt{770}i}{1232} & 0 & \frac{\sqrt{385}i}{112} & 0 & 0 & \frac{\sqrt{462}i}{1232} & 0 & -\frac{\sqrt{2310}i}{616} & 0 & -\frac{\sqrt{154}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{55}i}{176} & 0 & -\frac{3\sqrt{110}i}{176} & 0 & -\frac{\sqrt{11}i}{16} & -\frac{\sqrt{462}i}{1232} & 0 & -\frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{330}i}{176} & 0 & 0 & 0 \\ -\frac{\sqrt{33}i}{176} & 0 & -\frac{\sqrt{330}i}{176} & 0 & \frac{3\sqrt{165}i}{176} & 0 & 0 & \frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{110}i}{176} & 0 & 0 & 0 & \frac{\sqrt{154}i}{112} \\ 0 & \frac{5\sqrt{11}i}{176} & 0 & \frac{5\sqrt{22}i}{176} & 0 & -\frac{3\sqrt{55}i}{176} & \frac{\sqrt{2310}i}{616} & 0 & -\frac{\sqrt{110}i}{176} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{176} & 0 \\ \frac{3\sqrt{55}i}{176} & 0 & -\frac{5\sqrt{22}i}{176} & 0 & -\frac{5\sqrt{11}i}{176} & 0 & 0 & -\frac{\sqrt{330}i}{176} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{176} & 0 & \frac{\sqrt{2310}i}{616} \\ 0 & -\frac{3\sqrt{165}i}{176} & 0 & \frac{\sqrt{330}i}{176} & 0 & \frac{\sqrt{33}i}{176} & \frac{\sqrt{154}i}{112} & 0 & 0 & 0 & \frac{\sqrt{110}i}{176} & 0 & \frac{\sqrt{22}i}{88} & 0 \\ \frac{\sqrt{11}i}{16} & 0 & \frac{3\sqrt{110}i}{176} & 0 & -\frac{\sqrt{55}i}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{176} & 0 & -\frac{\sqrt{22}i}{88} & 0 & -\frac{\sqrt{462}i}{1232} \\ 0 & -\frac{\sqrt{385}i}{112} & 0 & -\frac{3\sqrt{770}i}{1232} & 0 & \frac{\sqrt{77}i}{1232} & 0 & 0 & -\frac{\sqrt{154}i}{112} & 0 & -\frac{\sqrt{2310}i}{616} & 0 & \frac{\sqrt{462}i}{1232} & 0 \end{bmatrix}$$

$$\boxed{x404} \quad \mathbb{Q}_{6,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{1232} & 0 & -\frac{\sqrt{33}}{176} & 0 & -\frac{3\sqrt{55}}{176} & 0 & \frac{\sqrt{11}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{176} & 0 & \frac{5\sqrt{11}}{176} & 0 & \frac{3\sqrt{165}}{176} & 0 & -\frac{\sqrt{385}}{112} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{1232} & 0 & \frac{\sqrt{330}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & -\frac{3\sqrt{110}}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & \frac{\sqrt{330}}{176} & 0 & \frac{3\sqrt{770}}{1232} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{112} & 0 & \frac{3\sqrt{165}}{176} & 0 & \frac{5\sqrt{11}}{176} & 0 & -\frac{\sqrt{55}}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{16} & 0 & -\frac{3\sqrt{55}}{176} & 0 & -\frac{\sqrt{33}}{176} & 0 & \frac{\sqrt{77}}{1232} \\ \frac{\sqrt{77}}{1232} & 0 & \frac{3\sqrt{770}}{1232} & 0 & -\frac{\sqrt{385}}{112} & 0 & 0 & -\frac{\sqrt{462}}{1232} & 0 & -\frac{\sqrt{2310}}{616} & 0 & \frac{\sqrt{154}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{55}}{176} & 0 & -\frac{3\sqrt{110}}{176} & 0 & \frac{\sqrt{11}}{16} & -\frac{\sqrt{462}}{1232} & 0 & \frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{176} & 0 & \frac{\sqrt{330}}{176} & 0 & \frac{3\sqrt{165}}{176} & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & -\frac{\sqrt{110}}{176} & 0 & 0 & 0 & -\frac{\sqrt{154}}{112} \\ 0 & \frac{5\sqrt{11}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & -\frac{3\sqrt{55}}{176} & -\frac{\sqrt{2310}}{616} & 0 & -\frac{\sqrt{110}}{176} & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & 0 \\ -\frac{3\sqrt{55}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & \frac{5\sqrt{11}}{176} & 0 & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & 0 & \frac{\sqrt{110}}{176} & 0 & \frac{\sqrt{2310}}{616} \\ 0 & \frac{3\sqrt{165}}{176} & 0 & \frac{\sqrt{330}}{176} & 0 & -\frac{\sqrt{33}}{176} & \frac{\sqrt{154}}{112} & 0 & 0 & 0 & \frac{\sqrt{110}}{176} & 0 & -\frac{\sqrt{22}}{88} & 0 \\ \frac{\sqrt{11}}{16} & 0 & -\frac{3\sqrt{110}}{176} & 0 & -\frac{\sqrt{55}}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & 0 & -\frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{462}}{1232} \\ 0 & -\frac{\sqrt{385}}{112} & 0 & \frac{3\sqrt{770}}{1232} & 0 & \frac{\sqrt{77}}{1232} & 0 & 0 & -\frac{\sqrt{154}}{112} & 0 & \frac{\sqrt{2310}}{616} & 0 & \frac{\sqrt{462}}{1232} & 0 \end{bmatrix}$$

$$\boxed{x405} \quad \mathbb{Q}_{6,1}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42i}}{448} & 0 & \frac{3\sqrt{2i}}{64} & 0 & \frac{\sqrt{30i}}{64} & 0 & \frac{\sqrt{6i}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30i}}{64} & 0 & -\frac{5\sqrt{6i}}{64} & 0 & -\frac{3\sqrt{10i}}{64} & 0 & -\frac{\sqrt{210i}}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105i}}{224} & 0 & \frac{3\sqrt{5i}}{32} & 0 & \frac{5\sqrt{3i}}{32} & 0 & \frac{\sqrt{15i}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15i}}{32} & 0 & -\frac{5\sqrt{3i}}{32} & 0 & -\frac{3\sqrt{5i}}{32} & 0 & -\frac{\sqrt{105i}}{224} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210i}}{448} & 0 & \frac{3\sqrt{10i}}{64} & 0 & \frac{5\sqrt{6i}}{64} & 0 & \frac{\sqrt{30i}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6i}}{64} & 0 & -\frac{\sqrt{30i}}{64} & 0 & -\frac{3\sqrt{2i}}{64} & 0 & -\frac{\sqrt{42i}}{448} \\ -\frac{\sqrt{42i}}{448} & 0 & -\frac{\sqrt{105i}}{224} & 0 & -\frac{\sqrt{210i}}{448} & 0 & 0 & \frac{3\sqrt{7i}}{224} & 0 & \frac{\sqrt{35i}}{112} & 0 & \frac{\sqrt{21i}}{224} & 0 & 0 \\ 0 & \frac{\sqrt{30i}}{64} & 0 & \frac{\sqrt{15i}}{32} & 0 & \frac{\sqrt{6i}}{64} & -\frac{3\sqrt{7i}}{224} & 0 & -\frac{\sqrt{3i}}{16} & 0 & -\frac{\sqrt{5i}}{32} & 0 & 0 & 0 \\ -\frac{3\sqrt{2i}}{64} & 0 & -\frac{3\sqrt{5i}}{32} & 0 & -\frac{3\sqrt{10i}}{64} & 0 & 0 & \frac{\sqrt{3i}}{16} & 0 & \frac{\sqrt{15i}}{32} & 0 & 0 & 0 & -\frac{\sqrt{21i}}{224} \\ 0 & \frac{5\sqrt{6i}}{64} & 0 & \frac{5\sqrt{3i}}{32} & 0 & \frac{\sqrt{30i}}{64} & -\frac{\sqrt{35i}}{112} & 0 & -\frac{\sqrt{15i}}{32} & 0 & 0 & 0 & \frac{\sqrt{5i}}{32} & 0 \\ -\frac{\sqrt{30i}}{64} & 0 & -\frac{5\sqrt{3i}}{32} & 0 & -\frac{5\sqrt{6i}}{64} & 0 & 0 & \frac{\sqrt{5i}}{32} & 0 & 0 & 0 & -\frac{\sqrt{15i}}{32} & 0 & -\frac{\sqrt{35i}}{112} \\ 0 & \frac{3\sqrt{10i}}{64} & 0 & \frac{3\sqrt{5i}}{32} & 0 & \frac{3\sqrt{2i}}{64} & -\frac{\sqrt{21i}}{224} & 0 & 0 & 0 & \frac{\sqrt{15i}}{32} & 0 & \frac{\sqrt{3i}}{16} & 0 \\ -\frac{\sqrt{6i}}{64} & 0 & -\frac{\sqrt{15i}}{32} & 0 & -\frac{\sqrt{30i}}{64} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5i}}{32} & 0 & -\frac{\sqrt{3i}}{16} & 0 & -\frac{3\sqrt{7i}}{224} \\ 0 & \frac{\sqrt{210i}}{448} & 0 & \frac{\sqrt{105i}}{224} & 0 & \frac{\sqrt{42i}}{448} & 0 & 0 & \frac{\sqrt{21i}}{224} & 0 & \frac{\sqrt{35i}}{112} & 0 & \frac{3\sqrt{7i}}{224} & 0 \end{bmatrix}$$

$$\boxed{x406} \quad \mathbb{Q}_{6,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{448} & 0 & -\frac{3\sqrt{2}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{6}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{64} & 0 & \frac{5\sqrt{6}}{64} & 0 & -\frac{3\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{\sqrt{15}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{\sqrt{105}}{224} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{448} & 0 & -\frac{3\sqrt{10}}{64} & 0 & \frac{5\sqrt{6}}{64} & 0 & -\frac{\sqrt{30}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{3\sqrt{2}}{64} & 0 & \frac{\sqrt{42}}{448} \\ \frac{\sqrt{42}}{448} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{210}}{448} & 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{21}}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{64} & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{6}}{64} & -\frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{3}}{16} & 0 & -\frac{\sqrt{5}}{32} & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{64} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{3\sqrt{10}}{64} & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & -\frac{\sqrt{15}}{32} & 0 & 0 & 0 & \frac{\sqrt{21}}{224} \\ 0 & \frac{5\sqrt{6}}{64} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{\sqrt{30}}{64} & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{15}}{32} & 0 & 0 & 0 & \frac{\sqrt{5}}{32} & 0 \\ \frac{\sqrt{30}}{64} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{5\sqrt{6}}{64} & 0 & 0 & -\frac{\sqrt{5}}{32} & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{35}}{112} \\ 0 & -\frac{3\sqrt{10}}{64} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{3\sqrt{2}}{64} & -\frac{\sqrt{21}}{224} & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{3}}{16} & 0 \\ -\frac{\sqrt{6}}{64} & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{30}}{64} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{32} & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{7}}{224} \\ 0 & \frac{\sqrt{210}}{448} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{42}}{448} & 0 & 0 & \frac{\sqrt{21}}{224} & 0 & -\frac{\sqrt{35}}{112} & 0 & \frac{3\sqrt{7}}{224} & 0 \end{bmatrix}$$

$$\boxed{x407} \quad \mathbb{Q}_{6,1}^{(a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{14784} & 0 & \frac{\sqrt{110}i}{704} & 0 & -\frac{9\sqrt{66}i}{704} & 0 & \frac{\sqrt{330}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & -\frac{5\sqrt{330}i}{2112} & 0 & \frac{27\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{462}i}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}i}{2464} & 0 & \frac{5\sqrt{11}i}{352} & 0 & \frac{5\sqrt{165}i}{1056} & 0 & -\frac{9\sqrt{33}i}{352} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{33}i}{352} & 0 & -\frac{5\sqrt{165}i}{1056} & 0 & -\frac{5\sqrt{11}i}{352} & 0 & \frac{9\sqrt{231}i}{2464} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{448} & 0 & -\frac{27\sqrt{22}i}{704} & 0 & \frac{5\sqrt{330}i}{2112} & 0 & \frac{5\sqrt{66}i}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{64} & 0 & \frac{9\sqrt{66}i}{704} & 0 & -\frac{\sqrt{110}i}{704} & 0 & -\frac{\sqrt{2310}i}{14784} \\ -\frac{\sqrt{2310}i}{14784} & 0 & \frac{9\sqrt{231}i}{2464} & 0 & -\frac{5\sqrt{462}i}{448} & 0 & 0 & \frac{\sqrt{385}i}{2464} & 0 & -\frac{9\sqrt{77}i}{1232} & 0 & \frac{\sqrt{1155}i}{224} & 0 & 0 \\ 0 & \frac{5\sqrt{66}i}{2112} & 0 & -\frac{9\sqrt{33}i}{352} & 0 & \frac{\sqrt{330}i}{64} & -\frac{\sqrt{385}i}{2464} & 0 & -\frac{\sqrt{165}i}{528} & 0 & \frac{9\sqrt{11}i}{352} & 0 & 0 & 0 \\ -\frac{\sqrt{110}i}{704} & 0 & -\frac{5\sqrt{11}i}{352} & 0 & \frac{27\sqrt{22}i}{704} & 0 & 0 & \frac{\sqrt{165}i}{528} & 0 & \frac{5\sqrt{33}i}{1056} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{224} \\ 0 & \frac{5\sqrt{330}i}{2112} & 0 & \frac{5\sqrt{165}i}{1056} & 0 & -\frac{9\sqrt{66}i}{704} & \frac{9\sqrt{77}i}{1232} & 0 & -\frac{5\sqrt{33}i}{1056} & 0 & 0 & 0 & -\frac{9\sqrt{11}i}{352} & 0 \\ \frac{9\sqrt{66}i}{704} & 0 & -\frac{5\sqrt{165}i}{1056} & 0 & -\frac{5\sqrt{330}i}{2112} & 0 & 0 & -\frac{9\sqrt{11}i}{352} & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{1056} & 0 & \frac{9\sqrt{77}i}{1232} \\ 0 & -\frac{27\sqrt{22}i}{704} & 0 & \frac{5\sqrt{11}i}{352} & 0 & \frac{\sqrt{110}i}{704} & -\frac{\sqrt{1155}i}{224} & 0 & 0 & 0 & \frac{5\sqrt{33}i}{1056} & 0 & \frac{\sqrt{165}i}{528} & 0 \\ -\frac{\sqrt{330}i}{64} & 0 & \frac{9\sqrt{33}i}{352} & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & 0 & 0 & 0 & \frac{9\sqrt{11}i}{352} & 0 & -\frac{\sqrt{165}i}{528} & 0 & -\frac{\sqrt{385}i}{2464} \\ 0 & \frac{5\sqrt{462}i}{448} & 0 & -\frac{9\sqrt{231}i}{2464} & 0 & \frac{\sqrt{2310}i}{14784} & 0 & 0 & \frac{\sqrt{1155}i}{224} & 0 & -\frac{9\sqrt{77}i}{1232} & 0 & \frac{\sqrt{385}i}{2464} & 0 \end{bmatrix}$$

$$\boxed{\text{x408}} \quad \mathbb{Q}_{6,2}^{(a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{14784} & 0 & -\frac{\sqrt{110}}{704} & 0 & -\frac{9\sqrt{66}}{704} & 0 & -\frac{\sqrt{330}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{2112} & 0 & \frac{5\sqrt{330}}{2112} & 0 & \frac{27\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{231}}{2464} & 0 & \frac{5\sqrt{11}}{352} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & -\frac{9\sqrt{33}}{352} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{33}}{352} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & \frac{5\sqrt{11}}{352} & 0 & \frac{9\sqrt{231}}{2464} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{448} & 0 & \frac{27\sqrt{22}}{704} & 0 & \frac{5\sqrt{330}}{2112} & 0 & -\frac{5\sqrt{66}}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{64} & 0 & -\frac{9\sqrt{66}}{704} & 0 & -\frac{\sqrt{110}}{704} & 0 & \frac{\sqrt{2310}}{14784} \\ \frac{\sqrt{2310}}{14784} & 0 & \frac{9\sqrt{231}}{2464} & 0 & \frac{5\sqrt{462}}{448} & 0 & 0 & -\frac{\sqrt{385}}{2464} & 0 & -\frac{9\sqrt{77}}{1232} & 0 & -\frac{\sqrt{1155}}{224} & 0 & 0 \\ 0 & -\frac{5\sqrt{66}}{2112} & 0 & -\frac{9\sqrt{33}}{352} & 0 & -\frac{\sqrt{330}}{64} & -\frac{\sqrt{385}}{2464} & 0 & \frac{\sqrt{165}}{528} & 0 & \frac{9\sqrt{11}}{352} & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{704} & 0 & \frac{5\sqrt{11}}{352} & 0 & \frac{27\sqrt{22}}{704} & 0 & 0 & \frac{\sqrt{165}}{528} & 0 & -\frac{5\sqrt{33}}{1056} & 0 & 0 & 0 & \frac{\sqrt{1155}}{224} \\ 0 & \frac{5\sqrt{330}}{2112} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & -\frac{9\sqrt{66}}{704} & -\frac{9\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{33}}{1056} & 0 & 0 & 0 & -\frac{9\sqrt{11}}{352} & 0 \\ -\frac{9\sqrt{66}}{704} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & \frac{5\sqrt{330}}{2112} & 0 & 0 & \frac{9\sqrt{11}}{352} & 0 & 0 & 0 & \frac{5\sqrt{33}}{1056} & 0 & \frac{9\sqrt{77}}{1232} \\ 0 & \frac{27\sqrt{22}}{704} & 0 & \frac{5\sqrt{11}}{352} & 0 & -\frac{\sqrt{110}}{704} & -\frac{\sqrt{1155}}{224} & 0 & 0 & 0 & \frac{5\sqrt{33}}{1056} & 0 & -\frac{\sqrt{165}}{528} & 0 \\ -\frac{\sqrt{330}}{64} & 0 & -\frac{9\sqrt{33}}{352} & 0 & -\frac{5\sqrt{66}}{2112} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{11}}{352} & 0 & -\frac{\sqrt{165}}{528} & 0 & \frac{\sqrt{385}}{2464} \\ 0 & \frac{5\sqrt{462}}{448} & 0 & \frac{9\sqrt{231}}{2464} & 0 & \frac{\sqrt{2310}}{14784} & 0 & 0 & \frac{\sqrt{1155}}{224} & 0 & \frac{9\sqrt{77}}{1232} & 0 & \frac{\sqrt{385}}{2464} & 0 \end{bmatrix}$$

$$\boxed{\text{x409}} \quad \mathbb{Q}_2^{(1,-1;a)}(A_{1g}) = \begin{bmatrix} -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \end{bmatrix}$$

$$\boxed{\text{x410}} \quad \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1) = \begin{bmatrix} \frac{1}{42} & 0 & 0 & 0 & \frac{\sqrt{5}}{42} & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & \frac{5\sqrt{2}}{84} & 0 & 0 \\ 0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{5}}{42} & 0 & 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 \\ 0 & 0 & \frac{1}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & \frac{\sqrt{105}}{84} \\ 0 & 0 & 0 & \frac{1}{21} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{42} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 \\ 0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & \frac{1}{42} & 0 & 0 & -\frac{5\sqrt{2}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{1}{6} & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 \\ \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & \frac{13}{42} & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & -\frac{5\sqrt{2}}{84} & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 \\ 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \\ 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 \\ \frac{5\sqrt{2}}{84} & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & \frac{13}{42} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & -\frac{1}{6} \end{bmatrix}$$

$$\boxed{\text{x411}} \quad \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2) = \begin{bmatrix} \frac{\sqrt{35}}{294} & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & -\frac{5\sqrt{14}}{147} & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{588} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{5\sqrt{105}}{588} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & \frac{5\sqrt{14}}{147} & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{294} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 \\ \frac{5\sqrt{210}}{588} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & \frac{13\sqrt{35}}{294} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & -\frac{5\sqrt{14}}{147} & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 \\ 0 & 0 & -\frac{5\sqrt{105}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & \frac{1}{6} \\ 0 & 0 & 0 & \frac{5\sqrt{105}}{588} & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & \frac{5\sqrt{14}}{147} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{13\sqrt{35}}{294} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \end{bmatrix}$$

$$\boxed{\text{x412}} \quad \mathbb{Q}_6^{(1,-1;a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{77}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{616} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & 0 & -\frac{\sqrt{385}}{616} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & \frac{\sqrt{231}}{1848} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & \frac{\sqrt{154}}{616} & 0 & 0 & 0 & -\frac{3\sqrt{110}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{385}}{616} & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & \frac{9\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{88} & 0 \\ 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & -\frac{3\sqrt{110}}{88} \\ 0 & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & -\frac{3\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 \\ \frac{\sqrt{77}}{88} & 0 & 0 & 0 & -\frac{\sqrt{385}}{616} & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 & 0 & \frac{9\sqrt{154}}{616} & 0 & 0 \\ 0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & 0 & \frac{\sqrt{231}}{1848} & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 \\ 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{88} & 0 & 0 & 0 & \frac{\sqrt{154}}{616} \end{bmatrix}$$

$$\boxed{\text{x413}} \quad \mathbb{Q}_6^{(1,-1;a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{11}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & \frac{\sqrt{165}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & \frac{\sqrt{33}}{264} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & 0 & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & \frac{9\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 \\ 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & \frac{3\sqrt{770}}{616} \\ 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{11}}{88} & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & \frac{9\sqrt{22}}{88} & 0 & 0 \\ 0 & \frac{\sqrt{165}}{264} & 0 & 0 & 0 & \frac{\sqrt{33}}{264} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 & \frac{\sqrt{22}}{88} \end{bmatrix}$$

$$\boxed{\text{x414}} \quad \mathbb{Q}_4^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x415}} \quad \mathbb{Q}_6^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x416}} \quad \mathbb{Q}_2^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 \\ 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & \frac{\sqrt{3}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x417}} \quad \mathbb{Q}_4^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & -\frac{9\sqrt{7}}{196} & 0 & 0 & 0 & \frac{17\sqrt{21}}{588} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & -\frac{17\sqrt{21}}{588} & 0 & 0 & 0 & \frac{9\sqrt{7}}{196} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{21}}{588} & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 \\ -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 \\ 0 & 0 & \frac{17\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 & \frac{\sqrt{5}}{7} \\ 0 & 0 & 0 & \frac{9\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x418}} \quad \mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & 0 & -\frac{\sqrt{5}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{24} & 0 & 0 & 0 & \frac{\sqrt{7}}{168} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & 0 & -\frac{1}{24} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & \frac{\sqrt{210}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 \\ \frac{1}{24} & 0 & 0 & 0 & \frac{\sqrt{5}}{24} & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{24} & 0 & 0 & 0 & -\frac{1}{24} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{56} & 0 & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x419}} \quad \mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{264} & 0 & 0 & 0 & -\frac{\sqrt{77}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{11}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & 0 & \frac{\sqrt{110}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{264} & 0 & 0 & 0 & -\frac{\sqrt{385}}{1848} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{56} & 0 & 0 & 0 & \frac{\sqrt{55}}{264} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{385}}{1848} & 0 & 0 & 0 & \frac{\sqrt{77}}{56} & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{462}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{55}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{11}}{264} & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 \\ 0 & \frac{5\sqrt{11}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{264} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & \frac{5\sqrt{154}}{616} \\ 0 & 0 & 0 & \frac{\sqrt{110}}{264} & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{77}}{56} & 0 & 0 & 0 & -\frac{\sqrt{385}}{1848} & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x420}} \quad \mathbb{Q}_2^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 \\ 0 & 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{196} & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 \\ 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x421}} \quad \mathbb{Q}_4^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{294} & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{98} & 0 & 0 & 0 & \frac{\sqrt{210}i}{294} & 0 & 0 & \frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & \frac{17\sqrt{21}i}{588} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{294} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{98} & 0 & 0 & \frac{17\sqrt{21}i}{588} & 0 & 0 & 0 & \frac{9\sqrt{7}i}{196} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{21}i}{588} & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 & 0 \\ \frac{5\sqrt{14}i}{98} & 0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & -\frac{\sqrt{21}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & \frac{5\sqrt{14}i}{98} & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & \frac{\sqrt{21}i}{147} & 0 \\ 0 & 0 & -\frac{17\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} \\ 0 & 0 & 0 & -\frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 \end{bmatrix}$$

x422

x423

$$\boxed{\text{x424}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{49} & 0 & \frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & \frac{11\sqrt{42}i}{392} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{49} & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}i}{392} & 0 & -\frac{\sqrt{210}i}{392} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}i}{392} & 0 & -\frac{5\sqrt{6}i}{56} \\ -\frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{14}i}{392} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{49} & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{42}i}{392} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{11\sqrt{42}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & \frac{2\sqrt{21}i}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{49} & 0 & \frac{3i}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x425}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{11\sqrt{42}}{392} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{49} & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}}{392} & 0 & \frac{\sqrt{210}}{392} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{392} & 0 & \frac{5\sqrt{6}}{56} \\ \frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{14}}{392} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{42}}{392} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{11\sqrt{42}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & -\frac{2\sqrt{21}}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & -\frac{3}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x426}} \quad \mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{42} & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{5\sqrt{30}i}{168} & 0 & \frac{5\sqrt{2}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{42} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & \frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{3}i}{336} & 0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{i}{112} & 0 \\ -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{3}i}{336} & 0 & -\frac{3\sqrt{7}i}{112} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{13\sqrt{2}i}{112} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{112} & 0 & -\frac{5\sqrt{30}i}{168} & 0 & -\frac{\sqrt{70}i}{112} \\ -\frac{\sqrt{70}i}{112} & 0 & -\frac{3\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{2}i}{112} & 0 & -\frac{i}{112} & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 \\ -\frac{5\sqrt{30}i}{168} & 0 & -\frac{\sqrt{3}i}{336} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{5\sqrt{2}i}{112} & \frac{\sqrt{21}i}{42} & 0 & -\frac{3i}{14} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 \\ -\frac{5\sqrt{2}i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{3i}{14} & 0 & \frac{\sqrt{21}i}{42} \\ 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{3}i}{336} & 0 & \frac{5\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & -\frac{\sqrt{5}i}{14} & 0 \\ 0 & 0 & \frac{i}{112} & 0 & -\frac{13\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{105}i}{42} \\ 0 & 0 & 0 & \frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{105}i}{42} & 0 \end{bmatrix}$$

$$\boxed{\text{x427}} \quad \mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{5\sqrt{30}}{168} & 0 & \frac{5\sqrt{2}}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & -\frac{3\sqrt{7}}{112} & 0 & \frac{\sqrt{3}}{336} & 0 & \frac{\sqrt{5}}{16} & 0 & -\frac{1}{112} & 0 \\ \frac{\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{1}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{3}}{336} & 0 & -\frac{3\sqrt{7}}{112} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{13\sqrt{2}}{112} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{112} & 0 & -\frac{5\sqrt{30}}{168} & 0 & \frac{\sqrt{70}}{112} \\ \frac{\sqrt{70}}{112} & 0 & -\frac{3\sqrt{7}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{2}}{112} & 0 & -\frac{1}{112} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 \\ -\frac{5\sqrt{30}}{168} & 0 & \frac{\sqrt{3}}{336} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{5\sqrt{2}}{112} & -\frac{\sqrt{21}}{42} & 0 & -\frac{3}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 \\ \frac{5\sqrt{2}}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{3}{14} & 0 & \frac{\sqrt{21}}{42} \\ 0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{3}}{336} & 0 & -\frac{5\sqrt{30}}{168} & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & \frac{\sqrt{5}}{14} & 0 \\ 0 & 0 & -\frac{1}{112} & 0 & -\frac{13\sqrt{2}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{42} \\ 0 & 0 & 0 & -\frac{3\sqrt{7}}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{105}}{42} & 0 \end{bmatrix}$$

$$\boxed{\text{x428}} \quad \mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{\sqrt{21}i}{294} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{5\sqrt{210}i}{1176} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{294} & 0 & -\frac{\sqrt{210}i}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}i}{784} & 0 & -\frac{\sqrt{70}i}{392} & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{588} & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & -\frac{3i}{16} & 0 & \frac{\sqrt{21}i}{2352} & 0 & -\frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 \\ \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{588} & 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{21}i}{2352} & 0 & \frac{3i}{16} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{588} & 0 & -\frac{\sqrt{21}i}{294} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{70}i}{392} & 0 & \frac{13\sqrt{14}i}{784} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{21}i}{294} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{210}i}{1176} & 0 & -\frac{\sqrt{10}i}{112} \\ -\frac{\sqrt{10}i}{112} & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{14}i}{784} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{35}i}{98} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 \\ -\frac{5\sqrt{210}i}{1176} & 0 & -\frac{\sqrt{21}i}{2352} & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & \frac{3\sqrt{7}i}{98} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{392} & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{5\sqrt{14}i}{112} & -\frac{\sqrt{3}i}{6} & 0 & -\frac{3\sqrt{7}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 \\ \frac{5\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{70}i}{392} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{98} & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{21}i}{2352} & 0 & \frac{5\sqrt{210}i}{1176} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{98} & 0 & -\frac{\sqrt{35}i}{98} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & -\frac{13\sqrt{14}i}{784} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & \frac{\sqrt{35}i}{98} & 0 & \frac{\sqrt{15}i}{42} \\ 0 & 0 & 0 & -\frac{3i}{16} & 0 & \frac{\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{15}i}{42} & 0 \end{bmatrix}$$

$$\boxed{\text{x429}} \quad \mathbb{Q}_{4,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{21}}{294} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{5\sqrt{210}}{1176} & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{294} & 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{784} & 0 & \frac{\sqrt{70}}{392} & 0 & -\frac{\sqrt{42}}{48} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & \frac{3}{16} & 0 & \frac{\sqrt{21}}{2352} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{7}}{112} & 0 \\ -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}}{588} & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{21}}{2352} & 0 & \frac{3}{16} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{588} & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{70}}{392} & 0 & -\frac{13\sqrt{14}}{784} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & -\frac{5\sqrt{210}}{1176} & 0 & \frac{\sqrt{10}}{112} \\ \frac{\sqrt{10}}{112} & 0 & \frac{3}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{14}}{784} & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{98} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 \\ -\frac{5\sqrt{210}}{1176} & 0 & \frac{\sqrt{21}}{2352} & 0 & -\frac{\sqrt{42}}{48} & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{392} & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{5\sqrt{14}}{112} & \frac{\sqrt{3}}{6} & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 \\ -\frac{5\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{70}}{392} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & -\frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{21}}{2352} & 0 & -\frac{5\sqrt{210}}{1176} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & \frac{\sqrt{35}}{98} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{112} & 0 & -\frac{13\sqrt{14}}{784} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{35}}{98} & 0 & -\frac{\sqrt{15}}{42} \\ 0 & 0 & 0 & \frac{3}{16} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{15}}{42} & 0 \end{bmatrix}$$

$$\boxed{\text{x430}} \quad \mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{7392} & 0 & \frac{\sqrt{22}i}{352} & 0 & -\frac{\sqrt{330}i}{352} & 0 & -\frac{\sqrt{66}i}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{1056} & 0 & -\frac{5\sqrt{66}i}{1056} & 0 & \frac{3\sqrt{110}i}{352} & 0 & \frac{\sqrt{2310}i}{672} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{1232} & 0 & \frac{\sqrt{55}i}{176} & 0 & \frac{5\sqrt{33}i}{528} & 0 & -\frac{\sqrt{165}i}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{5\sqrt{33}i}{528} & 0 & -\frac{\sqrt{55}i}{176} & 0 & \frac{\sqrt{1155}i}{1232} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{672} & 0 & -\frac{3\sqrt{110}i}{352} & 0 & \frac{5\sqrt{66}i}{1056} & 0 & \frac{\sqrt{330}i}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{96} & 0 & \frac{\sqrt{330}i}{352} & 0 & -\frac{\sqrt{22}i}{352} & 0 & -\frac{\sqrt{462}i}{7392} \\ -\frac{\sqrt{462}i}{7392} & 0 & \frac{\sqrt{1155}i}{1232} & 0 & \frac{\sqrt{2310}i}{672} & 0 & 0 & -\frac{3\sqrt{77}i}{616} & 0 & \frac{3\sqrt{385}i}{308} & 0 & \frac{\sqrt{231}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{330}i}{1056} & 0 & -\frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{66}i}{96} & \frac{3\sqrt{77}i}{616} & 0 & \frac{\sqrt{33}i}{44} & 0 & -\frac{3\sqrt{55}i}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{22}i}{352} & 0 & -\frac{\sqrt{55}i}{176} & 0 & \frac{3\sqrt{110}i}{352} & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & -\frac{\sqrt{165}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{56} \\ 0 & \frac{5\sqrt{66}i}{1056} & 0 & \frac{5\sqrt{33}i}{528} & 0 & -\frac{\sqrt{330}i}{352} & -\frac{3\sqrt{385}i}{308} & 0 & \frac{\sqrt{165}i}{88} & 0 & 0 & 0 & \frac{3\sqrt{55}i}{88} & 0 \\ \frac{\sqrt{330}i}{352} & 0 & -\frac{5\sqrt{33}i}{528} & 0 & -\frac{5\sqrt{66}i}{1056} & 0 & 0 & \frac{3\sqrt{55}i}{88} & 0 & 0 & 0 & \frac{\sqrt{165}i}{88} & 0 & -\frac{3\sqrt{385}i}{308} \\ 0 & -\frac{3\sqrt{110}i}{352} & 0 & \frac{\sqrt{55}i}{176} & 0 & \frac{\sqrt{22}i}{352} & -\frac{\sqrt{231}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{33}i}{44} & 0 \\ \frac{\sqrt{66}i}{96} & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{330}i}{1056} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{88} & 0 & \frac{\sqrt{33}i}{44} & 0 & \frac{3\sqrt{77}i}{616} \\ 0 & -\frac{\sqrt{2310}i}{672} & 0 & -\frac{\sqrt{1155}i}{1232} & 0 & \frac{\sqrt{462}i}{7392} & 0 & 0 & \frac{\sqrt{231}i}{56} & 0 & \frac{3\sqrt{385}i}{308} & 0 & -\frac{3\sqrt{77}i}{616} & 0 \end{bmatrix}$$

$$\boxed{\text{x431}} \quad \mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{7392} & 0 & -\frac{\sqrt{22}}{352} & 0 & -\frac{\sqrt{330}}{352} & 0 & \frac{\sqrt{66}}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{1056} & 0 & \frac{5\sqrt{66}}{1056} & 0 & \frac{3\sqrt{110}}{352} & 0 & -\frac{\sqrt{2310}}{672} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{1232} & 0 & \frac{\sqrt{55}}{176} & 0 & -\frac{5\sqrt{33}}{528} & 0 & -\frac{\sqrt{165}}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{176} & 0 & -\frac{5\sqrt{33}}{528} & 0 & \frac{\sqrt{55}}{176} & 0 & \frac{\sqrt{1155}}{1232} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{672} & 0 & \frac{3\sqrt{110}}{352} & 0 & \frac{5\sqrt{66}}{1056} & 0 & -\frac{\sqrt{330}}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{96} & 0 & -\frac{\sqrt{330}}{352} & 0 & -\frac{\sqrt{22}}{352} & 0 & \frac{\sqrt{462}}{7392} \\ \frac{\sqrt{462}}{7392} & 0 & \frac{\sqrt{1155}}{1232} & 0 & -\frac{\sqrt{2310}}{672} & 0 & 0 & \frac{3\sqrt{77}}{616} & 0 & \frac{3\sqrt{385}}{308} & 0 & -\frac{\sqrt{231}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{1056} & 0 & -\frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{66}}{96} & \frac{3\sqrt{77}}{616} & 0 & -\frac{\sqrt{33}}{44} & 0 & -\frac{3\sqrt{55}}{88} & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{352} & 0 & \frac{\sqrt{55}}{176} & 0 & \frac{3\sqrt{110}}{352} & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & \frac{\sqrt{231}}{56} \\ 0 & \frac{5\sqrt{66}}{1056} & 0 & -\frac{5\sqrt{33}}{528} & 0 & -\frac{\sqrt{330}}{352} & \frac{3\sqrt{385}}{308} & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & \frac{3\sqrt{55}}{88} & 0 \\ -\frac{\sqrt{330}}{352} & 0 & -\frac{5\sqrt{33}}{528} & 0 & \frac{5\sqrt{66}}{1056} & 0 & 0 & -\frac{3\sqrt{55}}{88} & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & -\frac{3\sqrt{385}}{308} \\ 0 & \frac{3\sqrt{110}}{352} & 0 & \frac{\sqrt{55}}{176} & 0 & -\frac{\sqrt{22}}{352} & -\frac{\sqrt{231}}{56} & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{33}}{44} & 0 \\ \frac{\sqrt{66}}{96} & 0 & -\frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{88} & 0 & \frac{\sqrt{33}}{44} & 0 & -\frac{3\sqrt{77}}{616} \\ 0 & -\frac{\sqrt{2310}}{672} & 0 & \frac{\sqrt{1155}}{1232} & 0 & \frac{\sqrt{462}}{7392} & 0 & 0 & \frac{\sqrt{231}}{56} & 0 & -\frac{3\sqrt{385}}{308} & 0 & -\frac{3\sqrt{77}}{616} & 0 \end{bmatrix}$$

$$\boxed{\text{x432}} \quad \mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{448} & 0 & \frac{\sqrt{3}i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 & \frac{i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{64} & 0 & -\frac{5i}{64} & 0 & -\frac{\sqrt{15}i}{64} & 0 & -\frac{\sqrt{35}i}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{448} & 0 & \frac{\sqrt{30}i}{64} & 0 & \frac{5\sqrt{2}i}{64} & 0 & \frac{\sqrt{10}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{64} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & -\frac{\sqrt{30}i}{64} & 0 & -\frac{\sqrt{70}i}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{448} & 0 & \frac{\sqrt{15}i}{64} & 0 & \frac{5i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & -\frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{3}i}{64} & 0 & -\frac{\sqrt{7}i}{448} \\ -\frac{\sqrt{7}i}{448} & 0 & -\frac{\sqrt{70}i}{448} & 0 & -\frac{\sqrt{35}i}{448} & 0 & 0 & -\frac{3\sqrt{42}i}{224} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{14}i}{224} & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{64} & 0 & \frac{\sqrt{10}i}{64} & 0 & \frac{i}{64} & \frac{3\sqrt{42}i}{224} & 0 & \frac{3\sqrt{2}i}{16} & 0 & \frac{\sqrt{30}i}{32} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{64} & 0 & -\frac{\sqrt{30}i}{64} & 0 & -\frac{\sqrt{15}i}{64} & 0 & 0 & -\frac{3\sqrt{2}i}{16} & 0 & -\frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{224} \\ 0 & \frac{5i}{64} & 0 & \frac{5\sqrt{2}i}{64} & 0 & \frac{\sqrt{5}i}{64} & \frac{\sqrt{210}i}{112} & 0 & \frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{32} & 0 \\ -\frac{\sqrt{5}i}{64} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & -\frac{5i}{64} & 0 & 0 & -\frac{\sqrt{30}i}{32} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{32} & 0 & \frac{\sqrt{210}i}{112} \\ 0 & \frac{\sqrt{15}i}{64} & 0 & \frac{\sqrt{30}i}{64} & 0 & \frac{\sqrt{3}i}{64} & \frac{3\sqrt{14}i}{224} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{32} & 0 & -\frac{3\sqrt{2}i}{16} & 0 \\ -\frac{i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 & -\frac{\sqrt{5}i}{64} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{32} & 0 & \frac{3\sqrt{2}i}{16} & 0 & \frac{3\sqrt{42}i}{224} \\ 0 & \frac{\sqrt{35}i}{448} & 0 & \frac{\sqrt{70}i}{448} & 0 & \frac{\sqrt{7}i}{448} & 0 & 0 & -\frac{3\sqrt{14}i}{224} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{42}i}{224} & 0 \end{bmatrix}$$

$$\boxed{\text{x433}} \quad \mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{448} & 0 & -\frac{\sqrt{3}}{64} & 0 & \frac{\sqrt{5}}{64} & 0 & -\frac{1}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{64} & 0 & \frac{5}{64} & 0 & -\frac{\sqrt{15}}{64} & 0 & \frac{\sqrt{35}}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{448} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{10}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{70}}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{448} & 0 & -\frac{\sqrt{15}}{64} & 0 & \frac{5}{64} & 0 & -\frac{\sqrt{5}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{64} & 0 & \frac{\sqrt{5}}{64} & 0 & -\frac{\sqrt{3}}{64} & 0 & \frac{\sqrt{7}}{448} \\ \frac{\sqrt{7}}{448} & 0 & -\frac{\sqrt{70}}{448} & 0 & \frac{\sqrt{35}}{448} & 0 & 0 & \frac{3\sqrt{42}}{224} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{3\sqrt{14}}{224} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{64} & 0 & \frac{\sqrt{10}}{64} & 0 & -\frac{1}{64} & \frac{3\sqrt{42}}{224} & 0 & -\frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{30}}{32} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{15}}{64} & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{224} \\ 0 & \frac{5}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{5}}{64} & -\frac{\sqrt{210}}{112} & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & -\frac{\sqrt{30}}{32} & 0 \\ \frac{\sqrt{5}}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{5}{64} & 0 & 0 & \frac{\sqrt{30}}{32} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & \frac{\sqrt{210}}{112} \\ 0 & -\frac{\sqrt{15}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{3}}{64} & \frac{3\sqrt{14}}{224} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & \frac{3\sqrt{2}}{16} & 0 \\ -\frac{1}{64} & 0 & \frac{\sqrt{10}}{64} & 0 & -\frac{\sqrt{5}}{64} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{32} & 0 & \frac{3\sqrt{2}}{16} & 0 & -\frac{3\sqrt{42}}{224} \\ 0 & \frac{\sqrt{35}}{448} & 0 & -\frac{\sqrt{70}}{448} & 0 & \frac{\sqrt{7}}{448} & 0 & 0 & -\frac{3\sqrt{14}}{224} & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{3\sqrt{42}}{224} & 0 \end{bmatrix}$$

$$\boxed{x434} \quad \mathbb{Q}_{6,1}^{(1,-1;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{14784} & 0 & \frac{\sqrt{165}i}{2112} & 0 & -\frac{9\sqrt{11}i}{704} & 0 & \frac{\sqrt{55}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{2112} & 0 & -\frac{5\sqrt{55}i}{2112} & 0 & \frac{9\sqrt{33}i}{704} & 0 & -\frac{5\sqrt{77}i}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}i}{4928} & 0 & \frac{5\sqrt{66}i}{2112} & 0 & \frac{5\sqrt{110}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{110}i}{2112} & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & \frac{9\sqrt{154}i}{4928} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{448} & 0 & -\frac{9\sqrt{33}i}{704} & 0 & \frac{5\sqrt{55}i}{2112} & 0 & \frac{5\sqrt{11}i}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{64} & 0 & \frac{9\sqrt{11}i}{704} & 0 & -\frac{\sqrt{165}i}{2112} & 0 & -\frac{\sqrt{385}i}{14784} \\ -\frac{\sqrt{385}i}{14784} & 0 & \frac{9\sqrt{154}i}{4928} & 0 & -\frac{5\sqrt{77}i}{448} & 0 & 0 & -\frac{\sqrt{2310}i}{2464} & 0 & \frac{9\sqrt{462}i}{1232} & 0 & -\frac{3\sqrt{770}i}{224} & 0 & 0 \\ 0 & \frac{5\sqrt{11}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} & 0 & \frac{\sqrt{55}i}{64} & \frac{\sqrt{2310}i}{2464} & 0 & \frac{\sqrt{110}i}{176} & 0 & -\frac{9\sqrt{66}i}{352} & 0 & 0 & 0 \\ -\frac{\sqrt{165}i}{2112} & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & \frac{9\sqrt{33}i}{704} & 0 & 0 & -\frac{\sqrt{110}i}{176} & 0 & -\frac{5\sqrt{22}i}{352} & 0 & 0 & 0 & \frac{3\sqrt{770}i}{224} \\ 0 & \frac{5\sqrt{55}i}{2112} & 0 & \frac{5\sqrt{110}i}{2112} & 0 & -\frac{9\sqrt{11}i}{704} & -\frac{9\sqrt{462}i}{1232} & 0 & \frac{5\sqrt{22}i}{352} & 0 & 0 & 0 & \frac{9\sqrt{66}i}{352} & 0 \\ \frac{9\sqrt{11}i}{704} & 0 & -\frac{5\sqrt{110}i}{2112} & 0 & -\frac{5\sqrt{55}i}{2112} & 0 & 0 & \frac{9\sqrt{66}i}{352} & 0 & 0 & 0 & \frac{5\sqrt{22}i}{352} & 0 & -\frac{9\sqrt{462}i}{1232} \\ 0 & -\frac{9\sqrt{33}i}{704} & 0 & \frac{5\sqrt{66}i}{2112} & 0 & \frac{\sqrt{165}i}{2112} & \frac{3\sqrt{770}i}{224} & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{352} & 0 & -\frac{\sqrt{110}i}{176} & 0 \\ -\frac{\sqrt{55}i}{64} & 0 & \frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{11}i}{2112} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{66}i}{352} & 0 & \frac{\sqrt{110}i}{176} & 0 & \frac{\sqrt{2310}i}{2464} \\ 0 & \frac{5\sqrt{77}i}{448} & 0 & -\frac{9\sqrt{154}i}{4928} & 0 & \frac{\sqrt{385}i}{14784} & 0 & 0 & -\frac{3\sqrt{770}i}{224} & 0 & \frac{9\sqrt{462}i}{1232} & 0 & -\frac{\sqrt{2310}i}{2464} & 0 \end{bmatrix}$$

$$\boxed{x435} \quad \mathbb{Q}_{6,2}^{(1,-1;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{14784} & 0 & -\frac{\sqrt{165}}{2112} & 0 & -\frac{9\sqrt{11}}{704} & 0 & -\frac{\sqrt{55}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{2112} & 0 & \frac{5\sqrt{55}}{2112} & 0 & \frac{9\sqrt{33}}{704} & 0 & \frac{5\sqrt{77}}{448} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{4928} & 0 & \frac{5\sqrt{66}}{2112} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & -\frac{9\sqrt{22}}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{704} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & \frac{5\sqrt{66}}{2112} & 0 & \frac{9\sqrt{154}}{4928} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{448} & 0 & \frac{9\sqrt{33}}{704} & 0 & \frac{5\sqrt{55}}{2112} & 0 & -\frac{5\sqrt{11}}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{64} & 0 & -\frac{9\sqrt{11}}{704} & 0 & -\frac{\sqrt{165}}{2112} & 0 & \frac{\sqrt{385}}{14784} \\ \frac{\sqrt{385}}{14784} & 0 & \frac{9\sqrt{154}}{4928} & 0 & \frac{5\sqrt{77}}{448} & 0 & 0 & \frac{\sqrt{2310}}{2464} & 0 & \frac{9\sqrt{462}}{1232} & 0 & \frac{3\sqrt{770}}{224} & 0 & 0 \\ 0 & -\frac{5\sqrt{11}}{2112} & 0 & -\frac{9\sqrt{22}}{704} & 0 & -\frac{\sqrt{55}}{64} & \frac{\sqrt{2310}}{2464} & 0 & -\frac{\sqrt{110}}{176} & 0 & -\frac{9\sqrt{66}}{352} & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{2112} & 0 & \frac{5\sqrt{66}}{2112} & 0 & \frac{9\sqrt{33}}{704} & 0 & 0 & -\frac{\sqrt{110}}{176} & 0 & \frac{5\sqrt{22}}{352} & 0 & 0 & 0 & -\frac{3\sqrt{770}}{224} \\ 0 & \frac{5\sqrt{55}}{2112} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & -\frac{9\sqrt{11}}{704} & \frac{9\sqrt{462}}{1232} & 0 & \frac{5\sqrt{22}}{352} & 0 & 0 & 0 & \frac{9\sqrt{66}}{352} & 0 \\ -\frac{9\sqrt{11}}{704} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & \frac{5\sqrt{55}}{2112} & 0 & 0 & -\frac{9\sqrt{66}}{352} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{352} & 0 & -\frac{9\sqrt{462}}{1232} \\ 0 & \frac{9\sqrt{33}}{704} & 0 & \frac{5\sqrt{66}}{2112} & 0 & -\frac{\sqrt{165}}{2112} & \frac{3\sqrt{770}}{224} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{352} & 0 & \frac{\sqrt{110}}{176} & 0 \\ -\frac{\sqrt{55}}{64} & 0 & -\frac{9\sqrt{22}}{704} & 0 & -\frac{5\sqrt{11}}{2112} & 0 & 0 & 0 & 0 & \frac{9\sqrt{66}}{352} & 0 & \frac{\sqrt{110}}{176} & 0 & -\frac{\sqrt{2310}}{2464} \\ 0 & \frac{5\sqrt{77}}{448} & 0 & \frac{9\sqrt{154}}{4928} & 0 & \frac{\sqrt{385}}{14784} & 0 & 0 & -\frac{3\sqrt{770}}{224} & 0 & -\frac{9\sqrt{462}}{1232} & 0 & -\frac{\sqrt{2310}}{2464} & 0 \end{bmatrix}$$

x436

x437

$$\boxed{\text{x438}} \quad \mathbb{Q}_4^{(1,1;a)}(A_{1g}, 1) = \begin{bmatrix} -\frac{\sqrt{110}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{84} & 0 & 0 & \frac{2\sqrt{165}}{231} & 0 & 0 & 0 & \frac{2\sqrt{55}}{231} & 0 & 0 \\ 0 & \frac{\sqrt{110}}{28} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{84} & 0 & 0 & -\frac{8\sqrt{11}}{231} & 0 & 0 & 0 & \frac{4\sqrt{33}}{231} & 0 \\ 0 & 0 & -\frac{\sqrt{110}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{231} & 0 & 0 & 0 & \frac{\sqrt{462}}{231} \\ 0 & 0 & 0 & -\frac{\sqrt{110}}{42} & 0 & 0 & -\frac{\sqrt{462}}{231} & 0 & 0 & 0 & \frac{\sqrt{330}}{231} & 0 & 0 & 0 \\ -\frac{5\sqrt{22}}{84} & 0 & 0 & 0 & \frac{\sqrt{110}}{28} & 0 & 0 & -\frac{4\sqrt{33}}{231} & 0 & 0 & 0 & \frac{8\sqrt{11}}{231} & 0 & 0 \\ 0 & -\frac{5\sqrt{22}}{84} & 0 & 0 & 0 & -\frac{\sqrt{110}}{84} & 0 & 0 & -\frac{2\sqrt{55}}{231} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{231} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{231} & 0 & 0 & \frac{\sqrt{110}}{264} & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 \\ \frac{2\sqrt{165}}{231} & 0 & 0 & 0 & -\frac{4\sqrt{33}}{231} & 0 & 0 & -\frac{13\sqrt{110}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 & 0 \\ 0 & -\frac{8\sqrt{11}}{231} & 0 & 0 & 0 & -\frac{2\sqrt{55}}{231} & 0 & 0 & -\frac{\sqrt{110}}{616} & 0 & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} \\ 0 & 0 & 0 & \frac{\sqrt{330}}{231} & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 \\ \frac{2\sqrt{55}}{231} & 0 & 0 & 0 & \frac{8\sqrt{11}}{231} & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{110}}{616} & 0 & 0 \\ 0 & \frac{4\sqrt{33}}{231} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{231} & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 & 0 & 0 & -\frac{13\sqrt{110}}{1848} & 0 \\ 0 & 0 & \frac{\sqrt{462}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & \frac{\sqrt{110}}{264} \end{bmatrix}$$

$$\boxed{\text{x439}} \quad \mathbb{Q}_4^{(1,1;a)}(A_{1g}, 2) = \begin{bmatrix} -\frac{5\sqrt{154}}{588} & 0 & 0 & 0 & \frac{\sqrt{770}}{84} & 0 & 0 & \frac{10\sqrt{231}}{1617} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{231} & 0 & 0 \\ 0 & \frac{5\sqrt{154}}{196} & 0 & 0 & 0 & \frac{\sqrt{770}}{84} & 0 & 0 & -\frac{8\sqrt{385}}{1617} & 0 & 0 & 0 & -\frac{4\sqrt{1155}}{1155} & 0 \\ 0 & 0 & -\frac{5\sqrt{154}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1617} & 0 & 0 & 0 & -\frac{\sqrt{330}}{165} \\ 0 & 0 & 0 & -\frac{5\sqrt{154}}{294} & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & \frac{5\sqrt{462}}{1617} & 0 & 0 & 0 \\ \frac{\sqrt{770}}{84} & 0 & 0 & 0 & \frac{5\sqrt{154}}{196} & 0 & 0 & \frac{4\sqrt{1155}}{1155} & 0 & 0 & 0 & \frac{8\sqrt{385}}{1617} & 0 & 0 \\ 0 & \frac{\sqrt{770}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{588} & 0 & 0 & \frac{2\sqrt{77}}{231} & 0 & 0 & 0 & -\frac{10\sqrt{231}}{1617} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 \\ \frac{10\sqrt{231}}{1617} & 0 & 0 & 0 & \frac{4\sqrt{1155}}{1155} & 0 & 0 & -\frac{65\sqrt{154}}{12936} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 \\ 0 & -\frac{8\sqrt{385}}{1617} & 0 & 0 & 0 & \frac{2\sqrt{77}}{231} & 0 & 0 & -\frac{5\sqrt{154}}{4312} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 \\ 0 & 0 & -\frac{5\sqrt{462}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{154}}{4312} & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} \\ 0 & 0 & 0 & \frac{5\sqrt{462}}{1617} & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & \frac{15\sqrt{154}}{4312} & 0 & 0 & 0 \\ -\frac{2\sqrt{77}}{231} & 0 & 0 & 0 & \frac{8\sqrt{385}}{1617} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{4312} & 0 & 0 \\ 0 & -\frac{4\sqrt{1155}}{1155} & 0 & 0 & 0 & -\frac{10\sqrt{231}}{1617} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & -\frac{65\sqrt{154}}{12936} & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} \end{bmatrix}$$

$$\boxed{\text{x440}} \quad \mathbb{Q}_4^{(1,1;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}i}{231} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}i}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x441}} \quad \mathbb{Q}_2^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & -\frac{4\sqrt{14}}{147} & 0 & 0 & 0 \\ \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 \\ 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 \\ 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{147} & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} \\ 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & \frac{4\sqrt{14}}{147} & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 \\ 0 & -\frac{4\sqrt{14}}{147} & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & -\frac{1}{14} \\ 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x442}} \quad \mathbb{Q}_4^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & \frac{6\sqrt{55}}{385} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 \\ \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & -\frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & \frac{17\sqrt{2310}}{8085} & 0 & 0 \\ 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & -\frac{17\sqrt{2310}}{8085} & 0 & 0 & 0 & \frac{9\sqrt{770}}{2695} & 0 \\ 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{6\sqrt{55}}{385} \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & 0 \\ 0 & \frac{6\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{2310}}{8085} & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 \\ -\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 \\ 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 \\ 0 & 0 & \frac{17\sqrt{2310}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} \\ 0 & 0 & 0 & \frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{6\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x443}} \quad \mathbb{Q}_2^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{42}i}{196} & 0 & 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & \frac{4\sqrt{14}i}{147} & 0 & 0 & 0 \\ \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{9\sqrt{42}i}{196} & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 \\ 0 & \frac{9\sqrt{42}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 \\ 0 & 0 & \frac{9\sqrt{42}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}i}{147} & 0 & 0 & 0 & \frac{\sqrt{10}i}{21} \\ 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{14}i}{147} & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 \\ 0 & -\frac{4\sqrt{14}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 \\ 0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & \frac{i}{14} \\ 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{x444} \quad \mathbb{Q}_4^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & -\frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 \\ -\frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & \frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & \frac{17\sqrt{2310}i}{8085} & 0 & 0 \\ 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{98} & 0 & 0 & \frac{17\sqrt{2310}i}{8085} & 0 & 0 & 0 & \frac{9\sqrt{770}i}{2695} & 0 \\ 0 & 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & -\frac{6\sqrt{55}i}{385} \\ 0 & 0 & 0 & -\frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{539} & 0 & 0 & 0 \\ 0 & \frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{2310}i}{8085} & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 & 0 \\ \frac{4\sqrt{385}i}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & \frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 \\ 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{4\sqrt{385}i}{539} & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{6468} & 0 \\ 0 & 0 & -\frac{17\sqrt{2310}i}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} \\ 0 & 0 & 0 & -\frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 \end{bmatrix}$$

$$\boxed{x445} \quad \mathbb{Q}_{2,1}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & \frac{5\sqrt{42}i}{294} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{105}i}{98} & 0 & -\frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{294} & 0 & \frac{11\sqrt{14}i}{294} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}i}{294} & 0 & -\frac{\sqrt{70}i}{294} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{294} & 0 & -\frac{5\sqrt{2}i}{42} \\ -\frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}i}{294} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{49} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{14}i}{294} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{11\sqrt{14}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{5\sqrt{42}i}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{\sqrt{3}i}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 \end{bmatrix}$$

$$\boxed{x446} \quad \mathbb{Q}_{2,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{105}}{98} & 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{294} & 0 & -\frac{11\sqrt{14}}{294} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{42}}{98} & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}}{294} & 0 & \frac{\sqrt{70}}{294} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{294} & 0 & \frac{5\sqrt{2}}{42} \\ \frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}}{294} & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{14}}{294} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & -\frac{11\sqrt{14}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & \frac{2\sqrt{7}}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & \frac{\sqrt{3}}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 \end{bmatrix}$$

$$\boxed{x447} \quad \mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{330}i}{84} & 0 & -\frac{\sqrt{165}i}{84} & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 & \frac{5\sqrt{33}i}{231} & 0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 \\ \frac{\sqrt{330}i}{84} & 0 & \frac{5\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{55}i}{770} & 0 & -\frac{\sqrt{11}i}{77} & 0 & \frac{\sqrt{165}i}{330} & 0 & 0 \\ 0 & -\frac{5\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{84} & \frac{3\sqrt{770}i}{1540} & 0 & \frac{\sqrt{330}i}{4620} & 0 & -\frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{110}i}{1540} & 0 \\ \frac{\sqrt{165}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{84} & 0 & 0 & \frac{\sqrt{110}i}{1540} & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{330}i}{4620} & 0 & -\frac{3\sqrt{770}i}{1540} \\ 0 & 0 & 0 & \frac{5\sqrt{33}i}{84} & 0 & \frac{\sqrt{330}i}{84} & 0 & 0 & -\frac{\sqrt{165}i}{330} & 0 & \frac{\sqrt{11}i}{77} & 0 & \frac{13\sqrt{55}i}{770} & 0 \\ 0 & 0 & -\frac{\sqrt{165}i}{84} & 0 & -\frac{\sqrt{330}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{154} & 0 & -\frac{5\sqrt{33}i}{231} & 0 & -\frac{\sqrt{77}i}{154} \\ -\frac{\sqrt{77}i}{154} & 0 & -\frac{3\sqrt{770}i}{1540} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{1848} & 0 & \frac{\sqrt{2310}i}{1848} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{55}i}{770} & 0 & -\frac{\sqrt{110}i}{1540} & 0 & 0 & -\frac{5\sqrt{462}i}{1848} & 0 & -\frac{5\sqrt{22}i}{616} & 0 & \frac{\sqrt{330}i}{924} & 0 & 0 & 0 \\ -\frac{5\sqrt{33}i}{231} & 0 & -\frac{\sqrt{330}i}{4620} & 0 & \frac{\sqrt{165}i}{330} & 0 & 0 & \frac{5\sqrt{22}i}{616} & 0 & -\frac{3\sqrt{110}i}{616} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}i}{77} & 0 & -\frac{\sqrt{22}i}{44} & 0 & \frac{\sqrt{55}i}{154} & -\frac{\sqrt{2310}i}{1848} & 0 & \frac{3\sqrt{110}i}{616} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 \\ -\frac{\sqrt{55}i}{154} & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{11}i}{77} & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & \frac{3\sqrt{110}i}{616} & 0 & -\frac{\sqrt{2310}i}{1848} \\ 0 & -\frac{\sqrt{165}i}{330} & 0 & \frac{\sqrt{330}i}{4620} & 0 & \frac{5\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{616} & 0 & \frac{5\sqrt{22}i}{616} & 0 \\ 0 & 0 & \frac{\sqrt{110}i}{1540} & 0 & -\frac{13\sqrt{55}i}{770} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{924} & 0 & -\frac{5\sqrt{22}i}{616} & 0 & -\frac{5\sqrt{462}i}{1848} \\ 0 & 0 & 0 & \frac{3\sqrt{770}i}{1540} & 0 & \frac{\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1848} & 0 & \frac{5\sqrt{462}i}{1848} & 0 \end{bmatrix}$$

$$\boxed{x448} \quad \mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{330}}{84} & 0 & -\frac{\sqrt{165}}{84} & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & -\frac{5\sqrt{33}}{231} & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 \\ \frac{\sqrt{330}}{84} & 0 & -\frac{5\sqrt{33}}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{55}}{770} & 0 & \frac{\sqrt{11}}{77} & 0 & \frac{\sqrt{165}}{330} & 0 & 0 \\ 0 & -\frac{5\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{84} & -\frac{3\sqrt{770}}{1540} & 0 & \frac{\sqrt{330}}{4620} & 0 & \frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{110}}{1540} & 0 \\ -\frac{\sqrt{165}}{84} & 0 & 0 & 0 & \frac{5\sqrt{33}}{84} & 0 & 0 & -\frac{\sqrt{110}}{1540} & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{330}}{4620} & 0 & -\frac{3\sqrt{770}}{1540} \\ 0 & 0 & 0 & \frac{5\sqrt{33}}{84} & 0 & -\frac{\sqrt{330}}{84} & 0 & 0 & \frac{\sqrt{165}}{330} & 0 & \frac{\sqrt{11}}{77} & 0 & -\frac{13\sqrt{55}}{770} & 0 \\ 0 & 0 & \frac{\sqrt{165}}{84} & 0 & -\frac{\sqrt{330}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & -\frac{5\sqrt{33}}{231} & 0 & \frac{\sqrt{77}}{154} \\ \frac{\sqrt{77}}{154} & 0 & -\frac{3\sqrt{770}}{1540} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{55}}{770} & 0 & -\frac{\sqrt{110}}{1540} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & \frac{5\sqrt{22}}{616} & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 \\ -\frac{5\sqrt{33}}{231} & 0 & \frac{\sqrt{330}}{4620} & 0 & \frac{\sqrt{165}}{330} & 0 & 0 & \frac{5\sqrt{22}}{616} & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}}{77} & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{55}}{154} & \frac{\sqrt{2310}}{1848} & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 & -\frac{\sqrt{330}}{924} & 0 \\ \frac{\sqrt{55}}{154} & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{11}}{77} & 0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 & -\frac{3\sqrt{110}}{616} & 0 & -\frac{\sqrt{2310}}{1848} \\ 0 & \frac{\sqrt{165}}{330} & 0 & \frac{\sqrt{330}}{4620} & 0 & -\frac{5\sqrt{33}}{231} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{616} & 0 & -\frac{5\sqrt{22}}{616} & 0 \\ 0 & 0 & -\frac{\sqrt{110}}{1540} & 0 & -\frac{13\sqrt{55}}{770} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{924} & 0 & -\frac{5\sqrt{22}}{616} & 0 & \frac{5\sqrt{462}}{1848} \\ 0 & 0 & 0 & -\frac{3\sqrt{770}}{1540} & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} & 0 & \frac{5\sqrt{462}}{1848} & 0 \end{bmatrix}$$

$$\boxed{x449} \quad \mathbb{Q}_{4,1}^{(1,1;a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{2310}i}{588} & 0 & \frac{\sqrt{1155}i}{84} & 0 & 0 & \frac{\sqrt{11}i}{154} & 0 & \frac{5\sqrt{231}i}{1617} & 0 & -\frac{\sqrt{385}i}{154} & 0 & 0 & 0 \\ \frac{\sqrt{2310}i}{588} & 0 & \frac{5\sqrt{231}i}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{385}i}{5390} & 0 & -\frac{\sqrt{77}i}{539} & 0 & -\frac{\sqrt{1155}i}{330} & 0 & 0 \\ 0 & -\frac{5\sqrt{231}i}{588} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{84} & -\frac{3\sqrt{110}i}{220} & 0 & \frac{\sqrt{2310}i}{32340} & 0 & -\frac{\sqrt{154}i}{308} & 0 & \frac{\sqrt{770}i}{1540} & 0 \\ -\frac{\sqrt{1155}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{588} & 0 & 0 & -\frac{\sqrt{770}i}{1540} & 0 & \frac{\sqrt{154}i}{308} & 0 & -\frac{\sqrt{2310}i}{32340} & 0 & \frac{3\sqrt{110}i}{220} \\ 0 & 0 & 0 & \frac{5\sqrt{231}i}{588} & 0 & \frac{\sqrt{2310}i}{588} & 0 & 0 & \frac{\sqrt{1155}i}{330} & 0 & \frac{\sqrt{77}i}{539} & 0 & \frac{13\sqrt{385}i}{5390} & 0 \\ 0 & 0 & \frac{\sqrt{1155}i}{84} & 0 & -\frac{\sqrt{2310}i}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{154} & 0 & -\frac{5\sqrt{231}i}{1617} & 0 & -\frac{\sqrt{11}i}{154} \\ -\frac{\sqrt{11}i}{154} & 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{1848} & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{385}i}{5390} & 0 & \frac{\sqrt{770}i}{1540} & 0 & 0 & -\frac{5\sqrt{66}i}{1848} & 0 & -\frac{5\sqrt{154}i}{4312} & 0 & -\frac{\sqrt{2310}i}{924} & 0 & 0 & 0 \\ -\frac{5\sqrt{231}i}{1617} & 0 & -\frac{\sqrt{2310}i}{32340} & 0 & -\frac{\sqrt{1155}i}{330} & 0 & 0 & \frac{5\sqrt{154}i}{4312} & 0 & -\frac{3\sqrt{770}i}{4312} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}i}{539} & 0 & -\frac{\sqrt{154}i}{308} & 0 & -\frac{\sqrt{385}i}{154} & \frac{\sqrt{330}i}{264} & 0 & \frac{3\sqrt{770}i}{4312} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 \\ \frac{\sqrt{385}i}{154} & 0 & \frac{\sqrt{154}i}{308} & 0 & -\frac{\sqrt{77}i}{539} & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & \frac{3\sqrt{770}i}{4312} & 0 & \frac{\sqrt{330}i}{264} \\ 0 & \frac{\sqrt{1155}i}{330} & 0 & \frac{\sqrt{2310}i}{32340} & 0 & \frac{5\sqrt{231}i}{1617} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}i}{4312} & 0 & \frac{5\sqrt{154}i}{4312} & 0 \\ 0 & 0 & -\frac{\sqrt{770}i}{1540} & 0 & -\frac{13\sqrt{385}i}{5390} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 & -\frac{5\sqrt{154}i}{4312} & 0 & -\frac{5\sqrt{66}i}{1848} \\ 0 & 0 & 0 & -\frac{3\sqrt{110}i}{220} & 0 & \frac{\sqrt{11}i}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & \frac{5\sqrt{66}i}{1848} & 0 \end{bmatrix}$$

$$\boxed{\text{x450}} \quad \mathbb{Q}_{4,2}^{(1,1;a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{\sqrt{2310}}{588} & 0 & \frac{\sqrt{1155}}{84} & 0 & 0 & \frac{\sqrt{11}}{154} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & 0 \\ \frac{\sqrt{2310}}{588} & 0 & -\frac{5\sqrt{231}}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{385}}{5390} & 0 & \frac{\sqrt{77}}{539} & 0 & -\frac{\sqrt{1155}}{330} & 0 & 0 \\ 0 & -\frac{5\sqrt{231}}{588} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{84} & \frac{3\sqrt{110}}{220} & 0 & \frac{\sqrt{2310}}{32340} & 0 & \frac{\sqrt{154}}{308} & 0 & \frac{\sqrt{770}}{1540} & 0 \\ \frac{\sqrt{1155}}{84} & 0 & 0 & 0 & \frac{5\sqrt{231}}{588} & 0 & 0 & \frac{\sqrt{770}}{1540} & 0 & \frac{\sqrt{154}}{308} & 0 & \frac{\sqrt{2310}}{32340} & 0 & \frac{3\sqrt{110}}{220} \\ 0 & 0 & 0 & \frac{5\sqrt{231}}{588} & 0 & -\frac{\sqrt{2310}}{588} & 0 & 0 & -\frac{\sqrt{1155}}{330} & 0 & \frac{\sqrt{77}}{539} & 0 & -\frac{13\sqrt{385}}{5390} & 0 \\ 0 & 0 & -\frac{\sqrt{1155}}{84} & 0 & -\frac{\sqrt{2310}}{588} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & \frac{\sqrt{11}}{154} \\ \frac{\sqrt{11}}{154} & 0 & \frac{3\sqrt{110}}{220} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{1848} & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{385}}{5390} & 0 & \frac{\sqrt{770}}{1540} & 0 & 0 & -\frac{5\sqrt{66}}{1848} & 0 & \frac{5\sqrt{154}}{4312} & 0 & -\frac{\sqrt{2310}}{924} & 0 & 0 & 0 \\ -\frac{5\sqrt{231}}{1617} & 0 & \frac{\sqrt{2310}}{32340} & 0 & -\frac{\sqrt{1155}}{330} & 0 & 0 & \frac{5\sqrt{154}}{4312} & 0 & \frac{3\sqrt{770}}{4312} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{539} & 0 & \frac{\sqrt{154}}{308} & 0 & -\frac{\sqrt{385}}{154} & -\frac{\sqrt{330}}{264} & 0 & \frac{3\sqrt{770}}{4312} & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 \\ -\frac{\sqrt{385}}{154} & 0 & \frac{\sqrt{154}}{308} & 0 & \frac{\sqrt{77}}{539} & 0 & 0 & -\frac{\sqrt{2310}}{924} & 0 & 0 & 0 & -\frac{3\sqrt{770}}{4312} & 0 & \frac{\sqrt{330}}{264} \\ 0 & -\frac{\sqrt{1155}}{330} & 0 & \frac{\sqrt{2310}}{32340} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{4312} & 0 & -\frac{5\sqrt{154}}{4312} & 0 \\ 0 & 0 & \frac{\sqrt{770}}{1540} & 0 & -\frac{13\sqrt{385}}{5390} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & -\frac{5\sqrt{154}}{4312} & 0 & \frac{5\sqrt{66}}{1848} \\ 0 & 0 & 0 & \frac{3\sqrt{110}}{220} & 0 & \frac{\sqrt{11}}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & \frac{5\sqrt{66}}{1848} & 0 \end{bmatrix}$$

$$\boxed{\text{x451}} \quad \mathbb{G}_5^{(1,0;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

x452

x453

x454

x455

$$\boxed{\text{x456}} \quad \mathbb{G}_3^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{12} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{1}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{14}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x457}} \quad \mathbb{G}_5^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{70}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x458}} \quad \mathbb{G}_3^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x459}} \quad \mathbb{G}_5^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{60} & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & \frac{\sqrt{70}i}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x460}} \quad \mathbb{G}_{1,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{210}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{6}i}{8} \\ -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x461}} \quad \mathbb{G}_{1,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{6}}{8} \\ \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x462}} \quad \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{48} & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{3}i}{48} & 0 & \frac{\sqrt{15}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{3}i}{48} & 0 & \frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{105}i}{48} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{30}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{42}i}{48} \\ -\frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{105}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{3}i}{48} & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{3}i}{48} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{48} & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x463}} \quad \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{48} & 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{3}}{48} & 0 & \frac{\sqrt{15}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{3}}{48} & 0 & -\frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{105}}{48} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{30}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{42}}{48} \\ \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{105}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{3}}{48} & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{48} & 0 & -\frac{\sqrt{3}}{48} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{16} & 0 & -\frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{48} & 0 & \frac{\sqrt{42}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x464}} \quad \mathbb{G}_{3,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{48} & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{16} & 0 & -\frac{5\sqrt{3}i}{48} & 0 & \frac{\sqrt{5}i}{48} & 0 & -\frac{3i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & -\frac{\sqrt{5}i}{48} & 0 & \frac{5\sqrt{3}i}{48} & 0 & -\frac{\sqrt{7}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{24} & 0 & \frac{5\sqrt{2}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{70}i}{48} \\ -\frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{7}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{2}i}{48} & 0 & -\frac{3i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{24} & 0 & \frac{5\sqrt{3}i}{48} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{5}i}{48} & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{48} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{5\sqrt{3}i}{48} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3i}{16} & 0 & -\frac{5\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{16} & 0 & \frac{\sqrt{70}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x465}} \quad \mathbb{G}_{3,2}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{48} & 0 & \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{48} & 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{16} & 0 & -\frac{5\sqrt{3}}{48} & 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{3}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{16} & 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{5\sqrt{3}}{48} & 0 & -\frac{\sqrt{7}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{5\sqrt{2}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{70}}{48} \\ \frac{\sqrt{70}}{48} & 0 & -\frac{\sqrt{7}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{2}}{48} & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{24} & 0 & -\frac{5\sqrt{3}}{48} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{16} & 0 & -\frac{\sqrt{5}}{48} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{5\sqrt{3}}{48} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{16} & 0 & -\frac{5\sqrt{2}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{16} & 0 & \frac{\sqrt{70}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x466}} \quad \mathbb{G}_{5,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{192} & 0 & -\frac{5\sqrt{70}i}{448} & 0 & \frac{5\sqrt{42}i}{192} & 0 & -\frac{\sqrt{210}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}i}{1344} & 0 & \frac{13\sqrt{210}i}{1344} & 0 & -\frac{\sqrt{14}i}{64} & 0 & -\frac{7\sqrt{6}i}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}i}{96} & 0 & \frac{11\sqrt{7}i}{224} & 0 & -\frac{\sqrt{105}i}{672} & 0 & -\frac{\sqrt{21}i}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{32} & 0 & \frac{\sqrt{105}i}{672} & 0 & -\frac{11\sqrt{7}i}{224} & 0 & \frac{7\sqrt{3}i}{96} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}i}{64} & 0 & \frac{\sqrt{14}i}{64} & 0 & -\frac{13\sqrt{210}i}{1344} & 0 & \frac{23\sqrt{42}i}{1344} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{5\sqrt{42}i}{192} & 0 & \frac{5\sqrt{70}i}{448} & 0 & -\frac{\sqrt{30}i}{192} \\ -\frac{\sqrt{30}i}{192} & 0 & \frac{7\sqrt{3}i}{96} & 0 & -\frac{7\sqrt{6}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{23\sqrt{42}i}{1344} & 0 & -\frac{\sqrt{21}i}{32} & 0 & -\frac{\sqrt{210}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{70}i}{448} & 0 & -\frac{11\sqrt{7}i}{224} & 0 & -\frac{\sqrt{14}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{210}i}{1344} & 0 & -\frac{\sqrt{105}i}{672} & 0 & \frac{5\sqrt{42}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}i}{192} & 0 & \frac{\sqrt{105}i}{672} & 0 & \frac{13\sqrt{210}i}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{64} & 0 & \frac{11\sqrt{7}i}{224} & 0 & -\frac{5\sqrt{70}i}{448} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{64} & 0 & \frac{\sqrt{21}i}{32} & 0 & -\frac{23\sqrt{42}i}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{6}i}{64} & 0 & -\frac{7\sqrt{3}i}{96} & 0 & \frac{\sqrt{30}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x467}} \quad \mathbb{G}_{5,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{192} & 0 & \frac{5\sqrt{70}}{448} & 0 & \frac{5\sqrt{42}}{192} & 0 & \frac{\sqrt{210}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{1344} & 0 & -\frac{13\sqrt{210}}{1344} & 0 & -\frac{\sqrt{14}}{64} & 0 & \frac{7\sqrt{6}}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{3}}{96} & 0 & \frac{11\sqrt{7}}{224} & 0 & \frac{\sqrt{105}}{672} & 0 & -\frac{\sqrt{21}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{32} & 0 & \frac{\sqrt{105}}{672} & 0 & \frac{11\sqrt{7}}{224} & 0 & \frac{7\sqrt{3}}{96} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}}{64} & 0 & -\frac{\sqrt{14}}{64} & 0 & -\frac{13\sqrt{210}}{1344} & 0 & -\frac{23\sqrt{42}}{1344} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{5\sqrt{42}}{192} & 0 & \frac{5\sqrt{70}}{448} & 0 & \frac{\sqrt{30}}{192} \\ \frac{\sqrt{30}}{192} & 0 & \frac{7\sqrt{3}}{96} & 0 & \frac{7\sqrt{6}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{23\sqrt{42}}{1344} & 0 & -\frac{\sqrt{21}}{32} & 0 & \frac{\sqrt{210}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{70}}{448} & 0 & \frac{11\sqrt{7}}{224} & 0 & -\frac{\sqrt{14}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{210}}{1344} & 0 & \frac{\sqrt{105}}{672} & 0 & \frac{5\sqrt{42}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{42}}{192} & 0 & \frac{\sqrt{105}}{672} & 0 & -\frac{13\sqrt{210}}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{64} & 0 & \frac{11\sqrt{7}}{224} & 0 & \frac{5\sqrt{70}}{448} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{21}}{32} & 0 & -\frac{23\sqrt{42}}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{6}}{64} & 0 & \frac{7\sqrt{3}}{96} & 0 & \frac{\sqrt{30}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x468}} \quad \mathbb{G}_{5,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{192} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & -\frac{3\sqrt{30}i}{64} & 0 & -\frac{5\sqrt{6}i}{192} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{30}i}{960} & 0 & \frac{13\sqrt{6}i}{192} & 0 & \frac{9\sqrt{10}i}{320} & 0 & -\frac{\sqrt{210}i}{192} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}i}{160} & 0 & \frac{11\sqrt{5}i}{160} & 0 & -\frac{\sqrt{3}i}{96} & 0 & \frac{9\sqrt{15}i}{160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{160} & 0 & \frac{\sqrt{3}i}{96} & 0 & -\frac{11\sqrt{5}i}{160} & 0 & -\frac{3\sqrt{105}i}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{192} & 0 & -\frac{9\sqrt{10}i}{320} & 0 & -\frac{13\sqrt{6}i}{192} & 0 & \frac{23\sqrt{30}i}{960} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{192} & 0 & \frac{3\sqrt{30}i}{64} & 0 & \frac{5\sqrt{2}i}{64} & 0 & -\frac{\sqrt{42}i}{192} \\ -\frac{\sqrt{42}i}{192} & 0 & -\frac{3\sqrt{105}i}{160} & 0 & -\frac{\sqrt{210}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{23\sqrt{30}i}{960} & 0 & \frac{9\sqrt{15}i}{160} & 0 & -\frac{5\sqrt{6}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{2}i}{64} & 0 & -\frac{11\sqrt{5}i}{160} & 0 & \frac{9\sqrt{10}i}{320} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{6}i}{192} & 0 & -\frac{\sqrt{3}i}{96} & 0 & -\frac{3\sqrt{30}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{64} & 0 & \frac{\sqrt{3}i}{96} & 0 & \frac{13\sqrt{6}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{10}i}{320} & 0 & \frac{11\sqrt{5}i}{160} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{6}i}{192} & 0 & -\frac{9\sqrt{15}i}{160} & 0 & -\frac{23\sqrt{30}i}{960} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{192} & 0 & \frac{3\sqrt{105}i}{160} & 0 & \frac{\sqrt{42}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x469}} \quad \mathbb{G}_{5,2}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{192} & 0 & \frac{5\sqrt{2}}{64} & 0 & -\frac{3\sqrt{30}}{64} & 0 & \frac{5\sqrt{6}}{192} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{30}}{960} & 0 & -\frac{13\sqrt{6}}{192} & 0 & \frac{9\sqrt{10}}{320} & 0 & \frac{\sqrt{210}}{192} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{160} & 0 & \frac{11\sqrt{5}}{160} & 0 & \frac{\sqrt{3}}{96} & 0 & \frac{9\sqrt{15}}{160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{15}}{160} & 0 & \frac{\sqrt{3}}{96} & 0 & \frac{11\sqrt{5}}{160} & 0 & -\frac{3\sqrt{105}}{160} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{192} & 0 & \frac{9\sqrt{10}}{320} & 0 & -\frac{13\sqrt{6}}{192} & 0 & -\frac{23\sqrt{30}}{960} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{192} & 0 & -\frac{3\sqrt{30}}{64} & 0 & \frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{42}}{192} \\ \frac{\sqrt{42}}{192} & 0 & -\frac{3\sqrt{105}}{160} & 0 & \frac{\sqrt{210}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{23\sqrt{30}}{960} & 0 & \frac{9\sqrt{15}}{160} & 0 & \frac{5\sqrt{6}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{2}}{64} & 0 & \frac{11\sqrt{5}}{160} & 0 & \frac{9\sqrt{10}}{320} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{6}}{192} & 0 & \frac{\sqrt{3}}{96} & 0 & -\frac{3\sqrt{30}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{30}}{64} & 0 & \frac{\sqrt{3}}{96} & 0 & -\frac{13\sqrt{6}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{10}}{320} & 0 & \frac{11\sqrt{5}}{160} & 0 & \frac{5\sqrt{2}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{6}}{192} & 0 & \frac{9\sqrt{15}}{160} & 0 & -\frac{23\sqrt{30}}{960} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{192} & 0 & -\frac{3\sqrt{105}}{160} & 0 & \frac{\sqrt{42}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x470}} \quad \mathbb{G}_{5,1}^{(1,0;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{96} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & \frac{\sqrt{10}i}{32} & 0 & \frac{5\sqrt{2}i}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}i}{480} & 0 & \frac{13\sqrt{2}i}{96} & 0 & -\frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{70}i}{32} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{80} & 0 & \frac{11\sqrt{15}i}{240} & 0 & -\frac{i}{48} & 0 & -\frac{3\sqrt{5}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{80} & 0 & \frac{i}{48} & 0 & -\frac{11\sqrt{15}i}{240} & 0 & \frac{\sqrt{35}i}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{32} & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{13\sqrt{2}i}{96} & 0 & \frac{23\sqrt{10}i}{480} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & -\frac{\sqrt{10}i}{32} & 0 & \frac{5\sqrt{6}i}{96} & 0 & -\frac{\sqrt{14}i}{96} \\ -\frac{\sqrt{14}i}{96} & 0 & \frac{\sqrt{35}i}{80} & 0 & \frac{\sqrt{70}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{23\sqrt{10}i}{480} & 0 & -\frac{3\sqrt{5}i}{80} & 0 & \frac{5\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{6}i}{96} & 0 & -\frac{11\sqrt{15}i}{240} & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{2}i}{96} & 0 & -\frac{i}{48} & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{32} & 0 & \frac{i}{48} & 0 & \frac{13\sqrt{2}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{11\sqrt{15}i}{240} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{2}i}{32} & 0 & \frac{3\sqrt{5}i}{80} & 0 & -\frac{23\sqrt{10}i}{480} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{32} & 0 & -\frac{\sqrt{35}i}{80} & 0 & \frac{\sqrt{14}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x471}} \quad \mathbb{G}_{5,2}^{(1,0;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{96} & 0 & \frac{5\sqrt{6}}{96} & 0 & \frac{\sqrt{10}}{32} & 0 & -\frac{5\sqrt{2}}{32} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}}{480} & 0 & -\frac{13\sqrt{2}}{96} & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{70}}{32} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{80} & 0 & \frac{11\sqrt{15}}{240} & 0 & \frac{1}{48} & 0 & -\frac{3\sqrt{5}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{80} & 0 & \frac{1}{48} & 0 & \frac{11\sqrt{15}}{240} & 0 & \frac{\sqrt{35}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{32} & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{13\sqrt{2}}{96} & 0 & -\frac{23\sqrt{10}}{480} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{32} & 0 & \frac{\sqrt{10}}{32} & 0 & \frac{5\sqrt{6}}{96} & 0 & \frac{\sqrt{14}}{96} \\ \frac{\sqrt{14}}{96} & 0 & \frac{\sqrt{35}}{80} & 0 & -\frac{\sqrt{70}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{23\sqrt{10}}{480} & 0 & -\frac{3\sqrt{5}}{80} & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{6}}{96} & 0 & \frac{11\sqrt{15}}{240} & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{2}}{96} & 0 & \frac{1}{48} & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{32} & 0 & \frac{1}{48} & 0 & -\frac{13\sqrt{2}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{160} & 0 & \frac{11\sqrt{15}}{240} & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{2}}{32} & 0 & -\frac{3\sqrt{5}}{80} & 0 & -\frac{23\sqrt{10}}{480} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{32} & 0 & \frac{\sqrt{35}}{80} & 0 & \frac{\sqrt{14}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x472}} \quad \mathbb{M}_5^{(a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x473}} \quad \mathbb{M}_1^{(a)}(A_{2g}) = \begin{bmatrix} \frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{28} \end{bmatrix}$$

$$\boxed{\text{x474}} \quad \mathbb{M}_3^{(a)}(A_{2g}) = \begin{bmatrix} -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$$

$$\boxed{\text{x475}} \quad \mathbb{M}_5^{(a)}(A_{2g}, 1) = \begin{bmatrix} \frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{42}}{588} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{42}}{588} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 \end{bmatrix}$$

$$\boxed{\text{x476}} \quad \mathbb{M}_5^{(a)}(A_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x477}} \quad \mathbb{M}_3^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 \\ -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} \\ 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{3i}{14} & 0 \\ 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x478}} \quad \mathbb{M}_5^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\ -\frac{i}{14} & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} \\ 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x479}} \quad \mathbb{M}_3^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 \\ -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{3}{14} & 0 \\ 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x480}} \quad \mathbb{M}_5^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ \frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 \\ 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 \\ \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} \\ 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x481}} \quad \mathbb{M}_{1,1}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{49} & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{196} & 0 & \frac{\sqrt{21}}{196} & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{7}}{49} & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & \frac{\sqrt{42}}{196} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{196} & 0 & \frac{\sqrt{70}}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{7}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{196} & 0 & \frac{\sqrt{105}}{196} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{196} & 0 & \frac{\sqrt{3}}{28} \\ -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{196} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{42}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{196} & 0 & -\frac{\sqrt{21}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & -\frac{\sqrt{7}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & \frac{3\sqrt{42}}{98} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{2}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x482}} \quad \mathbb{M}_{1,2}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{49} & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{196} & 0 & \frac{\sqrt{21}i}{196} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{7}i}{49} & 0 & \frac{3\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & \frac{\sqrt{42}i}{196} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{14}i}{49} & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{196} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{105}i}{196} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{196} & 0 & \frac{\sqrt{3}i}{28} \\ -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{196} & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{196} & 0 & -\frac{\sqrt{42}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & \frac{3\sqrt{14}i}{49} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{196} & 0 & -\frac{\sqrt{21}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{49} & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & -\frac{\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & \frac{3\sqrt{42}i}{98} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{2}i}{28} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x483}} \quad \mathbb{M}_{3,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{15}}{28} & 0 & -\frac{5\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{30}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{5\sqrt{30}}{168} & \frac{\sqrt{35}}{56} & 0 & \frac{\sqrt{15}}{56} & 0 & -\frac{1}{56} & 0 & -\frac{3\sqrt{5}}{56} & 0 \\ -\frac{5\sqrt{30}}{168} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{1}{56} & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{\sqrt{35}}{56} \\ 0 & -\frac{5\sqrt{3}}{42} & 0 & -\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 \\ 0 & 0 & -\frac{5\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{56} & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{28} & 0 & \frac{1}{56} & 0 & \frac{\sqrt{10}}{56} & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{15}}{21} & 0 \\ -\frac{\sqrt{10}}{56} & 0 & -\frac{1}{56} & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{105}}{84} \\ 0 & -\frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{1}{28} & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{56} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & \frac{1}{28} & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{21}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x484}} \quad \mathbb{M}_{3,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{5\sqrt{30}i}{168} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{5\sqrt{30}i}{168} & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{15}i}{56} & 0 & -\frac{i}{56} & 0 & \frac{3\sqrt{5}i}{56} & 0 \\ -\frac{5\sqrt{30}i}{168} & 0 & \frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & -\frac{i}{56} & 0 & -\frac{\sqrt{15}i}{56} & 0 & \frac{\sqrt{35}i}{56} \\ 0 & -\frac{5\sqrt{3}i}{42} & 0 & \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 \\ 0 & 0 & -\frac{5\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{14}i}{56} \\ -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{i}{28} & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{i}{56} & 0 & -\frac{\sqrt{10}i}{56} & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{15}i}{21} & 0 \\ -\frac{\sqrt{10}i}{56} & 0 & \frac{i}{56} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & \frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{105}i}{84} \\ 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{i}{28} & 0 \\ 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & -\frac{i}{28} & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{21}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x485}} \quad \mathbb{M}_{3,1}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{5}{28} & 0 & \frac{5\sqrt{2}}{56} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 \\ \frac{5}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & \frac{5\sqrt{6}}{168} & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{5\sqrt{2}}{56} & -\frac{\sqrt{21}}{56} & 0 & \frac{5}{56} & 0 & -\frac{\sqrt{15}}{168} & 0 & \frac{3\sqrt{3}}{56} & 0 \\ \frac{5\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & -\frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{168} & 0 & -\frac{5}{56} & 0 & \frac{\sqrt{21}}{56} \\ 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{5}{28} & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & -\frac{\sqrt{30}}{84} & 0 & -\frac{5\sqrt{6}}{168} & 0 \\ 0 & 0 & \frac{5\sqrt{2}}{56} & 0 & \frac{5}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{210}}{168} \\ -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{168} & 0 & -\frac{3\sqrt{3}}{56} & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{15}}{84} & 0 & \frac{1}{7} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{28} & 0 & \frac{5}{56} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{168} & 0 & -\frac{\sqrt{6}}{56} & \frac{\sqrt{7}}{28} & 0 & -\frac{5\sqrt{3}}{84} & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{1}{7} & 0 \\ \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{15}}{168} & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{1}{7} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{7}}{28} \\ 0 & \frac{3\sqrt{2}}{56} & 0 & -\frac{5}{56} & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{15}}{84} & 0 \\ 0 & 0 & \frac{3\sqrt{3}}{56} & 0 & -\frac{5\sqrt{6}}{168} & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & \frac{\sqrt{15}}{84} & 0 & \frac{\sqrt{35}}{28} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{35}}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x486}} \quad \mathbb{M}_{3,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{5i}{28} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 \\ -\frac{5i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & -\frac{5\sqrt{6}i}{168} & 0 & \frac{\sqrt{30}i}{84} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{5\sqrt{2}i}{56} & -\frac{\sqrt{21}i}{56} & 0 & -\frac{5i}{56} & 0 & -\frac{\sqrt{15}i}{168} & 0 & -\frac{3\sqrt{3}i}{56} & 0 \\ \frac{5\sqrt{2}i}{56} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & -\frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{168} & 0 & -\frac{5i}{56} & 0 & -\frac{\sqrt{21}i}{56} \\ 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{5i}{28} & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & \frac{\sqrt{30}i}{84} & 0 & -\frac{5\sqrt{6}i}{168} & 0 \\ 0 & 0 & \frac{5\sqrt{2}i}{56} & 0 & -\frac{5i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{210}i}{168} \\ -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}i}{168} & 0 & \frac{3\sqrt{3}i}{56} & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{15}i}{84} & 0 & -\frac{i}{7} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{28} & 0 & \frac{5i}{56} & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{84} & 0 & \frac{\sqrt{15}i}{168} & 0 & \frac{\sqrt{6}i}{56} & \frac{\sqrt{7}i}{28} & 0 & \frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{i}{7} & 0 \\ \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{168} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{7} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{7}i}{28} \\ 0 & \frac{3\sqrt{2}i}{56} & 0 & \frac{5i}{56} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{5\sqrt{3}i}{84} & 0 & \frac{\sqrt{15}i}{84} & 0 \\ 0 & 0 & \frac{3\sqrt{3}i}{56} & 0 & \frac{5\sqrt{6}i}{168} & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & -\frac{\sqrt{15}i}{84} & 0 & \frac{\sqrt{35}i}{28} \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{35}i}{28} & 0 \end{bmatrix}$$

$$\boxed{\text{x487}} \quad \mathbb{M}_{5,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{210}}{784} & 0 & -\frac{\sqrt{105}}{168} & 0 & \frac{3\sqrt{42}}{112} & -\frac{5}{224} & 0 & \frac{25\sqrt{21}}{1568} & 0 & -\frac{5\sqrt{35}}{224} & 0 & \frac{15\sqrt{7}}{224} & 0 \\ \frac{\sqrt{210}}{784} & 0 & -\frac{5\sqrt{21}}{392} & 0 & \frac{5\sqrt{42}}{336} & 0 & 0 & \frac{23\sqrt{35}}{1568} & 0 & -\frac{65\sqrt{7}}{1568} & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{3\sqrt{5}}{32} \\ 0 & -\frac{5\sqrt{21}}{392} & 0 & \frac{5\sqrt{42}}{392} & 0 & -\frac{\sqrt{105}}{168} & \frac{\sqrt{10}}{32} & 0 & -\frac{11\sqrt{210}}{1568} & 0 & \frac{5\sqrt{14}}{1568} & 0 & \frac{3\sqrt{70}}{224} & 0 \\ -\frac{\sqrt{105}}{168} & 0 & \frac{5\sqrt{42}}{392} & 0 & -\frac{5\sqrt{21}}{392} & 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 & -\frac{5\sqrt{14}}{1568} & 0 & \frac{11\sqrt{210}}{1568} & 0 & -\frac{\sqrt{10}}{32} \\ 0 & \frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{21}}{392} & 0 & \frac{\sqrt{210}}{784} & -\frac{3\sqrt{5}}{32} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{65\sqrt{7}}{1568} & 0 & -\frac{23\sqrt{35}}{1568} & 0 \\ \frac{3\sqrt{42}}{112} & 0 & -\frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{210}}{784} & 0 & 0 & -\frac{15\sqrt{7}}{224} & 0 & \frac{5\sqrt{35}}{224} & 0 & -\frac{25\sqrt{21}}{1568} & 0 & \frac{5}{224} \\ -\frac{5}{224} & 0 & \frac{\sqrt{10}}{32} & 0 & -\frac{3\sqrt{5}}{32} & 0 & 0 & \frac{5\sqrt{6}}{224} & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{3\sqrt{2}}{32} & 0 & 0 \\ 0 & \frac{23\sqrt{35}}{1568} & 0 & -\frac{3\sqrt{70}}{224} & 0 & -\frac{15\sqrt{7}}{224} & \frac{5\sqrt{6}}{224} & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{3\sqrt{42}}{112} & 0 \\ \frac{25\sqrt{21}}{1568} & 0 & -\frac{11\sqrt{210}}{1568} & 0 & -\frac{\sqrt{105}}{224} & 0 & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{70}}{1568} & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{3\sqrt{2}}{32} \\ 0 & -\frac{65\sqrt{7}}{1568} & 0 & -\frac{5\sqrt{14}}{1568} & 0 & \frac{5\sqrt{35}}{224} & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{70}}{1568} & 0 & \frac{5\sqrt{42}}{392} & 0 & \frac{\sqrt{210}}{672} & 0 \\ -\frac{5\sqrt{35}}{224} & 0 & \frac{5\sqrt{14}}{1568} & 0 & \frac{65\sqrt{7}}{1568} & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{5\sqrt{42}}{392} & 0 & \frac{\sqrt{70}}{1568} & 0 & -\frac{\sqrt{30}}{48} \\ 0 & \frac{\sqrt{105}}{224} & 0 & \frac{11\sqrt{210}}{1568} & 0 & -\frac{25\sqrt{21}}{1568} & \frac{3\sqrt{2}}{32} & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{70}}{1568} & 0 & -\frac{5\sqrt{14}}{196} & 0 \\ \frac{15\sqrt{7}}{224} & 0 & \frac{3\sqrt{70}}{224} & 0 & -\frac{23\sqrt{35}}{1568} & 0 & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{5\sqrt{6}}{224} \\ 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{\sqrt{10}}{32} & 0 & \frac{5}{224} & 0 & 0 & \frac{3\sqrt{2}}{32} & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{5\sqrt{6}}{224} & 0 \end{bmatrix}$$

$$\boxed{\text{x488}} \quad \mathbb{M}_{5,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{210}i}{784} & 0 & \frac{\sqrt{105}i}{168} & 0 & \frac{3\sqrt{42}i}{112} & \frac{5i}{224} & 0 & \frac{25\sqrt{21}i}{1568} & 0 & \frac{5\sqrt{35}i}{224} & 0 & \frac{15\sqrt{7}i}{224} & 0 \\ -\frac{\sqrt{210}i}{784} & 0 & -\frac{5\sqrt{21}i}{392} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & -\frac{23\sqrt{35}i}{1568} & 0 & -\frac{65\sqrt{7}i}{1568} & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{32} \\ 0 & \frac{5\sqrt{21}i}{392} & 0 & \frac{5\sqrt{42}i}{392} & 0 & \frac{\sqrt{105}i}{168} & \frac{\sqrt{10}i}{32} & 0 & \frac{11\sqrt{210}i}{1568} & 0 & \frac{5\sqrt{14}i}{1568} & 0 & -\frac{3\sqrt{70}i}{224} & 0 \\ -\frac{\sqrt{105}i}{168} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & -\frac{5\sqrt{21}i}{392} & 0 & 0 & -\frac{3\sqrt{70}i}{224} & 0 & \frac{5\sqrt{14}i}{1568} & 0 & \frac{11\sqrt{210}i}{1568} & 0 & \frac{\sqrt{10}i}{32} \\ 0 & \frac{5\sqrt{42}i}{336} & 0 & \frac{5\sqrt{21}i}{392} & 0 & \frac{\sqrt{210}i}{784} & \frac{3\sqrt{5}i}{32} & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{65\sqrt{7}i}{1568} & 0 & -\frac{23\sqrt{35}i}{1568} & 0 \\ -\frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{210}i}{784} & 0 & 0 & \frac{15\sqrt{7}i}{224} & 0 & \frac{5\sqrt{35}i}{224} & 0 & \frac{25\sqrt{21}i}{1568} & 0 & \frac{5i}{224} \\ -\frac{5i}{224} & 0 & -\frac{\sqrt{10}i}{32} & 0 & -\frac{3\sqrt{5}i}{32} & 0 & 0 & \frac{5\sqrt{6}i}{224} & 0 & \frac{\sqrt{30}i}{48} & 0 & \frac{3\sqrt{2}i}{32} & 0 & 0 \\ 0 & \frac{23\sqrt{35}i}{1568} & 0 & \frac{3\sqrt{70}i}{224} & 0 & -\frac{15\sqrt{7}i}{224} & -\frac{5\sqrt{6}i}{224} & 0 & -\frac{5\sqrt{14}i}{196} & 0 & -\frac{\sqrt{210}i}{672} & 0 & \frac{3\sqrt{42}i}{112} & 0 \\ -\frac{25\sqrt{21}i}{1568} & 0 & -\frac{11\sqrt{210}i}{1568} & 0 & \frac{\sqrt{105}i}{224} & 0 & 0 & \frac{5\sqrt{14}i}{196} & 0 & \frac{\sqrt{70}i}{1568} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & \frac{3\sqrt{2}i}{32} \\ 0 & \frac{65\sqrt{7}i}{1568} & 0 & -\frac{5\sqrt{14}i}{1568} & 0 & -\frac{5\sqrt{35}i}{224} & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{70}i}{1568} & 0 & \frac{5\sqrt{42}i}{392} & 0 & -\frac{\sqrt{210}i}{672} & 0 \\ -\frac{5\sqrt{35}i}{224} & 0 & -\frac{5\sqrt{14}i}{1568} & 0 & \frac{65\sqrt{7}i}{1568} & 0 & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & \frac{\sqrt{70}i}{1568} & 0 & \frac{\sqrt{30}i}{48} \\ 0 & \frac{\sqrt{105}i}{224} & 0 & -\frac{11\sqrt{210}i}{1568} & 0 & -\frac{25\sqrt{21}i}{1568} & -\frac{3\sqrt{2}i}{32} & 0 & \frac{5\sqrt{42}i}{336} & 0 & -\frac{\sqrt{70}i}{1568} & 0 & -\frac{5\sqrt{14}i}{196} & 0 \\ -\frac{15\sqrt{7}i}{224} & 0 & \frac{3\sqrt{70}i}{224} & 0 & \frac{23\sqrt{35}i}{1568} & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & \frac{\sqrt{210}i}{672} & 0 & \frac{5\sqrt{14}i}{196} & 0 & \frac{5\sqrt{6}i}{224} \\ 0 & -\frac{3\sqrt{5}i}{32} & 0 & -\frac{\sqrt{10}i}{32} & 0 & -\frac{5i}{224} & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{5\sqrt{6}i}{224} & 0 \end{bmatrix}$$

$$\boxed{\text{x489}} \quad \mathbb{M}_{5,1}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{112} & 0 & \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{30}}{112} & -\frac{\sqrt{35}}{224} & 0 & \frac{5\sqrt{15}}{224} & 0 & \frac{45}{224} & 0 & \frac{5\sqrt{5}}{224} & 0 \\ \frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{3\sqrt{30}}{112} & 0 & 0 & \frac{23}{224} & 0 & -\frac{13\sqrt{5}}{224} & 0 & -\frac{9\sqrt{3}}{224} & 0 & \frac{5\sqrt{7}}{224} \\ 0 & -\frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{3\sqrt{3}}{56} & -\frac{9\sqrt{14}}{224} & 0 & -\frac{11\sqrt{6}}{224} & 0 & \frac{\sqrt{10}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 \\ \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & \frac{27\sqrt{2}}{224} & 0 & -\frac{\sqrt{10}}{224} & 0 & \frac{11\sqrt{6}}{224} & 0 & \frac{9\sqrt{14}}{224} \\ 0 & -\frac{3\sqrt{30}}{112} & 0 & -\frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{6}}{112} & -\frac{5\sqrt{7}}{224} & 0 & \frac{9\sqrt{3}}{224} & 0 & \frac{13\sqrt{5}}{224} & 0 & -\frac{23}{224} & 0 \\ \frac{\sqrt{30}}{112} & 0 & \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{6}}{112} & 0 & 0 & -\frac{5\sqrt{5}}{224} & 0 & -\frac{45}{224} & 0 & -\frac{5\sqrt{15}}{224} & 0 & \frac{\sqrt{35}}{224} \\ -\frac{\sqrt{35}}{224} & 0 & -\frac{9\sqrt{14}}{224} & 0 & -\frac{5\sqrt{7}}{224} & 0 & 0 & \frac{\sqrt{210}}{224} & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{224} & 0 & 0 \\ 0 & \frac{23}{224} & 0 & \frac{27\sqrt{2}}{224} & 0 & -\frac{5\sqrt{5}}{224} & \frac{\sqrt{210}}{224} & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 \\ \frac{5\sqrt{15}}{224} & 0 & -\frac{11\sqrt{6}}{224} & 0 & \frac{9\sqrt{3}}{224} & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{3\sqrt{30}}{112} & 0 & \frac{\sqrt{70}}{224} \\ 0 & -\frac{13\sqrt{5}}{224} & 0 & -\frac{\sqrt{10}}{224} & 0 & -\frac{45}{224} & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{3\sqrt{6}}{224} & 0 \\ \frac{45}{224} & 0 & \frac{\sqrt{10}}{224} & 0 & \frac{13\sqrt{5}}{224} & 0 & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{3\sqrt{42}}{112} \\ 0 & -\frac{9\sqrt{3}}{224} & 0 & \frac{11\sqrt{6}}{224} & 0 & -\frac{5\sqrt{15}}{224} & \frac{\sqrt{70}}{224} & 0 & -\frac{3\sqrt{30}}{112} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{\sqrt{10}}{28} & 0 \\ \frac{5\sqrt{5}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & -\frac{23}{224} & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{3\sqrt{6}}{224} & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{210}}{224} \\ 0 & \frac{5\sqrt{7}}{224} & 0 & \frac{9\sqrt{14}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{210}}{224} & 0 \end{bmatrix}$$

$$\boxed{\text{x490}} \quad \mathbb{M}_{5,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{\sqrt{6}i}{112} & 0 & -\frac{3\sqrt{3}i}{56} & 0 & \frac{\sqrt{30}i}{112} & \frac{\sqrt{35}i}{224} & 0 & \frac{5\sqrt{15}i}{224} & 0 & -\frac{45i}{224} & 0 & \frac{5\sqrt{5}i}{224} & 0 \\ -\frac{\sqrt{6}i}{112} & 0 & -\frac{\sqrt{15}i}{56} & 0 & \frac{3\sqrt{30}i}{112} & 0 & 0 & -\frac{23i}{224} & 0 & -\frac{13\sqrt{5}i}{224} & 0 & \frac{9\sqrt{3}i}{224} & 0 & \frac{5\sqrt{7}i}{224} \\ 0 & \frac{\sqrt{15}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 & -\frac{3\sqrt{3}i}{56} & -\frac{9\sqrt{14}i}{224} & 0 & \frac{11\sqrt{6}i}{224} & 0 & \frac{\sqrt{10}i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 \\ \frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & \frac{27\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{224} & 0 & \frac{11\sqrt{6}i}{224} & 0 & -\frac{9\sqrt{14}i}{224} \\ 0 & -\frac{3\sqrt{30}i}{112} & 0 & \frac{\sqrt{15}i}{56} & 0 & \frac{\sqrt{6}i}{112} & \frac{5\sqrt{7}i}{224} & 0 & \frac{9\sqrt{3}i}{224} & 0 & -\frac{13\sqrt{5}i}{224} & 0 & -\frac{23i}{224} & 0 \\ -\frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{6}i}{112} & 0 & 0 & \frac{5\sqrt{5}i}{224} & 0 & -\frac{45i}{224} & 0 & \frac{5\sqrt{15}i}{224} & 0 & \frac{\sqrt{35}i}{224} \\ -\frac{\sqrt{35}i}{224} & 0 & \frac{9\sqrt{14}i}{224} & 0 & -\frac{5\sqrt{7}i}{224} & 0 & 0 & \frac{\sqrt{210}i}{224} & 0 & -\frac{3\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 \\ 0 & \frac{23i}{224} & 0 & -\frac{27\sqrt{2}i}{224} & 0 & -\frac{5\sqrt{5}i}{224} & -\frac{\sqrt{210}i}{224} & 0 & -\frac{\sqrt{10}i}{28} & 0 & \frac{3\sqrt{6}i}{224} & 0 & \frac{\sqrt{30}i}{112} & 0 \\ -\frac{5\sqrt{15}i}{224} & 0 & -\frac{11\sqrt{6}i}{224} & 0 & -\frac{9\sqrt{3}i}{224} & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{2}i}{224} & 0 & \frac{3\sqrt{30}i}{112} & 0 & \frac{\sqrt{70}i}{224} \\ 0 & \frac{13\sqrt{5}i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{45i}{224} & \frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{\sqrt{30}i}{56} & 0 & \frac{3\sqrt{6}i}{224} & 0 \\ \frac{45i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{13\sqrt{5}i}{224} & 0 & 0 & -\frac{3\sqrt{6}i}{224} & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{2}i}{224} & 0 & -\frac{3\sqrt{42}i}{112} \\ 0 & -\frac{9\sqrt{3}i}{224} & 0 & -\frac{11\sqrt{6}i}{224} & 0 & -\frac{5\sqrt{15}i}{224} & -\frac{\sqrt{70}i}{224} & 0 & -\frac{3\sqrt{30}i}{112} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{\sqrt{10}i}{28} & 0 \\ -\frac{5\sqrt{5}i}{224} & 0 & -\frac{27\sqrt{2}i}{224} & 0 & \frac{23i}{224} & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{210}i}{224} \\ 0 & -\frac{5\sqrt{7}i}{224} & 0 & \frac{9\sqrt{14}i}{224} & 0 & -\frac{\sqrt{35}i}{224} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{210}i}{224} & 0 \end{bmatrix}$$

$$\boxed{\text{x491}} \quad \mathbb{M}_{5,1}^{(a)}(E_g, 3) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{1}{28} & 0 & -\frac{3\sqrt{10}}{56} & -\frac{\sqrt{105}}{336} & 0 & \frac{5\sqrt{5}}{112} & 0 & -\frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{15}}{112} & 0 \\ \frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & \frac{23\sqrt{3}}{336} & 0 & -\frac{13\sqrt{15}}{336} & 0 & \frac{3}{112} & 0 & -\frac{5\sqrt{21}}{112} \\ 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & -\frac{1}{28} & \frac{\sqrt{42}}{112} & 0 & -\frac{11\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{3\sqrt{6}}{112} & 0 \\ -\frac{1}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 & -\frac{\sqrt{30}}{336} & 0 & \frac{11\sqrt{2}}{112} & 0 & -\frac{\sqrt{42}}{112} \\ 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & \frac{5\sqrt{21}}{112} & 0 & -\frac{3}{112} & 0 & \frac{13\sqrt{15}}{336} & 0 & -\frac{23\sqrt{3}}{336} & 0 \\ -\frac{3\sqrt{10}}{56} & 0 & -\frac{1}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{5\sqrt{15}}{112} & 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{5}}{112} & 0 & \frac{\sqrt{105}}{336} \\ -\frac{\sqrt{105}}{336} & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{5\sqrt{21}}{112} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 \\ 0 & \frac{23\sqrt{3}}{336} & 0 & -\frac{3\sqrt{6}}{112} & 0 & \frac{5\sqrt{15}}{112} & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{2}}{112} & 0 & -\frac{3\sqrt{10}}{56} & 0 \\ \frac{5\sqrt{5}}{112} & 0 & -\frac{11\sqrt{2}}{112} & 0 & -\frac{3}{112} & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{6}}{336} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{112} \\ 0 & -\frac{13\sqrt{15}}{336} & 0 & -\frac{\sqrt{30}}{336} & 0 & \frac{5\sqrt{3}}{112} & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{6}}{336} & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{2}}{112} & 0 \\ -\frac{5\sqrt{3}}{112} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{13\sqrt{15}}{336} & 0 & 0 & \frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{336} & 0 & -\frac{\sqrt{14}}{56} \\ 0 & \frac{3}{112} & 0 & \frac{11\sqrt{2}}{112} & 0 & -\frac{5\sqrt{5}}{112} & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{336} & 0 & -\frac{\sqrt{30}}{42} & 0 \\ -\frac{5\sqrt{15}}{112} & 0 & \frac{3\sqrt{6}}{112} & 0 & -\frac{23\sqrt{3}}{336} & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & \frac{\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{70}}{112} \\ 0 & -\frac{5\sqrt{21}}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{105}}{336} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 \end{bmatrix}$$

$$\boxed{\text{x492}} \quad \mathbb{M}_{5,2}^{(a)}(E_g, 3) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{i}{28} & 0 & -\frac{3\sqrt{10}i}{56} & \frac{\sqrt{105}i}{336} & 0 & \frac{5\sqrt{5}i}{112} & 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{15}i}{112} & 0 \\ -\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & -\frac{23\sqrt{3}i}{336} & 0 & -\frac{13\sqrt{15}i}{336} & 0 & -\frac{3i}{112} & 0 & -\frac{5\sqrt{21}i}{112} \\ 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{i}{28} & \frac{\sqrt{42}i}{112} & 0 & \frac{11\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{336} & 0 & -\frac{3\sqrt{6}i}{112} & 0 \\ -\frac{i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & -\frac{3\sqrt{6}i}{112} & 0 & \frac{\sqrt{30}i}{336} & 0 & \frac{11\sqrt{2}i}{112} & 0 & \frac{\sqrt{42}i}{112} \\ 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{2}i}{56} & -\frac{5\sqrt{21}i}{112} & 0 & -\frac{3i}{112} & 0 & -\frac{13\sqrt{15}i}{336} & 0 & -\frac{23\sqrt{3}i}{336} & 0 \\ \frac{3\sqrt{10}i}{56} & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{5\sqrt{15}i}{112} & 0 & \frac{5\sqrt{3}i}{112} & 0 & \frac{5\sqrt{5}i}{112} & 0 & \frac{\sqrt{105}i}{336} \\ -\frac{\sqrt{105}i}{336} & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{5\sqrt{21}i}{112} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \\ 0 & \frac{23\sqrt{3}i}{336} & 0 & \frac{3\sqrt{6}i}{112} & 0 & \frac{5\sqrt{15}i}{112} & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{30}i}{42} & 0 & -\frac{\sqrt{2}i}{112} & 0 & -\frac{3\sqrt{10}i}{56} & 0 \\ -\frac{5\sqrt{5}i}{112} & 0 & -\frac{11\sqrt{2}i}{112} & 0 & \frac{3i}{112} & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{112} \\ 0 & \frac{13\sqrt{15}i}{336} & 0 & -\frac{\sqrt{30}i}{336} & 0 & -\frac{5\sqrt{3}i}{112} & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{6}i}{336} & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{2}i}{112} & 0 \\ -\frac{5\sqrt{3}i}{112} & 0 & -\frac{\sqrt{30}i}{336} & 0 & \frac{13\sqrt{15}i}{336} & 0 & 0 & \frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{6}i}{336} & 0 & \frac{\sqrt{14}i}{56} \\ 0 & \frac{3i}{112} & 0 & -\frac{11\sqrt{2}i}{112} & 0 & -\frac{5\sqrt{5}i}{112} & \frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{30}i}{42} & 0 \\ \frac{5\sqrt{15}i}{112} & 0 & \frac{3\sqrt{6}i}{112} & 0 & \frac{23\sqrt{3}i}{336} & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & \frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{70}i}{112} \\ 0 & \frac{5\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{105}i}{336} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 \end{bmatrix}$$

$$\boxed{\text{x493}} \quad \mathbb{M}_5^{(1,-1;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}i}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$$

x494

x495

$$\boxed{\text{x496}} \quad \mathbb{M}_3^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \end{bmatrix}$$

$$\boxed{\text{x497}} \quad \mathbb{M}_5^{(1,-1;a)}(A_{2g}, 1) = \begin{bmatrix} -\frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{385}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{385}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{385}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 \end{bmatrix}$$

$$\boxed{\text{x498}} \quad \mathbb{M}_5^{(1,-1;a)}(A_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x499}} \quad \mathbb{M}_7^{(1,-1;a)}(A_{2g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{572} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{35\sqrt{858}}{1716} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{35\sqrt{858}}{1716} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{572} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{1716} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{1716} & 0 \end{bmatrix}$$

$$\boxed{\text{x500}} \quad \mathbb{M}_7^{(1,-1;a)}(A_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}}{52} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}}{52} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{52} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}}{52} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}}{52} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x501}} \quad \mathbb{M}_3^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{98} & 0 & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & -\frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & \frac{\sqrt{21}i}{98} & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{98} & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{21} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & -\frac{10\sqrt{7}i}{147} & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 \\ 0 & \frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x502}} \quad \mathbb{M}_5^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{66}i}{924} & 0 & 0 & -\frac{\sqrt{770}i}{385} & 0 & 0 & 0 & \frac{2\sqrt{22}i}{77} & 0 & 0 & 0 \\ \frac{\sqrt{330}i}{924} & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{924} & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & \frac{2\sqrt{165}i}{385} & 0 & 0 \\ 0 & -\frac{5\sqrt{66}i}{924} & 0 & 0 & 0 & \frac{\sqrt{330}i}{924} & 0 & 0 & -\frac{2\sqrt{165}i}{385} & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 \\ 0 & 0 & \frac{5\sqrt{66}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{22}i}{77} & 0 & 0 & 0 & \frac{\sqrt{770}i}{385} \\ 0 & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{165}i}{385} & 0 & 0 & -\frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{77} & 0 & 0 & 0 \\ \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & \frac{2\sqrt{22}i}{77} & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & \frac{2\sqrt{55}i}{77} & 0 & 0 \\ 0 & -\frac{2\sqrt{22}i}{77} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & \frac{2\sqrt{55}i}{77} & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 \\ 0 & 0 & -\frac{2\sqrt{165}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{77} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{154} \\ 0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{154} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x503}} \quad \mathbb{M}_7^{(1,-1;a)}(B_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 & \frac{\sqrt{66}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & -\frac{\sqrt{66}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{88} & 0 & 0 & 0 & \frac{\sqrt{770}i}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{88} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{88} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{24} & 0 & 0 & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x504}} \quad \mathbb{M}_7^{(1,-1;a)}(B_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{26}i}{104} & 0 & 0 & 0 & -\frac{\sqrt{78}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2730}i}{312} & 0 & 0 & 0 & \frac{\sqrt{78}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{26}i}{104} & 0 & 0 & 0 & \frac{\sqrt{910}i}{104} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2730}i}{312} & 0 & 0 & 0 & -\frac{\sqrt{910}i}{104} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{910}i}{104} & 0 & 0 & 0 & \frac{\sqrt{2730}i}{312} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{910}i}{104} & 0 & 0 & 0 & -\frac{\sqrt{26}i}{104} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{78}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{2730}i}{312} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}i}{24} & 0 & 0 & 0 & \frac{\sqrt{26}i}{104} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x505}} \quad \mathbb{M}_3^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}}{98} & 0 & 0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 \\ 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x506}} \quad \mathbb{M}_5^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{330}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{66}}{924} & 0 & 0 & \frac{\sqrt{770}}{385} & 0 & 0 & 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{924} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{924} & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 \\ 0 & \frac{5\sqrt{66}}{924} & 0 & 0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 \\ 0 & 0 & -\frac{5\sqrt{66}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & 0 & \frac{\sqrt{770}}{385} \\ 0 & 0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 & \frac{5\sqrt{77}}{154} & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 \\ 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 \\ 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} \\ 0 & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x507}} \quad \mathbb{M}_7^{(1,-1;a)}(B_{2g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x508}} \quad \mathbb{M}_7^{(1,-1;a)}(B_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{286} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{286} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{286} & 0 & 0 & 0 & -\frac{\sqrt{15015}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{286} & 0 & 0 & 0 & \frac{\sqrt{15015}}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}}{286} & 0 & 0 & 0 & -\frac{\sqrt{5005}}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{286} & 0 & 0 & 0 & \frac{\sqrt{429}}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}}{286} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x509}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & -\frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{49} & 0 & -\frac{\sqrt{3}}{7} & 0 \\ \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{49} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{49} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{2}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x510}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & -\frac{\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & -\frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{49} & 0 & -\frac{\sqrt{105}i}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{49} & 0 & -\frac{\sqrt{3}i}{7} \\ \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{49} & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{210}i}{98} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{49} & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{98} & 0 & \frac{2\sqrt{14}i}{49} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{49} & 0 & \frac{\sqrt{210}i}{98} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & \frac{\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{98} & 0 & \frac{\sqrt{42}i}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{2}i}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 \end{bmatrix}$$

$$\boxed{\text{x511}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{3\sqrt{14}}{196} & 0 & \frac{5\sqrt{7}}{196} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{49} & 0 & \frac{5\sqrt{21}}{294} & 0 & 0 & 0 \\ -\frac{3\sqrt{14}}{196} & 0 & \frac{3\sqrt{35}}{980} & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & -\frac{5\sqrt{21}}{294} & 0 & -\frac{\sqrt{105}}{147} & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 \\ 0 & \frac{3\sqrt{35}}{980} & 0 & \frac{3\sqrt{70}}{490} & 0 & \frac{5\sqrt{7}}{196} & -\frac{5\sqrt{6}}{84} & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{210}}{588} & 0 & \frac{5\sqrt{42}}{196} & 0 \\ \frac{5\sqrt{7}}{196} & 0 & \frac{3\sqrt{70}}{490} & 0 & \frac{3\sqrt{35}}{980} & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & -\frac{\sqrt{210}}{588} & 0 & \frac{5\sqrt{14}}{196} & 0 & \frac{5\sqrt{6}}{84} \\ 0 & \frac{\sqrt{70}}{98} & 0 & \frac{3\sqrt{35}}{980} & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & \frac{\sqrt{105}}{147} & 0 & \frac{5\sqrt{21}}{294} & 0 \\ 0 & 0 & \frac{5\sqrt{7}}{196} & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{294} & 0 & \frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{15}}{42} \\ \frac{\sqrt{15}}{42} & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & -\frac{5\sqrt{2}}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}}{294} & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{49} & 0 & -\frac{5\sqrt{14}}{196} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{5\sqrt{42}}{392} & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{147} & 0 & -\frac{\sqrt{210}}{588} & 0 & -\frac{5\sqrt{21}}{294} & -\frac{5\sqrt{2}}{56} & 0 & -\frac{5\sqrt{42}}{392} & 0 & -\frac{3\sqrt{70}}{196} & 0 & -\frac{5\sqrt{14}}{98} & 0 \\ \frac{5\sqrt{21}}{294} & 0 & \frac{\sqrt{210}}{588} & 0 & \frac{\sqrt{105}}{147} & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & -\frac{3\sqrt{70}}{196} & 0 & -\frac{5\sqrt{42}}{392} & 0 & -\frac{5\sqrt{2}}{56} \\ 0 & \frac{5\sqrt{7}}{98} & 0 & \frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & -\frac{5\sqrt{42}}{392} & 0 & \frac{\sqrt{210}}{392} & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & \frac{5\sqrt{21}}{294} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & \frac{\sqrt{210}}{392} & 0 & \frac{3\sqrt{10}}{56} \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{56} & 0 & \frac{3\sqrt{10}}{56} & 0 \end{bmatrix}$$

$$\boxed{\text{x512}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{3\sqrt{14}i}{196} & 0 & -\frac{5\sqrt{7}i}{196} & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{35}i}{49} & 0 & -\frac{5\sqrt{21}i}{294} & 0 & 0 & 0 \\ \frac{3\sqrt{14}i}{196} & 0 & \frac{3\sqrt{35}i}{980} & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & \frac{5\sqrt{21}i}{294} & 0 & -\frac{\sqrt{105}i}{147} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 \\ 0 & -\frac{3\sqrt{35}i}{980} & 0 & \frac{3\sqrt{70}i}{490} & 0 & -\frac{5\sqrt{7}i}{196} & -\frac{5\sqrt{6}i}{84} & 0 & \frac{5\sqrt{14}i}{196} & 0 & \frac{\sqrt{210}i}{588} & 0 & -\frac{5\sqrt{42}i}{196} & 0 \\ \frac{5\sqrt{7}i}{196} & 0 & -\frac{3\sqrt{70}i}{490} & 0 & \frac{3\sqrt{35}i}{980} & 0 & 0 & -\frac{5\sqrt{42}i}{196} & 0 & \frac{\sqrt{210}i}{588} & 0 & \frac{5\sqrt{14}i}{196} & 0 & -\frac{5\sqrt{6}i}{84} \\ 0 & \frac{\sqrt{70}i}{98} & 0 & -\frac{3\sqrt{35}i}{980} & 0 & -\frac{3\sqrt{14}i}{196} & 0 & 0 & -\frac{5\sqrt{7}i}{98} & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{5\sqrt{21}i}{294} & 0 \\ 0 & 0 & \frac{5\sqrt{7}i}{196} & 0 & \frac{3\sqrt{14}i}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{294} & 0 & -\frac{\sqrt{35}i}{49} & 0 & -\frac{\sqrt{15}i}{42} \\ \frac{\sqrt{15}i}{42} & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & \frac{5\sqrt{2}i}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{294} & 0 & \frac{5\sqrt{42}i}{196} & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{392} & 0 & \frac{5\sqrt{14}i}{98} & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{49} & 0 & -\frac{5\sqrt{14}i}{196} & 0 & \frac{5\sqrt{7}i}{98} & 0 & 0 & -\frac{\sqrt{210}i}{392} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & \frac{5\sqrt{70}i}{196} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{147} & 0 & -\frac{\sqrt{210}i}{588} & 0 & \frac{5\sqrt{21}i}{294} & -\frac{5\sqrt{2}i}{56} & 0 & \frac{5\sqrt{42}i}{392} & 0 & -\frac{3\sqrt{70}i}{196} & 0 & \frac{5\sqrt{14}i}{98} & 0 \\ \frac{5\sqrt{21}i}{294} & 0 & -\frac{\sqrt{210}i}{588} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & \frac{3\sqrt{70}i}{196} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & \frac{5\sqrt{2}i}{56} \\ 0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{5\sqrt{14}i}{196} & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & \frac{5\sqrt{42}i}{392} & 0 & \frac{\sqrt{210}i}{392} & 0 \\ 0 & 0 & \frac{5\sqrt{42}i}{196} & 0 & -\frac{5\sqrt{21}i}{294} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & -\frac{\sqrt{210}i}{392} & 0 & \frac{3\sqrt{10}i}{56} \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{10}i}{56} & 0 \end{bmatrix}$$

$$\boxed{\text{x513}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{210}}{196} & 0 & -\frac{\sqrt{105}}{196} & 0 & 0 & \frac{5}{42} & 0 & -\frac{5\sqrt{21}}{147} & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{196} & 0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & -\frac{5\sqrt{35}}{294} & 0 & -\frac{5\sqrt{7}}{147} & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 \\ 0 & \frac{\sqrt{21}}{196} & 0 & \frac{\sqrt{42}}{98} & 0 & -\frac{\sqrt{105}}{196} & \frac{\sqrt{10}}{28} & 0 & -\frac{5\sqrt{210}}{588} & 0 & \frac{5\sqrt{14}}{588} & 0 & -\frac{3\sqrt{70}}{196} & 0 \\ -\frac{\sqrt{105}}{196} & 0 & \frac{\sqrt{42}}{98} & 0 & \frac{\sqrt{21}}{196} & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & -\frac{5\sqrt{14}}{588} & 0 & \frac{5\sqrt{210}}{588} & 0 & -\frac{\sqrt{10}}{28} \\ 0 & -\frac{\sqrt{42}}{98} & 0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & \frac{5\sqrt{7}}{147} & 0 & \frac{5\sqrt{35}}{294} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{196} & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & \frac{5\sqrt{21}}{147} & 0 & -\frac{5}{42} \\ \frac{5}{42} & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{35}}{294} & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & \frac{5\sqrt{14}}{392} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 \\ -\frac{5\sqrt{21}}{147} & 0 & -\frac{5\sqrt{210}}{588} & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & \frac{5\sqrt{14}}{392} & 0 & -\frac{5\sqrt{70}}{392} & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 \\ 0 & -\frac{5\sqrt{7}}{147} & 0 & -\frac{5\sqrt{14}}{588} & 0 & \frac{\sqrt{35}}{98} & \frac{\sqrt{30}}{56} & 0 & -\frac{5\sqrt{70}}{392} & 0 & -\frac{5\sqrt{42}}{196} & 0 & \frac{\sqrt{210}}{98} & 0 \\ -\frac{\sqrt{35}}{98} & 0 & \frac{5\sqrt{14}}{588} & 0 & \frac{5\sqrt{7}}{147} & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & -\frac{5\sqrt{42}}{196} & 0 & -\frac{5\sqrt{70}}{392} & 0 & \frac{\sqrt{30}}{56} \\ 0 & -\frac{\sqrt{105}}{98} & 0 & \frac{5\sqrt{210}}{588} & 0 & \frac{5\sqrt{21}}{147} & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & -\frac{5\sqrt{70}}{392} & 0 & \frac{5\sqrt{14}}{392} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & \frac{5\sqrt{35}}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{5\sqrt{14}}{392} & 0 & \frac{5\sqrt{6}}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{5\sqrt{6}}{56} & 0 \end{bmatrix}$$

$$\boxed{\text{x514}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{210}i}{196} & 0 & \frac{\sqrt{105}i}{196} & 0 & 0 & -\frac{5i}{42} & 0 & -\frac{5\sqrt{21}i}{147} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{196} & 0 & \frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{42}i}{98} & 0 & 0 & \frac{5\sqrt{35}i}{294} & 0 & -\frac{5\sqrt{7}i}{147} & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{42}i}{98} & 0 & \frac{\sqrt{105}i}{196} & \frac{\sqrt{10}i}{28} & 0 & \frac{5\sqrt{210}i}{588} & 0 & \frac{5\sqrt{14}i}{588} & 0 & \frac{3\sqrt{70}i}{196} & 0 \\ -\frac{\sqrt{105}i}{196} & 0 & -\frac{\sqrt{42}i}{98} & 0 & \frac{\sqrt{21}i}{196} & 0 & 0 & \frac{3\sqrt{70}i}{196} & 0 & \frac{5\sqrt{14}i}{588} & 0 & \frac{5\sqrt{210}i}{588} & 0 & \frac{\sqrt{10}i}{28} \\ 0 & -\frac{\sqrt{42}i}{98} & 0 & -\frac{\sqrt{21}i}{196} & 0 & -\frac{\sqrt{210}i}{196} & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & -\frac{5\sqrt{7}i}{147} & 0 & \frac{5\sqrt{35}i}{294} & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{196} & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{98} & 0 & -\frac{5\sqrt{21}i}{147} & 0 & -\frac{5i}{42} \\ \frac{5i}{42} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{35}i}{294} & 0 & -\frac{3\sqrt{70}i}{196} & 0 & 0 & -\frac{5\sqrt{6}i}{56} & 0 & \frac{5\sqrt{14}i}{392} & 0 & -\frac{\sqrt{210}i}{98} & 0 & 0 & 0 \\ \frac{5\sqrt{21}i}{147} & 0 & -\frac{5\sqrt{210}i}{588} & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & -\frac{5\sqrt{14}i}{392} & 0 & -\frac{5\sqrt{70}i}{392} & 0 & -\frac{5\sqrt{42}i}{196} & 0 & 0 \\ 0 & \frac{5\sqrt{7}i}{147} & 0 & -\frac{5\sqrt{14}i}{588} & 0 & -\frac{\sqrt{35}i}{98} & \frac{\sqrt{30}i}{56} & 0 & \frac{5\sqrt{70}i}{392} & 0 & -\frac{5\sqrt{42}i}{196} & 0 & -\frac{\sqrt{210}i}{98} & 0 \\ -\frac{\sqrt{35}i}{98} & 0 & -\frac{5\sqrt{14}i}{588} & 0 & \frac{5\sqrt{7}i}{147} & 0 & 0 & \frac{\sqrt{210}i}{98} & 0 & \frac{5\sqrt{42}i}{196} & 0 & -\frac{5\sqrt{70}i}{392} & 0 & -\frac{\sqrt{30}i}{56} \\ 0 & -\frac{\sqrt{105}i}{98} & 0 & -\frac{5\sqrt{210}i}{588} & 0 & \frac{5\sqrt{21}i}{147} & 0 & 0 & \frac{5\sqrt{42}i}{196} & 0 & \frac{5\sqrt{70}i}{392} & 0 & \frac{5\sqrt{14}i}{392} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}i}{196} & 0 & -\frac{5\sqrt{35}i}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{98} & 0 & -\frac{5\sqrt{14}i}{392} & 0 & \frac{5\sqrt{6}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & -\frac{5\sqrt{6}i}{56} & 0 \end{bmatrix}$$

$$\boxed{\text{x515}} \quad \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{5\sqrt{77}}{8624} & 0 & \frac{5\sqrt{154}}{3696} & 0 & -\frac{3\sqrt{385}}{1232} & \frac{\sqrt{330}}{1232} & 0 & -\frac{15\sqrt{770}}{8624} & 0 & \frac{5\sqrt{462}}{1232} & 0 & -\frac{3\sqrt{2310}}{1232} & 0 \\ -\frac{5\sqrt{77}}{8624} & 0 & \frac{5\sqrt{770}}{8624} & 0 & -\frac{5\sqrt{385}}{3696} & 0 & 0 & -\frac{23\sqrt{462}}{8624} & 0 & \frac{13\sqrt{2310}}{8624} & 0 & -\frac{3\sqrt{154}}{1232} & 0 & -\frac{3\sqrt{66}}{176} \\ 0 & \frac{5\sqrt{770}}{8624} & 0 & -\frac{5\sqrt{385}}{4312} & 0 & \frac{5\sqrt{154}}{3696} & -\frac{\sqrt{33}}{88} & 0 & \frac{3\sqrt{77}}{392} & 0 & -\frac{\sqrt{1155}}{4312} & 0 & -\frac{3\sqrt{231}}{616} & 0 \\ \frac{5\sqrt{154}}{3696} & 0 & -\frac{5\sqrt{385}}{4312} & 0 & \frac{5\sqrt{770}}{8624} & 0 & 0 & \frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{1155}}{4312} & 0 & -\frac{3\sqrt{77}}{392} & 0 & \frac{\sqrt{33}}{88} \\ 0 & -\frac{5\sqrt{385}}{3696} & 0 & \frac{5\sqrt{770}}{8624} & 0 & -\frac{5\sqrt{77}}{8624} & \frac{3\sqrt{66}}{176} & 0 & \frac{3\sqrt{154}}{1232} & 0 & -\frac{13\sqrt{2310}}{8624} & 0 & \frac{23\sqrt{462}}{8624} & 0 \\ -\frac{3\sqrt{385}}{1232} & 0 & \frac{5\sqrt{154}}{3696} & 0 & -\frac{5\sqrt{77}}{8624} & 0 & 0 & \frac{3\sqrt{2310}}{1232} & 0 & -\frac{5\sqrt{462}}{1232} & 0 & \frac{15\sqrt{770}}{8624} & 0 & -\frac{\sqrt{330}}{1232} \\ \frac{\sqrt{330}}{1232} & 0 & -\frac{\sqrt{33}}{88} & 0 & \frac{3\sqrt{66}}{176} & 0 & 0 & \frac{15\sqrt{55}}{1232} & 0 & -\frac{5\sqrt{11}}{88} & 0 & \frac{3\sqrt{165}}{176} & 0 & 0 \\ 0 & -\frac{23\sqrt{462}}{8624} & 0 & \frac{3\sqrt{231}}{616} & 0 & \frac{3\sqrt{2310}}{1232} & \frac{15\sqrt{55}}{1232} & 0 & -\frac{5\sqrt{1155}}{1078} & 0 & \frac{5\sqrt{77}}{1232} & 0 & \frac{9\sqrt{385}}{616} & 0 \\ -\frac{15\sqrt{770}}{8624} & 0 & \frac{3\sqrt{77}}{392} & 0 & \frac{3\sqrt{154}}{1232} & 0 & 0 & -\frac{5\sqrt{1155}}{1078} & 0 & \frac{5\sqrt{231}}{8624} & 0 & \frac{5\sqrt{385}}{616} & 0 & \frac{3\sqrt{165}}{176} \\ 0 & \frac{13\sqrt{2310}}{8624} & 0 & \frac{\sqrt{1155}}{4312} & 0 & -\frac{5\sqrt{462}}{1232} & -\frac{5\sqrt{11}}{88} & 0 & \frac{5\sqrt{231}}{8624} & 0 & \frac{15\sqrt{385}}{2156} & 0 & \frac{5\sqrt{77}}{1232} & 0 \\ \frac{5\sqrt{462}}{1232} & 0 & -\frac{\sqrt{1155}}{4312} & 0 & -\frac{13\sqrt{2310}}{8624} & 0 & 0 & \frac{5\sqrt{77}}{1232} & 0 & \frac{15\sqrt{385}}{2156} & 0 & \frac{5\sqrt{231}}{8624} & 0 & -\frac{5\sqrt{11}}{88} \\ 0 & -\frac{3\sqrt{154}}{1232} & 0 & -\frac{3\sqrt{77}}{392} & 0 & \frac{15\sqrt{770}}{8624} & \frac{3\sqrt{165}}{176} & 0 & \frac{5\sqrt{385}}{616} & 0 & \frac{5\sqrt{231}}{8624} & 0 & -\frac{5\sqrt{1155}}{1078} & 0 \\ -\frac{3\sqrt{2310}}{1232} & 0 & -\frac{3\sqrt{231}}{616} & 0 & \frac{23\sqrt{462}}{8624} & 0 & 0 & \frac{9\sqrt{385}}{616} & 0 & \frac{5\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{1155}}{1078} & 0 & \frac{15\sqrt{55}}{1232} \\ 0 & -\frac{3\sqrt{66}}{176} & 0 & \frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{330}}{1232} & 0 & 0 & \frac{3\sqrt{165}}{176} & 0 & -\frac{5\sqrt{11}}{88} & 0 & \frac{15\sqrt{55}}{1232} & 0 \end{bmatrix}$$

x516	$\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 1)$													
=	0	$-\frac{5\sqrt{77}i}{8624}$	0	$-\frac{5\sqrt{154}i}{3696}$	0	$-\frac{3\sqrt{385}i}{1232}$	$-\frac{\sqrt{330}i}{1232}$	0	$-\frac{15\sqrt{770}i}{8624}$	0	$-\frac{5\sqrt{462}i}{1232}$	0	$-\frac{3\sqrt{2310}i}{1232}$	0
	$\frac{5\sqrt{77}i}{8624}$	0	$\frac{5\sqrt{770}i}{8624}$	0	$\frac{5\sqrt{385}i}{3696}$	0	0	$\frac{23\sqrt{462}i}{8624}$	0	$\frac{13\sqrt{2310}i}{8624}$	0	$\frac{3\sqrt{154}i}{1232}$	0	$-\frac{3\sqrt{66}i}{176}$
	0	$-\frac{5\sqrt{770}i}{8624}$	0	$-\frac{5\sqrt{385}i}{4312}$	0	$-\frac{5\sqrt{154}i}{3696}$	$-\frac{\sqrt{33}i}{88}$	0	$-\frac{3\sqrt{77}i}{392}$	0	$-\frac{\sqrt{1155}i}{4312}$	0	$\frac{3\sqrt{231}i}{616}$	0
	$\frac{5\sqrt{154}i}{3696}$	0	$\frac{5\sqrt{385}i}{4312}$	0	$\frac{5\sqrt{770}i}{8624}$	0	0	$\frac{3\sqrt{231}i}{616}$	0	$-\frac{\sqrt{1155}i}{4312}$	0	$-\frac{3\sqrt{77}i}{392}$	0	$-\frac{\sqrt{33}i}{88}$
	0	$-\frac{5\sqrt{385}i}{3696}$	0	$-\frac{5\sqrt{770}i}{8624}$	0	$-\frac{5\sqrt{77}i}{8624}$	$-\frac{3\sqrt{66}i}{176}$	0	$\frac{3\sqrt{154}i}{1232}$	0	$\frac{13\sqrt{2310}i}{8624}$	0	$\frac{23\sqrt{462}i}{8624}$	0
	$\frac{3\sqrt{385}i}{1232}$	0	$\frac{5\sqrt{154}i}{3696}$	0	$\frac{5\sqrt{77}i}{8624}$	0	0	$-\frac{3\sqrt{2310}i}{1232}$	0	$-\frac{5\sqrt{462}i}{1232}$	0	$-\frac{15\sqrt{770}i}{8624}$	0	$-\frac{\sqrt{330}i}{1232}$
	$\frac{\sqrt{330}i}{1232}$	0	$\frac{\sqrt{33}i}{88}$	0	$\frac{3\sqrt{66}i}{176}$	0	0	$\frac{15\sqrt{55}i}{1232}$	0	$\frac{5\sqrt{11}i}{88}$	0	$\frac{3\sqrt{165}i}{176}$	0	0
	0	$-\frac{23\sqrt{462}i}{8624}$	0	$-\frac{3\sqrt{231}i}{616}$	0	$\frac{3\sqrt{2310}i}{1232}$	$-\frac{15\sqrt{55}i}{1232}$	0	$-\frac{5\sqrt{1155}i}{1078}$	0	$-\frac{5\sqrt{77}i}{1232}$	0	$\frac{9\sqrt{385}i}{616}$	0
	$\frac{15\sqrt{770}i}{8624}$	0	$\frac{3\sqrt{77}i}{392}$	0	$-\frac{3\sqrt{154}i}{1232}$	0	0	$\frac{5\sqrt{1155}i}{1078}$	0	$\frac{5\sqrt{231}i}{8624}$	0	$-\frac{5\sqrt{385}i}{616}$	0	$\frac{3\sqrt{165}i}{176}$
	0	$-\frac{13\sqrt{2310}i}{8624}$	0	$\frac{\sqrt{1155}i}{4312}$	0	$\frac{5\sqrt{462}i}{1232}$	$-\frac{5\sqrt{11}i}{88}$	0	$-\frac{5\sqrt{231}i}{8624}$	0	$\frac{15\sqrt{385}i}{2156}$	0	$-\frac{5\sqrt{77}i}{1232}$	0
	$\frac{5\sqrt{462}i}{1232}$	0	$\frac{\sqrt{1155}i}{4312}$	0	$-\frac{13\sqrt{2310}i}{8624}$	0	0	$\frac{5\sqrt{77}i}{1232}$	0	$-\frac{15\sqrt{385}i}{2156}$	0	$\frac{5\sqrt{231}i}{8624}$	0	$\frac{5\sqrt{11}i}{88}$
	0	$-\frac{3\sqrt{154}i}{1232}$	0	$\frac{3\sqrt{77}i}{392}$	0	$\frac{15\sqrt{770}i}{8624}$	$-\frac{3\sqrt{165}i}{176}$	0	$\frac{5\sqrt{385}i}{616}$	0	$-\frac{5\sqrt{231}i}{8624}$	0	$-\frac{5\sqrt{1155}i}{1078}$	0
	$\frac{3\sqrt{2310}i}{1232}$	0	$-\frac{3\sqrt{231}i}{616}$	0	$-\frac{23\sqrt{462}i}{8624}$	0	0	$-\frac{9\sqrt{385}i}{616}$	0	$\frac{5\sqrt{77}i}{1232}$	0	$\frac{5\sqrt{1155}i}{1078}$	0	$\frac{15\sqrt{55}i}{1232}$
	0	$\frac{3\sqrt{66}i}{176}$	0	$\frac{\sqrt{33}i}{88}$	0	$\frac{\sqrt{330}i}{1232}$	0	0	$-\frac{3\sqrt{165}i}{176}$	0	$-\frac{5\sqrt{11}i}{88}$	0	$-\frac{15\sqrt{55}i}{1232}$	0

$$\boxed{\text{x517}} \quad \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix}
0 & -\frac{\sqrt{55}}{1232} & 0 & -\frac{3\sqrt{110}}{1232} & 0 & -\frac{5\sqrt{11}}{1232} & \frac{\sqrt{462}}{1232} & 0 & -\frac{15\sqrt{22}}{1232} & 0 & -\frac{9\sqrt{330}}{1232} & 0 & -\frac{5\sqrt{66}}{1232} & 0 \\
-\frac{\sqrt{55}}{1232} & 0 & \frac{5\sqrt{22}}{1232} & 0 & \frac{15\sqrt{11}}{1232} & 0 & 0 & -\frac{23\sqrt{330}}{6160} & 0 & \frac{13\sqrt{66}}{1232} & 0 & \frac{27\sqrt{110}}{6160} & 0 & -\frac{\sqrt{2310}}{1232} \\
0 & \frac{5\sqrt{22}}{1232} & 0 & -\frac{5\sqrt{11}}{616} & 0 & -\frac{3\sqrt{110}}{1232} & \frac{9\sqrt{1155}}{3080} & 0 & \frac{3\sqrt{55}}{280} & 0 & -\frac{\sqrt{33}}{616} & 0 & \frac{27\sqrt{165}}{3080} & 0 \\
-\frac{3\sqrt{110}}{1232} & 0 & -\frac{5\sqrt{11}}{616} & 0 & \frac{5\sqrt{22}}{1232} & 0 & 0 & -\frac{27\sqrt{165}}{3080} & 0 & \frac{\sqrt{33}}{616} & 0 & -\frac{3\sqrt{55}}{280} & 0 & -\frac{9\sqrt{1155}}{3080} \\
0 & \frac{15\sqrt{11}}{1232} & 0 & \frac{5\sqrt{22}}{1232} & 0 & -\frac{\sqrt{55}}{1232} & \frac{\sqrt{2310}}{1232} & 0 & -\frac{27\sqrt{110}}{6160} & 0 & -\frac{13\sqrt{66}}{1232} & 0 & \frac{23\sqrt{330}}{6160} & 0 \\
-\frac{5\sqrt{11}}{1232} & 0 & -\frac{3\sqrt{110}}{1232} & 0 & -\frac{\sqrt{55}}{1232} & 0 & 0 & \frac{5\sqrt{66}}{1232} & 0 & \frac{9\sqrt{330}}{1232} & 0 & \frac{15\sqrt{22}}{1232} & 0 & -\frac{\sqrt{462}}{1232} \\
\frac{\sqrt{462}}{1232} & 0 & \frac{9\sqrt{1155}}{3080} & 0 & \frac{\sqrt{2310}}{1232} & 0 & 0 & \frac{15\sqrt{77}}{1232} & 0 & \frac{9\sqrt{385}}{616} & 0 & \frac{5\sqrt{231}}{1232} & 0 & 0 \\
0 & -\frac{23\sqrt{330}}{6160} & 0 & -\frac{27\sqrt{165}}{3080} & 0 & \frac{5\sqrt{66}}{1232} & \frac{15\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{33}}{154} & 0 & -\frac{9\sqrt{55}}{1232} & 0 & \frac{15\sqrt{11}}{616} & 0 \\
-\frac{15\sqrt{22}}{1232} & 0 & \frac{3\sqrt{55}}{280} & 0 & -\frac{27\sqrt{110}}{6160} & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 & \frac{\sqrt{165}}{1232} & 0 & -\frac{45\sqrt{11}}{616} & 0 & \frac{5\sqrt{231}}{1232} \\
0 & \frac{13\sqrt{66}}{1232} & 0 & \frac{\sqrt{33}}{616} & 0 & \frac{9\sqrt{330}}{1232} & \frac{9\sqrt{385}}{616} & 0 & \frac{\sqrt{165}}{1232} & 0 & \frac{15\sqrt{11}}{308} & 0 & -\frac{9\sqrt{55}}{1232} & 0 \\
-\frac{9\sqrt{330}}{1232} & 0 & -\frac{\sqrt{33}}{616} & 0 & -\frac{13\sqrt{66}}{1232} & 0 & 0 & -\frac{9\sqrt{55}}{1232} & 0 & \frac{15\sqrt{11}}{308} & 0 & \frac{\sqrt{165}}{1232} & 0 & \frac{9\sqrt{385}}{616} \\
0 & \frac{27\sqrt{110}}{6160} & 0 & -\frac{3\sqrt{55}}{280} & 0 & \frac{15\sqrt{22}}{1232} & \frac{5\sqrt{231}}{1232} & 0 & -\frac{45\sqrt{11}}{616} & 0 & \frac{\sqrt{165}}{1232} & 0 & -\frac{5\sqrt{33}}{154} & 0 \\
-\frac{5\sqrt{66}}{1232} & 0 & \frac{27\sqrt{165}}{3080} & 0 & \frac{23\sqrt{330}}{6160} & 0 & 0 & \frac{15\sqrt{11}}{616} & 0 & -\frac{9\sqrt{55}}{1232} & 0 & -\frac{5\sqrt{33}}{154} & 0 & \frac{15\sqrt{77}}{1232} \\
0 & -\frac{\sqrt{2310}}{1232} & 0 & -\frac{9\sqrt{1155}}{3080} & 0 & -\frac{\sqrt{462}}{1232} & 0 & 0 & \frac{5\sqrt{231}}{1232} & 0 & \frac{9\sqrt{385}}{616} & 0 & \frac{15\sqrt{77}}{1232} & 0
\end{bmatrix}$$

$$\begin{aligned}
& \boxed{x518} \quad \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2) \\
& = \begin{bmatrix}
0 & -\frac{\sqrt{55}i}{1232} & 0 & \frac{3\sqrt{110}i}{1232} & 0 & -\frac{5\sqrt{11}i}{1232} & -\frac{\sqrt{462}i}{1232} & 0 & -\frac{15\sqrt{22}i}{1232} & 0 & \frac{9\sqrt{330}i}{1232} & 0 & -\frac{5\sqrt{66}i}{1232} & 0 \\
\frac{\sqrt{55}i}{1232} & 0 & \frac{5\sqrt{22}i}{1232} & 0 & -\frac{15\sqrt{11}i}{1232} & 0 & 0 & \frac{23\sqrt{330}i}{6160} & 0 & \frac{13\sqrt{66}i}{1232} & 0 & -\frac{27\sqrt{110}i}{6160} & 0 & -\frac{\sqrt{2310}i}{1232} \\
0 & -\frac{5\sqrt{22}i}{1232} & 0 & -\frac{5\sqrt{11}i}{616} & 0 & \frac{3\sqrt{110}i}{1232} & \frac{9\sqrt{1155}i}{3080} & 0 & -\frac{3\sqrt{55}i}{280} & 0 & -\frac{\sqrt{33}i}{616} & 0 & -\frac{27\sqrt{165}i}{3080} & 0 \\
-\frac{3\sqrt{110}i}{1232} & 0 & \frac{5\sqrt{11}i}{616} & 0 & \frac{5\sqrt{22}i}{1232} & 0 & 0 & -\frac{27\sqrt{165}i}{3080} & 0 & -\frac{\sqrt{33}i}{616} & 0 & -\frac{3\sqrt{55}i}{280} & 0 & \frac{9\sqrt{1155}i}{3080} \\
0 & \frac{15\sqrt{11}i}{1232} & 0 & -\frac{5\sqrt{22}i}{1232} & 0 & -\frac{\sqrt{55}i}{1232} & -\frac{\sqrt{2310}i}{1232} & 0 & -\frac{27\sqrt{110}i}{6160} & 0 & \frac{13\sqrt{66}i}{1232} & 0 & \frac{23\sqrt{330}i}{6160} & 0 \\
\frac{5\sqrt{11}i}{1232} & 0 & -\frac{3\sqrt{110}i}{1232} & 0 & \frac{\sqrt{55}i}{1232} & 0 & 0 & -\frac{5\sqrt{66}i}{1232} & 0 & \frac{9\sqrt{330}i}{1232} & 0 & -\frac{15\sqrt{22}i}{1232} & 0 & -\frac{\sqrt{462}i}{1232} \\
\frac{\sqrt{462}i}{1232} & 0 & -\frac{9\sqrt{1155}i}{3080} & 0 & \frac{\sqrt{2310}i}{1232} & 0 & 0 & \frac{15\sqrt{77}i}{1232} & 0 & -\frac{9\sqrt{385}i}{616} & 0 & \frac{5\sqrt{231}i}{1232} & 0 & 0 \\
0 & -\frac{23\sqrt{330}i}{6160} & 0 & \frac{27\sqrt{165}i}{3080} & 0 & \frac{5\sqrt{66}i}{1232} & -\frac{15\sqrt{77}i}{1232} & 0 & -\frac{5\sqrt{33}i}{154} & 0 & \frac{9\sqrt{55}i}{1232} & 0 & \frac{15\sqrt{11}i}{616} & 0 \\
\frac{15\sqrt{22}i}{1232} & 0 & \frac{3\sqrt{55}i}{280} & 0 & \frac{27\sqrt{110}i}{6160} & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & \frac{\sqrt{165}i}{1232} & 0 & \frac{45\sqrt{11}i}{616} & 0 & \frac{5\sqrt{231}i}{1232} \\
0 & -\frac{13\sqrt{66}i}{1232} & 0 & \frac{\sqrt{33}i}{616} & 0 & -\frac{9\sqrt{330}i}{1232} & \frac{9\sqrt{385}i}{616} & 0 & -\frac{\sqrt{165}i}{1232} & 0 & \frac{15\sqrt{11}i}{308} & 0 & \frac{9\sqrt{55}i}{1232} & 0 \\
-\frac{9\sqrt{330}i}{1232} & 0 & \frac{\sqrt{33}i}{616} & 0 & -\frac{13\sqrt{66}i}{1232} & 0 & 0 & -\frac{9\sqrt{55}i}{1232} & 0 & -\frac{15\sqrt{11}i}{308} & 0 & \frac{\sqrt{165}i}{1232} & 0 & -\frac{9\sqrt{385}i}{616} \\
0 & \frac{27\sqrt{110}i}{6160} & 0 & \frac{3\sqrt{55}i}{280} & 0 & \frac{15\sqrt{22}i}{1232} & -\frac{5\sqrt{231}i}{1232} & 0 & -\frac{45\sqrt{11}i}{616} & 0 & -\frac{\sqrt{165}i}{1232} & 0 & -\frac{5\sqrt{33}i}{154} & 0 \\
\frac{5\sqrt{66}i}{1232} & 0 & \frac{27\sqrt{165}i}{3080} & 0 & -\frac{23\sqrt{330}i}{6160} & 0 & 0 & -\frac{15\sqrt{11}i}{616} & 0 & -\frac{9\sqrt{55}i}{1232} & 0 & \frac{5\sqrt{33}i}{154} & 0 & \frac{15\sqrt{77}i}{1232} \\
0 & \frac{\sqrt{2310}i}{1232} & 0 & -\frac{9\sqrt{1155}i}{3080} & 0 & \frac{\sqrt{462}i}{1232} & 0 & 0 & -\frac{5\sqrt{231}i}{1232} & 0 & \frac{9\sqrt{385}i}{616} & 0 & -\frac{15\sqrt{77}i}{1232} & 0
\end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x519}} \quad \mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3) = \begin{bmatrix} 0 & -\frac{\sqrt{165}}{1848} & 0 & \frac{\sqrt{330}}{1848} & 0 & \frac{5\sqrt{33}}{616} & \frac{\sqrt{154}}{616} & 0 & -\frac{5\sqrt{66}}{616} & 0 & \frac{3\sqrt{110}}{616} & 0 & \frac{15\sqrt{22}}{616} & 0 \\ -\frac{\sqrt{165}}{1848} & 0 & \frac{5\sqrt{66}}{1848} & 0 & -\frac{5\sqrt{33}}{1848} & 0 & 0 & -\frac{23\sqrt{110}}{3080} & 0 & \frac{13\sqrt{22}}{616} & 0 & -\frac{3\sqrt{330}}{3080} & 0 & \frac{3\sqrt{770}}{616} \\ 0 & \frac{5\sqrt{66}}{1848} & 0 & -\frac{5\sqrt{33}}{924} & 0 & \frac{\sqrt{330}}{1848} & -\frac{3\sqrt{385}}{1540} & 0 & \frac{\sqrt{165}}{140} & 0 & -\frac{\sqrt{11}}{308} & 0 & -\frac{9\sqrt{55}}{1540} & 0 \\ \frac{\sqrt{330}}{1848} & 0 & -\frac{5\sqrt{33}}{924} & 0 & \frac{5\sqrt{66}}{1848} & 0 & 0 & \frac{9\sqrt{55}}{1540} & 0 & \frac{\sqrt{11}}{308} & 0 & -\frac{\sqrt{165}}{140} & 0 & \frac{3\sqrt{385}}{1540} \\ 0 & -\frac{5\sqrt{33}}{1848} & 0 & \frac{5\sqrt{66}}{1848} & 0 & -\frac{\sqrt{165}}{1848} & -\frac{3\sqrt{770}}{616} & 0 & \frac{3\sqrt{330}}{3080} & 0 & -\frac{13\sqrt{22}}{616} & 0 & \frac{23\sqrt{110}}{3080} & 0 \\ \frac{5\sqrt{33}}{616} & 0 & \frac{\sqrt{330}}{1848} & 0 & -\frac{\sqrt{165}}{1848} & 0 & 0 & -\frac{15\sqrt{22}}{616} & 0 & -\frac{3\sqrt{110}}{616} & 0 & \frac{5\sqrt{66}}{616} & 0 & -\frac{\sqrt{154}}{616} \\ \frac{\sqrt{154}}{616} & 0 & -\frac{3\sqrt{385}}{1540} & 0 & -\frac{3\sqrt{770}}{616} & 0 & 0 & \frac{5\sqrt{231}}{616} & 0 & -\frac{\sqrt{1155}}{308} & 0 & -\frac{15\sqrt{77}}{616} & 0 & 0 \\ 0 & -\frac{23\sqrt{110}}{3080} & 0 & \frac{9\sqrt{55}}{1540} & 0 & -\frac{15\sqrt{22}}{616} & \frac{5\sqrt{231}}{616} & 0 & -\frac{5\sqrt{11}}{77} & 0 & \frac{\sqrt{165}}{616} & 0 & -\frac{15\sqrt{33}}{308} & 0 \\ -\frac{5\sqrt{66}}{616} & 0 & \frac{\sqrt{165}}{140} & 0 & \frac{3\sqrt{330}}{3080} & 0 & 0 & -\frac{5\sqrt{11}}{77} & 0 & \frac{\sqrt{55}}{616} & 0 & \frac{5\sqrt{33}}{308} & 0 & -\frac{15\sqrt{77}}{616} \\ 0 & \frac{13\sqrt{22}}{616} & 0 & \frac{\sqrt{11}}{308} & 0 & -\frac{3\sqrt{110}}{616} & -\frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{55}}{616} & 0 & \frac{5\sqrt{33}}{154} & 0 & \frac{\sqrt{165}}{616} & 0 \\ \frac{3\sqrt{110}}{616} & 0 & -\frac{\sqrt{11}}{308} & 0 & -\frac{13\sqrt{22}}{616} & 0 & 0 & \frac{\sqrt{165}}{616} & 0 & \frac{5\sqrt{33}}{154} & 0 & \frac{\sqrt{55}}{616} & 0 & -\frac{\sqrt{1155}}{308} \\ 0 & -\frac{3\sqrt{330}}{3080} & 0 & -\frac{\sqrt{165}}{140} & 0 & \frac{5\sqrt{66}}{616} & -\frac{15\sqrt{77}}{616} & 0 & \frac{5\sqrt{33}}{308} & 0 & \frac{\sqrt{55}}{616} & 0 & -\frac{5\sqrt{11}}{77} & 0 \\ \frac{15\sqrt{22}}{616} & 0 & -\frac{9\sqrt{55}}{1540} & 0 & \frac{23\sqrt{110}}{3080} & 0 & 0 & -\frac{15\sqrt{33}}{308} & 0 & \frac{\sqrt{165}}{616} & 0 & -\frac{5\sqrt{11}}{77} & 0 & \frac{5\sqrt{231}}{616} \\ 0 & \frac{3\sqrt{770}}{616} & 0 & \frac{3\sqrt{385}}{1540} & 0 & -\frac{\sqrt{154}}{616} & 0 & 0 & -\frac{15\sqrt{77}}{616} & 0 & -\frac{\sqrt{1155}}{308} & 0 & \frac{5\sqrt{231}}{616} & 0 \end{bmatrix}$$

$$\boxed{\text{x520}} \quad \mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3) = \begin{bmatrix} 0 & -\frac{\sqrt{165}i}{1848} & 0 & -\frac{\sqrt{330}i}{1848} & 0 & \frac{5\sqrt{33}i}{616} & -\frac{\sqrt{154}i}{616} & 0 & -\frac{5\sqrt{66}i}{616} & 0 & -\frac{3\sqrt{110}i}{616} & 0 & \frac{15\sqrt{22}i}{616} & 0 \\ \frac{\sqrt{165}i}{1848} & 0 & \frac{5\sqrt{66}i}{1848} & 0 & \frac{5\sqrt{33}i}{1848} & 0 & 0 & \frac{23\sqrt{110}i}{3080} & 0 & \frac{13\sqrt{22}i}{616} & 0 & \frac{3\sqrt{330}i}{3080} & 0 & \frac{3\sqrt{770}i}{616} \\ 0 & -\frac{5\sqrt{66}i}{1848} & 0 & -\frac{5\sqrt{33}i}{924} & 0 & -\frac{\sqrt{330}i}{1848} & -\frac{3\sqrt{385}i}{1540} & 0 & -\frac{\sqrt{165}i}{140} & 0 & -\frac{\sqrt{11}i}{308} & 0 & \frac{9\sqrt{55}i}{1540} & 0 \\ \frac{\sqrt{330}i}{1848} & 0 & \frac{5\sqrt{33}i}{924} & 0 & \frac{5\sqrt{66}i}{1848} & 0 & 0 & \frac{9\sqrt{55}i}{1540} & 0 & -\frac{\sqrt{11}i}{308} & 0 & -\frac{\sqrt{165}i}{140} & 0 & -\frac{3\sqrt{385}i}{1540} \\ 0 & -\frac{5\sqrt{33}i}{1848} & 0 & -\frac{5\sqrt{66}i}{1848} & 0 & -\frac{\sqrt{165}i}{1848} & \frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{330}i}{3080} & 0 & \frac{13\sqrt{22}i}{616} & 0 & \frac{23\sqrt{110}i}{3080} & 0 \\ -\frac{5\sqrt{33}i}{616} & 0 & \frac{\sqrt{330}i}{1848} & 0 & \frac{\sqrt{165}i}{1848} & 0 & 0 & \frac{15\sqrt{22}i}{616} & 0 & -\frac{3\sqrt{110}i}{616} & 0 & -\frac{5\sqrt{66}i}{616} & 0 & -\frac{\sqrt{154}i}{616} \\ \frac{\sqrt{154}i}{616} & 0 & \frac{3\sqrt{385}i}{1540} & 0 & -\frac{3\sqrt{770}i}{616} & 0 & 0 & \frac{5\sqrt{231}i}{616} & 0 & \frac{\sqrt{1155}i}{308} & 0 & -\frac{15\sqrt{77}i}{616} & 0 & 0 \\ 0 & -\frac{23\sqrt{110}i}{3080} & 0 & -\frac{9\sqrt{55}i}{1540} & 0 & -\frac{15\sqrt{22}i}{616} & -\frac{5\sqrt{231}i}{616} & 0 & -\frac{5\sqrt{11}i}{77} & 0 & -\frac{\sqrt{165}i}{616} & 0 & -\frac{15\sqrt{33}i}{308} & 0 \\ \frac{5\sqrt{66}i}{616} & 0 & \frac{\sqrt{165}i}{140} & 0 & -\frac{3\sqrt{330}i}{3080} & 0 & 0 & \frac{5\sqrt{11}i}{77} & 0 & \frac{\sqrt{55}i}{616} & 0 & -\frac{5\sqrt{33}i}{308} & 0 & -\frac{15\sqrt{77}i}{616} \\ 0 & -\frac{13\sqrt{22}i}{616} & 0 & \frac{\sqrt{11}i}{308} & 0 & \frac{3\sqrt{110}i}{616} & -\frac{\sqrt{1155}i}{308} & 0 & -\frac{\sqrt{55}i}{616} & 0 & \frac{5\sqrt{33}i}{154} & 0 & -\frac{\sqrt{165}i}{616} & 0 \\ \frac{3\sqrt{110}i}{616} & 0 & \frac{\sqrt{11}i}{308} & 0 & -\frac{13\sqrt{22}i}{616} & 0 & 0 & \frac{\sqrt{165}i}{616} & 0 & -\frac{5\sqrt{33}i}{154} & 0 & \frac{\sqrt{55}i}{616} & 0 & \frac{\sqrt{1155}i}{308} \\ 0 & -\frac{3\sqrt{330}i}{3080} & 0 & \frac{\sqrt{165}i}{140} & 0 & \frac{5\sqrt{66}i}{616} & \frac{15\sqrt{77}i}{616} & 0 & \frac{5\sqrt{33}i}{308} & 0 & -\frac{\sqrt{55}i}{616} & 0 & -\frac{5\sqrt{11}i}{77} & 0 \\ -\frac{15\sqrt{22}i}{616} & 0 & -\frac{9\sqrt{55}i}{1540} & 0 & -\frac{23\sqrt{110}i}{3080} & 0 & 0 & \frac{15\sqrt{33}i}{308} & 0 & \frac{\sqrt{165}i}{616} & 0 & \frac{5\sqrt{11}i}{77} & 0 & \frac{5\sqrt{231}i}{616} \\ 0 & -\frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{385}i}{1540} & 0 & \frac{\sqrt{154}i}{616} & 0 & 0 & \frac{15\sqrt{77}i}{616} & 0 & -\frac{\sqrt{1155}i}{308} & 0 & -\frac{5\sqrt{231}i}{616} & 0 \end{bmatrix}$$

x521

x522

x523

x524

x525

x526

x527

x528

$$\boxed{\text{x529}} \quad \mathbb{M}_5^{(1,1;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}i}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}i}{2002} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x530}} \quad \mathbb{M}_1^{(1,1;a)}(A_{2g}) = \begin{bmatrix} \frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{294} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 \end{bmatrix}$$

$$\boxed{\text{x531}} \quad \mathbb{M}_3^{(1,1;a)}(A_{2g}) = \begin{bmatrix}
-\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{539} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77}
\end{bmatrix}$$

$$\begin{aligned}
& \boxed{x532} \quad \mathbb{M}_5^{(1,1;a)}(A_{2g}, 1) \\
& = \begin{bmatrix}
\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1001}}{14014} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1001}}{14014} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{2002}
\end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x533}} \quad \mathbb{M}_5^{(1,1;a)}(A_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{2002} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x534}} \quad \mathbb{M}_3^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{5\sqrt{462i}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154i}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310i}}{294} & 0 & 0 & -\frac{\sqrt{22i}}{28} & 0 & 0 & 0 & -\frac{\sqrt{770i}}{196} & 0 & 0 & 0 \\ -\frac{5\sqrt{462i}}{294} & 0 & 0 & 0 & -\frac{\sqrt{2310i}}{294} & 0 & 0 & -\frac{\sqrt{77i}}{196} & 0 & 0 & 0 & -\frac{\sqrt{231i}}{196} & 0 & 0 \\ 0 & -\frac{\sqrt{2310i}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{462i}}{294} & 0 & 0 & \frac{\sqrt{231i}}{196} & 0 & 0 & 0 & \frac{\sqrt{77i}}{196} & 0 \\ 0 & 0 & \frac{\sqrt{2310i}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770i}}{196} & 0 & 0 & 0 & \frac{\sqrt{22i}}{28} \\ 0 & 0 & 0 & \frac{5\sqrt{462i}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154i}}{98} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{22i}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55i}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{77i}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231i}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{231i}}{196} & 0 & 0 & \frac{\sqrt{55i}}{77} & 0 & 0 & 0 & -\frac{2\sqrt{77i}}{539} & 0 & 0 & 0 \\ \frac{\sqrt{154i}}{98} & 0 & 0 & 0 & -\frac{\sqrt{770i}}{196} & 0 & 0 & \frac{3\sqrt{231i}}{539} & 0 & 0 & 0 & \frac{2\sqrt{77i}}{539} & 0 & 0 \\ 0 & \frac{\sqrt{770i}}{196} & 0 & 0 & 0 & -\frac{\sqrt{154i}}{98} & 0 & 0 & \frac{2\sqrt{77i}}{539} & 0 & 0 & 0 & \frac{3\sqrt{231i}}{539} & 0 \\ 0 & 0 & \frac{\sqrt{231i}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77i}}{539} & 0 & 0 & 0 & \frac{\sqrt{55i}}{77} \\ 0 & 0 & 0 & -\frac{\sqrt{77i}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231i}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22i}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55i}}{77} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x535}} \quad \mathbb{M}_5^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{858}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}i}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{4290}i}{154} & 0 & 0 & -\frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}i}{462} & 0 & 0 & 0 \\ -\frac{\sqrt{858}i}{154} & 0 & 0 & 0 & \frac{\sqrt{4290}i}{154} & 0 & 0 & \frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & \frac{\sqrt{429}i}{462} & 0 & 0 \\ 0 & \frac{\sqrt{4290}i}{154} & 0 & 0 & 0 & -\frac{\sqrt{858}i}{154} & 0 & 0 & -\frac{\sqrt{429}i}{462} & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{231} & 0 \\ 0 & 0 & -\frac{\sqrt{4290}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{462} & 0 & 0 & 0 & \frac{\sqrt{2002}i}{924} \\ 0 & 0 & 0 & \frac{\sqrt{858}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{286}i}{924} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{429}i}{462} & 0 & 0 & \frac{\sqrt{5005}i}{2002} & 0 & 0 & 0 & \frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 \\ \frac{5\sqrt{286}i}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}i}{462} & 0 & 0 & -\frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{1001} & 0 & 0 \\ 0 & -\frac{\sqrt{1430}i}{462} & 0 & 0 & 0 & -\frac{5\sqrt{286}i}{924} & 0 & 0 & -\frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 & -\frac{3\sqrt{429}i}{2002} & 0 \\ 0 & 0 & -\frac{\sqrt{429}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 & \frac{\sqrt{5005}i}{2002} \\ 0 & 0 & 0 & \frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{2002} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x536}} \quad \mathbb{M}_3^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 \\ -\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & \frac{\sqrt{2310}}{294} & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 \\ 0 & 0 & \frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & -\frac{\sqrt{22}}{28} \\ 0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & \frac{\sqrt{55}}{77} & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 \\ \frac{\sqrt{154}}{98} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & \frac{3\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 \\ 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{231}}{539} & 0 \\ 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{\sqrt{55}}{77} \\ 0 & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{77} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x537}} \quad \mathbb{M}_5^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{858}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{4290}}{154} & 0 & 0 & \frac{\sqrt{2002}}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & 0 \\ \frac{\sqrt{858}}{154} & 0 & 0 & 0 & \frac{\sqrt{4290}}{154} & 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 & 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 \\ 0 & -\frac{\sqrt{4290}}{154} & 0 & 0 & 0 & -\frac{\sqrt{858}}{154} & 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 \\ 0 & 0 & \frac{\sqrt{4290}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & 0 & \frac{\sqrt{2002}}{924} \\ 0 & 0 & 0 & -\frac{\sqrt{858}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}}{924} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2002}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 & -\frac{\sqrt{5005}}{2002} & 0 & 0 & 0 & \frac{2\sqrt{143}}{1001} & 0 & 0 & 0 \\ -\frac{5\sqrt{286}}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & \frac{3\sqrt{429}}{2002} & 0 & 0 & 0 & -\frac{2\sqrt{143}}{1001} & 0 & 0 \\ 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & 0 & -\frac{5\sqrt{286}}{924} & 0 & 0 & \frac{2\sqrt{143}}{1001} & 0 & 0 & 0 & -\frac{3\sqrt{429}}{2002} & 0 \\ 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{143}}{1001} & 0 & 0 & 0 & \frac{\sqrt{5005}}{2002} \\ 0 & 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{429}}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2002}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{2002} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x538}} \quad \mathbb{M}_{1,1}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{21}}{49} & 0 & \frac{4\sqrt{210}}{245} & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{392} & 0 & -\frac{3\sqrt{70}}{392} & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{210}}{245} & 0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{196} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 \\ 0 & 0 & \frac{6\sqrt{105}}{245} & 0 & \frac{4\sqrt{210}}{245} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & -\frac{5\sqrt{21}}{196} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{210}}{245} & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{392} & 0 & -\frac{15\sqrt{14}}{392} & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{3\sqrt{10}}{56} \\ \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{392} & 0 & \frac{5\sqrt{21}}{196} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{70}}{392} & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & \frac{3\sqrt{70}}{392} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{21}}{196} & 0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & -\frac{\sqrt{35}}{49} & 0 \\ 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{15}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x539}} \quad \mathbb{M}_{1,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{21}i}{49} & 0 & \frac{4\sqrt{210}i}{245} & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}i}{392} & 0 & -\frac{3\sqrt{70}i}{392} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{210}i}{245} & 0 & \frac{6\sqrt{105}i}{245} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{196} & 0 & -\frac{3\sqrt{35}i}{196} & 0 & 0 & 0 \\ 0 & 0 & -\frac{6\sqrt{105}i}{245} & 0 & \frac{4\sqrt{210}i}{245} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & -\frac{5\sqrt{21}i}{196} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{210}i}{245} & 0 & \frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{392} & 0 & -\frac{15\sqrt{14}i}{392} & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{392} & 0 & -\frac{3\sqrt{10}i}{56} \\ \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{392} & 0 & \frac{5\sqrt{21}i}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{70}i}{392} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{35}i}{196} & 0 & \frac{3\sqrt{70}i}{392} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{21}i}{196} & 0 & \frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{\sqrt{35}i}{49} & 0 \\ 0 & 0 & 0 & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & -\frac{\sqrt{15}i}{42} \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 \end{bmatrix}$$

$$\boxed{\text{x540}} \quad \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{385}}{98} & 0 & -\frac{5\sqrt{770}}{588} & 0 & 0 & \frac{\sqrt{66}}{112} & 0 & -\frac{3\sqrt{154}}{392} & 0 & \frac{\sqrt{2310}}{784} & 0 & 0 & 0 \\ \frac{\sqrt{385}}{98} & 0 & -\frac{\sqrt{154}}{196} & 0 & -\frac{5\sqrt{77}}{147} & 0 & 0 & -\frac{\sqrt{2310}}{784} & 0 & -\frac{\sqrt{462}}{392} & 0 & \frac{3\sqrt{770}}{784} & 0 & 0 \\ 0 & -\frac{\sqrt{154}}{196} & 0 & -\frac{\sqrt{77}}{49} & 0 & -\frac{5\sqrt{770}}{588} & -\frac{\sqrt{165}}{112} & 0 & -\frac{3\sqrt{385}}{784} & 0 & \frac{\sqrt{231}}{784} & 0 & \frac{3\sqrt{1155}}{784} & 0 \\ -\frac{5\sqrt{770}}{588} & 0 & -\frac{\sqrt{77}}{49} & 0 & -\frac{\sqrt{154}}{196} & 0 & 0 & -\frac{3\sqrt{1155}}{784} & 0 & -\frac{\sqrt{231}}{784} & 0 & \frac{3\sqrt{385}}{784} & 0 & \frac{\sqrt{165}}{112} \\ 0 & -\frac{5\sqrt{77}}{147} & 0 & -\frac{\sqrt{154}}{196} & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & -\frac{3\sqrt{770}}{784} & 0 & \frac{\sqrt{462}}{392} & 0 & \frac{\sqrt{2310}}{784} & 0 \\ 0 & 0 & -\frac{5\sqrt{770}}{588} & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{784} & 0 & \frac{3\sqrt{154}}{392} & 0 & -\frac{\sqrt{66}}{112} \\ \frac{\sqrt{66}}{112} & 0 & -\frac{\sqrt{165}}{112} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{784} & 0 & -\frac{3\sqrt{1155}}{784} & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 & -\frac{\sqrt{231}}{1078} & 0 & \frac{2\sqrt{385}}{539} & 0 & 0 & 0 \\ -\frac{3\sqrt{154}}{392} & 0 & -\frac{3\sqrt{385}}{784} & 0 & -\frac{3\sqrt{770}}{784} & 0 & 0 & -\frac{\sqrt{231}}{1078} & 0 & \frac{\sqrt{1155}}{1078} & 0 & \frac{5\sqrt{77}}{539} & 0 & 0 \\ 0 & -\frac{\sqrt{462}}{392} & 0 & -\frac{\sqrt{231}}{784} & 0 & -\frac{\sqrt{2310}}{784} & \frac{\sqrt{55}}{154} & 0 & \frac{\sqrt{1155}}{1078} & 0 & \frac{3\sqrt{77}}{539} & 0 & \frac{2\sqrt{385}}{539} & 0 \\ \frac{\sqrt{2310}}{784} & 0 & \frac{\sqrt{231}}{784} & 0 & \frac{\sqrt{462}}{392} & 0 & 0 & \frac{2\sqrt{385}}{539} & 0 & \frac{3\sqrt{77}}{539} & 0 & \frac{\sqrt{1155}}{1078} & 0 & \frac{\sqrt{55}}{154} \\ 0 & \frac{3\sqrt{770}}{784} & 0 & \frac{3\sqrt{385}}{784} & 0 & \frac{3\sqrt{154}}{392} & 0 & 0 & \frac{5\sqrt{77}}{539} & 0 & \frac{\sqrt{1155}}{1078} & 0 & -\frac{\sqrt{231}}{1078} & 0 \\ 0 & 0 & \frac{3\sqrt{1155}}{784} & 0 & \frac{\sqrt{2310}}{784} & 0 & 0 & 0 & \frac{2\sqrt{385}}{539} & 0 & -\frac{\sqrt{231}}{1078} & 0 & -\frac{3\sqrt{11}}{154} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{165}}{112} & 0 & -\frac{\sqrt{66}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & -\frac{3\sqrt{11}}{154} & 0 \end{bmatrix}$$

$$\boxed{\text{x541}} \quad \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{385}i}{98} & 0 & \frac{5\sqrt{770}i}{588} & 0 & 0 & -\frac{\sqrt{66}i}{112} & 0 & -\frac{3\sqrt{154}i}{392} & 0 & -\frac{\sqrt{2310}i}{784} & 0 & 0 & 0 \\ -\frac{\sqrt{385}i}{98} & 0 & -\frac{\sqrt{154}i}{196} & 0 & \frac{5\sqrt{77}i}{147} & 0 & 0 & \frac{\sqrt{2310}i}{784} & 0 & -\frac{\sqrt{462}i}{392} & 0 & -\frac{3\sqrt{770}i}{784} & 0 & 0 \\ 0 & \frac{\sqrt{154}i}{196} & 0 & -\frac{\sqrt{77}i}{49} & 0 & \frac{5\sqrt{770}i}{588} & -\frac{\sqrt{165}i}{112} & 0 & \frac{3\sqrt{385}i}{784} & 0 & \frac{\sqrt{231}i}{784} & 0 & -\frac{3\sqrt{1155}i}{784} & 0 \\ -\frac{5\sqrt{770}i}{588} & 0 & \frac{\sqrt{77}i}{49} & 0 & -\frac{\sqrt{154}i}{196} & 0 & 0 & -\frac{3\sqrt{1155}i}{784} & 0 & \frac{\sqrt{231}i}{784} & 0 & \frac{3\sqrt{385}i}{784} & 0 & -\frac{\sqrt{165}i}{112} \\ 0 & -\frac{5\sqrt{77}i}{147} & 0 & \frac{\sqrt{154}i}{196} & 0 & \frac{\sqrt{385}i}{98} & 0 & 0 & -\frac{3\sqrt{770}i}{784} & 0 & -\frac{\sqrt{462}i}{392} & 0 & \frac{\sqrt{2310}i}{784} & 0 \\ 0 & 0 & -\frac{5\sqrt{770}i}{588} & 0 & -\frac{\sqrt{385}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{784} & 0 & -\frac{3\sqrt{154}i}{392} & 0 & -\frac{\sqrt{66}i}{112} \\ \frac{\sqrt{66}i}{112} & 0 & \frac{\sqrt{165}i}{112} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{154} & 0 & -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}i}{784} & 0 & \frac{3\sqrt{1155}i}{784} & 0 & 0 & \frac{3\sqrt{11}i}{154} & 0 & -\frac{\sqrt{231}i}{1078} & 0 & -\frac{2\sqrt{385}i}{539} & 0 & 0 & 0 \\ \frac{3\sqrt{154}i}{392} & 0 & -\frac{3\sqrt{385}i}{784} & 0 & \frac{3\sqrt{770}i}{784} & 0 & 0 & \frac{\sqrt{231}i}{1078} & 0 & \frac{\sqrt{1155}i}{1078} & 0 & -\frac{5\sqrt{77}i}{539} & 0 & 0 \\ 0 & \frac{\sqrt{462}i}{392} & 0 & -\frac{\sqrt{231}i}{784} & 0 & \frac{\sqrt{2310}i}{784} & \frac{\sqrt{55}i}{154} & 0 & -\frac{\sqrt{1155}i}{1078} & 0 & \frac{3\sqrt{77}i}{539} & 0 & -\frac{2\sqrt{385}i}{539} & 0 \\ \frac{\sqrt{2310}i}{784} & 0 & -\frac{\sqrt{231}i}{784} & 0 & \frac{\sqrt{462}i}{392} & 0 & 0 & \frac{2\sqrt{385}i}{539} & 0 & -\frac{3\sqrt{77}i}{539} & 0 & \frac{\sqrt{1155}i}{1078} & 0 & -\frac{\sqrt{55}i}{154} \\ 0 & \frac{3\sqrt{770}i}{784} & 0 & -\frac{3\sqrt{385}i}{784} & 0 & \frac{3\sqrt{154}i}{392} & 0 & 0 & \frac{5\sqrt{77}i}{539} & 0 & -\frac{\sqrt{1155}i}{1078} & 0 & -\frac{\sqrt{231}i}{1078} & 0 \\ 0 & 0 & \frac{3\sqrt{1155}i}{784} & 0 & -\frac{\sqrt{2310}i}{784} & 0 & 0 & 0 & 0 & \frac{2\sqrt{385}i}{539} & 0 & \frac{\sqrt{231}i}{1078} & 0 & -\frac{3\sqrt{11}i}{154} \\ 0 & 0 & 0 & \frac{\sqrt{165}i}{112} & 0 & \frac{\sqrt{66}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{154} & 0 & \frac{3\sqrt{11}i}{154} & 0 \end{bmatrix}$$

$$\boxed{\text{x542}} \quad \mathbb{M}_{3,1}^{(1,1;a)}(E_g, 2) = \begin{bmatrix} 0 & \frac{5\sqrt{231}}{294} & 0 & \frac{5\sqrt{462}}{588} & 0 & 0 & \frac{\sqrt{110}}{112} & 0 & -\frac{\sqrt{2310}}{392} & 0 & -\frac{3\sqrt{154}}{784} & 0 & 0 & 0 \\ \frac{5\sqrt{231}}{294} & 0 & -\frac{\sqrt{2310}}{588} & 0 & \frac{\sqrt{1155}}{147} & 0 & 0 & -\frac{5\sqrt{154}}{784} & 0 & -\frac{\sqrt{770}}{392} & 0 & -\frac{3\sqrt{462}}{784} & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{588} & 0 & -\frac{\sqrt{1155}}{147} & 0 & \frac{5\sqrt{462}}{588} & \frac{3\sqrt{11}}{112} & 0 & -\frac{5\sqrt{231}}{784} & 0 & \frac{\sqrt{385}}{784} & 0 & -\frac{9\sqrt{77}}{784} & 0 \\ \frac{5\sqrt{462}}{588} & 0 & -\frac{\sqrt{1155}}{147} & 0 & -\frac{\sqrt{2310}}{588} & 0 & 0 & \frac{9\sqrt{77}}{784} & 0 & -\frac{\sqrt{385}}{784} & 0 & \frac{5\sqrt{231}}{784} & 0 & -\frac{3\sqrt{11}}{112} \\ 0 & \frac{\sqrt{1155}}{147} & 0 & -\frac{\sqrt{2310}}{588} & 0 & \frac{5\sqrt{231}}{294} & 0 & 0 & \frac{3\sqrt{462}}{784} & 0 & \frac{\sqrt{770}}{392} & 0 & \frac{5\sqrt{154}}{784} & 0 \\ 0 & 0 & \frac{5\sqrt{462}}{588} & 0 & \frac{5\sqrt{231}}{294} & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{784} & 0 & \frac{\sqrt{2310}}{392} & 0 & -\frac{\sqrt{110}}{112} \\ \frac{\sqrt{110}}{112} & 0 & \frac{3\sqrt{11}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{154} & 0 & -\frac{\sqrt{33}}{154} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{154}}{784} & 0 & \frac{9\sqrt{77}}{784} & 0 & 0 & -\frac{\sqrt{165}}{154} & 0 & -\frac{\sqrt{385}}{1078} & 0 & -\frac{2\sqrt{231}}{539} & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{392} & 0 & -\frac{5\sqrt{231}}{784} & 0 & \frac{3\sqrt{462}}{784} & 0 & 0 & -\frac{\sqrt{385}}{1078} & 0 & \frac{5\sqrt{77}}{1078} & 0 & -\frac{\sqrt{1155}}{539} & 0 & 0 \\ 0 & -\frac{\sqrt{770}}{392} & 0 & -\frac{\sqrt{385}}{784} & 0 & \frac{3\sqrt{154}}{784} & -\frac{\sqrt{33}}{154} & 0 & \frac{5\sqrt{77}}{1078} & 0 & \frac{\sqrt{1155}}{539} & 0 & -\frac{2\sqrt{231}}{539} & 0 \\ -\frac{3\sqrt{154}}{784} & 0 & \frac{\sqrt{385}}{784} & 0 & \frac{\sqrt{770}}{392} & 0 & 0 & -\frac{2\sqrt{231}}{539} & 0 & \frac{\sqrt{1155}}{539} & 0 & \frac{5\sqrt{77}}{1078} & 0 & -\frac{\sqrt{33}}{154} \\ 0 & -\frac{3\sqrt{462}}{784} & 0 & \frac{5\sqrt{231}}{784} & 0 & \frac{\sqrt{2310}}{392} & 0 & 0 & -\frac{\sqrt{1155}}{539} & 0 & \frac{5\sqrt{77}}{1078} & 0 & -\frac{\sqrt{385}}{1078} & 0 \\ 0 & 0 & -\frac{9\sqrt{77}}{784} & 0 & \frac{5\sqrt{154}}{784} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}}{539} & 0 & -\frac{\sqrt{385}}{1078} & 0 & -\frac{\sqrt{165}}{154} \\ 0 & 0 & 0 & -\frac{3\sqrt{11}}{112} & 0 & -\frac{\sqrt{110}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{154} & 0 & -\frac{\sqrt{165}}{154} & 0 \end{bmatrix}$$

$$\boxed{x543} \quad \mathbb{M}_{3,2}^{(1,1;a)}(E_g, 2) = \begin{bmatrix}
0 & \frac{5\sqrt{231}i}{294} & 0 & -\frac{5\sqrt{462}i}{588} & 0 & 0 & -\frac{\sqrt{110}i}{112} & 0 & -\frac{\sqrt{2310}i}{392} & 0 & \frac{3\sqrt{154}i}{784} & 0 & 0 & 0 \\
-\frac{5\sqrt{231}i}{294} & 0 & -\frac{\sqrt{2310}i}{588} & 0 & -\frac{\sqrt{1155}i}{147} & 0 & 0 & \frac{5\sqrt{154}i}{784} & 0 & -\frac{\sqrt{770}i}{392} & 0 & \frac{3\sqrt{462}i}{784} & 0 & 0 \\
0 & \frac{\sqrt{2310}i}{588} & 0 & -\frac{\sqrt{1155}i}{147} & 0 & -\frac{5\sqrt{462}i}{588} & \frac{3\sqrt{11}i}{112} & 0 & \frac{5\sqrt{231}i}{784} & 0 & \frac{\sqrt{385}i}{784} & 0 & \frac{9\sqrt{77}i}{784} & 0 \\
\frac{5\sqrt{462}i}{588} & 0 & \frac{\sqrt{1155}i}{147} & 0 & -\frac{\sqrt{2310}i}{588} & 0 & 0 & \frac{9\sqrt{77}i}{784} & 0 & \frac{\sqrt{385}i}{784} & 0 & \frac{5\sqrt{231}i}{784} & 0 & \frac{3\sqrt{11}i}{112} \\
0 & \frac{\sqrt{1155}i}{147} & 0 & \frac{\sqrt{2310}i}{588} & 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & \frac{3\sqrt{462}i}{784} & 0 & -\frac{\sqrt{770}i}{392} & 0 & \frac{5\sqrt{154}i}{784} & 0 \\
0 & 0 & \frac{5\sqrt{462}i}{588} & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}i}{784} & 0 & -\frac{\sqrt{2310}i}{392} & 0 & -\frac{\sqrt{110}i}{112} \\
\frac{\sqrt{110}i}{112} & 0 & -\frac{3\sqrt{11}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{154} & 0 & \frac{\sqrt{33}i}{154} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{154}i}{784} & 0 & -\frac{9\sqrt{77}i}{784} & 0 & 0 & \frac{\sqrt{165}i}{154} & 0 & -\frac{\sqrt{385}i}{1078} & 0 & \frac{2\sqrt{231}i}{539} & 0 & 0 & 0 \\
\frac{\sqrt{2310}i}{392} & 0 & -\frac{5\sqrt{231}i}{784} & 0 & -\frac{3\sqrt{462}i}{784} & 0 & 0 & \frac{\sqrt{385}i}{1078} & 0 & \frac{5\sqrt{77}i}{1078} & 0 & \frac{\sqrt{1155}i}{539} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{392} & 0 & -\frac{\sqrt{385}i}{784} & 0 & -\frac{3\sqrt{154}i}{784} & -\frac{\sqrt{33}i}{154} & 0 & -\frac{5\sqrt{77}i}{1078} & 0 & \frac{\sqrt{1155}i}{539} & 0 & \frac{2\sqrt{231}i}{539} & 0 \\
-\frac{3\sqrt{154}i}{784} & 0 & -\frac{\sqrt{385}i}{784} & 0 & \frac{\sqrt{770}i}{392} & 0 & 0 & -\frac{2\sqrt{231}i}{539} & 0 & -\frac{\sqrt{1155}i}{539} & 0 & \frac{5\sqrt{77}i}{1078} & 0 & \frac{\sqrt{33}i}{154} \\
0 & -\frac{3\sqrt{462}i}{784} & 0 & -\frac{5\sqrt{231}i}{784} & 0 & \frac{\sqrt{2310}i}{392} & 0 & 0 & -\frac{\sqrt{1155}i}{539} & 0 & -\frac{5\sqrt{77}i}{1078} & 0 & -\frac{\sqrt{385}i}{1078} & 0 \\
0 & 0 & -\frac{9\sqrt{77}i}{784} & 0 & -\frac{5\sqrt{154}i}{784} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{539} & 0 & \frac{\sqrt{385}i}{1078} & 0 & -\frac{\sqrt{165}i}{154} \\
0 & 0 & 0 & -\frac{3\sqrt{11}i}{112} & 0 & \frac{\sqrt{110}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{154} & 0 & \frac{\sqrt{165}i}{154} & 0
\end{bmatrix}$$

$$\begin{array}{c}
\boxed{x544} \quad \mathbb{M}_{5,1}^{(1,1;a)}(E_g, 1) \\
= \left[\begin{array}{cccccccccccccccc}
0 & \frac{3\sqrt{5005}}{4312} & 0 & -\frac{\sqrt{10010}}{616} & 0 & \frac{9\sqrt{1001}}{616} & \frac{5\sqrt{858}}{14784} & 0 & -\frac{25\sqrt{2002}}{34496} & 0 & \frac{5\sqrt{30030}}{14784} & 0 & -\frac{5\sqrt{6006}}{4928} & 0 \\
\frac{3\sqrt{5005}}{4312} & 0 & -\frac{15\sqrt{2002}}{4312} & 0 & \frac{5\sqrt{1001}}{616} & 0 & 0 & -\frac{23\sqrt{30030}}{103488} & 0 & \frac{65\sqrt{6006}}{103488} & 0 & -\frac{\sqrt{10010}}{4928} & 0 & -\frac{\sqrt{4290}}{704} \\
0 & -\frac{15\sqrt{2002}}{4312} & 0 & \frac{15\sqrt{1001}}{2156} & 0 & -\frac{\sqrt{10010}}{616} & -\frac{\sqrt{2145}}{1056} & 0 & \frac{\sqrt{5005}}{1568} & 0 & -\frac{5\sqrt{3003}}{51744} & 0 & -\frac{\sqrt{15015}}{2464} & 0 \\
-\frac{\sqrt{10010}}{616} & 0 & \frac{15\sqrt{1001}}{2156} & 0 & -\frac{15\sqrt{2002}}{4312} & 0 & 0 & \frac{\sqrt{15015}}{2464} & 0 & \frac{5\sqrt{3003}}{51744} & 0 & -\frac{\sqrt{5005}}{1568} & 0 & \frac{\sqrt{2145}}{1056} \\
0 & \frac{5\sqrt{1001}}{616} & 0 & -\frac{15\sqrt{2002}}{4312} & 0 & \frac{3\sqrt{5005}}{4312} & \frac{\sqrt{4290}}{704} & 0 & \frac{\sqrt{10010}}{4928} & 0 & -\frac{65\sqrt{6006}}{103488} & 0 & \frac{23\sqrt{30030}}{103488} & 0 \\
\frac{9\sqrt{1001}}{616} & 0 & -\frac{\sqrt{10010}}{616} & 0 & \frac{3\sqrt{5005}}{4312} & 0 & 0 & \frac{5\sqrt{6006}}{4928} & 0 & -\frac{5\sqrt{30030}}{14784} & 0 & \frac{25\sqrt{2002}}{34496} & 0 & -\frac{5\sqrt{858}}{14784} \\
\frac{5\sqrt{858}}{14784} & 0 & -\frac{\sqrt{2145}}{1056} & 0 & \frac{\sqrt{4290}}{704} & 0 & 0 & -\frac{15\sqrt{143}}{16016} & 0 & \frac{\sqrt{715}}{1144} & 0 & -\frac{3\sqrt{429}}{2288} & 0 & 0 \\
0 & -\frac{23\sqrt{30030}}{103488} & 0 & \frac{\sqrt{15015}}{2464} & 0 & \frac{5\sqrt{6006}}{4928} & -\frac{15\sqrt{143}}{16016} & 0 & \frac{5\sqrt{3003}}{14014} & 0 & -\frac{\sqrt{5005}}{16016} & 0 & -\frac{9\sqrt{1001}}{8008} & 0 \\
-\frac{25\sqrt{2002}}{34496} & 0 & \frac{\sqrt{5005}}{1568} & 0 & \frac{\sqrt{10010}}{4928} & 0 & 0 & \frac{5\sqrt{3003}}{14014} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & -\frac{5\sqrt{1001}}{8008} & 0 & -\frac{3\sqrt{429}}{2288} \\
0 & \frac{65\sqrt{6006}}{103488} & 0 & \frac{5\sqrt{3003}}{51744} & 0 & -\frac{5\sqrt{30030}}{14784} & \frac{\sqrt{715}}{1144} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & -\frac{15\sqrt{1001}}{28028} & 0 & -\frac{\sqrt{5005}}{16016} & 0 \\
\frac{5\sqrt{30030}}{14784} & 0 & -\frac{5\sqrt{3003}}{51744} & 0 & -\frac{65\sqrt{6006}}{103488} & 0 & 0 & -\frac{\sqrt{5005}}{16016} & 0 & -\frac{15\sqrt{1001}}{28028} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & \frac{\sqrt{715}}{1144} \\
0 & -\frac{\sqrt{10010}}{4928} & 0 & -\frac{\sqrt{5005}}{1568} & 0 & \frac{25\sqrt{2002}}{34496} & -\frac{3\sqrt{429}}{2288} & 0 & -\frac{5\sqrt{1001}}{8008} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & \frac{5\sqrt{3003}}{14014} & 0 \\
-\frac{5\sqrt{6006}}{4928} & 0 & -\frac{\sqrt{15015}}{2464} & 0 & \frac{23\sqrt{30030}}{103488} & 0 & 0 & -\frac{9\sqrt{1001}}{8008} & 0 & -\frac{\sqrt{5005}}{16016} & 0 & \frac{5\sqrt{3003}}{14014} & 0 & -\frac{15\sqrt{143}}{16016} \\
0 & -\frac{\sqrt{4290}}{704} & 0 & \frac{\sqrt{2145}}{1056} & 0 & -\frac{5\sqrt{858}}{14784} & 0 & 0 & -\frac{3\sqrt{429}}{2288} & 0 & \frac{\sqrt{715}}{1144} & 0 & -\frac{15\sqrt{143}}{16016} & 0
\end{array} \right]
\end{array}$$

$$\begin{array}{c}
\boxed{x545} \quad \mathbb{M}_{5,2}^{(1,1;a)}(E_g, 1) \\
= \left[\begin{array}{cccccccccccccc}
0 & \frac{3\sqrt{5005}i}{4312} & 0 & \frac{\sqrt{10010}i}{616} & 0 & \frac{9\sqrt{1001}i}{616} & -\frac{5\sqrt{858}i}{14784} & 0 & -\frac{25\sqrt{2002}i}{34496} & 0 & -\frac{5\sqrt{30030}i}{14784} & 0 & -\frac{5\sqrt{6006}i}{4928} & 0 \\
-\frac{3\sqrt{5005}i}{4312} & 0 & -\frac{15\sqrt{2002}i}{4312} & 0 & -\frac{5\sqrt{1001}i}{616} & 0 & 0 & \frac{23\sqrt{30030}i}{103488} & 0 & \frac{65\sqrt{6006}i}{103488} & 0 & \frac{\sqrt{10010}i}{4928} & 0 & -\frac{\sqrt{4290}i}{704} \\
0 & \frac{15\sqrt{2002}i}{4312} & 0 & \frac{15\sqrt{1001}i}{2156} & 0 & \frac{\sqrt{10010}i}{616} & -\frac{\sqrt{2145}i}{1056} & 0 & -\frac{\sqrt{5005}i}{1568} & 0 & -\frac{5\sqrt{3003}i}{51744} & 0 & \frac{\sqrt{15015}i}{2464} & 0 \\
-\frac{\sqrt{10010}i}{616} & 0 & -\frac{15\sqrt{1001}i}{2156} & 0 & -\frac{15\sqrt{2002}i}{4312} & 0 & 0 & \frac{\sqrt{15015}i}{2464} & 0 & -\frac{5\sqrt{3003}i}{51744} & 0 & -\frac{\sqrt{5005}i}{1568} & 0 & -\frac{\sqrt{2145}i}{1056} \\
0 & \frac{5\sqrt{1001}i}{616} & 0 & \frac{15\sqrt{2002}i}{4312} & 0 & \frac{3\sqrt{5005}i}{4312} & -\frac{\sqrt{4290}i}{704} & 0 & \frac{\sqrt{10010}i}{4928} & 0 & \frac{65\sqrt{6006}i}{103488} & 0 & \frac{23\sqrt{30030}i}{103488} & 0 \\
-\frac{9\sqrt{1001}i}{616} & 0 & -\frac{\sqrt{10010}i}{616} & 0 & -\frac{3\sqrt{5005}i}{4312} & 0 & 0 & -\frac{5\sqrt{6006}i}{4928} & 0 & -\frac{5\sqrt{30030}i}{14784} & 0 & -\frac{25\sqrt{2002}i}{34496} & 0 & -\frac{5\sqrt{858}i}{14784} \\
\frac{5\sqrt{858}i}{14784} & 0 & \frac{\sqrt{2145}i}{1056} & 0 & \frac{\sqrt{4290}i}{704} & 0 & 0 & -\frac{15\sqrt{143}i}{16016} & 0 & -\frac{\sqrt{715}i}{1144} & 0 & -\frac{3\sqrt{429}i}{2288} & 0 & 0 \\
0 & -\frac{23\sqrt{30030}i}{103488} & 0 & -\frac{\sqrt{15015}i}{2464} & 0 & \frac{5\sqrt{6006}i}{4928} & \frac{15\sqrt{143}i}{16016} & 0 & \frac{5\sqrt{3003}i}{14014} & 0 & \frac{\sqrt{5005}i}{16016} & 0 & -\frac{9\sqrt{1001}i}{8008} & 0 \\
\frac{25\sqrt{2002}i}{34496} & 0 & \frac{\sqrt{5005}i}{1568} & 0 & -\frac{\sqrt{10010}i}{4928} & 0 & 0 & -\frac{5\sqrt{3003}i}{14014} & 0 & -\frac{\sqrt{15015}i}{112112} & 0 & \frac{5\sqrt{1001}i}{8008} & 0 & -\frac{3\sqrt{429}i}{2288} \\
0 & -\frac{65\sqrt{6006}i}{103488} & 0 & \frac{5\sqrt{3003}i}{51744} & 0 & \frac{5\sqrt{30030}i}{14784} & \frac{\sqrt{715}i}{1144} & 0 & \frac{\sqrt{15015}i}{112112} & 0 & -\frac{15\sqrt{1001}i}{28028} & 0 & \frac{\sqrt{5005}i}{16016} & 0 \\
\frac{5\sqrt{30030}i}{14784} & 0 & \frac{5\sqrt{3003}i}{51744} & 0 & -\frac{65\sqrt{6006}i}{103488} & 0 & 0 & -\frac{\sqrt{5005}i}{16016} & 0 & \frac{15\sqrt{1001}i}{28028} & 0 & -\frac{\sqrt{15015}i}{112112} & 0 & -\frac{\sqrt{715}i}{1144} \\
0 & -\frac{\sqrt{10010}i}{4928} & 0 & \frac{\sqrt{5005}i}{1568} & 0 & \frac{25\sqrt{2002}i}{34496} & \frac{3\sqrt{429}i}{2288} & 0 & -\frac{5\sqrt{1001}i}{8008} & 0 & \frac{\sqrt{15015}i}{112112} & 0 & \frac{5\sqrt{3003}i}{14014} & 0 \\
\frac{5\sqrt{6006}i}{4928} & 0 & -\frac{\sqrt{15015}i}{2464} & 0 & -\frac{23\sqrt{30030}i}{103488} & 0 & 0 & \frac{9\sqrt{1001}i}{8008} & 0 & -\frac{\sqrt{5005}i}{16016} & 0 & -\frac{5\sqrt{3003}i}{14014} & 0 & -\frac{15\sqrt{143}i}{16016} \\
0 & \frac{\sqrt{4290}i}{704} & 0 & \frac{\sqrt{2145}i}{1056} & 0 & \frac{5\sqrt{858}i}{14784} & 0 & 0 & \frac{3\sqrt{429}i}{2288} & 0 & \frac{\sqrt{715}i}{1144} & 0 & \frac{15\sqrt{143}i}{16016} & 0
\end{array} \right]
\end{array}$$

x546	$\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 2)$													
=	0	$\frac{3\sqrt{143}}{616}$	0	$\frac{9\sqrt{286}}{616}$	0	$\frac{3\sqrt{715}}{616}$	$\frac{\sqrt{30030}}{14784}$	0	$-\frac{5\sqrt{1430}}{4928}$	0	$-\frac{15\sqrt{858}}{4928}$	0	$-\frac{5\sqrt{4290}}{14784}$	0
	$\frac{3\sqrt{143}}{616}$	0	$-\frac{3\sqrt{1430}}{616}$	0	$-\frac{9\sqrt{715}}{616}$	0	0	$-\frac{23\sqrt{858}}{14784}$	0	$\frac{13\sqrt{4290}}{14784}$	0	$\frac{9\sqrt{286}}{4928}$	0	$-\frac{5\sqrt{6006}}{14784}$
	0	$-\frac{3\sqrt{1430}}{616}$	0	$\frac{3\sqrt{715}}{308}$	0	$\frac{9\sqrt{286}}{616}$	$\frac{3\sqrt{3003}}{2464}$	0	$\frac{\sqrt{143}}{224}$	0	$-\frac{\sqrt{2145}}{7392}$	0	$\frac{9\sqrt{429}}{2464}$	0
	$\frac{9\sqrt{286}}{616}$	0	$\frac{3\sqrt{715}}{308}$	0	$-\frac{3\sqrt{1430}}{616}$	0	0	$-\frac{9\sqrt{429}}{2464}$	0	$\frac{\sqrt{2145}}{7392}$	0	$-\frac{\sqrt{143}}{224}$	0	$-\frac{3\sqrt{3003}}{2464}$
	0	$-\frac{9\sqrt{715}}{616}$	0	$-\frac{3\sqrt{1430}}{616}$	0	$\frac{3\sqrt{143}}{616}$	$\frac{5\sqrt{6006}}{14784}$	0	$-\frac{9\sqrt{286}}{4928}$	0	$-\frac{13\sqrt{4290}}{14784}$	0	$\frac{23\sqrt{858}}{14784}$	0
	$\frac{3\sqrt{715}}{616}$	0	$\frac{9\sqrt{286}}{616}$	0	$\frac{3\sqrt{143}}{616}$	0	0	$\frac{5\sqrt{4290}}{14784}$	0	$\frac{15\sqrt{858}}{4928}$	0	$\frac{5\sqrt{1430}}{4928}$	0	$-\frac{\sqrt{30030}}{14784}$
	$\frac{\sqrt{30030}}{14784}$	0	$\frac{3\sqrt{3003}}{2464}$	0	$\frac{5\sqrt{6006}}{14784}$	0	0	$-\frac{3\sqrt{5005}}{16016}$	0	$-\frac{9\sqrt{1001}}{8008}$	0	$-\frac{\sqrt{15015}}{16016}$	0	0
	0	$-\frac{23\sqrt{858}}{14784}$	0	$-\frac{9\sqrt{429}}{2464}$	0	$\frac{5\sqrt{4290}}{14784}$	$-\frac{3\sqrt{5005}}{16016}$	0	$\frac{\sqrt{2145}}{2002}$	0	$\frac{9\sqrt{143}}{16016}$	0	$-\frac{3\sqrt{715}}{8008}$	0
	$-\frac{5\sqrt{1430}}{4928}$	0	$\frac{\sqrt{143}}{224}$	0	$-\frac{9\sqrt{286}}{4928}$	0	0	$\frac{\sqrt{2145}}{2002}$	0	$-\frac{\sqrt{429}}{16016}$	0	$\frac{9\sqrt{715}}{8008}$	0	$-\frac{\sqrt{15015}}{16016}$
	0	$\frac{13\sqrt{4290}}{14784}$	0	$\frac{\sqrt{2145}}{7392}$	0	$\frac{15\sqrt{858}}{4928}$	$-\frac{9\sqrt{1001}}{8008}$	0	$-\frac{\sqrt{429}}{16016}$	0	$-\frac{3\sqrt{715}}{4004}$	0	$\frac{9\sqrt{143}}{16016}$	0
	$-\frac{15\sqrt{858}}{4928}$	0	$-\frac{\sqrt{2145}}{7392}$	0	$-\frac{13\sqrt{4290}}{14784}$	0	0	$\frac{9\sqrt{143}}{16016}$	0	$-\frac{3\sqrt{715}}{4004}$	0	$-\frac{\sqrt{429}}{16016}$	0	$-\frac{9\sqrt{1001}}{8008}$
	0	$\frac{9\sqrt{286}}{4928}$	0	$-\frac{\sqrt{143}}{224}$	0	$\frac{5\sqrt{1430}}{4928}$	$-\frac{\sqrt{15015}}{16016}$	0	$\frac{9\sqrt{715}}{8008}$	0	$-\frac{\sqrt{429}}{16016}$	0	$\frac{\sqrt{2145}}{2002}$	0
	$-\frac{5\sqrt{4290}}{14784}$	0	$\frac{9\sqrt{429}}{2464}$	0	$\frac{23\sqrt{858}}{14784}$	0	0	$-\frac{3\sqrt{715}}{8008}$	0	$\frac{9\sqrt{143}}{16016}$	0	$\frac{\sqrt{2145}}{2002}$	0	$-\frac{3\sqrt{5005}}{16016}$
	0	$-\frac{5\sqrt{6006}}{14784}$	0	$-\frac{3\sqrt{3003}}{2464}$	0	$-\frac{\sqrt{30030}}{14784}$	0	0	$-\frac{\sqrt{15015}}{16016}$	0	$-\frac{9\sqrt{1001}}{8008}$	0	$-\frac{3\sqrt{5005}}{16016}$	0

x547	$\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 2)$																																																																																																																																																																																																				
=	<table border="0" style="width: 100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{143i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{286i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{715i}}{616}$</td> <td style="text-align: center;">$-\frac{\sqrt{30030i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{5\sqrt{1430i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{15\sqrt{858i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{5\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">$-\frac{3\sqrt{143i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{1430i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{715i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{23\sqrt{858i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{13\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{286i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{5\sqrt{6006i}}{14784}$</td> </tr> <tr> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{1430i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{715i}}{308}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{286i}}{616}$</td> <td style="text-align: center;">$\frac{3\sqrt{3003i}}{2464}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{143i}}{224}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{2145i}}{7392}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{429i}}{2464}$</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">$\frac{9\sqrt{286i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{715i}}{308}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{1430i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{429i}}{2464}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{2145i}}{7392}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{143i}}{224}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{3003i}}{2464}$</td> </tr> <tr> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{715i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{1430i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{143i}}{616}$</td> <td style="text-align: center;">$-\frac{5\sqrt{6006i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{286i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{13\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{23\sqrt{858i}}{14784}$</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">$-\frac{3\sqrt{715i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{286i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{143i}}{616}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{5\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{15\sqrt{858i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{5\sqrt{1430i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{30030i}}{14784}$</td> </tr> <tr> <td style="text-align: center;">$\frac{\sqrt{30030i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{3003i}}{2464}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{5\sqrt{6006i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{5005i}}{16016}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{1001i}}{8008}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{15015i}}{16016}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{23\sqrt{858i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{429i}}{2464}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{5\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">$\frac{3\sqrt{5005i}}{16016}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{\sqrt{2145i}}{2002}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{143i}}{16016}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{715i}}{8008}$</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">$\frac{5\sqrt{1430i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{\sqrt{143i}}{224}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{286i}}{4928}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{2145i}}{2002}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{429i}}{16016}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{9\sqrt{715i}}{8008}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{15015i}}{16016}$</td> </tr> <tr> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{13\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: 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style="text-align: center;">$\frac{\sqrt{2145i}}{2002}$</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">$\frac{5\sqrt{4290i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{429i}}{2464}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{23\sqrt{858i}}{14784}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{3\sqrt{715i}}{8008}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{9\sqrt{143i}}{16016}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{\sqrt{2145i}}{2002}$</td> <td style="text-align: center;">0</td> <td style="text-align: center;">$-\frac{3\sqrt{5005i}}{16016}$</td> </tr> <tr> <td style="text-align: center;">0</td> <td style="text-align: center;">$\frac{5\sqrt{6006i}}{14784}$</td> <td style="text-align: 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</table>	0	$\frac{3\sqrt{143i}}{616}$	0	$-\frac{9\sqrt{286i}}{616}$	0	$\frac{3\sqrt{715i}}{616}$	$-\frac{\sqrt{30030i}}{14784}$	0	$-\frac{5\sqrt{1430i}}{4928}$	0	$\frac{15\sqrt{858i}}{4928}$	0	$-\frac{5\sqrt{4290i}}{14784}$	0	$-\frac{3\sqrt{143i}}{616}$	0	$-\frac{3\sqrt{1430i}}{616}$	0	$\frac{9\sqrt{715i}}{616}$	0	0	$\frac{23\sqrt{858i}}{14784}$	0	$\frac{13\sqrt{4290i}}{14784}$	0	$-\frac{9\sqrt{286i}}{4928}$	0	$-\frac{5\sqrt{6006i}}{14784}$	0	$\frac{3\sqrt{1430i}}{616}$	0	$\frac{3\sqrt{715i}}{308}$	0	$-\frac{9\sqrt{286i}}{616}$	$\frac{3\sqrt{3003i}}{2464}$	0	$-\frac{\sqrt{143i}}{224}$	0	$-\frac{\sqrt{2145i}}{7392}$	0	$-\frac{9\sqrt{429i}}{2464}$	0	$\frac{9\sqrt{286i}}{616}$	0	$-\frac{3\sqrt{715i}}{308}$	0	$-\frac{3\sqrt{1430i}}{616}$	0	0	$-\frac{9\sqrt{429i}}{2464}$	0	$-\frac{\sqrt{2145i}}{7392}$	0	$-\frac{\sqrt{143i}}{224}$	0	$\frac{3\sqrt{3003i}}{2464}$	0	$-\frac{9\sqrt{715i}}{616}$	0	$\frac{3\sqrt{1430i}}{616}$	0	$\frac{3\sqrt{143i}}{616}$	$-\frac{5\sqrt{6006i}}{14784}$	0	$-\frac{9\sqrt{286i}}{4928}$	0	$\frac{13\sqrt{4290i}}{14784}$	0	$\frac{23\sqrt{858i}}{14784}$	0	$-\frac{3\sqrt{715i}}{616}$	0	$\frac{9\sqrt{286i}}{616}$	0	$-\frac{3\sqrt{143i}}{616}$	0	0	$-\frac{5\sqrt{4290i}}{14784}$	0	$\frac{15\sqrt{858i}}{4928}$	0	$-\frac{5\sqrt{1430i}}{4928}$	0	$-\frac{\sqrt{30030i}}{14784}$	$\frac{\sqrt{30030i}}{14784}$	0	$-\frac{3\sqrt{3003i}}{2464}$	0	$\frac{5\sqrt{6006i}}{14784}$	0	0	$-\frac{3\sqrt{5005i}}{16016}$	0	$\frac{9\sqrt{1001i}}{8008}$	0	$-\frac{\sqrt{15015i}}{16016}$	0	0	0	$-\frac{23\sqrt{858i}}{14784}$	0	$\frac{9\sqrt{429i}}{2464}$	0	$\frac{5\sqrt{4290i}}{14784}$	$\frac{3\sqrt{5005i}}{16016}$	0	$\frac{\sqrt{2145i}}{2002}$	0	$-\frac{9\sqrt{143i}}{16016}$	0	$-\frac{3\sqrt{715i}}{8008}$	0	$\frac{5\sqrt{1430i}}{4928}$	0	$\frac{\sqrt{143i}}{224}$	0	$\frac{9\sqrt{286i}}{4928}$	0	0	$-\frac{\sqrt{2145i}}{2002}$	0	$-\frac{\sqrt{429i}}{16016}$	0	$-\frac{9\sqrt{715i}}{8008}$	0	$-\frac{\sqrt{15015i}}{16016}$	0	$-\frac{13\sqrt{4290i}}{14784}$	0	$\frac{\sqrt{2145i}}{7392}$	0	$-\frac{15\sqrt{858i}}{4928}$	$-\frac{9\sqrt{1001i}}{8008}$	0	$\frac{\sqrt{429i}}{16016}$	0	$-\frac{3\sqrt{715i}}{4004}$	0	$-\frac{9\sqrt{143i}}{16016}$	0	$-\frac{15\sqrt{858i}}{4928}$	0	$\frac{\sqrt{2145i}}{7392}$	0	$-\frac{13\sqrt{4290i}}{14784}$	0	0	$\frac{9\sqrt{143i}}{16016}$	0	$\frac{3\sqrt{715i}}{4004}$	0	$-\frac{\sqrt{429i}}{16016}$	0	$\frac{9\sqrt{1001i}}{8008}$	0	$\frac{9\sqrt{286i}}{4928}$	0	$\frac{\sqrt{143i}}{224}$	0	$\frac{5\sqrt{1430i}}{4928}$	$\frac{\sqrt{15015i}}{16016}$	0	$\frac{9\sqrt{715i}}{8008}$	0	$\frac{\sqrt{429i}}{16016}$	0	$\frac{\sqrt{2145i}}{2002}$	0	$\frac{5\sqrt{4290i}}{14784}$	0	$\frac{9\sqrt{429i}}{2464}$	0	$-\frac{23\sqrt{858i}}{14784}$	0	0	$\frac{3\sqrt{715i}}{8008}$	0	$\frac{9\sqrt{143i}}{16016}$	0	$-\frac{\sqrt{2145i}}{2002}$	0	$-\frac{3\sqrt{5005i}}{16016}$	0	$\frac{5\sqrt{6006i}}{14784}$	0	$-\frac{3\sqrt{3003i}}{2464}$	0	$\frac{\sqrt{30030i}}{14784}$	0	0	$\frac{\sqrt{15015i}}{16016}$	0	$-\frac{9\sqrt{1001i}}{8008}$	0	$\frac{3\sqrt{5005i}}{16016}$	0
0	$\frac{3\sqrt{143i}}{616}$	0	$-\frac{9\sqrt{286i}}{616}$	0	$\frac{3\sqrt{715i}}{616}$	$-\frac{\sqrt{30030i}}{14784}$	0	$-\frac{5\sqrt{1430i}}{4928}$	0	$\frac{15\sqrt{858i}}{4928}$	0	$-\frac{5\sqrt{4290i}}{14784}$	0																																																																																																																																																																																								
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x548

$\mathbb{M}_{5,1}^{(1,1;a)}(E_g, 3) =$

$$\begin{bmatrix}
 0 & \frac{\sqrt{429}}{308} & 0 & -\frac{\sqrt{858}}{308} & 0 & -\frac{3\sqrt{2145}}{308} & \frac{\sqrt{10010}}{7392} & 0 & -\frac{5\sqrt{4290}}{7392} & 0 & \frac{5\sqrt{286}}{2464} & 0 & \frac{5\sqrt{1430}}{2464} & 0 \\
 \frac{\sqrt{429}}{308} & 0 & -\frac{\sqrt{4290}}{308} & 0 & \frac{\sqrt{2145}}{308} & 0 & 0 & -\frac{23\sqrt{286}}{7392} & 0 & \frac{13\sqrt{1430}}{7392} & 0 & -\frac{\sqrt{858}}{2464} & 0 & \frac{5\sqrt{2002}}{2464} \\
 0 & -\frac{\sqrt{4290}}{308} & 0 & \frac{\sqrt{2145}}{154} & 0 & -\frac{\sqrt{858}}{308} & -\frac{\sqrt{1001}}{1232} & 0 & \frac{\sqrt{429}}{336} & 0 & -\frac{\sqrt{715}}{3696} & 0 & -\frac{3\sqrt{143}}{1232} & 0 \\
 -\frac{\sqrt{858}}{308} & 0 & \frac{\sqrt{2145}}{154} & 0 & -\frac{\sqrt{4290}}{308} & 0 & 0 & \frac{3\sqrt{143}}{1232} & 0 & \frac{\sqrt{715}}{3696} & 0 & -\frac{\sqrt{429}}{336} & 0 & \frac{\sqrt{1001}}{1232} \\
 0 & \frac{\sqrt{2145}}{308} & 0 & -\frac{\sqrt{4290}}{308} & 0 & \frac{\sqrt{429}}{308} & -\frac{5\sqrt{2002}}{2464} & 0 & \frac{\sqrt{858}}{2464} & 0 & -\frac{13\sqrt{1430}}{7392} & 0 & \frac{23\sqrt{286}}{7392} & 0 \\
 -\frac{3\sqrt{2145}}{308} & 0 & -\frac{\sqrt{858}}{308} & 0 & \frac{\sqrt{429}}{308} & 0 & 0 & -\frac{5\sqrt{1430}}{2464} & 0 & -\frac{5\sqrt{286}}{2464} & 0 & \frac{5\sqrt{4290}}{7392} & 0 & -\frac{\sqrt{10010}}{7392} \\
 \frac{\sqrt{10010}}{7392} & 0 & -\frac{\sqrt{1001}}{1232} & 0 & -\frac{5\sqrt{2002}}{2464} & 0 & 0 & -\frac{\sqrt{15015}}{8008} & 0 & \frac{\sqrt{3003}}{4004} & 0 & \frac{3\sqrt{5005}}{8008} & 0 & 0 \\
 0 & -\frac{23\sqrt{286}}{7392} & 0 & \frac{3\sqrt{143}}{1232} & 0 & -\frac{5\sqrt{1430}}{2464} & -\frac{\sqrt{15015}}{8008} & 0 & \frac{\sqrt{715}}{1001} & 0 & -\frac{\sqrt{429}}{8008} & 0 & \frac{3\sqrt{2145}}{4004} & 0 \\
 -\frac{5\sqrt{4290}}{7392} & 0 & \frac{\sqrt{429}}{336} & 0 & \frac{\sqrt{858}}{2464} & 0 & 0 & \frac{\sqrt{715}}{1001} & 0 & -\frac{\sqrt{143}}{8008} & 0 & -\frac{\sqrt{2145}}{4004} & 0 & \frac{3\sqrt{5005}}{8008} \\
 0 & \frac{13\sqrt{1430}}{7392} & 0 & \frac{\sqrt{715}}{3696} & 0 & -\frac{5\sqrt{286}}{2464} & \frac{\sqrt{3003}}{4004} & 0 & -\frac{\sqrt{143}}{8008} & 0 & -\frac{\sqrt{2145}}{2002} & 0 & -\frac{\sqrt{429}}{8008} & 0 \\
 \frac{5\sqrt{286}}{2464} & 0 & -\frac{\sqrt{715}}{3696} & 0 & -\frac{13\sqrt{1430}}{7392} & 0 & 0 & -\frac{\sqrt{429}}{8008} & 0 & -\frac{\sqrt{2145}}{2002} & 0 & -\frac{\sqrt{143}}{8008} & 0 & \frac{\sqrt{3003}}{4004} \\
 0 & -\frac{\sqrt{858}}{2464} & 0 & -\frac{\sqrt{429}}{336} & 0 & \frac{5\sqrt{4290}}{7392} & \frac{3\sqrt{5005}}{8008} & 0 & -\frac{\sqrt{2145}}{4004} & 0 & -\frac{\sqrt{143}}{8008} & 0 & \frac{\sqrt{715}}{1001} & 0 \\
 \frac{5\sqrt{1430}}{2464} & 0 & -\frac{3\sqrt{143}}{1232} & 0 & \frac{23\sqrt{286}}{7392} & 0 & 0 & \frac{3\sqrt{2145}}{4004} & 0 & -\frac{\sqrt{429}}{8008} & 0 & \frac{\sqrt{715}}{1001} & 0 & -\frac{\sqrt{15015}}{8008} \\
 0 & \frac{5\sqrt{2002}}{2464} & 0 & \frac{\sqrt{1001}}{1232} & 0 & -\frac{\sqrt{10010}}{7392} & 0 & 0 & \frac{3\sqrt{5005}}{8008} & 0 & \frac{\sqrt{3003}}{4004} & 0 & -\frac{\sqrt{15015}}{8008} & 0
 \end{bmatrix}$$

x549	$\mathbb{M}_{5,2}^{(1,1;a)}(E_g, 3)$																																																																																																																																																																																																				
=	<table border="0" style="width: 100%; border-collapse: collapse;"> <tr> <td>0</td><td>$\frac{\sqrt{429}i}{308}$</td><td>0</td><td>$\frac{\sqrt{858}i}{308}$</td><td>0</td><td>$-\frac{3\sqrt{2145}i}{308}$</td><td>$-\frac{\sqrt{10010}i}{7392}$</td><td>$0$</td><td>$-\frac{5\sqrt{4290}i}{7392}$</td><td>$0$</td><td>$-\frac{5\sqrt{286}i}{2464}$</td><td>$0$</td><td>$\frac{5\sqrt{1430}i}{2464}$</td><td>$0$</td></tr> <tr> <td>$-\frac{\sqrt{429}i}{308}$</td><td>$0$</td><td>$-\frac{\sqrt{4290}i}{308}$</td><td>$0$</td><td>$-\frac{\sqrt{2145}i}{308}$</td><td>$0$</td><td>$0$</td><td>$\frac{23\sqrt{286}i}{7392}$</td><td>$0$</td><td>$\frac{13\sqrt{1430}i}{7392}$</td><td>$0$</td><td>$\frac{\sqrt{858}i}{2464}$</td><td>$0$</td><td>$\frac{5\sqrt{2002}i}{2464}$</td></tr> <tr> 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<td>$\frac{5\sqrt{286}i}{2464}$</td><td>$0$</td><td>$\frac{\sqrt{715}i}{3696}$</td><td>$0$</td><td>$-\frac{13\sqrt{1430}i}{7392}$</td><td>$0$</td><td>$0$</td><td>$-\frac{\sqrt{429}i}{8008}$</td><td>$0$</td><td>$\frac{\sqrt{2145}i}{2002}$</td><td>$0$</td><td>$-\frac{\sqrt{143}i}{8008}$</td><td>$0$</td><td>$-\frac{\sqrt{3003}i}{4004}$</td></tr> <tr> <td>$0$</td><td>$-\frac{\sqrt{858}i}{2464}$</td><td>$0$</td><td>$\frac{\sqrt{429}i}{336}$</td><td>$0$</td><td>$\frac{5\sqrt{4290}i}{7392}$</td><td>$-\frac{3\sqrt{5005}i}{8008}$</td><td>$0$</td><td>$-\frac{\sqrt{2145}i}{4004}$</td><td>$0$</td><td>$\frac{\sqrt{143}i}{8008}$</td><td>$0$</td><td>$\frac{\sqrt{715}i}{1001}$</td><td>$0$</td></tr> <tr> <td>$-\frac{5\sqrt{1430}i}{2464}$</td><td>$0$</td><td>$-\frac{3\sqrt{143}i}{1232}$</td><td>$0$</td><td>$-\frac{23\sqrt{286}i}{7392}$</td><td>$0$</td><td>$0$</td><td>$-\frac{3\sqrt{2145}i}{4004}$</td><td>$0$</td><td>$-\frac{\sqrt{429}i}{8008}$</td><td>$0$</td><td>$-\frac{\sqrt{715}i}{1001}$</td><td>$0$</td><td>$-\frac{\sqrt{15015}i}{8008}$</td></tr> <tr> <td>$0$</td><td>$-\frac{5\sqrt{2002}i}{2464}$</td><td>$0$</td><td>$\frac{\sqrt{1001}i}{1232}$</td><td>$0$</td><td>$\frac{\sqrt{10010}i}{7392}$</td><td>$0$</td><td>$0$</td><td>$-\frac{3\sqrt{5005}i}{8008}$</td><td>$0$</td><td>$\frac{\sqrt{3003}i}{4004}$</td><td>$0$</td><td>$\frac{\sqrt{15015}i}{8008}$</td><td>$0$</td></tr> </table>	0	$\frac{\sqrt{429}i}{308}$	0	$\frac{\sqrt{858}i}{308}$	0	$-\frac{3\sqrt{2145}i}{308}$	$-\frac{\sqrt{10010}i}{7392}$	0	$-\frac{5\sqrt{4290}i}{7392}$	0	$-\frac{5\sqrt{286}i}{2464}$	0	$\frac{5\sqrt{1430}i}{2464}$	0	$-\frac{\sqrt{429}i}{308}$	0	$-\frac{\sqrt{4290}i}{308}$	0	$-\frac{\sqrt{2145}i}{308}$	0	0	$\frac{23\sqrt{286}i}{7392}$	0	$\frac{13\sqrt{1430}i}{7392}$	0	$\frac{\sqrt{858}i}{2464}$	0	$\frac{5\sqrt{2002}i}{2464}$	0	$\frac{\sqrt{4290}i}{308}$	0	$\frac{\sqrt{2145}i}{154}$	0	$\frac{\sqrt{858}i}{308}$	$-\frac{\sqrt{1001}i}{1232}$	0	$-\frac{\sqrt{429}i}{336}$	0	$-\frac{\sqrt{715}i}{3696}$	0	$\frac{3\sqrt{143}i}{1232}$	0	$-\frac{\sqrt{858}i}{308}$	0	$-\frac{\sqrt{2145}i}{154}$	0	$-\frac{\sqrt{4290}i}{308}$	0	0	$\frac{3\sqrt{143}i}{1232}$	0	$-\frac{\sqrt{715}i}{3696}$	0	$-\frac{\sqrt{429}i}{336}$	0	$-\frac{\sqrt{1001}i}{1232}$	0	$\frac{\sqrt{2145}i}{308}$	0	$\frac{\sqrt{4290}i}{308}$	0	$\frac{\sqrt{429}i}{308}$	$\frac{5\sqrt{2002}i}{2464}$	0	$\frac{\sqrt{858}i}{2464}$	0	$\frac{13\sqrt{1430}i}{7392}$	0	$\frac{23\sqrt{286}i}{7392}$	0	$\frac{3\sqrt{2145}i}{308}$	0	$-\frac{\sqrt{858}i}{308}$	0	$-\frac{\sqrt{429}i}{308}$	0	0	$\frac{5\sqrt{1430}i}{2464}$	0	$-\frac{5\sqrt{286}i}{2464}$	0	$-\frac{5\sqrt{4290}i}{7392}$	0	$-\frac{\sqrt{10010}i}{7392}$	$\frac{\sqrt{10010}i}{7392}$	0	$\frac{\sqrt{1001}i}{1232}$	0	$-\frac{5\sqrt{2002}i}{2464}$	0	0	$-\frac{\sqrt{15015}i}{8008}$	0	$-\frac{\sqrt{3003}i}{4004}$	0	$\frac{3\sqrt{5005}i}{8008}$	0	0	0	$-\frac{23\sqrt{286}i}{7392}$	0	$-\frac{3\sqrt{143}i}{1232}$	0	$-\frac{5\sqrt{1430}i}{2464}$	$\frac{\sqrt{15015}i}{8008}$	0	$\frac{\sqrt{715}i}{1001}$	0	$\frac{\sqrt{429}i}{8008}$	0	$\frac{3\sqrt{2145}i}{4004}$	0	$\frac{5\sqrt{4290}i}{7392}$	0	$\frac{\sqrt{429}i}{336}$	0	$-\frac{\sqrt{858}i}{2464}$	0	0	$-\frac{\sqrt{715}i}{1001}$	0	$-\frac{\sqrt{143}i}{8008}$	0	$\frac{\sqrt{2145}i}{4004}$	0	$\frac{3\sqrt{5005}i}{8008}$	0	$-\frac{13\sqrt{1430}i}{7392}$	0	$\frac{\sqrt{715}i}{3696}$	0	$\frac{5\sqrt{286}i}{2464}$	$\frac{\sqrt{3003}i}{4004}$	0	$\frac{\sqrt{143}i}{8008}$	0	$-\frac{\sqrt{2145}i}{2002}$	0	$\frac{\sqrt{429}i}{8008}$	0	$\frac{5\sqrt{286}i}{2464}$	0	$\frac{\sqrt{715}i}{3696}$	0	$-\frac{13\sqrt{1430}i}{7392}$	0	0	$-\frac{\sqrt{429}i}{8008}$	0	$\frac{\sqrt{2145}i}{2002}$	0	$-\frac{\sqrt{143}i}{8008}$	0	$-\frac{\sqrt{3003}i}{4004}$	0	$-\frac{\sqrt{858}i}{2464}$	0	$\frac{\sqrt{429}i}{336}$	0	$\frac{5\sqrt{4290}i}{7392}$	$-\frac{3\sqrt{5005}i}{8008}$	0	$-\frac{\sqrt{2145}i}{4004}$	0	$\frac{\sqrt{143}i}{8008}$	0	$\frac{\sqrt{715}i}{1001}$	0	$-\frac{5\sqrt{1430}i}{2464}$	0	$-\frac{3\sqrt{143}i}{1232}$	0	$-\frac{23\sqrt{286}i}{7392}$	0	0	$-\frac{3\sqrt{2145}i}{4004}$	0	$-\frac{\sqrt{429}i}{8008}$	0	$-\frac{\sqrt{715}i}{1001}$	0	$-\frac{\sqrt{15015}i}{8008}$	0	$-\frac{5\sqrt{2002}i}{2464}$	0	$\frac{\sqrt{1001}i}{1232}$	0	$\frac{\sqrt{10010}i}{7392}$	0	0	$-\frac{3\sqrt{5005}i}{8008}$	0	$\frac{\sqrt{3003}i}{4004}$	0	$\frac{\sqrt{15015}i}{8008}$	0
0	$\frac{\sqrt{429}i}{308}$	0	$\frac{\sqrt{858}i}{308}$	0	$-\frac{3\sqrt{2145}i}{308}$	$-\frac{\sqrt{10010}i}{7392}$	0	$-\frac{5\sqrt{4290}i}{7392}$	0	$-\frac{5\sqrt{286}i}{2464}$	0	$\frac{5\sqrt{1430}i}{2464}$	0																																																																																																																																																																																								
$-\frac{\sqrt{429}i}{308}$	0	$-\frac{\sqrt{4290}i}{308}$	0	$-\frac{\sqrt{2145}i}{308}$	0	0	$\frac{23\sqrt{286}i}{7392}$	0	$\frac{13\sqrt{1430}i}{7392}$	0	$\frac{\sqrt{858}i}{2464}$	0	$\frac{5\sqrt{2002}i}{2464}$																																																																																																																																																																																								
0	$\frac{\sqrt{4290}i}{308}$	0	$\frac{\sqrt{2145}i}{154}$	0	$\frac{\sqrt{858}i}{308}$	$-\frac{\sqrt{1001}i}{1232}$	0	$-\frac{\sqrt{429}i}{336}$	0	$-\frac{\sqrt{715}i}{3696}$	0	$\frac{3\sqrt{143}i}{1232}$	0																																																																																																																																																																																								
$-\frac{\sqrt{858}i}{308}$	0	$-\frac{\sqrt{2145}i}{154}$	0	$-\frac{\sqrt{4290}i}{308}$	0	0	$\frac{3\sqrt{143}i}{1232}$	0	$-\frac{\sqrt{715}i}{3696}$	0	$-\frac{\sqrt{429}i}{336}$	0	$-\frac{\sqrt{1001}i}{1232}$																																																																																																																																																																																								
0	$\frac{\sqrt{2145}i}{308}$	0	$\frac{\sqrt{4290}i}{308}$	0	$\frac{\sqrt{429}i}{308}$	$\frac{5\sqrt{2002}i}{2464}$	0	$\frac{\sqrt{858}i}{2464}$	0	$\frac{13\sqrt{1430}i}{7392}$	0	$\frac{23\sqrt{286}i}{7392}$	0																																																																																																																																																																																								
$\frac{3\sqrt{2145}i}{308}$	0	$-\frac{\sqrt{858}i}{308}$	0	$-\frac{\sqrt{429}i}{308}$	0	0	$\frac{5\sqrt{1430}i}{2464}$	0	$-\frac{5\sqrt{286}i}{2464}$	0	$-\frac{5\sqrt{4290}i}{7392}$	0	$-\frac{\sqrt{10010}i}{7392}$																																																																																																																																																																																								
$\frac{\sqrt{10010}i}{7392}$	0	$\frac{\sqrt{1001}i}{1232}$	0	$-\frac{5\sqrt{2002}i}{2464}$	0	0	$-\frac{\sqrt{15015}i}{8008}$	0	$-\frac{\sqrt{3003}i}{4004}$	0	$\frac{3\sqrt{5005}i}{8008}$	0	0																																																																																																																																																																																								
0	$-\frac{23\sqrt{286}i}{7392}$	0	$-\frac{3\sqrt{143}i}{1232}$	0	$-\frac{5\sqrt{1430}i}{2464}$	$\frac{\sqrt{15015}i}{8008}$	0	$\frac{\sqrt{715}i}{1001}$	0	$\frac{\sqrt{429}i}{8008}$	0	$\frac{3\sqrt{2145}i}{4004}$	0																																																																																																																																																																																								
$\frac{5\sqrt{4290}i}{7392}$	0	$\frac{\sqrt{429}i}{336}$	0	$-\frac{\sqrt{858}i}{2464}$	0	0	$-\frac{\sqrt{715}i}{1001}$	0	$-\frac{\sqrt{143}i}{8008}$	0	$\frac{\sqrt{2145}i}{4004}$	0	$\frac{3\sqrt{5005}i}{8008}$																																																																																																																																																																																								
0	$-\frac{13\sqrt{1430}i}{7392}$	0	$\frac{\sqrt{715}i}{3696}$	0	$\frac{5\sqrt{286}i}{2464}$	$\frac{\sqrt{3003}i}{4004}$	0	$\frac{\sqrt{143}i}{8008}$	0	$-\frac{\sqrt{2145}i}{2002}$	0	$\frac{\sqrt{429}i}{8008}$	0																																																																																																																																																																																								
$\frac{5\sqrt{286}i}{2464}$	0	$\frac{\sqrt{715}i}{3696}$	0	$-\frac{13\sqrt{1430}i}{7392}$	0	0	$-\frac{\sqrt{429}i}{8008}$	0	$\frac{\sqrt{2145}i}{2002}$	0	$-\frac{\sqrt{143}i}{8008}$	0	$-\frac{\sqrt{3003}i}{4004}$																																																																																																																																																																																								
0	$-\frac{\sqrt{858}i}{2464}$	0	$\frac{\sqrt{429}i}{336}$	0	$\frac{5\sqrt{4290}i}{7392}$	$-\frac{3\sqrt{5005}i}{8008}$	0	$-\frac{\sqrt{2145}i}{4004}$	0	$\frac{\sqrt{143}i}{8008}$	0	$\frac{\sqrt{715}i}{1001}$	0																																																																																																																																																																																								
$-\frac{5\sqrt{1430}i}{2464}$	0	$-\frac{3\sqrt{143}i}{1232}$	0	$-\frac{23\sqrt{286}i}{7392}$	0	0	$-\frac{3\sqrt{2145}i}{4004}$	0	$-\frac{\sqrt{429}i}{8008}$	0	$-\frac{\sqrt{715}i}{1001}$	0	$-\frac{\sqrt{15015}i}{8008}$																																																																																																																																																																																								
0	$-\frac{5\sqrt{2002}i}{2464}$	0	$\frac{\sqrt{1001}i}{1232}$	0	$\frac{\sqrt{10010}i}{7392}$	0	0	$-\frac{3\sqrt{5005}i}{8008}$	0	$\frac{\sqrt{3003}i}{4004}$	0	$\frac{\sqrt{15015}i}{8008}$	0																																																																																																																																																																																								

$$\boxed{\text{x550}} \quad \mathbb{T}_2^{(1,0;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x551}} \quad \mathbb{T}_4^{(1,0;a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{33} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & -\frac{\sqrt{66}i}{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{110}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}i}{33} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & \frac{\sqrt{66}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x552}} \quad \mathbb{T}_4^{(1,0;a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{308} & 0 & 0 & 0 & \frac{\sqrt{462}i}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{231} & 0 & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & \frac{7\sqrt{55}i}{220} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{110} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{231} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{132} & 0 & 0 & 0 & \frac{5\sqrt{154}i}{308} & 0 \\ 0 & 0 & 0 & \frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{154}i}{308} & 0 & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}i}{231} & 0 & 0 & 0 & \frac{\sqrt{462}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{462}i}{132} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{770}i}{110} & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x553}} \quad \mathbb{T}_6^{(1,0;a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{264} & 0 & 0 & 0 & -\frac{7\sqrt{11}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 & \frac{7\sqrt{165}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{264} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}i}{264} & 0 & 0 & 0 & \frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}i}{88} & 0 & 0 & 0 & -\frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & \frac{\sqrt{33}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x554}} \quad \mathbb{T}_6^{(1,0;a)}(A_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{264} & 0 & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & \frac{\sqrt{330}i}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & \frac{\sqrt{385}i}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{264} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{231}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{385}i}{88} & 0 & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & \frac{\sqrt{231}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x555}} \quad \mathbb{T}_4^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x556}} \quad \mathbb{T}_6^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x557}} \quad \mathbb{T}_2^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & \frac{\sqrt{35}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x558}} \quad \mathbb{T}_4^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{220} & 0 & 0 & 0 & \frac{\sqrt{462}i}{308} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{1155}i}{1540} & 0 & 0 & 0 & -\frac{17\sqrt{385}i}{1540} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}i}{1540} & 0 & 0 & 0 & -\frac{9\sqrt{1155}i}{1540} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{308} & 0 & 0 & 0 & \frac{3\sqrt{330}i}{220} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{330}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{1155}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & \frac{\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{462}i}{308} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{17\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{1155}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x559}} \quad \mathbb{T}_6^{(1,0;a)}(B_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & 0 & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & \frac{\sqrt{14}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{24} & 0 & 0 & 0 & -\frac{i}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & 0 \\ 0 & -\frac{i}{24} & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{24} & 0 & 0 & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x560}} \quad \mathbb{T}_6^{(1,0;a)}(B_{1g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{264} & 0 & 0 & 0 & \frac{\sqrt{11}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{264} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}i}{264} & 0 & 0 & 0 & \frac{\sqrt{11}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{770}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{385}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & \frac{\sqrt{385}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{11}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x561}} \quad \mathbb{T}_2^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{84} & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

[illegible]

[illegible]

$$\boxed{\text{x564}} \quad \mathbb{T}_6^{(1,0;a)}(B_{2g}, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{11}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x565}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{24} & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & \frac{11\sqrt{14}}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{12} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}}{168} & 0 & -\frac{\sqrt{70}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{2}}{24} \\ \frac{5\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{11\sqrt{14}}{168} & 0 & -\frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{12} & 0 & -\frac{11\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x566}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{24} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & \frac{11\sqrt{14}i}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{7}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{12} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{14}i}{168} & 0 & -\frac{\sqrt{70}i}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{2}i}{24} \\ \frac{5\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}i}{168} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{14}i}{168} & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{12} & 0 & -\frac{11\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x567}} \quad \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{176} & 0 & \frac{5\sqrt{22}}{88} & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{330}}{880} & 0 & -\frac{\sqrt{66}}{88} & 0 & \frac{7\sqrt{110}}{880} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}}{880} & 0 & \frac{\sqrt{55}}{880} & 0 & -\frac{7\sqrt{33}}{176} & 0 & -\frac{\sqrt{165}}{880} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{880} & 0 & \frac{7\sqrt{33}}{176} & 0 & -\frac{\sqrt{55}}{880} & 0 & -\frac{3\sqrt{1155}}{880} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{110}}{880} & 0 & \frac{\sqrt{66}}{88} & 0 & \frac{13\sqrt{330}}{880} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & 0 & -\frac{5\sqrt{22}}{88} & 0 & -\frac{\sqrt{462}}{176} \\ \frac{\sqrt{462}}{176} & 0 & \frac{3\sqrt{1155}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{330}}{880} & 0 & \frac{\sqrt{165}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{22}}{88} & 0 & \frac{\sqrt{55}}{880} & 0 & -\frac{7\sqrt{110}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}}{88} & 0 & \frac{7\sqrt{33}}{176} & 0 & -\frac{\sqrt{330}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{330}}{176} & 0 & -\frac{7\sqrt{33}}{176} & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{110}}{880} & 0 & -\frac{\sqrt{55}}{880} & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{165}}{880} & 0 & \frac{13\sqrt{330}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{1155}}{880} & 0 & -\frac{\sqrt{462}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x568}} \quad \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{176} & 0 & \frac{5\sqrt{22}i}{88} & 0 & -\frac{\sqrt{330}i}{176} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{330}i}{880} & 0 & -\frac{\sqrt{66}i}{88} & 0 & -\frac{7\sqrt{110}i}{880} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}i}{880} & 0 & -\frac{\sqrt{55}i}{880} & 0 & -\frac{7\sqrt{33}i}{176} & 0 & \frac{\sqrt{165}i}{880} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{880} & 0 & -\frac{7\sqrt{33}i}{176} & 0 & -\frac{\sqrt{55}i}{880} & 0 & \frac{3\sqrt{1155}i}{880} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{110}i}{880} & 0 & -\frac{\sqrt{66}i}{88} & 0 & \frac{13\sqrt{330}i}{880} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{176} & 0 & \frac{5\sqrt{22}i}{88} & 0 & -\frac{\sqrt{462}i}{176} \\ \frac{\sqrt{462}i}{176} & 0 & -\frac{3\sqrt{1155}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{330}i}{880} & 0 & -\frac{\sqrt{165}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{22}i}{88} & 0 & \frac{\sqrt{55}i}{880} & 0 & \frac{7\sqrt{110}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{66}i}{88} & 0 & \frac{7\sqrt{33}i}{176} & 0 & \frac{\sqrt{330}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{330}i}{176} & 0 & \frac{7\sqrt{33}i}{176} & 0 & \frac{\sqrt{66}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{110}i}{880} & 0 & \frac{\sqrt{55}i}{880} & 0 & -\frac{5\sqrt{22}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{165}i}{880} & 0 & -\frac{13\sqrt{330}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{1155}i}{880} & 0 & \frac{\sqrt{462}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x569}} \quad \mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{176} & 0 & \frac{5\sqrt{154}}{616} & 0 & -\frac{\sqrt{2310}}{176} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}}{6160} & 0 & -\frac{\sqrt{462}}{616} & 0 & -\frac{7\sqrt{770}}{880} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{21\sqrt{165}}{880} & 0 & \frac{\sqrt{385}}{6160} & 0 & -\frac{\sqrt{231}}{176} & 0 & \frac{\sqrt{1155}}{880} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{880} & 0 & \frac{\sqrt{231}}{176} & 0 & -\frac{\sqrt{385}}{6160} & 0 & \frac{21\sqrt{165}}{880} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{770}}{880} & 0 & \frac{\sqrt{462}}{616} & 0 & \frac{13\sqrt{2310}}{6160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{176} & 0 & -\frac{5\sqrt{154}}{616} & 0 & -\frac{\sqrt{66}}{176} \\ \frac{\sqrt{66}}{176} & 0 & -\frac{21\sqrt{165}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{2310}}{6160} & 0 & -\frac{\sqrt{1155}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{154}}{616} & 0 & \frac{\sqrt{385}}{6160} & 0 & \frac{7\sqrt{770}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{462}}{616} & 0 & \frac{\sqrt{231}}{176} & 0 & \frac{\sqrt{2310}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{176} & 0 & -\frac{\sqrt{231}}{176} & 0 & \frac{\sqrt{462}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{770}}{880} & 0 & -\frac{\sqrt{385}}{6160} & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{880} & 0 & \frac{13\sqrt{2310}}{6160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{21\sqrt{165}}{880} & 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x570}} \quad \mathbb{T}_{4,2}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{176} & 0 & \frac{5\sqrt{154}i}{616} & 0 & \frac{\sqrt{2310}i}{176} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{2310}i}{6160} & 0 & -\frac{\sqrt{462}i}{616} & 0 & \frac{7\sqrt{770}i}{880} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{21\sqrt{165}i}{880} & 0 & -\frac{\sqrt{385}i}{6160} & 0 & -\frac{\sqrt{231}i}{176} & 0 & -\frac{\sqrt{1155}i}{880} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{880} & 0 & -\frac{\sqrt{231}i}{176} & 0 & -\frac{\sqrt{385}i}{6160} & 0 & -\frac{21\sqrt{165}i}{880} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{770}i}{880} & 0 & -\frac{\sqrt{462}i}{616} & 0 & \frac{13\sqrt{2310}i}{6160} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{176} & 0 & \frac{5\sqrt{154}i}{616} & 0 & -\frac{\sqrt{66}i}{176} \\ \frac{\sqrt{66}i}{176} & 0 & \frac{21\sqrt{165}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{2310}i}{6160} & 0 & \frac{\sqrt{1155}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{154}i}{616} & 0 & \frac{\sqrt{385}i}{6160} & 0 & -\frac{7\sqrt{770}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{462}i}{616} & 0 & \frac{\sqrt{231}i}{176} & 0 & -\frac{\sqrt{2310}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}i}{176} & 0 & \frac{\sqrt{231}i}{176} & 0 & \frac{\sqrt{462}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{770}i}{880} & 0 & \frac{\sqrt{385}i}{6160} & 0 & -\frac{5\sqrt{154}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}i}{880} & 0 & -\frac{13\sqrt{2310}i}{6160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{21\sqrt{165}i}{880} & 0 & \frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x571}} \quad \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{1056} & 0 & \frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{462}}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1056} & 0 & -\frac{5\sqrt{462}}{1056} & 0 & \frac{3\sqrt{770}}{352} & 0 & \frac{\sqrt{330}}{96} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{385}}{176} & 0 & \frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{1155}}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{385}}{176} & 0 & \frac{\sqrt{165}}{176} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{96} & 0 & -\frac{3\sqrt{770}}{352} & 0 & \frac{5\sqrt{462}}{1056} & 0 & \frac{\sqrt{2310}}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{96} & 0 & \frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{66}}{1056} \\ \frac{\sqrt{66}}{1056} & 0 & -\frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{330}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{1056} & 0 & \frac{\sqrt{1155}}{176} & 0 & \frac{\sqrt{462}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{154}}{352} & 0 & \frac{\sqrt{385}}{176} & 0 & -\frac{3\sqrt{770}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{462}}{1056} & 0 & -\frac{5\sqrt{231}}{528} & 0 & \frac{\sqrt{2310}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{352} & 0 & \frac{5\sqrt{231}}{528} & 0 & \frac{5\sqrt{462}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{770}}{352} & 0 & -\frac{\sqrt{385}}{176} & 0 & -\frac{\sqrt{154}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{462}}{96} & 0 & -\frac{\sqrt{1155}}{176} & 0 & \frac{\sqrt{2310}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{96} & 0 & \frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{66}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x572}} \quad \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{1056} & 0 & \frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{2310}i}{352} & 0 & -\frac{\sqrt{462}i}{96} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1056} & 0 & -\frac{5\sqrt{462}i}{1056} & 0 & -\frac{3\sqrt{770}i}{352} & 0 & \frac{\sqrt{330}i}{96} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{385}i}{176} & 0 & \frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{1155}i}{176} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{176} & 0 & \frac{5\sqrt{231}i}{528} & 0 & -\frac{\sqrt{385}i}{176} & 0 & -\frac{\sqrt{165}i}{176} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{96} & 0 & -\frac{3\sqrt{770}i}{352} & 0 & -\frac{5\sqrt{462}i}{1056} & 0 & \frac{\sqrt{2310}i}{1056} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{96} & 0 & \frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{154}i}{352} & 0 & -\frac{\sqrt{66}i}{1056} \\ \frac{\sqrt{66}i}{1056} & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{330}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}i}{1056} & 0 & -\frac{\sqrt{1155}i}{176} & 0 & \frac{\sqrt{462}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{385}i}{176} & 0 & \frac{3\sqrt{770}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{462}i}{1056} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}i}{352} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{770}i}{352} & 0 & \frac{\sqrt{385}i}{176} & 0 & -\frac{\sqrt{154}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{462}i}{96} & 0 & -\frac{\sqrt{1155}i}{176} & 0 & -\frac{\sqrt{2310}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}i}{96} & 0 & \frac{\sqrt{165}i}{176} & 0 & \frac{\sqrt{66}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x573}} \quad \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & \frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & \frac{\sqrt{7}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{5}}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & -\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{10}}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & -\frac{1}{64} \\ \frac{1}{64} & 0 & \frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{5}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{70}}{64} & 0 & -\frac{\sqrt{7}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{64} & 0 & \frac{5\sqrt{14}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{64} & 0 & -\frac{\sqrt{10}}{64} & 0 & -\frac{1}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
\boxed{x574} \quad \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 2) = & \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{7}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & -\frac{\sqrt{5}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{64} & 0 & -\frac{\sqrt{210}i}{64} & 0 & \frac{5\sqrt{14}i}{64} & 0 & -\frac{\sqrt{70}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{64} & 0 & \frac{5\sqrt{14}i}{64} & 0 & -\frac{\sqrt{210}i}{64} & 0 & \frac{\sqrt{10}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & -\frac{i}{64} \\
\frac{i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{70}i}{64} & 0 & -\frac{\sqrt{7}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{64} & 0 & \frac{\sqrt{70}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 & \frac{i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix} \\
\\
\boxed{x575} \quad \mathbb{T}_{6,1}^{(1,0;a)}(E_g, 3) = & \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{2112} & 0 & \frac{\sqrt{1155}}{2112} & 0 & -\frac{9\sqrt{77}}{704} & 0 & \frac{\sqrt{385}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{2112} & 0 & -\frac{5\sqrt{385}}{2112} & 0 & \frac{9\sqrt{231}}{704} & 0 & -\frac{5\sqrt{11}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{2112} & 0 & \frac{5\sqrt{770}}{2112} & 0 & -\frac{9\sqrt{154}}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{704} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & -\frac{5\sqrt{462}}{2112} & 0 & \frac{9\sqrt{22}}{704} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{64} & 0 & -\frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{385}}{2112} & 0 & \frac{5\sqrt{77}}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{64} & 0 & \frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & -\frac{\sqrt{55}}{2112} \\
\frac{\sqrt{55}}{2112} & 0 & -\frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{11}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{77}}{2112} & 0 & \frac{9\sqrt{154}}{704} & 0 & -\frac{\sqrt{385}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{1155}}{2112} & 0 & \frac{5\sqrt{462}}{2112} & 0 & -\frac{9\sqrt{231}}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{385}}{2112} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & \frac{9\sqrt{77}}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{9\sqrt{77}}{704} & 0 & \frac{5\sqrt{770}}{2112} & 0 & \frac{5\sqrt{385}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{231}}{704} & 0 & -\frac{5\sqrt{462}}{2112} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{385}}{64} & 0 & -\frac{9\sqrt{154}}{704} & 0 & \frac{5\sqrt{77}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{11}}{64} & 0 & \frac{9\sqrt{22}}{704} & 0 & -\frac{\sqrt{55}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\end{aligned}$$

$$\boxed{\text{x576}} \quad \mathbb{T}_{6,2}^{(1,0;a)}(E_g, 3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{2112} & 0 & \frac{\sqrt{1155}i}{2112} & 0 & \frac{9\sqrt{77}i}{704} & 0 & \frac{\sqrt{385}i}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{2112} & 0 & -\frac{5\sqrt{385}i}{2112} & 0 & -\frac{9\sqrt{231}i}{704} & 0 & -\frac{5\sqrt{11}i}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{462}i}{2112} & 0 & \frac{5\sqrt{770}i}{2112} & 0 & \frac{9\sqrt{154}i}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}i}{704} & 0 & \frac{5\sqrt{770}i}{2112} & 0 & -\frac{5\sqrt{462}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{64} & 0 & -\frac{9\sqrt{231}i}{704} & 0 & -\frac{5\sqrt{385}i}{2112} & 0 & \frac{5\sqrt{77}i}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{64} & 0 & \frac{9\sqrt{77}i}{704} & 0 & \frac{\sqrt{1155}i}{2112} & 0 & -\frac{\sqrt{55}i}{2112} \\ \frac{\sqrt{55}i}{2112} & 0 & \frac{9\sqrt{22}i}{704} & 0 & \frac{5\sqrt{11}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{77}i}{2112} & 0 & -\frac{9\sqrt{154}i}{704} & 0 & -\frac{\sqrt{385}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{1155}i}{2112} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & \frac{9\sqrt{231}i}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{385}i}{2112} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{9\sqrt{77}i}{704} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{385}i}{64} & 0 & -\frac{9\sqrt{154}i}{704} & 0 & -\frac{5\sqrt{77}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{11}i}{64} & 0 & \frac{9\sqrt{22}i}{704} & 0 & \frac{\sqrt{55}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: 2a

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_{1g}) = [1]$$

- Bond cluster

** Wyckoff: 4b@4c

$$\boxed{\text{y2}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_2^{(b)}(B_{1g}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y4}} \quad \mathbb{T}_{1,1}^{(b)}(E_u) = [0, -i]$$

$$\boxed{\text{y5}} \quad \mathbb{T}_{1,2}^{(b)}(E_u) = [i, 0]$$

— Site and Bond —

Table 4: Orbital of each site

#	site	orbital
1	U	$ \frac{3}{2}, \frac{3}{2}; d\rangle, \frac{3}{2}, \frac{1}{2}; d\rangle, \frac{3}{2}, -\frac{1}{2}; d\rangle, \frac{3}{2}, -\frac{3}{2}; d\rangle, \frac{5}{2}, \frac{5}{2}; d\rangle, \frac{5}{2}, \frac{3}{2}; d\rangle, \frac{5}{2}, \frac{1}{2}; d\rangle, \frac{5}{2}, -\frac{1}{2}; d\rangle, \frac{5}{2}, -\frac{3}{2}; d\rangle, \frac{5}{2}, -\frac{5}{2}; d\rangle, \frac{5}{2}, \frac{5}{2}; f\rangle, \frac{5}{2}, \frac{3}{2}; f\rangle, \frac{5}{2}, \frac{1}{2}; f\rangle, \frac{5}{2}, -\frac{1}{2}; f\rangle, \frac{5}{2}, -\frac{3}{2}; f\rangle, \frac{5}{2}, -\frac{5}{2}; f\rangle$ $ \frac{7}{2}, \frac{7}{2}; f\rangle, \frac{7}{2}, \frac{5}{2}; f\rangle, \frac{7}{2}, \frac{3}{2}; f\rangle, \frac{7}{2}, \frac{1}{2}; f\rangle, \frac{7}{2}, -\frac{1}{2}; f\rangle, \frac{7}{2}, -\frac{3}{2}; f\rangle, \frac{7}{2}, -\frac{5}{2}; f\rangle, \frac{7}{2}, -\frac{7}{2}; f\rangle$

Table 5: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	U	U	[1]	[d,f]	[d,f]

— Site in Unit Cell —

Sites in (conventional) cell (no plus set), SL = sublattice

Table 6: 'U' (#1) site cluster (2a), 4/mmm

SL	position (s)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 7: 1-th 'U'-'U' [1] (#1) bond cluster (4b@4c), ND, $|v|=4.126$ (cartesian)

SL	vector (v)	center (c)	mapping	head	tail	R (primitive)
1	[0.00000, 1.00000, 0.00000]	[0.00000, 0.50000, 0.00000]	[1,-2,5,-6,-9,10,-13,14]	(1,1)	(1,1)	[-1,0,-1]
2	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.00000]	[3,-4,-7,8,-11,12,15,-16]	(1,1)	(1,1)	[0,1,1]